

# How a Golf Swing Carries Energy

A Book-Length Visual Journey Through Motion, Forces, Timing, Models, and Uncertainty

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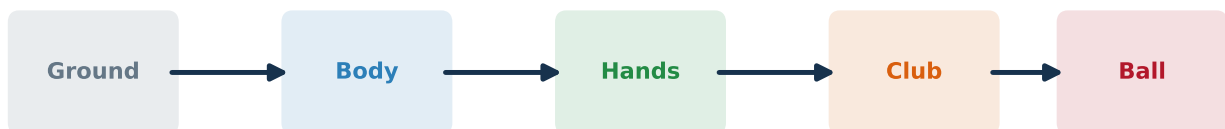
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This is a patient exploration of a moving system. It follows forces, motion and energy accounting through ground contact, the body and hands, a flexible club, and the ball. An ideal stationary ground contact can supply impulse without doing work; it is not a universal energy source. Ordinary scenes make the mechanics visible; equations and experiments keep the scenes honest. No model result becomes swing instruction merely because it is easy to picture. The purpose is to make the scientific questions understandable enough that a reader can examine, challenge, and improve them.

## Follow the Interactions and the Energy Ledger



Ground contact can redirect momentum without supplying work.  
Other interfaces may transfer, store or dissipate mechanical energy.

Figure 1: A Five-Stage Map Connects Ground Contact, Body, Hands, Club, and Ball Without Treating Every Interaction as an Energy Supply.

# How to Read This Book

The book is organized as a long walk rather than a stack of definitions. Part I builds the language of motion. Part II follows forces through a double pendulum. Part III opens the two-hand and flexible-shaft mechanisms. Part IV shows how models grow without pretending that the largest model is automatically the best. Part V asks what is identifiable, robust, measured, and falsifiable. The final part turns the argument into a reviewer path and a neutral practical synthesis.

Every chapter has four recurring landmarks:

- **A Concrete Picture** begins with an ordinary situation that can be held in the mind while the mechanics develops.
- **How the Mechanism Works** supplies the sustained explanation, equations, figures, and worked reasoning.
- **Where the Picture Breaks** states exactly where the analogy or reduced model stops earning conclusions.
- **Go Deeper** links to the corresponding technical chapter, executable evidence, and real literature.

Claim labels recur throughout. **Model Result** means an executable mathematical system produced a result under declared assumptions. **Human Evidence** means people were measured. **Hypothesis** means a proposed explanation awaits a decisive test. **Practical Interpretation** translates a question into ordinary language without declaring a universal technique.

Readers who want to change the models can use the [interactive proximal-distal workbench](#). Readers who want every equation and numerical contract can open the [governed scientific monograph](#) or [download its exact PDF](#). Its models, claims, and evidence are pinned to [UpstreamDrift commit 85cce4d3](#).

This book is **not a coaching instruction**. It explains testable mechanics, evidence boundaries, and open questions. The preload discussion also links to the machine-readable [transmission robustness study](#) used by the technical analysis.

# Chapter 1

## Follow the Energy

### 1.1 A Concrete Picture

Imagine standing above a busy railway yard at dusk. A locomotive enters with fuel in its tanks and motion in its wheels. It couples to a line of cars. Some cars begin moving, one spring-loaded coupling compresses, brakes warm on another car, and the last car eventually rolls faster than the first. If we watched only the peak speed of each car, we would miss nearly everything that made the event interesting. We would not know which engine supplied work, which coupling stored energy for a moment, which brake removed it, or which force merely kept a car on the curved track.

A golf swing is smaller and faster, but the accounting problem is similar. The pelvis, trunk, arms, hands, shaft, and clubhead do not behave as a row of rigid blocks passing a package along a conveyor. They are linked bodies whose geometry changes while forces act. A connection can transmit power in either direction. A segment can slow while still doing positive work on its neighbor. A large force can redirect motion while transferring almost no energy. A flexible shaft can accept energy, keep it briefly, and return some later. Gravity can help one part of the path and oppose another. Damping quietly turns organized mechanical energy into heat.

The first discipline of this book is therefore simple to state and surprisingly hard to maintain: follow the complete energy ledger, not just the visible order of speed peaks.

The familiar phrase *proximal to distal* means, roughly, from nearer the center of the body toward farther-away segments. In golf, researchers often describe a sequence in which pelvis speed peaks, then trunk speed, then arm or hand speed, and finally club speed. Ordered peaks have been reported, but the reported order depends on which segments and angular-velocity quantities are measured ([Tinmark et al. 2010](#); [Marsan et al. 2019](#)). It is an observation about motion. It is not yet an explanation of energy flow. The explanation requires forces, moments, velocities, and the boundaries across which power is computed.

### 1.2 How the Mechanism Works

Begin with a modest definition. Mechanical energy is a bookkeeping quantity for a chosen system. A moving rigid body has translational and rotational kinetic energy. A raised body has gravitational potential energy. A bent shaft or stretched tendon may have elastic potential energy. When external

## Follow the Interactions and the Energy Ledger



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forces do work, the system's mechanical energy can change. Internal interactions can redistribute mechanical energy, convert elastic storage to kinetic energy, or dissipate mechanical energy into heat. Internal muscles can convert chemical energy into mechanical work. Conservation of total energy does not imply that the mechanical energy of a golfer-plus-club system can increase only through external work.

That last sentence is the heart of the matter. Suppose we draw one boundary around the golfer and club together. The forces between a hand and the grip are internal to that combined system. The forces are equal and opposite. At an ideal shared no-slip contact their material-point velocities agree, so the two contact powers cancel in the combined ledger. If slip, deformation or an intervening compliant element is retained, its power, storage and dissipation require explicit accounting. Merely reporting the same physical contact in two ways cannot create an unexplained power gap. Now draw a smaller boundary around the club alone. The same hand forces become external inputs to the club. They can change its kinetic energy, bend its shaft, or redirect its motion. Neither boundary is more correct. They answer different questions.

For a force  $\mathbf{F}$  acting at a point moving with velocity  $\mathbf{v}$ , the instantaneous power delivered through that point is

$$P_F = \mathbf{F} \cdot \mathbf{v}.$$

For an applied free moment  $\tau$  acting on a rigid body with angular velocity  $\omega$  in the chosen observer frame, moment power is

$$P_\tau = \tau \cdot \omega.$$

For several forces on one rigid body, reduce their resultant force and moment to one declared point. Pair the force with the rigid velocity field evaluated at that point and the moment with the body's angular velocity. The total power is the sum of the two terms; a moment arm must not be counted again. A deforming club instead needs its contact/deformation velocities. At a rotational joint, paired moment power uses relative angular velocity between the bodies. Net joint power is not identified individual-muscle power.

These dot products are more informative than force or torque magnitude alone. When force points generally with velocity, power is positive. When it points against velocity, power is negative. When it points perpendicular to velocity, it may bend the path strongly while doing no instantaneous work. A centripetal force in uniform circular motion is the clean textbook example: it is essential to the curved path, yet its dot product with tangential velocity is zero.

This gives us several distinct verbs for the swing. An actuator can **supply** energy by positive work. An interaction can **transport** energy across a chosen boundary. A spring can **store** and later **return** energy. Damping can **dissipate** it. A perpendicular force can **redirect** motion. Constraint reactions can **redirect** motion or exchange power between subsystems. A prescribed moving constraint can also supply external power; its actuator must be included when the accounting boundary includes the mechanism that moves it. Saying only that energy “flows down the chain” compresses all those verbs into one and can hide the mechanism we most need to understand.

Consider a relay race, another useful picture. The order in which runners reach their top speed resembles a kinematic sequence. The baton resembles a physical object crossing a boundary. But a golf swing has no single baton of energy. Energy is a scalar assigned to states, and power is the rate at which work crosses a chosen boundary. What passes through the wrist is not a glowing substance. It is a calculable mechanical interaction whose sign and magnitude depend on the force, moment, motion, reference point, and frame.

The technical study uses an energy ledger precisely because several mechanisms can overlap. In a double-pendulum model, shoulder torque can add energy while the arm-club interaction redistributes it. In the two-hand model, a pair of opposed hand forces can create a club moment even when their net force is small. In the flexible-shaft model, interface power can move into elastic storage and return later. In an ideal stationary no-slip ground contact, reactions constrain motion and provide impulse while doing zero boundary work. Deforming, slipping or moving contacts require a different power calculation. The ledger asks each contribution to identify itself rather than accepting one story for all of them.

### 1.2.1 A Small Numerical Ledger

Suppose a simplified club receives 500 watts of positive hand power for 0.10 seconds. If the power were constant, the hands would deliver 50 joules of work. During the same interval, imagine gravity contributes 3 joules, damping removes 5 joules, and the shaft finishes with 8 joules more elastic energy than it began. Treat these as synthetic values and include all retained club kinetic modes in  $K$ . Count gravity as external work here, rather than also including its potential-energy change. The remaining increase in total kinetic energy is

$$\Delta K = 50 + 3 - 5 - 8 = 40 \text{ J.}$$

Nothing here says the shaft “lost” eight joules. At the chosen final time, those joules are stored rather than expressed as total kinetic energy. If the shaft subsequently loses eight joules of storage, two are dissipated and there is no other work or storage change, total kinetic energy rises by six joules. This does not say that all six reach the clubhead. Timing and interface powers decide which part holds the energy near impact.

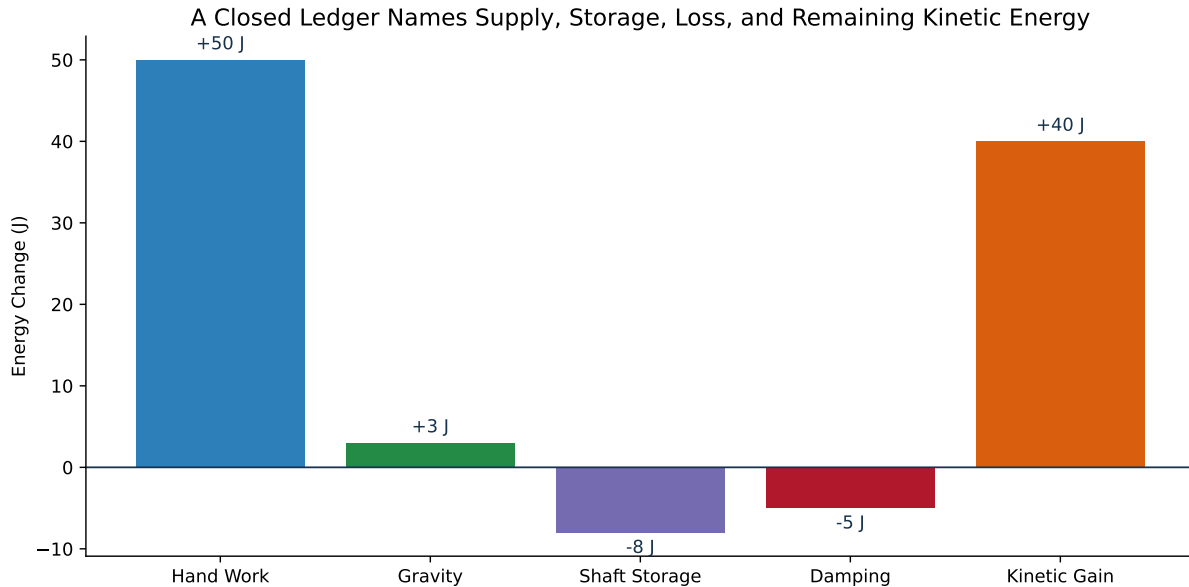


Figure 1.2: A Worked Energy Ledger Separates Hand Work, Gravity, Shaft Storage, Damping Loss, and the Remaining Kinetic-Energy Gain.

The same care is needed when a proximal segment slows. A slowing arm may be losing kinetic energy, but that loss is not automatically equal to the club’s gain. The shoulder may still be doing work. Gravity and damping may contribute. Interaction power may move energy between arm and club. Their kinetic energies also depend on configuration because the inertia matrix of a linked system is not a set of independent constants. The only safe claim comes from a closed balance in a declared model or from measurements rich enough to estimate one.

**Model Result:** The [archived forward two-hand study](#) reports decreasing, nonzero work-energy residuals under timestep refinement. Those numerical checks support bounded internal consistency, not exact trajectories or biological validation. Chapter 29 distinguishes these residuals from tighter algebraic wrench-power identities.

**Human Evidence:** Tinmark and colleagues reported ordered pelvis, torso and hand peak angular-speed magnitudes in their 45-golfer study ([Tinmark et al. 2010](#)). Marsan and colleagues found strong dependence of the identified sequence on angular-velocity definition ([Marsan et al. 2019](#)). These kinematic results do not measure joint work or identify a causal energy-transfer route.

**Hypothesis:** A useful late-downswing strategy may depend on arranging the state so passive and interaction terms assist the club while direct distal actuation is delayed or even briefly opposing. The rest of the book makes this proposal precise enough to challenge.

**Practical Interpretation:** Instead of asking which body part “creates all the speed,” ask what each boundary is doing during each phase: supplying, carrying, storing, returning, dissipating, or redirecting

mechanical energy.

### 1.2.2 A Swing-Phase Walkthrough

At address, the club already has a state and a preparation history. Its gravitational potential energy is defined relative to a chosen reference height. Raising it requires work against gravity; gravity has not supplied that positive energy change. Holding a pose may require muscle activity even when the net mechanical work is zero. Every later claim must name its ledger. During the backswing, muscles do work to raise and rotate body segments and the club. Some of that work becomes gravitational potential energy, some becomes kinetic energy, some is stored in deformed tissue or the shaft, and some is dissipated. Saying that the backswing “loads” the swing is therefore reasonable only after “load” is divided into measurable forms.

Around transition, club speed may be low while forces and internal activity are not. The direction of motion is changing, the golfer is reorganizing the state, and compliant elements may carry load. This is a useful antidote to the idea that low visible speed means nothing mechanical is happening. Early in the downswing, proximal work can increase the energy of the whole system while the club is still largely being carried. The club’s own speed need not immediately mirror the rate at which energy enters elsewhere.

In the middle and late downswing, the ledger becomes especially interesting. For the whole club, hand power and gravity are external entries, while shaft return is internal conversion from stored energy. Coupling to the body shapes the hand power and trajectory; it is not a third external power to add to the same ledger. At the same time, one generalized torque can have negative power and still participate in a state evolution that produces greater later club speed. Near impact, energy in translation, rotation, and shaft deformation all affect the collision. After impact, substantial energy remains in the body and club or is dissipated; the ball never receives the entire pre-impact system energy.

### 1.2.3 A Numerical Thought Experiment

Suppose two simulated swings both deliver a clubhead at 45 metres per second. For a deliberately reduced example, represent each head as a point mass moving at that speed: 0.20 kg in Swing A and 0.22 kg in Swing B. Their kinetic energies are  $\frac{1}{2}mv^2 = 202.5$  J and 222.75 J. These are not whole-club energy estimates. A rigid head requires center-of-mass speed and a separate rotational term; a flexible club also carries distributed motion. Impact effective mass is a directional contact-response property and must not be substituted for physical mass in this energy calculation. Equal speed need not mean equal energy. Now suppose Swing A reaches that speed with 400 watts crossing an interface for half a second, while Swing B has a 1,000-watt peak lasting only a few hundredths of a second. The 200 J of interface work in Swing A is only one ledger entry; initial energy, gravity, storage changes and losses must close its complete balance. Peak power alone still does not say which boundary supplied more work. Integrating the complete histories does.

The numbers are illustrative, not measurements of a golfer. Their job is to force the vocabulary apart. Speed is a state variable, energy is a ledger quantity, power is a rate, and work is an accumulation. When a result is surprising, returning to those four definitions usually reveals whether the surprise is physical or merely verbal.

### 1.2.4 Questions for a Skeptical Reader

Before accepting any energy-transfer plot, ask whether all external powers were included; whether the sign convention is stated; whether the shaft was modeled as rigid or flexible; whether joint power was assigned consistently; and whether numerical energy residuals are small relative to the reported effect. Then ask the separate experimental questions: Were inertial parameters measured or assumed? Were force and motion synchronized? Does uncertainty in those inputs change the conclusion? A closed ledger earns the right to continue the argument. It does not finish it.

## 1.3 Where the Picture Breaks

Energy is not a liquid, electricity in a wire, traffic on a road, or a baton. Those images help us remember conservation and boundaries, but they can mislead us into imagining a material substance with a unique route. In a multibody system, the decomposition itself depends on coordinates and system boundaries. Some internal transfers can be described in more than one mathematically valid way. Equivalent coordinate descriptions must recover the same physical motion and consistent whole-system balance in the same observer frame. Moving to another observer can change kinetic energy and power values without violating mechanics. Individual coordinate-term labels are not invariant physical transfers.

The railway picture also makes couplings look sequential. In a swing, forces act throughout the system at the same time. The distal club can influence the proximal arms immediately through action-reaction. A shaft mode can affect grip loads before its stored energy is returned. The correct picture is a coupled network with time-varying geometry, not a one-way bucket brigade.

Finally, a closed simulated ledger is necessary but not sufficient. A wrong model can conserve its own energy perfectly. Numerical closure shows that the implemented equations agree with their accounting. Experimental agreement is a separate question requiring measured motion, wrenches, shaft deformation, ground reactions, and uncertainty.

**Go Deeper:** Read the scientific monograph's [mechanics chapter](#), the detailed [interaction-force decomposition](#), and Robertson and Winter's segment-energy framework ([Robertson and Winter 1980](#)). The [open evidence directory](#) contains the machine-readable ledgers used by the technical study.

## Chapter 2

# Choose the System Before Counting

### 2.1 A Concrete Picture

Picture a household checking its finances. A payment to a spouse leaves one person's wallet but stays inside the household. The money has crossed one boundary, not the other. A mechanical energy account also needs an inventory of what is inside. The club, the hands and club, and the entire golfer and club are different systems, even when we observe the same swing.

There is one crucial limit to the analogy: a force is not an energy payment. A golfer can push against a stationary surface and change whole-body momentum while the contact does no mechanical work. Muscles can convert chemical energy, and tissues can release stored energy, while the ground supplies the reaction that makes the motion possible. The reaction matters without being the energy source. A deforming surface or moving support requires a different account.

Before saying that the arms, wrists, shaft or ground supplied energy, ask which bodies and interface elements are included, which observer measures velocity, and which forms of energy are being counted. Those choices make a claim testable. They also explain why a force, a negative moment and a positive energy transfer are three different observations.

### 2.2 How the Mechanism Works

#### 2.2.1 Draw the Boundary Through Named Bodies

For the **club alone**, include the head, shaft and grip material. The hands, air and ball can apply external loads. Earth's gravity is also an external force. A flexible shaft stores strain energy and has moving mass; its energy is not just a massless spring's storage. Grip or contact compliance belongs where the model explicitly places it, rather than disappearing between two subsystem drawings.

Expanding the inventory changes the ports at which loads and motion must be paired. Here, a *port* means an identified interface across which mechanical power is evaluated, not an additional energy source.

## Choose the Boundary Before Naming the Transfer

| Interaction / Conversion | Club              | Golfer + Club  | Golfer + Club + Earth |
|--------------------------|-------------------|----------------|-----------------------|
| Hand Contact             | External          | Internal       | Internal              |
| Foot-Ground              | Outside Inventory | External       | Internal              |
| Gravity from Earth       | External Force    | External Force | Internal Interaction  |
| Muscle Conversion        | Outside Inventory | Internal       | Internal              |

Effective Gravitational Potential Can Represent an External Field

Interaction Classification Does Not Determine Work

Figure 2.1: The Chosen Body Inventory Determines Which Interactions Cross the Boundary.

| Included Bodies                           | Interfaces That Still Cross the Boundary                                                |
|-------------------------------------------|-----------------------------------------------------------------------------------------|
| Head, shaft and grip                      | Hand contact, air and ball contact; gravity remains an external force                   |
| Both hands and the club                   | Wrist interfaces with the forearms; grip contact is internal                            |
| Both upper arms, forearms, hands and club | Shoulder interfaces with the torso; elbows and wrists are internal                      |
| The golfer's body and club                | Foot-ground, air and ball interfaces; gravity remains external unless Earth is included |

These are model inventories, not sharp anatomical compartments. A muscle or tendon spanning an interface needs an explicit allocation. For the golfer-club mechanical account, muscle actuation converts energy within the included biological system. It can increase mechanical energy without external work. If chemical and thermal energy are also tracked, conversion and dissipation move energy between forms instead of creating or destroying total energy.

Adding Earth internalizes the golfer-ground interaction and the mutual gravitational interaction. It does not automatically include the air, ball, Sun or every environmental interaction, and does not by itself make the enlarged system isolated. Keep the actual inventory, not merely the phrase “whole system,” beside the balance.

### 2.2.2 Ground Reaction Is Not Ground Work

For an applied contact force, the instantaneous power delivered to the chosen body is

$$P_{\text{contact}} = \mathbf{F} \cdot \mathbf{v}_{\text{contact}}.$$

The velocity is that of the body's material point where the force acts, in the declared observer frame. A contact moment adds its power-conjugate angular term. For distributed contact, sum or integrate the local load-velocity pairings. A stationary, ideal sticking contact has zero local velocity and zero force power; a stationary support that also prevents rotation has zero contact-moment power. These are kinematic assumptions, not universal properties of shoes, turf or force plates ([Widnall and Peraire 2008](#)).

For a fixed-mass system in an inertial frame, Newton's equation also gives

$$\frac{dK_G}{dt} = \mathbf{F}_{\text{ext}} \cdot \mathbf{v}_G, \quad K_G = \frac{1}{2}M\|\mathbf{v}_G\|^2,$$

where  $G$  is the system center of mass and  $\mathbf{F}_{\text{ext}}$  is the resultant external force. This is a balance for **COM translational kinetic energy**, sometimes called a pseudowork relation. The actual application points need not move with  $G$ . Consequently, its right-hand side is not generally the external mechanical power of an extended or deforming system.

Consider a deliberately simple manufactured example, with gravity omitted. Two masses of 2 kg are connected by an internal actuator. The lower mass stays against a stationary unilateral support. An actuator force of 40 N pushes the upper mass through 0.5 m from rest and pushes the lower mass into the support. The support supplies 40 N in the opposite direction but its point moves zero distance, so its work is zero. The actuator converts 20 J into motion. The upper mass reaches  $\sqrt{20}$  m/s, giving 20 J total kinetic energy, of which 10 J is COM kinetic energy and 10 J is motion relative to the COM. An ideal internal energy reservoir falls by 20 J.

The COM moves 0.25 m: support force times COM displacement is 10 J, consistent with the COM equation, but it is not 10 J supplied by the support. Removing the support produces a different problem: the two initially resting free masses move apart while their combined momentum remains zero. This is why internal actuation, external reaction and energy transfer must be kept together without identifying them with one another. The example illustrates accounting; it is not a quantitative model of a golfer's push-off.

### 2.2.3 Internal Contact Power Cancels Only Under the Right Conditions

Let body 1 receive contact force  $\mathbf{F}$  and contact moment  $\mathbf{M}$ , and body 2 receive their opposites at the same contact location. Express all vectors in common axes. Their combined mechanical power is

$$P_1 + P_2 = \mathbf{F} \cdot (\mathbf{v}_1 - \mathbf{v}_2) + \mathbf{M} \cdot (\omega_1 - \omega_2).$$

Equal and opposite loads cancel in a force or moment sum; power also needs the local motions. Coincident spatial contact positions need not have equal material-point velocities during slip. Compatible

motion in an ideal passive constraint makes the load-relative-motion pairing zero. Equality of every velocity component is sufficient but not necessary: a force perpendicular to relative motion also contributes zero power.

For example, with force on body 1 equal to  $(-4, 0, 10)$  N and contact velocities  $(5, 1, 0)$  and  $(2, 1, 0)$  m/s, the two force powers are  $-20$  W and  $+8$  W. Their sum is  $-12$  W. This is consistent with sliding friction opposing the relative tangential motion; the normal component does no work here. The mechanical loss needs a thermal or other nonrecoverable-energy account. The power calculation alone does not identify how that loss is distributed.

A compliant interface can also store energy. For an explicitly massless, passive interface with storage  $U_c$  and nonnegative dissipation  $D_c$ , its powers on the adjoining bodies satisfy  $P_1 + P_2 = -\dot{U}_c - D_c$ . Include interface kinetic energy if it has inertia, and an actuator conversion term if it is powered. A joint actuator's torque pairs with **relative** angular velocity when computing its combined mechanical contribution; a torque's power on one segment is a different quantity.

Two skaters pushing apart therefore do not establish that their mutual contact must supply positive external work to each skater. Internal deformation and chemical conversion can raise their kinetic energies even with a stationary common contact. For the golfer, draw separate arm-side, club-side and interface accounts before assigning an energy source. Constraint reaction, transmission, storage, actuation and loss can all participate in the same motion.

#### 2.2.4 Count Gravity Once

Gravity can remain an external force for a club-only body inventory while its prescribed, time-independent field is represented by an effective potential. For uniform downward gravity, let  $U_g = \sum_i m_i g z_i$ ; then

$$P_g = -\dot{U}_g.$$

Earth need not have simulated degrees of freedom inside the body inventory for this representation to be useful. Distinguish that effective external-field potential from the mutual interaction energy of a dynamical Earth-club model. With  $U_e$  denoting included elastic storage, two equivalent mechanical accounts are

$$\frac{d(K + U_e)}{dt} = P_g + P_{\text{other}} + P_{\text{act}} - D,$$

$$\frac{d(K + U_e + U_g)}{dt} = P_{\text{other}} + P_{\text{act}} - D.$$

Here  $P_{\text{other}}$  is external mechanical power excluding gravity,  $P_{\text{act}}$  is included actuation's mechanical conversion, and  $D$  is the modeled loss from recoverable mechanical energy. Do not put  $U_g$  on the left while also counting  $P_g$  on the right. A falling body makes the error obvious: its kinetic increase balances its potential decrease without a second gravitational deposit. A time-dependent field needs the additional explicit time-dependence term; the displayed equivalence assumes a static potential.

For an unpowered club-only inventory,  $P_{\text{act}} = 0$ : the golfer's work already enters through the hand-contact port. Adding wrist-actuator work again would count the same input twice under that boundary.

The zero of potential may be shifted by a fixed constant without changing these balances. That freedom does not change the body inventory or permit dropping a changing energy term.

Earth's recoil requires an equally explicit frame. If a constant mass  $M_E$  has initial COM velocity  $\mathbf{V}$  and receives total impulse  $\mathbf{J}$ , its translational kinetic change in that inertial frame is

$$\Delta K_E = \mathbf{V} \cdot \mathbf{J} + \frac{\|\mathbf{J}\|^2}{2M_E}.$$

For bounded impulse and an initially resting receiver, large mass makes the quadratic recoil small. It does not eliminate the first term in another observer's account. Rotation, local ground deformation and slip are separate quantities; negligible COM recoil does not prove zero foot-ground work. For a short golf-swing calculation, state the laboratory-frame approximation and support assumptions rather than relying on mass alone.

### 2.2.5 Change a Reporting Point Without Changing the Observer

For a rigid body, resultant force  $\mathbf{R}$ , moment about  $A$  and the rigid velocity field give total wrench power

$$P = \mathbf{R} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \boldsymbol{\omega}.$$

Moving the reporting point to  $B$  requires both transformations:

$$\mathbf{M}_B = \mathbf{M}_A - (\mathbf{r}_B - \mathbf{r}_A) \times \mathbf{R}, \quad \mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times (\mathbf{r}_B - \mathbf{r}_A).$$

The two extra scalar-triple-product terms cancel in power. Consistently rotating the component axes also preserves the scalar. These operations represent the **same physical motion for the same observer**;  $\mathbf{v}_B$  is the rigid velocity field evaluated at  $B$ , not an unrelated anatomical point's velocity (Lynch and Park, n.d.). Only a pure couple, with zero resultant, has a moment independent of reporting point.

Changing to a nonrotating inertial observer moving at constant velocity  $\mathbf{U}$  instead changes every material-point velocity by  $-\mathbf{U}$ . For the same Newtonian applied loads,

$$P' = P - \mathbf{R} \cdot \mathbf{U}.$$

Work and kinetic energy change together; the work-energy equation retains its form, but those quantities need not have the same numerical values (Diaz et al. 2009). A zero-resultant couple has no boost term. Neither operation changes the physical swing. A reader comparing two reports must first align the body inventory, observer, moment point, signs and interval.

### 2.2.6 What a Net Club Wrench Can Hide

For a rigid club, the same complete applied-wrench history, inertial parameters, other loads and initial state give the same trajectory under the same well-posed dynamics. That does not identify how the two hands or the muscles produced those loads. Opposing hand forces can partly cancel in the resultant while generating a couple; an opposing pair along the line joining the contacts can also have zero resultant moment. The latter can change grip loading without appearing in the net wrench.

Flexibility adds another limitation. At positions  $(-1, 0, 0)$  and  $(1, 0, 0)$  m, apply outward axial forces  $(-1, 0, 0)$  and  $(1, 0, 0)$  N. The resultant force and moment are both zero. If the material points move outward at 1 m/s each, the loads nevertheless supply 2 W of deformation power. Zero loads give the same resultant wrench but zero power. For one rigid velocity field, that self-equilibrated pair would give zero power instead. A net-wrench measurement therefore does not determine all flexible-body power, deformation or internal load allocation. This manufactured contrast establishes a possibility, not its magnitude in a human grip.

These distinctions connect the later two-hand and shaft chapters. A negative projected couple is not automatically negative total power. Positive hand power is not automatically the power of a particular muscle. Shaft energy release is not automatically an equal increase in clubhead energy: handle exchange, shaft kinetic energy and dissipation also enter. Different boundary accounts are useful because they expose these simultaneous pathways.

### 2.2.7 Audit the Ledger Before Explaining the Swing

Suppose the club gains 40 J while an arm loses 12 J. Those endpoint changes do not establish that the arm supplied exactly 12 J to the club. Integrate the appropriate interface powers and account for actuation, gravity, storage, other segment exchanges and losses over the same interval. A closed total energy account still does not reveal a unique anatomical pathway.

**Model Result:** The pinned technical treatment separates two-sided interface power, elastic storage and whole-system residuals. It also records differences between reconstructed and archived power columns. Algebraic closure or solver convergence verifies a declared calculation; it does not independently validate anatomy, contact laws or angular-momentum closure. Each balance needs its own check.

**Human Evidence:** Nesbit and Serrano’s four-subject analysis combines measured kinematics with full-body and flexible-club models. Its compliant ground-contact model differs from the stationary-support example above. Comparing modeled vertical ground reactions with force-plate data does not directly measure all joint loads or individual muscle work ([Nesbit and Serrano 2005](#)). Net joint power remains a model-dependent quantity, with anatomical allocation requiring additional evidence ([Winter 2009](#)).

**Hypothesis:** Simultaneous handle, shaft and segment accounts may reveal a transfer pathway that segment-speed plots miss. Testing it requires compatible measurements or model states, explicit interface storage, matched intervals and a declared endpoint; agreement with one energy ledger is insufficient.

**Practical Interpretation:** Write down the included bodies, observer and stored forms. Locate each load’s material-point motion, count gravity once, then check subsystem and total balances. Only then ask which mechanism the evidence identifies and which competing allocations remain possible.

## 2.3 Where the Picture Breaks

Unlike a household payment, a segment-energy change need not have a unique anatomical payer. Coordinate choices redistribute inertial terms; muscles can span joints; soft tissue and contact layers store and dissipate energy. A modeled wrist can aggregate carpal motion, forearm rotation, grip deformation and translation. A modeled shoulder can aggregate several anatomical motions. The body inventory and power convention remain useful only when those approximations are explicit.

Equal energy does not imply equal tissue loading, comfort, accuracy, control effort or repeatability. Energy accounting supplies consistency constraints and exposes possible pathways. Those other outcomes need their own quantities and evidence. The goal is a connected explanation of the swing, with each ledger answering its own question without claiming to answer all of them.

**Go Deeper:** The pinned monograph’s [whole-system and interface-power treatment](#) develops the rigid and flexible accounts. Its [two-hand wrench chapter](#) separates reference transport, allocation and contact-deformation power. The manufactured examples here check mechanics identities; they do not reproduce those archived simulations or validate a human intervention.

## Chapter 3

# The State Is the Complete Snapshot

### 3.1 A Concrete Picture

Two photographs can look identical and conceal opposite futures. In the first, a playground swing has reached its highest point and is about to fall. In the second, it is moving upward through the same angle and is about to slow. The seat, chains, and child occupy nearly the same places in both photographs. Yet one moment leads downward and the other upward because velocity is missing from the picture.

Now photograph a flexible beam only at its two ends. Different bends inside the span can leave those endpoints in the same places while storing different elastic energy. The photograph has missed deformation. Changing the material or rest shape is a different comparison: it changes the model parameters. For a fixed elastic law, complete deformation determines strain energy; energy is not an independently adjustable hidden quantity. Two muscles at the same joint angle can likewise have different activation or internal tissue states and respond differently to a disturbance.

A useful dynamics model gathers everything it needs to continue the motion into a **state**. For a rigid double pendulum, the state may consist of two angles and two angular velocities. For a flexible club, it also includes shaft deflection and deflection rate. For an actuator with first-order lag, it may include an activation state. A pure time delay instead requires relevant past history, or a declared finite-dimensional approximation. A compliant contact may require interface deformation, friction memory, and a contact mode. The complete snapshot is more than a pose.

### 3.2 How the Mechanism Works

The formal state is commonly written  $\mathbf{x}$ . For a two-link planar model, one convenient choice is

$$\mathbf{x} = [q_1, q_2, \dot{q}_1, \dot{q}_2]^T,$$

where  $q_1$  and  $q_2$  describe configuration and the dotted variables describe angular velocity. The equations of motion define a rule

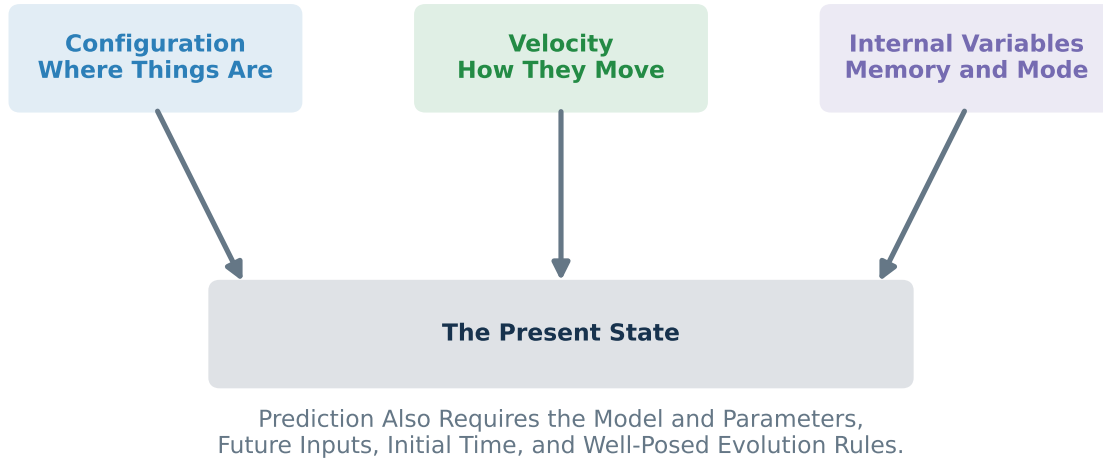


Figure 3.1: State Components and the Additional Conditions Required for Prediction.

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}, t),$$

which says: given the present state, the present control  $\mathbf{u}$ , and any explicit time dependence, compute the rate at which the state changes. A numerical integrator advances that rule through many small steps.

The word *state* has a strict purpose. Within a declared model, it contains the information about the past needed to evolve from the present. Identical states produce identical futures **when the same model, parameters, initial time, future inputs, and well-posed evolution rules are used**. State alone does not supply tomorrow's commands. In a nonautonomous model, starting the same state at a different clock time can also change the future.

Well-posedness matters. A single-valued deterministic right-hand side does not by itself guarantee uniqueness:  $\dot{x} = \sqrt{x}$  for  $x \geq 0$ , starting at zero, admits both a permanently zero solution and a solution that waits before growing. A common sufficient uniqueness condition is local Lipschitz continuity in state, with appropriate regularity of the inputs; it is not a necessary condition in every model. The claim concerns times for which the solution exists in the model's valid domain, not an unconditional forecast forever.

Contacts and impacts may require more than the displayed smooth ODE. Specify constraint consistency, discrete modes, event guards, and reset rules, and verify that they define a unique admissible evolution for the comparison. An impact can reset velocity while leaving configuration continuous. Numerical time stepping and tolerances approximate this declared evolution; disagreement between simulations can reveal numerical error or ill-posedness as well as missing state.

### 3.2.1 Configuration Is More Than a Photograph

Configuration describes where the modeled bodies are within their allowed geometry. In a planar double pendulum, two angles may be enough. In a spatial golfer model, configuration can include pelvis position and orientation, spinal and shoulder rotations, elbow and wrist coordinates, club pose, and perhaps shaft modes. Rotations require care because three-dimensional orientation is not just three independent slider positions. Different angle sequences can describe the same physical orientation, and some parameterizations contain singularities. With redundant orientation coordinates, coordinate rates and physical angular velocity are different objects: a kinematic map can relate  $\dot{q}$  to a velocity vector  $v$ . Normalization and closed-chain constraints restrict admissible states; arbitrary coordinate and velocity arrays need not describe a possible motion.

Configuration also determines mechanical advantage. The mass matrix, moment arms, constraint Jacobian, generalized gravity forces, and mapping from joint torques to club acceleration can change with posture. Gravity need not change direction in an inertial frame; its body-frame components and generalized projections depend on configuration. The same torque vector applied at two configurations can produce different accelerations. State-dependent geometry is therefore not scenery around the mechanism; it is part of the mechanism.

### 3.2.2 Velocity Gives the Snapshot a Direction

Velocity tells how configuration is changing. For ordinary rigid-body mechanics in independent coordinates, write the mechanical part as

$$M(q)\ddot{q} = \tau - c(q, \dot{q}) - g(q) - D\dot{q},$$

Here  $q$  and  $\tau$  are coordinate and generalized-torque vectors. With positive inertias and a regular independent-coordinate description,  $M$  is positive definite and can be inverted to obtain acceleration. The vector  $c$  collects quadratic Coriolis and centrifugal terms,  $g$  is the potential-force term, and  $D\dot{q}$  illustrates viscous resistance. At the same configuration,  $M(q)$  is the **same matrix**. Reversing every velocity reverses momentum  $M\dot{q}$  and, for a fixed torque, power  $\tau^T\dot{q}$ , but leaves quadratic  $c(q, \dot{q})$  unchanged: each product contains two velocity factors. The matrix factorization  $C(q, \dot{q})$  can change sign while its product  $c = C(q, \dot{q})\dot{q}$  stays the same. Viscous resistance changes sign instead. These statements apply to this model class; rotating-frame, magnetic, or other odd-in-velocity forces need separate treatment.

Consequently, in the undamped model, opposite velocities can yield **equal instantaneous accelerations** under the same posture and torque. Their futures still differ because their initial velocities differ. Acceleration is only the velocity derivative, not the full state derivative. Reversing just one joint's velocity, or changing speed magnitudes, can change quadratic coupling terms. Large velocity-dependent effects therefore do not imply that every reversal changes every term.

Velocity is also needed to evaluate power. A 100-newton force at a stationary point delivers zero instantaneous translational power. The same force at a point moving 5 metres per second in its direction delivers 500 watts. Reverse the velocity and the power is  $-500$  watts. Force without velocity cannot tell the energy story.

### 3.2.3 Stored Variables Carry History Into the Present

Rigid-body coordinates and velocities are not always enough. A flexible shaft has modal displacement and velocity. A series-elastic actuator may have tendon stretch. A first-order activation model has a state that lags its neural command. A transmission model with an evolving deflection can use that state to determine engagement. A memoryless dead-zone map alone does not acquire an independent state merely because it is nonlinear. These variables are **history summarized**: they contain the past information needed to continue under specified future inputs. Once two histories lead to the same sufficient state, their earlier histories need not remain separate predictors in that model.

This matters directly to preload. In the finite-preparation experiment, both programs begin relaxed, build their own transmission deflections for 180 ms, and carry those states continuously through transition. Continuity holds **within each branch**; the two strategies need not have the same internal state at transition. Comparing their futures tests the effect of preparation and its resulting state under the declared later commands, not the isolated effect of a new command from identical states. Chapter 14 of this companion gives the model and its limits. Finite preparation also need not reach equilibrium to avoid a state reset.

Pure delay is a different kind of memory. For a scalar channel  $y(t) = u(t - d)$ , two past input histories can have the same current value  $u(0)$  and identical future commands, but yield different outputs during the next  $d$  seconds. One must retain the relevant history over  $[-d, 0]$ , not only the current command. Replacing that history with a first-order lag is a modeling approximation with different transient behavior; assess it over the frequencies and time scales used in the claim.

### 3.2.4 State and Input Must Not Be Confused

An input is something the model treats as externally prescribed for the future: a joint torque command, a desired actuator level, or a measured base trajectory. A state is something the model evolves. The distinction can change between model tiers. A prescribed hand path can be a kinematic input to an inverse-dynamics model; it does not determine every joint motion or internal force without additional measurements and constraints. In a forward model driven by joint torques, the hand path can instead be an outcome. A prescribed pelvis motion can hide the reaction of the club on the body; a dynamically coupled moving-base model can let that feedback alter the state. A base trajectory prescribed in advance still prevents that response.

If a controller generates commands from measurements, its memory also belongs in the closed-loop state, or its complete future input must be prescribed. Calling a signal an input does not mean it is independent of state in the physical experiment.

This difference explains why two models can answer different questions even if they draw the same stick figure. An inverse model asks what forces are compatible with measured motion. A forward model asks what motion results from forces and an initial state. Neither is automatically superior. They have different estimands and different opportunities for circular reasoning.

### 3.2.5 A Matched-State Intervention

Suppose we want to ask what the current shoulder torque contributes to club acceleration. One useful experiment freezes the state at an instant and solves the acceleration twice: once with the recorded torque and once with that torque set to zero. Hold the entire relevant state, time, parameters,

constraints, and other inputs fixed. Matching configuration and velocity is sufficient for the simplest rigid model, but not for models with additional actuator or contact states. The acceleration difference is then a pointwise intervention contrast. If acceleration is affine in the selected torque at that state and contact mode, the contrast equals its linear control contribution. For nonlinear control maps, the contrast remains defined but need not form an additive partition across actuators.

If we then integrate the zero-torque branch forward, its state begins to diverge. After a few milliseconds, force and power differences include both the missing torque and the changed trajectory it caused. That forward counterfactual answers a persistence question rather than a pure instantaneous attribution question. Both are legitimate if their contracts are stated. An instantaneous torque step changes acceleration without itself causing a velocity jump; an impulse or a contact reset is a different intervention. Zeroing a neural or actuator command also need not instantly remove transmitted torque when activation or transmission states persist.

The initial state difference is zero by construction. Fix other inputs, parameters, disturbances, and either the future open-loop program or the feedback policy being compared. Numerical divergence beyond tolerance then represents the declared intervention and its later interactions within that model. Reoptimizing each future or resetting an internal state changes the question and requires a new contract.

### 3.2.6 Phase Landmarks Need an Event Contract

Golf language uses transition, early downswing, delivery, and impact. A clock time alone may not align these phases across participants or simulations. An event can use a position or velocity crossing, a peak criterion, a contact guard, or a measurement history. Specify the quantity, direction, admissible time window, which crossing counts, and whether the event must be detected causally. A velocity peak requires a sign change or comparison over a neighborhood; retrospective peak labeling differs from real-time detection. Freezing a criterion in advance limits outcome-driven selection but does not guarantee that it exists, is unique, or is robust to noise.

Comparing two runs at their respective impact events also differs from comparing them at a common elapsed time: the intervention may shift the event itself. Report that timing difference and handle missing events explicitly. Otherwise, aligning curves can hide part of the effect one intended to measure.

### 3.2.7 State-Based Timing Versus Clock-Based Timing

A command labeled 120 milliseconds before impact is not necessarily equivalent across golfers or clubs. If impact time is known only after the swing, this is retrospective labeling; a real-time controller needs an estimator or a causally detectable trigger. One system may already have a small arm–club angle and high relative speed; another may still be carried with substantial lag. A state-based rule might refer to relative angle, angular velocity, shaft mode, or the sign of an interaction term. It may adapt across tempo changes if that variable remains observable and its threshold predicts the desired response. This advantage needs testing: a noisy or poorly estimated state trigger can be less reliable than a clock rule.

The backswing-preload proposal belongs here. Its plausible benefit is not that one direction of effort is inherently superior. It is that continuous loading might deliver a transition state with less force-development delay than a full role reversal. To test that claim, the state must include the modeled transmission or activation variable. Without those variables, it cannot attribute an effect to that particular transmission-memory pathway. Other modeled pathways, such as changed velocity or

geometry, could still produce an allocation difference and must be separated.

### 3.2.8 What an Incomplete Observation Can Miss

Now give the club a flexible shaft. The grip and clubhead may occupy the same visible positions in two trials while unresolved shaft modes or modal rates differ. For example, a scalar observation  $y = a + b$  cannot distinguish modal amplitudes  $(a, b) = (0, 0)$  from  $(1, -1)$ . With unit modal stiffnesses, their elastic energies  $\frac{1}{2}(a^2 + b^2)$  differ despite equal  $y$ . This is an abstract observability example, not a calibrated shaft. Complete deformation and modal rates, together with the constitutive law and loading, determine storage and exchange; visible endpoints alone do not. Add muscle activation and short-range stiffness, and two outwardly similar states can also resist perturbation differently. “Same position” is therefore far weaker than “same state.”

### 3.2.9 What Must Be Measured

Motion capture provides observations from which model coordinates and velocities are estimated. Filtering and differentiation affect noise, phase, and uncertainty near transition. Strain gauges observe strain at selected locations; recovering shaft modes requires a model and enough independent measurements. Bilateral grip sensors measure the wrench components they are designed and calibrated to resolve. EMG records electrical activity; inferred activation timing and force require additional dynamics and assumptions.

Neither a plausible fit nor a larger sensor count guarantees observability. For a linear model, observability depends on the state-to-output map and dynamics; for a nonlinear model it can also depend on trajectory, excitation, operating point, and parameters. Different states can produce indistinguishable outputs within measurement precision. A single measured output at one instant is also different from a time series under known excitation.

Represent uncertainty in states, parameters, sensor calibration, and model choice as far as the evidence allows. A distribution is conditional on the estimator and its assumptions, not automatically a calibrated statement of truth. Repeating counterfactuals across plausible reconstructions tests sensitivity to those assumptions; it cannot establish a causal human effect or cure a structurally unidentifiable state by itself.

## 3.3 Where the Picture Breaks

The “complete snapshot” phrase can suggest that a finite model state contains all physical reality. It does not. A model state is complete only relative to the equations chosen. A two-link rigid model omits shaft vibrations, soft tissue, muscle activation, grip deformation, aerodynamics, and neural feedback. Its four-number state, specified future inputs, and well-posed rules can determine a unique future of that reduced model while remaining a sparse description of a human swing.

Biological systems may also contain memory that simple first-order states do not capture: muscle history dependence, reflex delays, thixotropy, fatigue, and learning. Adding every possible state would make estimation impossible. The scientific task is to add a state when it changes the question or survives a falsification test, not because complexity sounds realistic.

Finally, measurements never reveal state without uncertainty. Velocities are often derived from noisy

positions. Internal force and activation states may be only indirectly observed. A responsible model carries those uncertainties into its conclusions instead of treating one reconstructed trajectory as exact.

**Model Result:** In the registered rigid double-pendulum counterfactual study, each commanded/kill-switch pair begins with the same four-component angle/velocity state, at the same cut time. The commanded branch continues its declared program; the other sets both generalized torques to zero. The finite-preparation study instead preserves each transmission branch’s state through its command switch and deliberately allows preparation to create different states. These are different experimental contracts, with different causal questions.

**Measurement Boundary:** Motion capture estimates visible kinematics, not a complete biological state. Inferring velocities, activation, deformation modes, or contact memory requires a measurement model and uncertainty assumptions. McPhee’s review surveys prescribed-motion, torque-driven, predictive, and contact models in golf ([McPhee 2022](#)); that survey does not certify a particular sensor set as a complete state reconstruction.

**Hypothesis:** A golfer may use backswing loading and early-downswing motion to arrive at a state from which interaction dynamics assist late club acceleration. That is a state-arrangement hypothesis, not merely a claim about waiting a fixed number of milliseconds.

**Practical Interpretation:** A pose is the cover of the book, not the page you are on. To understand what comes next, ask how everything is moving and what internal elements are already loaded.

**Go Deeper:** See the monograph’s [matched-state persistence chapter](#), the [continuous preparation experiment](#), and the [interactive workbench](#) for guided state-versus-clock interventions.

## Chapter 4

# Geometry Is Part of the Machine

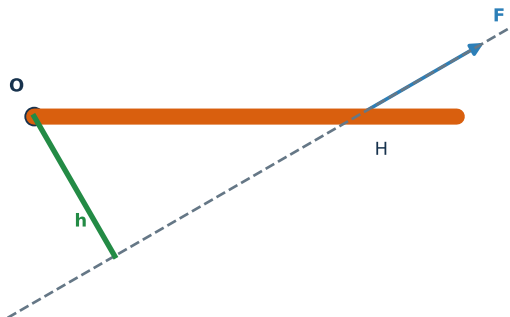
### 4.1 A Concrete Picture

Push a door near its hinge, then with the same force near its handle. The turning effect changes because the force has a different moment arm. Push directly toward the hinge and the moment about that hinge is zero in the ideal model. Your hand can still exert force and load the structure. Moment, force, motion, and effort are related, but they are not interchangeable.

A swing has several moving versions of this geometry. Joint axes, hand positions, club orientation, and shaft deformation change how forces map into moments and how commanded torques produce acceleration. The useful question is not whether one posture has more “leverage” in general. It is leverage for which motion or force, about which reference, under which constraints.

**Moment About O: Force Magnitude  $\times$  Perpendicular Distance  $h$**

**Shorter Moment Arm**



**Perpendicular Force**

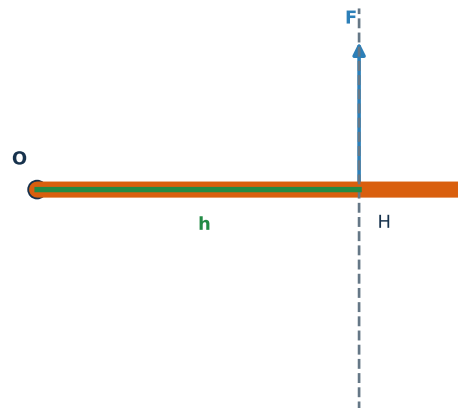


Figure 4.1: The Moment Arm Is the Perpendicular Distance From O to the Force's Line of Action.

## 4.2 How the Mechanism Works

### 4.2.1 Name the Point, Plane, and Direction

For a force on the club applied at point  $H$ , its moment about  $O$  is

$$\mathbf{M}_O = (\mathbf{r}_H - \mathbf{r}_O) \times \mathbf{F}.$$

This instantaneous definition does not require  $O$  to be fixed or inertial. Using that moment in an angular-momentum balance about a moving point requires the appropriate transport terms; the definition alone is not that balance. An additional applied free moment must be included separately.

The magnitude of a point-force moment about  $O$  is force magnitude times the perpendicular distance from  $O$  to its line of action. It is not generally the distance to the arrow tip, the contact point, or the end of the club. A 100 N force with a 0.04 m perpendicular moment arm gives 4 N m. Doubling that distance at the same force gives 8 N m. Passing the line through  $O$  gives exactly zero.

For a planar idealization, let  $\mathbf{n}$  be the declared unit normal to the plane, let the unit vector  $\mathbf{e}_r$  point along the straight club, and define the **in-plane** transverse direction  $\mathbf{e}_t = \mathbf{n} \times \mathbf{e}_r$ . The transverse direction is not the normal to the swing plane. If the force and club lie in the plane, write  $\mathbf{F} = F_r \mathbf{e}_r + F_t \mathbf{e}_t$ . When  $\mathbf{r}_H - \mathbf{r}_O = \ell \mathbf{e}_r$ , the signed moment about  $\mathbf{n}$  is  $\ell F_t$ . The signed distance  $\ell$  matters: changing which side of  $O$  contains the contact can change the sign.

The radial term has zero moment under those collinearity assumptions. It can nevertheless accelerate the club center, load the shaft, or transfer power when the contact moves along its force. A curved shaft, an offset reference, or an out-of-plane force needs the full geometry. “Radial” alone does not specify a zero moment about every point.

### 4.2.2 Changing the Reference Preserves a Contract, Not Every Number

A wrench combines resultant force  $\mathbf{R}$  and total moment about a declared point. Moving the reference from  $A$  to  $B$ , with  $\mathbf{d} = \mathbf{r}_B - \mathbf{r}_A$ , gives

$$\mathbf{M}_B = \mathbf{M}_A - \mathbf{d} \times \mathbf{R}.$$

A pure couple has  $\mathbf{R} = 0$  and the same moment about every point. For a nearly opposed force pair, the moment change is bounded by  $\|\mathbf{d}\|\|\mathbf{R}\|$ . A small resultant supports a small-change claim only with a bounded reference displacement and a stated tolerance. It does not make a nonzero-resultant wrench a pure couple.

Power provides a second check. On a rigid body, use the velocity of the same reference point as the moment:

$$\begin{aligned} P &= \mathbf{R} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \boldsymbol{\omega}, \\ \mathbf{v}_B &= \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{d}. \end{aligned}$$

Transporting both moment and velocity leaves  $P$  unchanged. Transporting only one creates a false power

difference. For a deforming shaft or distributed contact, a single rigid-body twist need not describe every loaded point; retain the individual force–velocity pairings or the corresponding distributed power integral.

Expressing the same vectors in another orthonormal basis changes their components while preserving dot products. Changing the physical observer is a different operation: an observer translating at velocity  $\mathbf{U}$  measures force power shifted by  $-\mathbf{R} \cdot \mathbf{U}$  when the same physical forces are used. Wrenches and trajectories have transformation rules; their component lists are not universally frame-invariant observables. The [Modern Robotics wrench discussion](#) develops the paired force/motion representation.

### 4.2.3 The Jacobian Maps Velocities and Their Dual Forces

Suppose a hand position is  $\mathbf{x}_H = \mathbf{f}(\mathbf{q}, t)$ . Differentiating gives a velocity contribution from the modeled coordinates and another from any explicit prescribed motion:

$$\begin{aligned}\mathbf{v}_H &= \mathbf{J}_H \dot{\mathbf{q}} + \mathbf{b}_H, \\ \mathbf{J}_H &= \partial \mathbf{f} / \partial \mathbf{q}, \quad \mathbf{b}_H = \partial \mathbf{f} / \partial t, \\ \mathbf{Q}_H &= \mathbf{J}_H^\top \mathbf{F}_H.\end{aligned}$$

Here  $\mathbf{F}_H$  acts on the modeled hand. The force that hand exerts on the club has the opposite sign; choose the system before applying the mapping. The last line maps this applied force into generalized force; it does not assert static equilibrium or identify the total actuator torque. Force and Jacobian must use compatible components and the same point. Generalized force is a torque for an angular coordinate and a force for a length coordinate; its units and components change with coordinate scaling.

The power identity is  $\mathbf{F}_H \cdot \mathbf{v}_H = \mathbf{Q}_H \cdot \dot{\mathbf{q}} + \mathbf{F}_H \cdot \mathbf{b}_H$ . Ignoring the last term can hide the power associated with prescribed base motion. If base coordinates are included in the dynamical system instead, their rates and generalized forces belong in the larger mapping. A force-plus-free-moment contact uses the corresponding linear and angular velocity Jacobians.

Coordinate derivatives are not always the software’s generalized speeds. Quaternion orientation, for example, uses four constrained coordinates but three rotational speed components. Write  $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\nu$  and use  $\mathbf{J}_H \mathbf{N}$  when mapping those speeds. The dual generalized force is then  $\mathbf{N}^\top \mathbf{Q}_H$ . [Tedrake’s multibody notes](#) explain this distinction. Neither a particular coordinate count nor a matrix entry by itself identifies an anatomical actuator.

### 4.2.4 Near a Singularity, Motion and Force Have Different Scalings

A Jacobian becomes singular for its declared task when its rank falls below the maximum attainable rank. A robot with fewer joints than task dimensions can already lack some instantaneous motions without being at a special singular posture. Instantaneous inability to move in one direction also does not prove that a finite path to another configuration is impossible.

Consider a dimensionless, consistently scaled example  $\mathbf{J} = \text{diag}(1, \varepsilon)$  with  $\varepsilon > 0$ . Unit task velocity in the second direction requires joint rate  $1/\varepsilon$  in that direction. But a specified task-force component of 10 gives generalized force  $10\varepsilon$ , not  $10/\varepsilon$ . Conversely, solving for task force from a fixed generalized force gives  $1/\varepsilon$  scaling under this ideal invertible mapping.

At  $\varepsilon = 0$ , no finite joint rate gives the missing task velocity, while that task-force component contributes zero generalized force. Its magnitude is therefore not determined by joint torque through this mapping. Structure, contact, compliance, strength limits, and dynamics still matter; the equation does not grant unlimited physical force or independent force control. The [Modern Robotics singularity examples](#) illustrate the difference between active motion and structural force support.

For multiple coordinates, the analogous statements apply along singular-vector directions with declared units and scaling. Distinguish motions in the Jacobian's nullspace from force components in the transpose nullspace. Neither algebraic ambiguity alone proves that a biological force is unidentifiable from every possible sensor, dynamical model, and experiment.

#### 4.2.5 Inertia Couples the Response Through the Equations of Motion

In independent coordinates, a reduced equation can be written  $\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h} = \mathbf{Q}$ . At fixed state and fixed non-control terms, the acceleration change is  $\delta\ddot{\mathbf{q}} = \mathbf{M}^{-1}\delta\mathbf{Q}$ . It is the inverse mapping, not one off-diagonal mass entry alone, that answers which coordinate accelerates under an added generalized force.

For example, in consistent normalized units, let  $\mathbf{M} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ . An added force vector  $(1, 0)$  gives acceleration change  $(2/3, -1/3)$ . The off-diagonal mass entry is positive while the cross-acceleration response is negative. A linear coordinate transformation can diagonalize this matrix; transforming the force and reconstructing the motion gives the same physical response. Removing an off-diagonal entry by relabeling coordinates does not remove the original endpoint response.

In a constrained two-hand model, admissible acceleration changes also satisfy the linearized acceleration constraint at the fixed state. Reactions change with the command, so solve the coupled constrained equations or use a valid reduced-coordinate model. Applying an unconstrained inverse mass matrix and ignoring the changed reactions answers another question. Near a constraint-rank change, check the solve's validity before interpreting large numerical values.

Posture affects these mappings, while velocity affects the remaining dynamics. A same-state control comparison holds both fixed. Comparing club angles is a comparison across different configurations, each with its own same-state allocation experiment; it is not one unchanged state with several angles.

#### 4.2.6 A Two-Hand Couple Has Exact, Conditional Controls

Let the contact separation be  $\mathbf{d} = \mathbf{r}_T - \mathbf{r}_L$ . If the trail contact applies  $+\mathbf{F}$  and the lead contact applies  $-\mathbf{F}$ , the resultant is zero and the couple is exactly  $\mathbf{d} \times \mathbf{F}$ . Its magnitude is  $dF$  only when the vectors are perpendicular. Collapsing separation or making  $\mathbf{F}$  parallel to  $\mathbf{d}$  removes this point-force couple. Applied free moments can remain.

Reversing the force with the separation held fixed reverses the couple. Reversing the separation while retaining the same force assignment also reverses it. Rigidly rotating the entire force system rotates its moment; it does not necessarily reverse a consistently transformed scalar component. Merely swapping contact labels changes no physical force system at all.

For arbitrary forces, define the common component  $\mathbf{F}_c = (\mathbf{F}_L + \mathbf{F}_T)/2$  and differential component  $\mathbf{F}_d = (\mathbf{F}_L - \mathbf{F}_T)/2$ . The resultant is  $2\mathbf{F}_c$ , and the point-force moment about the midpoint is  $-\mathbf{d} \times \mathbf{F}_d$ . For the pure pair above,  $\mathbf{F}_d = -\mathbf{F}$ , recovering the same positive  $\mathbf{d} \times \mathbf{F}$  expression. A common component

can contribute a moment about another reference. Direct free moments must still be added. This decomposition is useful bookkeeping, not a measurement of separate muscles.

These geometric controls require declared held-fixed forces. If changing grip separation also changes solved reactions, the resulting couple need not scale linearly with separation. A measured or simulated reversal test must report both the changed geometry and force response. Otherwise a failed simple scaling law need not be a failed cross-product implementation.

#### 4.2.7 What the Allocation Study Actually Holds Fixed

The pinned provider uses a planar rigid-club model with two constrained arms. At each of 19 club angles from  $-12^\circ$  to  $24^\circ$ , it constructs a constraint-consistent configuration with club center at  $(0, -0.50)$  m and zero generalized velocity. This is an instantaneous dynamical comparison; zero initial velocity does not imply static equilibrium.

For each configuration it evaluates commanded and zero-command dynamics at the same state, then subtracts them. The fixed task is the club’s rotational inertia times the **control-only angular-acceleration difference**, set to 8 N m. It is not the full net wrench, every acceleration component, or a complete trajectory. The force resultant is allowed to change.

One control subspace uses four shoulder/elbow torques with both wrist commands zero; another uses the two wrist torques with proximal commands zero. In each, the implementation chooses the minimum unweighted Euclidean norm of actuator commands that satisfies the scalar task. It blends these two normalized controls, scaled to the common task, at 21 fractions from zero to one. Linearity of the fixed-state constrained solve preserves the scalar task across that blend; it is not a general superposition claim for finite nonlinear swings.

The blending fraction is **not** the fraction of net club moment supplied by direct wrist moments. At the wrist-only endpoint, direct moments sum to about 8.13–10.02 N m across the angles; opposing contact-force moments bring the control-only total back to 8 N m. The archive’s variable called a grip-force “couple” is the contact-force moment about the club center. With a nonzero resultant, it need not be a pure couple of equal opposite forces.

The reported hand-force RMS is the root mean square of the two **control-only force-vector norms at that instant**. It is not a time RMS or the norm of total physical contact loads. A control-only force is the difference between two same-state solutions; the norm of that difference is not the difference of their norms. The torque metric is the unweighted norm of six actuator commands, not muscle force, metabolic cost, tissue loading, work, or a coordinate-free measure of effort.

Independent recomputation of the archived arrays reproduces the scalar task and moment closure with residuals below  $6 \times 10^{-15}$  N m. The hand-force metric ranges from 7.58 to 91.51 N and the torque norm from 5.74 to 25.45 N m. Among the sampled blend fractions, hand-force minima occur from 0.75 to 0.90, whereas torque-norm minima occur from 0.85 to 0.95, depending on angle. These are grid minima along one restricted blend of two subspace solutions. They do not establish global optima over every admissible allocation, a physiological optimum, or an optimal human grip. Archived summaries were recomputed; the constrained model was not rerun for this chapter review.

### 4.2.8 Connect Geometry to the Question Being Tested

Geometry, inertia, actuation, constraints, and objectives enter the same chain of reasoning. Geometry maps contact motion and forces; inertia and constraints determine acceleration response; a chosen objective prices the resulting delivery and internal demands. Changing one of these can change a preferred allocation without proving that the original physical mechanism was absent.

For a geometry hypothesis, first declare the system, reference, coordinates, contact model, and task. Then distinguish a bookkeeping transformation from a physical intervention. Check algebraic identities at fixed forces, and report the dynamically recomputed forces for changed geometry. Finally evaluate the outcomes and constraints that matter to the proposed use. Chapter 3 explains state matching, Chapters 9–10 connect force to work, and Chapter 23 treats uncertainty and competing objectives.

Human motion and contact are more complicated than this planar illustration. Kinematic measurements can describe geometry, but they do not alone select the bilateral force allocation that produced it. McPhee’s review provides context for the range and limitations of golf models (McPhee 2022); it is not direct validation of this allocation surface or a technique prescription.

## 4.3 Where the Picture Breaks

The door has a fixed hinge; the golfer, compliant grip, and shaft do not reduce to that single joint. A contact’s moment about one point is not automatically its contribution to every generalized acceleration. Likewise, favorable force transmission in one direction can coincide with poor velocity authority, limited motion range, or unacceptable loading.

**Model Result:** The archived reduced model admits different actuator blends with the same scalar control-moment task and different internal metrics. Geometry changes the preferred sampled blend for each declared metric.

**Measurement Boundary:** Coordinate-consistent wrenches and motion are needed to compare physical actions. Independent information is needed to constrain individual hand forces and biological actuator allocation; small numerical closure residuals supply no human validation.

**Hypothesis:** An opposing club-moment component can arise from two-hand force geometry. Whether it persists or improves delivery under a changed grip must be tested with the changed reactions and full dynamical state, not inferred from one negative projection.

**Practical Interpretation:** Ask where the force acts, which direction it points, what motion it accompanies, and which task it helps accomplish. Those questions turn a vague claim about leverage into a comparison that can fail.

**Go Deeper:** Read the pinned [two-hand wrench derivation](#), [allocation implementation](#), [archived allocation record](#), and [wrench–power frame audit](#).

## Chapter 5

# Speed, Acceleration, Energy, and Power

### 5.1 A Concrete Picture

A car dashboard separates quantities that casual speech blends together. The speedometer reports how fast the car is moving now. Acceleration describes how its velocity changes, including a change of direction. The fuel gauge concerns stored chemical energy, whereas power describes a rate of energy conversion or transfer. A heavy truck and a bicycle can have the same speed and very different kinetic energies. These quantities answer different questions even when they rise together.

The same distinction matters in a swing. Clubhead speed describes one point, while the club's kinetic energy includes motion throughout the club. Pelvis angular speed alone does not determine pelvis energy, and a slowing pelvis does not identify where energy went. A large power peak may deliver little work if it is brief or followed by negative power. Before interpreting a transfer, name the body or collection of bodies, the point whose motion is reported, the time interval, and the observer frame. In this chapter, velocities are measured in a declared inertial laboratory frame unless a relative velocity is explicitly named. Changing the direction of coordinate axes in that frame does not change energy; changing to a physically moving observer can.

### 5.2 How the Mechanism Works

#### 5.2.1 The Point, the Body, and the System

Velocity includes direction; speed is its magnitude. For a rigid club, the velocity of a head point H is

$$\mathbf{v}_H = \mathbf{v}_C + \omega \times \mathbf{r}_{H/C},$$

Here C is the club's center of mass,  $\mathbf{v}_C$  its velocity,  $\omega$  the club's absolute angular velocity, and  $\mathbf{r}_{H/C}$  the vector from C to H. Translation and rotation combine as vectors. A wrist rate multiplied by club length omits motion of the wrist and may also confuse a relative joint rate with the club's absolute angular velocity. A flexible club requires additional deformation velocities; one rigid-body formula cannot describe every material point exactly.

Acceleration is the time derivative of velocity. When speed is nonzero, its rate of change is

## A Speedometer Is Not an Energy Meter

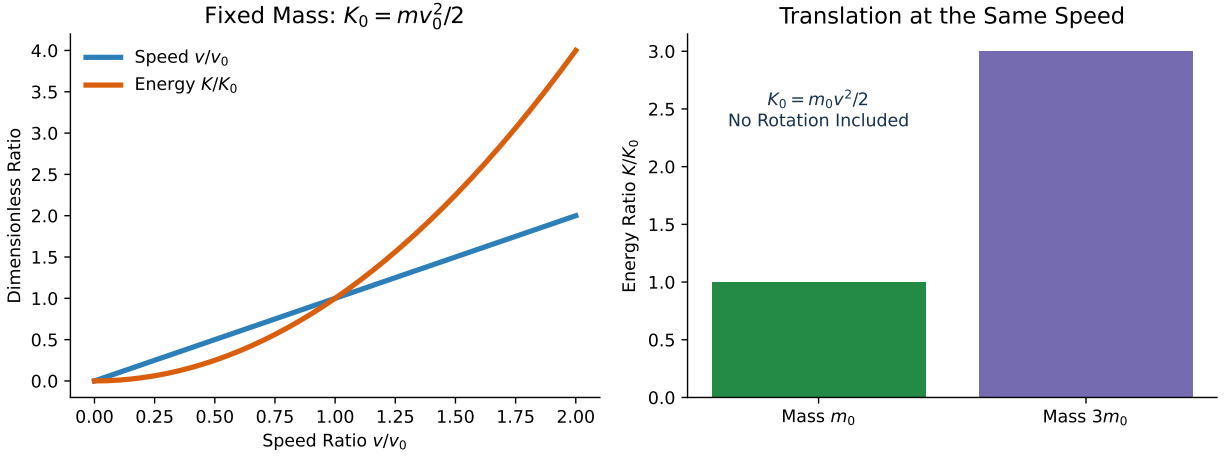


Figure 5.1: Normalized Speed and Kinetic Energy, and the Effect of Mass at a Fixed Translational Speed.

$$\frac{d|\mathbf{v}_H|}{dt} = \frac{\mathbf{v}_H \cdot \mathbf{a}_H}{|\mathbf{v}_H|}.$$

A Cartesian acceleration component may change sign while speed rises; the tangential component in this equation determines the speed change. Normal acceleration bends the path and is present even at constant speed. For a particle moving around a fixed circle, an exactly radial force is perpendicular to velocity and has zero instantaneous power. A force directed along a club toward moving hands need not be perpendicular to the head's laboratory-frame velocity. Calling it radial does not establish zero work.

For a translating particle of mass  $m$ ,  $K = \frac{1}{2}m|\mathbf{v}|^2$ . For a rigid segment with arbitrary three-dimensional motion,

$$K = \frac{1}{2}m|\mathbf{v}_C|^2 + \frac{1}{2}\boldsymbol{\omega}^T \mathbf{I}_C \boldsymbol{\omega}.$$

The inertia tensor  $\mathbf{I}_C$  and angular velocity must be represented in the same axes. The familiar  $\frac{1}{2}I\omega^2$  is the fixed-axis specialization, with  $I$  taken about that axis. For a translating club, adding a rotational term about the moving grip to a center-of-mass translational term is not generally this decomposition; the reference shift needs its accompanying cross-term. Using center-of-mass translation and center-of-mass rotational inertia avoids that particular double count.

In generalized coordinates for a system with time-independent position maps,

$$K = \frac{1}{2}\dot{\mathbf{q}}^T \mathbf{M}(\mathbf{q}) \dot{\mathbf{q}}.$$

Here  $\mathbf{q}$  contains the generalized coordinates,  $\dot{\mathbf{q}}$  their rates, and  $\mathbf{M}(\mathbf{q})$  is the symmetric generalized mass matrix. Configuration-dependent cross-terms describe shared physical motion. They do not make

the energy of a specified rigid segment arbitrary: its center-of-mass and rotational terms still define it in the chosen observer frame. What can be arbitrary is assigning each generalized coordinate its own independent energy share while distributing the cross-terms by convention. Prescribed moving supports add explicit time dependence unless their motion is included among the coordinates; then the simple quadratic expression needs additional terms. The moving-base chapter returns to that distinction.

### 5.2.2 What a Speed Increase Costs

Compare equal-mass idealized clubheads moving at 40 and 44 m/s. The second is 10% faster and has 21% more **translational** kinetic energy because  $(44/40)^2 = 1.21$ . For a 0.20 kg point-mass representation, those energies are 160 and 193.6 J. The extra 33.6 J is a useful accounting requirement for that model, not a measurement of extra muscular work. It could reflect different external work, release of stored energy, lower dissipation, or different initial energy. A full club also has rotational and possibly elastic energy.

Likewise, a 0.20 kg head at 50 m/s has 250 J of translational energy, while a 70 kg golfer's center-of-mass translation at 1 m/s contributes 35 J. The latter calculation omits all motion relative to the golfer's center of mass. It therefore does not say that the golfer's entire kinetic energy is 35 J or that the club contains more energy than the body. Comparisons are useful only when their included motions are explicit.

A point can speed up while its extended body loses kinetic energy. As a manufactured example, take a uniform 2 m rod of mass 3 kg with center-of-mass planar inertia  $1 \text{ kg m}^2$ . At one instant let its center move at  $(1, 0) \text{ m/s}$  with acceleration  $(-3, 0) \text{ m/s}^2$ , its angular velocity be  $2 \text{ rad/s}$ , and its angular acceleration be  $4 \text{ rad/s}^2$ . For the endpoint 1 m below the center at that instant, rigid-body kinematics gives velocity  $(3, 0) \text{ m/s}$  and acceleration  $(1, 4) \text{ m/s}^2$ . Its speed increases at  $1 \text{ m/s}^2$ , but the rod's kinetic-energy rate is  $3(1)(-3) + 1(2)(4) = -1 \text{ W}$ . The center's translational energy falls faster than rotational energy rises. This is a consistent instantaneous mechanics example, not measured golf behavior. For a single particle, by contrast, increasing speed at fixed mass requires positive **net** power  $m\mathbf{v} \cdot \mathbf{a}$ . Changing coordinates cannot evade that identity.

### 5.2.3 Power Needs a Named Interaction

Power is work per unit time. A force acting at a material point supplies  $P = \mathbf{F} \cdot \mathbf{v}$  using that point's velocity. For a rigid body, a resultant force and moment represented at C supply

$$P = \mathbf{F} \cdot \mathbf{v}_C + \mathbf{M}_C \cdot \boldsymbol{\omega}.$$

Moment and point velocity must be shifted together if another rigid-body reference is used. For several deforming contacts, retain their separate force-velocity and moment-angular-velocity pairings unless a common rigid motion has been justified. One interaction's positive power does not require positive net power after all other interactions are included. Nor does power at an external boundary always equal total mechanical-energy rate: internal elastic storage, dissipation and, for a biological system, chemical-to-mechanical conversion must be accounted for.

A constant 1,000 W input for 0.05 s supplies 50 J. A 5,000 W input for 0.002 s supplies 10 J. For a variable signal the work is its signed integral,  $W = \int P dt$ . Peak height alone supplies neither that

integral nor its sign. Positive and negative portions can nearly cancel even when both peaks are large.

#### 5.2.4 Joint Actuator Power and Segment Power

Torque sign is a direction statement, not an energy verdict. In a planar model, a moment  $\tau$  acting on a segment contributes  $\tau\omega$  using that segment's absolute angular velocity in the declared frame. If  $\tau$  is instead the actuator moment conjugate to a **relative** joint angle  $q$ , its actuator power is  $\tau\dot{q}$  using relative angular velocity. These are different quantities. In three dimensions, Euler-angle rates are not generally angular-velocity components; their generalized moments must be transformed to preserve power.

For an ideal joint joining two bodies, an actuator applies equal and opposite moments. If the distal and proximal angular velocities are  $\omega_d$  and  $\omega_p$ , their moment powers are  $\tau\omega_d$  and  $-\tau\omega_p$ . Their sum is actuator power  $\tau(\omega_d - \omega_p)$ . For example,  $\tau = 2$  N m,  $\omega_d = 1$  rad/s and  $\omega_p = 3$  rad/s give moment-power contributions of +2 W on the distal body and -6 W on the proximal body. Of the 6 W removed from the proximal body through this moment channel, 2 W enters the distal body and 4 W is absorbed by the actuator. This example isolates the moment channel; any joint-force power must be included separately when finding the total power on either segment. This does not imply anything about muscle activation or metabolic cost. It shows why a joint-power label needs its velocity convention.

For a negative moment acting along a declared swing-plane normal, the sign of its contribution also depends on angular velocity along that normal. Even when that moment channel removes energy from the club, hand forces or other channels can supply more energy than it removes. Do not add “wrist power” to already complete hand-wrench power if both represent the same transmitted moment. The final speed depends on the coupled trajectory, rather than the sign of one moment channel.

#### 5.2.5 A Worked Whole-Club Ledger

Choose the **whole flexible club**, including its shaft, as the system. Write its mechanical storage as  $E = K + V + U_s$ , where  $V$  includes gravitational potential and  $U_s$  is shaft strain energy. Assume no internal actuator, no shaft damping, and no other omitted work channels for this illustration. Then

$$\Delta(K + V + U_s) = W_{\text{hands}} + W_{\text{aero}}.$$

Gravity is represented through  $V$  and must not also be added as gravitational work on the right. Shaft release changes internal storage; it is not another external supply. Real calculations must add their actual losses and other interactions instead of inheriting these illustrative exclusions.

Consider two intervals of 0.05 s. In the first, total hand-wrench power averages 800 W, supplying 40 J. In the second it peaks at 1,500 W but averages only 300 W, supplying 15 J. Let aerodynamic work be -2 J in each, and let shaft strain energy remain unchanged in the first interval but fall by 8 J in the second. The ledgers are:

| Quantity                      | First Interval | Second Interval |
|-------------------------------|----------------|-----------------|
| Hand work                     | 40 J           | 15 J            |
| Aerodynamic work              | -2 J           | -2 J            |
| Change in shaft strain energy | 0 J            | -8 J            |

| Quantity                               | First Interval | Second Interval |
|----------------------------------------|----------------|-----------------|
| Change in total club mechanical energy | 38 J           | 13 J            |
| Change in club $K + V$                 | 38 J           | 21 J            |

The second whole-club energy gain is **13 joules**, while its  $K + V$  gain is 21 J because strain energy decreases. Adding the 8 J to the whole-club gain would count an internal conversion as new energy. If instead the system is just a rigid head, excluding the shaft, shaft-to-head interface work is external to that subsystem. To split the second interval consistently, add two illustrative assumptions: the shaft's own  $K + V$  rises by 5 J, and all -2 J of aerodynamic work acts on the head. The shaft's mechanical-energy change is then  $5 - 8 = -3$  J, so 15 J entering from the hands and 18 J leaving through the head interface close its ledger. The head receives those 18 J and loses 2 J aerodynamically, gaining 16 J. The head's 16 J and shaft's -3 J sum to the whole club's 13 J gain. These assigned storage and drag partitions are not measurements: without them or corresponding observations, whole-club hand work alone does not determine head-interface work. Adding proximal hand work again to that head ledger would count a transmission path twice.

Equal terminal head speeds therefore do not establish equal work histories or equal total club energy. The trials may begin with different storage, distribute energy differently among translation, rotation and shaft deformation, or produce different face and path outcomes. For a specified rigid club, mass and body-frame inertia do not change just because its orientation changes; changing orientation alters projections and the relation between endpoint speed and the rest of its motion.

### 5.2.6 Sequence, Acceleration, and Impulse

An ordered sequence of pelvis, trunk, arm and club speed peaks describes coordination. It does not demonstrate that one energy package passed along those peaks. A lower angular speed does not alone prove lower segment energy: translation, axis direction and the inertia tensor still matter. Track the segment's actual energy change, the work through its boundaries, and any storage or conversion within the selected system (Robertson and Winter 1980; Winter 2009). Ideal joint reactions can transmit power between individual segments while their net power cancels for the combined system. Zero net work at that larger boundary does not make a force dynamically irrelevant.

For a fixed collection of material bodies, net external force impulse equals the change in total linear momentum in an inertial frame. Net external angular impulse equals the angular-momentum change about a fixed point in that frame; moving-reference formulas require their appropriate transport terms. Work instead pairs forces with displacement and moments with angular motion. A perpendicular force may redirect momentum substantially while doing no instantaneous work. Momentum, energy and speed therefore need separate graphs.

At a fixed state and fixed contact/constraint mode, suitable equations can split generalized acceleration into named force contributions. These are not automatically separate physical energy supplies. Power requires the appropriate force-velocity pairing, and coordinate velocity-bias terms must be interpreted together with the changing mass matrix. Comparing instantaneous contributions also differs from changing an input and following the resulting trajectory. The force/work and counterfactual chapters develop those distinctions.

### 5.2.7 What the Evidence Supports

**Model Result:** In the pinned double-pendulum timing sweep, selected later-drive schedules produce more delivery speed than early drive or the passive baseline under the stated input bounds and event rules. A separate coupled uncertainty study reports higher held-out 10th-percentile delivery speed for early restraint than late drive, but worse mean planar face/path error. These are different model comparisons, not one universal optimum or a demonstration of human preactivation. The [timing results](#) and [uncertainty/control study](#) specify the respective assumptions and outcomes.

**Human and Methodological Evidence:** Tinmark and colleagues measured pelvis, upper-torso and hand kinematics in 45 skilled golfers, using full and partial shots; their reported peak sequence is kinematic evidence, not a direct measurement of the segment energy route ([Tinmark et al. 2010](#)). Robertson and Winter’s segment-power study used a sagittal linked-body model and data from two walking participants; it supports an accounting methodology rather than a golf-specific transfer magnitude ([Robertson and Winter 1980](#)). McPhee’s narrative review is a map of golf models and measurement methods, not an additional intervention experiment ([McPhee 2022](#)). Keep those evidence roles distinct when reading a claim about how a golfer generated speed.

### 5.2.8 Reader Lab: Diagnose a Misleading Result

Suppose a report favors one strategy because head speed rises more steeply late in the swing and wrist power has a larger peak. First identify the speed point, observer frame, joint-rate convention, system boundary and integration interval. Compare initial storage and integrate distinct work channels exactly once. For a whole-club ledger include strain-energy change inside storage; for a head-only ledger use shaft-to-head work at the boundary. Check the closure residual against the size of the proposed effect, with consistent event detection and numerical resolution.

Then compare the actual task outputs: face orientation, path, impact location, variability and modeled loads. A speed improvement may be real while “better strategy” remains unsupported. A large force with little mechanical work may still matter for tissue loading, and mechanical work is not metabolic effort.

Finally, compare schedules from the same admissible state, including controller or actuator memory where the model has it. Hold parameter choices, event rules and other inputs fixed for that comparison. If the ranking survives timestep refinement and specified uncertainty tests, it has support within those tested conditions, rather than universal robustness. If it disappears, investigate numerical error, changed event selection, initial preparation, parameter choices or other altered conditions; disappearance alone does not identify which explanation is responsible. A simulation intervention does not by itself establish what a human can execute.

## 5.3 Where the Picture Breaks

The dashboard analogy separates the questions, but the swing couples their answers. Head velocity combines translation and rotation; joint power depends on relative motion; physical segment energies differ from arbitrary coordinate energy shares. Changing the system boundary moves an interaction between internal conversion and external transfer without creating or destroying energy. Those relationships are why apparently contradictory graphs can coexist.

Biological systems also contain chemical energy, active muscle processes and losses that a rigid mechanical model does not resolve. An internally driven system can increase mechanical energy without positive external wrench work. Conversely, high muscular effort may coexist with little external mechanical work. Do not infer muscle action or a coaching prescription merely from a closed rigid-body ledger.

Use “head speed increased,” “club mechanical energy increased,” and “the hands supplied positive work” as separate statements until the accounting connects them. Avoid “power was stored,” because power is a rate, and “speed was transferred,” because speed is not a conserved substance. A useful explanation lets readers follow the connection without substituting one quantity for another.

**Go Deeper:** See the monograph’s [mechanics and operational energy definitions](#), [phase-resolved results](#), and [coupled uncertainty study](#). The manufactured examples here test mathematical consistency; they are not measurements of golfers.

## Chapter 6

# Constraints Push Back

### 6.1 A Concrete Picture

A key on a taut cord cannot follow an arbitrary path relative to your hand. The cord pulls along its length and changes the key's acceleration. It cannot push: if the motion would require compression, the cord goes slack and the taut constraint stops applying. Cut it without an impulse and the key retains its instantaneous velocity, tangent to the former circle for a stationary hand. Gravity then bends its subsequent trajectory. The tangent is an initial direction, not a promise of straight-line flight.

A rigid pin can transmit forces in either direction; a cord cannot. A shoe can push against the ground but normally cannot pull the ground upward, and its resistance to slipping is limited by contact conditions. These distinctions matter before solving any equations. A mathematically enforceable motion need not be physically feasible for the chosen connection.

The shared lesson is useful in the golf swing: **zero selected motor torque does not imply zero connection force**. A wrist can transmit force while its modeled torque command is zero. Two separated hand forces can oppose one another yet produce a club moment. The ground reaction depends on the complete constrained motion, not on a single local command. None of those observations identifies which human muscles are active.

## Connection Force and Release Velocity

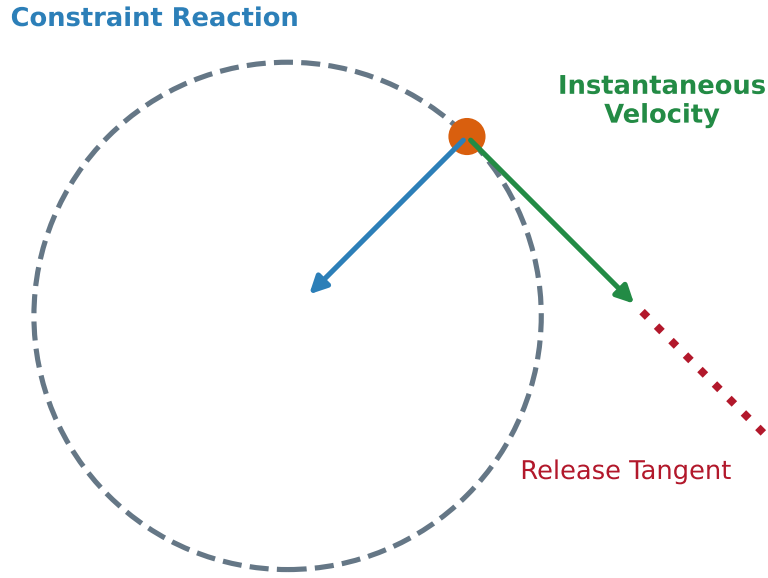


Figure 6.1: An Ideal Circular Guide Changes Acceleration; the Tangent Marks Release Velocity, Not a Gravity-Affected Future Trajectory.

## 6.2 How the Mechanism Works

### 6.2.1 Geometry and Dynamics Must Agree

For a smooth, time-independent holonomic constraint, write  $\phi(\mathbf{q}) = 0$ . Here  $\mathbf{q}$  contains configuration coordinates,  $\mathbf{J} = \partial\phi/\partial\mathbf{q}$  maps their rates into constraint rates, and dots mean time derivatives along the motion. The compatible velocity and acceleration satisfy  $\mathbf{J}\dot{\mathbf{q}} = 0$  and  $\mathbf{J}\ddot{\mathbf{q}} + \dot{\mathbf{J}}\dot{\mathbf{q}} = 0$ . The second term is essential: a key moving around a circle has inward acceleration even at constant speed.

For the coordinate convention in which generalized velocity is  $\dot{\mathbf{q}}$ , one common simultaneous solve is

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^\top \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \ddot{\mathbf{q}} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{Q} - \mathbf{h} \\ -\dot{\mathbf{J}}\dot{\mathbf{q}} \end{bmatrix}.$$

$\mathbf{M}$  is the inertia matrix;  $\mathbf{Q}$  contains declared applied generalized loads, including controls;  $\mathbf{h}$  contains the remaining bias terms under the chosen gravity convention. The generalized constraint load is  $\mathbf{Q}_c = \mathbf{J}^\top \lambda$ . A multiplier is not automatically a Cartesian force in newtons: its meaning and units depend on how the constraint equation is defined and scaled. In formulations using generalized speeds  $\mathbf{v}$  with  $\dot{\mathbf{q}} = \mathbf{N}(\mathbf{q})\mathbf{v}$ , the corresponding velocity Jacobian and dynamics must be used consistently. Here  $\mathbf{N}$  maps generalized speeds into coordinate rates.

Positive-definite  $\mathbf{M}$  and independent rows of  $\mathbf{J}$  make this fixed-state bilateral system nonsingular. It requires compatible initial position and velocity. Integrating only the acceleration equation can still accumulate position or velocity drift; numerical projection or stabilization must be audited separately from physical forces and work. These distinctions follow the [multibody formulation](#) and are developed further in the book’s forward-model discussion.

The solve is valid within its declared contact mode. A solution demanding negative cord tension, tensile ground contact, or excessive sticking friction cannot be rescued by calling its multiplier a reaction. The mode or model must change. Impacts additionally require an impulse/contact treatment; this smooth acceleration equation is not an impact law.

### 6.2.2 A Circle Checks the Signs

At the top of a one-metre circle, a two-kilogram key has required vertical acceleration  $-v^2/(1\text{ m})$ , with upward positive. Use terrestrial gravity  $g = 9.81\text{ m s}^{-2}$  and let  $\lambda$  denote the guide’s upward reaction in newtons. Newton’s law gives  $\lambda = mg - mv^2/(1\text{ m})$ . At  $v = 1\text{ m s}^{-1}$ , the required reaction is  $+17.62\text{ N}$ : an enclosing bilateral guide could push upward, but a cord from the centre could not. At  $v = 4\text{ m s}^{-1}$ , it is  $-12.38\text{ N}$ , so the required inward tension is feasible in this idealized example.

This is a manufactured sign-and-feasibility check, not a golfer parameter estimate. It shows why solving an equality and checking the permitted force direction are distinct operations.

### 6.2.3 Zero Virtual Work Is Not Always Zero Actual Work

An ideal constraint is defined through virtual work: its generalized load pairs to zero with every allowed virtual displacement at a fixed instant. That is a force–displacement pairing, not necessarily Euclidean perpendicularity in a mixed set of angular and translational coordinates. For stationary ideal constraints, the actual compatible velocity is also in the allowed tangent space, giving zero total constraint power.

A prescribed moving support changes the result. If  $\phi(\mathbf{q}, t) = 0$ , then  $\mathbf{J}\dot{\mathbf{q}} + \phi_t = 0$ , and therefore

$$P_c = \mathbf{Q}_c^T \dot{\mathbf{q}} = \lambda^T \mathbf{J}\dot{\mathbf{q}} = -\lambda^T \phi_t.$$

This can be nonzero even for an ideal constraint. The support’s drive supplies or absorbs that power. Here  $\phi_t$  is the partial time derivative at fixed configuration. The acceleration condition must also retain the full time derivative of the moving-constraint velocity condition; the autonomous formula above cannot simply be reused with its explicit-time terms omitted.

An existing manufactured example makes the accounting concrete. A two-kilogram mass follows the prescribed guide  $x(t) = \alpha t^2$ , with  $\alpha = 1\text{ m s}^{-2}$ . At one second,  $v = 2\text{ m s}^{-1}$  and  $a = 2\text{ m s}^{-2}$ . A separate applied force of  $3\text{ N}$  leaves a  $1\text{ N}$  guide reaction. Their powers are  $6\text{ W}$  and  $2\text{ W}$ , summing to the  $8\text{ W}$  kinetic-energy rate. The moving guide’s work must appear in the selected system’s ledger.

This is the same boundary issue encountered with a prescribed torso or hand path. If the support is outside the model, its reaction power is external. If its driven body is included, account for that body’s actuator, storage, and other interactions instead of treating the imposed motion as free energy.

### 6.2.4 Zero Total Constraint Power Can Hide Transfer Between Bodies

Now consider a passive ideal pin joining two bodies. At their common point, compatible velocities and equal-and-opposite connection forces give opposite one-sided powers. The sum is zero, but either body's contribution can be large. A powered wagon can pull an unpowered wagon through such a connection, raising the latter's kinetic energy without a motor on it.

For a connection that also transmits a couple, include the moment–angular- velocity pairing. Total cancellation requires the corresponding compatible relative motion; force cancellation by itself is insufficient. Slip, deformation, interface inertia, or actuation changes the ledger. An ideal rigid constraint has no elastic storage state. A compliant element replacing it may store strain energy, while damping dissipates energy.

This explains how a wrist reaction can contribute to club power without creating energy for the complete chain. A rigid foot held stationary against rigid ground has zero contact-wrench power because its contact twist vanishes, while the support still enables internal actuators to accelerate the golfer and club. The contact conditions and system boundary must be stated before either claim is applied.

### 6.2.5 Three Different Questions About Reaction Uniqueness

**A forward model with known inputs.** Given its complete state, applied loads, inertia, and independent bilateral constraints, the nonsingular system above determines acceleration and multipliers uniquely. Having two hands or a closed loop does not, by itself, make that solve indeterminate. The pinned provider's planar two-hand model has seven coordinates and four independent scalar point-coincidence constraints; it rejects rank loss rather than silently choosing a different allocation.

**A redundant description.** If rows of  $\mathbf{J}$  repeat the same restriction, different multipliers can produce the same generalized load. For example, with rows  $(1, 0)$  and  $(2, 0)$ , multipliers  $(-2, 0)$  and  $(-0.4, -0.8)$  both give  $\mathbf{J}^T \lambda = (-2, 0)$ . The latter has smaller Euclidean multiplier norm. Rescaling the second equation changes the minimum-norm numbers again without changing the physical restriction. For positive-definite inertia and compatible redundant acceleration equations, acceleration can remain unique while multipliers are not. Inconsistent equations instead have no exact solution. A pseudoinverse does not turn an inconsistent residual into physical evidence.

**Inference from incomplete observations.** Measuring the club's resultant wrench is a different problem. It supplies fewer conditions than a detailed description of two hand-contact wrenches, and the required biological inputs are usually unknown. A complete model can provide additional equations or constitutive assumptions; sensors can provide additional observations. Identifiability then depends on what those additions actually determine. Neither a solver's preferred allocation nor a club trajectory alone measures muscle recruitment. Chapter 25 follows this inference problem inward from hand loads to joint demand and musculotendon forces.

### 6.2.6 Opposing Forces Do Not Necessarily Cancel the Wrench

Let hand-force increments be  $+\mathbf{f}$  and  $-\mathbf{f}$  at fixed club points  $\mathbf{r}_L$  and  $\mathbf{r}_R$ . Their net-force change is zero, but their moment change about the same origin is

$$\Delta \mathbf{M} = (\mathbf{r}_L - \mathbf{r}_R) \times \mathbf{f}.$$

For points 0.2 m apart along the club’s local  $x$  axis, opposite 1 N transverse forces produce a 0.2 N m couple. Opposite axial increments along the separation vector instead give zero complete-wrench change. Contact couples, if permitted, must also be included when testing a wrench-neutral allocation.

Such an algebraic load change need not be feasible under friction, strength, joint, or constitutive constraints. Nor does zero resultant wrench guarantee zero deformation or identical tissue demand. It leaves the rigid club’s instantaneous Newton–Euler load unchanged under the stated geometry and other loads; it does not by itself establish stability or a beneficial squeeze. Two-foot load inference has analogous limits. Separate six-axis force plates can add bilateral wrench information. Ordinary pressure-only insoles measure normal pressure; their shear and full-wrench estimates require additional information or modeling.

### 6.2.7 Radial Force, Moment, and Power Ask Different Questions

For a straight rigid club, a wrist force directed along the line from wrist to club centre of mass has zero moment about that centre. A transverse component has a lever arm and contributes to the rotational balance. The complete acceleration still depends on translation, other loads, any applied couple, and the connected bodies. A component’s name does not isolate a causal driver.

Force power uses the force’s application-point velocity. A force radial to the club can therefore perform hand-point work when that point moves along it. By contrast, the proximal centripetal term in the provider’s fixed-hub planar decomposition is radial to the *arm* and perpendicular to the wrist velocity, so that particular term has zero wrist-force power. Those two uses of “radial” are not interchangeable. The signed dot product resolves the issue.

### 6.2.8 Remove an Input Without Counting the Drift Twice

A pointwise counterfactual copies the full state and time, removes specified commands, and solves again with the other declared loads and constraints retained. The resulting zero-command reaction is not a force-free wrist, a measurement of passive tissue, or a statement about human effort.

For a fixed smooth contact mode and dynamics affine in two declared inputs (proximal  $P$  and distal  $D$ ), let  $\mathbf{y}_{PD}$  denote the response with both inputs,  $\mathbf{y}_P$  and  $\mathbf{y}_D$  the respective single-input cases, and  $\mathbf{y}_0$  the zero-input case. The response can include acceleration and reactions. Then

$$\mathbf{y}_{PD} = \mathbf{y}_0 + (\mathbf{y}_P - \mathbf{y}_0) + (\mathbf{y}_D - \mathbf{y}_0) = \mathbf{y}_P + \mathbf{y}_D - \mathbf{y}_0.$$

Simply adding the two single-input runs counts shared gravity, velocity, external-load, and constraint-bias effects twice. For endpoint position  $\mathbf{x}(\mathbf{q})$ , the kinematic Jacobian  $\mathbf{J}_x = \partial \mathbf{x} / \partial \mathbf{q}$  gives  $\mathbf{a}_x = \mathbf{J}_x \ddot{\mathbf{q}} + \dot{\mathbf{J}}_x \dot{\mathbf{q}}$ . That changing-Jacobian contribution belongs in the fixed-state baseline too. The identity does not generally survive a contact-mode change, a nonlinear input map, or comparison of different states.

A forward killswitch begins from the same state but removes a command over an interval. The trajectories then diverge, and later differences include that changed history. Retaining an open-loop

command history is different from retaining a feedback policy whose outputs change with the state. Report the chosen continuation.

### 6.2.9 Compliance Is a Model With Its Own Dynamics

A force–extension law stores strain energy; a static stiffness value is not a time delay. Inertia, damping, relaxation, activation, and wave propagation can introduce timescales in a more complete model. Specify which mechanism is present instead of attributing all timing to compliance.

Making a penalty spring stiffer also creates faster dynamics. For a one-kilogram undamped mass with initial extension  $F_0/k$  and zero velocity, where  $F_0 = 1$  N, the extension is  $\delta(t) = (F_0/k) \cos(\sqrt{k/m} t)$ . Its amplitude and initial stored energy  $F_0^2/(2k)$  tend to zero as  $k$  increases, yet the spring-force amplitude remains 1 N. Vanishing displacement alone therefore does not establish uniform convergence of reaction forces to the rigid model.

A useful rigid/compliant comparison declares initial compatibility, damping, contact law, integration accuracy, and the observable whose convergence is claimed. Resolve the fast modes or justify a reduced description. Track constraint residuals and numerical energy changes alongside the physical work ledger. Successful convergence of one observable is not human validation of the chosen compliance.

#### 6.2.10 What the Evidence Establishes

**Model Result:** The pinned double-pendulum source derives nonzero wrist reactions after specified torque removal and separates force magnitude from signed wrist-force power. Its forward two-hand extension reports a finite negative-couple persistence interval after all six applied commands are cut. Those are declared model results, not new simulations performed for this chapter, universal persistence, or measurements of biological passivity.

**Human Evidence:** Koike’s 2016 instrumented-grip report describes a professional instructor and a design with twelve sets of strain gauges (Koike 2016). Choi and Park’s 2020 study used a six-axis sensor between sections of a customized grip in nine right-handed male professional golfers, combined with motion capture and inverse dynamics (Choi and Park 2020). The internal sensor wrench is measured; the remaining joint loads are inferred through the model. The latter study reports limitations including altered instrument mass, hand-contact coupling, foam-ball impact, and rigid-segment assumptions. Its measurements improve load inference without identifying individual muscle forces or establishing an optimal intervention.

**Hypothesis:** An achieved late-downswing state may permit favorable club acceleration or power through connection forces while selected distal commands are small or opposing. Specify the observable, sign, interval, load limits, and alternative models before testing it. Large reactions or a negative couple alone do not establish that hypothesis.

#### 6.2.11 Reader Lab and Ways the Claim Could Fail

Run all-input, proximal-only, distal-only, and zero-input cases from one copied state. Compare the reactions and endpoint accelerations using the shared-drift identity above. Check the physical force directions and contact limits before accepting a branch. Then integrate explicitly defined continuations and compare power, work, delivery events, and trajectory divergence.

Change geometry through constraint-consistent states. A force component's projection may reverse under a specified geometric transformation, but the recomputed reaction can also change; an arbitrary angle reversal need not reverse total transfer. Changing coordinate signs consistently must leave physical predictions unchanged. Removing a force lever arm has a sharper null prediction: its moment contribution vanishes, while separately applied couples must remain separately accounted for.

A claimed mechanism is undermined by failure of its registered counterfactual effect, force/moment balance, work ledger, admissible contact conditions, or observable convergence. Agreement between independent solvers strengthens the numerical result only to the extent their methods and assumptions are actually independent. Human relevance additionally needs synchronized load and motion measurements, uncertainty, participant-held-out predictions, and evidence that the model represents the tested task.

### 6.2.12 One Sentence to Retain

A connection can transmit force and transfer energy between bodies without being an independent energy source; its physical feasibility, power, and inferred biological meaning are three separate questions.

## 6.3 Where the Picture Breaks

The circular guide is a deliberately simple example. A human wrist has multiple degrees of freedom, deformable tissues, muscle action, and distributed contact; the shoulder girdle and grip can move. Ideal point constraints omit those features. Their multipliers cannot be read directly as tissue stresses, muscle forces, injury estimates, or coaching instructions.

Keep three boundaries visible: the permitted motion and contact mode; the complete system whose energy is counted; and the measurements or assumptions used to identify loads. A plausible mechanism can survive those checks and remain an untested hypothesis about how a particular golfer produces it.

**Go Deeper:** Read the pinned [wrist-reaction decomposition](#), [forward two-hand constraints](#), and [ground-reaction attribution](#). Chapters 2, 17, and 25 connect these reactions to system boundaries, prescribed versus moving supports, and biological load inference.

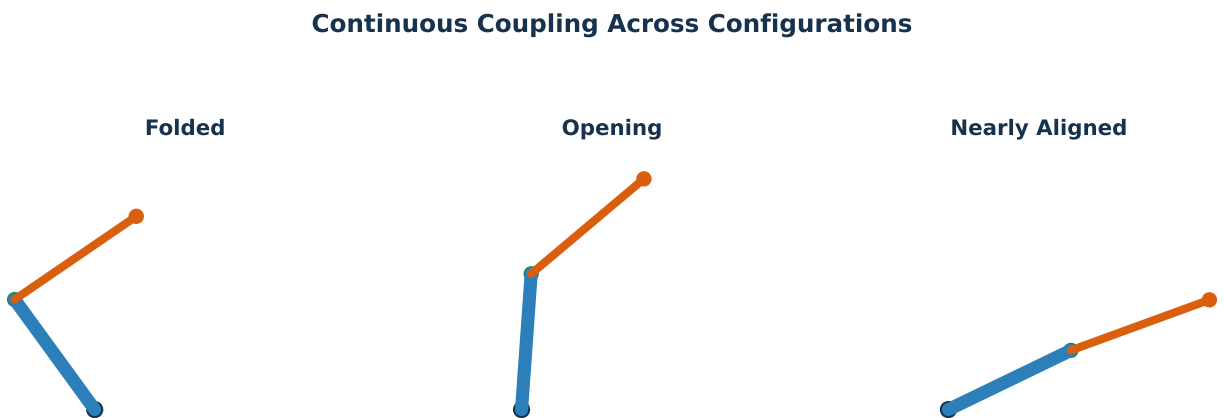
## Chapter 7

# From One Pendulum to Two

### 7.1 A Concrete Picture

A single playground swing has one dominant rhythm. Add a loose second link at the seat and the motion changes character. The outer link can trail, fold, accelerate relative to the first, and feed reactions back into it. There is no moment when the first link acts alone and then politely hands control to the second. Both influence one another throughout.

The golf double pendulum replaces the inner link with an arm-like segment and the outer link with a club-like segment. It is deliberately spare. That is its strength: it is the smallest familiar model that contains changing geometry, inertial coupling, interaction force, relative-angle retention, and distal release. The diagram below specifies configurations, not a measured or simulated downswing; geometry alone supplies neither velocities nor forces.



Illustrative Geometry; Not a Simulated Trajectory

Figure 7.1: Illustrative Folded, Opening, and Nearly Aligned Configurations; These Are Not Frames From a Simulated Swing.

## 7.2 How the Mechanism Works

### 7.2.1 Angles and Torques Must Use Compatible Coordinates

Start with two rigid planar links, a fixed shoulder pivot, ideal revolute joints, and no shaft spring or joint friction. Let  $\theta_1$  and  $\theta_2$  be the **absolute** arm and club angles, measured counterclockwise from downward vertical. A shoulder actuator applies  $\tau_s$  to the arm. An internal wrist actuator applies  $+\tau_w$  to the club and  $-\tau_w$  to the arm. The equations are

$$\mathbf{M}_\theta \ddot{\theta} + \mathbf{h}_\theta = \begin{bmatrix} \tau_s - \tau_w \\ \tau_w \end{bmatrix}.$$

The right side contains **generalized loads**, not simply a list of actuator names. Write the relative wrist angle as  $\phi = \theta_2 - \theta_1$  and use  $\mathbf{q} = (\theta_1, \phi)^T$  instead. Then  $\theta = \mathbf{A}\mathbf{q}$ , with  $\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$ . Virtual work gives  $\mathbf{Q}_q = \mathbf{A}^T \mathbf{Q}_\theta = (\tau_s, \tau_w)^T$ ; kinetic energy gives  $\mathbf{M}_q = \mathbf{A}^T \mathbf{M}_\theta \mathbf{A}$ . Either convention works when every term is transformed. In particular, actuator power must agree:

$$P = \tau_s \dot{\theta}_1 + \tau_w (\dot{\theta}_2 - \dot{\theta}_1) = \tau_s \dot{q}_1 + \tau_w \dot{q}_2.$$

For absolute rates  $(2, 5)$  rad/s and actuator torques  $(3, 4)$  N m, this is 18 W. Multiplying the actuator torques directly by the two absolute rates gives 26 W: it omits the wrist actuator's  $-8$  W contribution on the arm. A bookkeeping choice has changed the apparent work without changing the machine.

### 7.2.2 Relative Angle Changes the Coupling

Let  $m_i$  and  $I_i$  be each link's mass and center-of-mass inertia,  $l_1$  the arm length, and  $c_i$  the distance from each link's proximal joint to its center of mass. Define  $a = I_1 + m_1 c_1^2 + m_2 l_1^2$ ,  $b = m_2 l_1 c_2$ , and  $c = I_2 + m_2 c_2^2$ . For  $\delta = \theta_1 - \theta_2 = -\phi$ , kinetic energy is

$$T = \frac{1}{2} a \dot{\theta}_1^2 + b \cos \delta \dot{\theta}_1 \dot{\theta}_2 + \frac{1}{2} c \dot{\theta}_2^2, \quad \mathbf{M}_\theta = \begin{bmatrix} a & b \cos \delta \\ b \cos \delta & c \end{bmatrix}.$$

Even a diagonal entry includes transported distal mass; it is not simply one link's own center-of-mass inertia. The bias  $\mathbf{h}_\theta$  contains gravity and velocity-dependent terms. In this convention the latter are  $(b \sin \delta \dot{\theta}_2^2, -b \sin \delta \dot{\theta}_1^2)^T$ . Thus coupling can persist when  $\cos \delta = 0$ : a zero off-diagonal entry at one configuration does not make two independent pendulums.

For example, take  $(a, b, c) = (2, 1, 1)$  kg m<sup>2</sup>,  $\delta = +\pi/2$ , absolute rates  $(2, 0)$  rad/s, and no gravity or actuator torque. The mass matrix is diagonal at that instant, but the club acceleration is 4 rad/s<sup>2</sup> because the velocity-dependent bias is nonzero. Deleting an off-diagonal entry without consistently changing the kinetic energy, bias and force mapping is not a physical decoupling experiment.

### 7.2.3 Carrying Is Not Wasted Motion

Early in many downswing patterns, the arm and club rotate more nearly together while the wrist angle remains relatively closed. The hand translates, and the club center of mass can gain motion before

large relative club rotation. Calling this only “holding lag” misses the transport: the distal body is being carried by a moving base. Whether its energy rises still requires a work and energy balance, not a visual impression.

As geometry changes, coupling can favor rapid relative rotation. Distinguish the wrist rate  $\dot{\phi} = \dot{\theta}_2 - \dot{\theta}_1$  from the club’s absolute angular velocity  $\omega_c = \dot{\theta}_2 \hat{\mathbf{z}}$ . If  $H$  is the hand and  $C$  the clubhead, rigid-body kinematics gives  $\mathbf{v}_C = \mathbf{v}_H + \omega_c \times \mathbf{r}_{C/H}$ . These are **vectors**; their scalar speeds do not generally add. Neither relative wrist rate nor absolute club rate alone specifies clubhead speed.

The visible release is not energy appearing at the wrist. In this rigid benchmark, stored mechanical energy is kinetic or gravitational potential energy. Boundary work changes the total; internal exchanges redistribute energy between the links and between kinetic and potential forms. Elastic shaft storage belongs to a model that actually includes deformation. The rigid-link calculation cannot silently borrow that mechanism.

#### 7.2.4 The Club Pushes Back

Action-reaction runs both ways. The arm affects the club, and the club’s motion affects shoulder and wrist reactions. Depending on the state and inputs, the club can slow the arm or demand additional proximal torque. A “passive” club is not dynamically invisible; specify whether passive means zero commanded couple, a spring-and-damper joint, or an absent actuator.

Resolve the hand force along and transverse to  $\mathbf{r}_{G/H}$ , where  $G$  is the club center of mass. **Both components applied at  $H$  have zero moment about  $H$ .** The transverse component can have a moment about  $G$ . The hand accelerates, so it cannot be treated as a stationary inertial pivot. For a rigid planar club and body-fixed hand point,

$$\sum M_H = I_H \ddot{\theta}_2 + m_2 (\mathbf{r}_{G/H} \times \mathbf{a}_H)_z, \quad I_H = I_2 + m_2 \|\mathbf{r}_{G/H}\|^2.$$

The left side includes all external moments about  $H$ , including gravity and any applied couple. The additional acceleration term explains how the club can rotate without an applied hand couple. Contact power on the club is  $\mathbf{F}_H \cdot \mathbf{v}_H + \tau_w \dot{\theta}_2$ ; the wrist actuator’s **whole-system** power instead uses the relative rate. Zero moment about a force’s application point does not make that force workless.

A manufactured example makes this concrete. Omit gravity and take  $m_2 = 2$  kg,  $I_2 = 0.5$  kg m<sup>2</sup>,  $\mathbf{r}_{G/H} = (1, 0)$  m, zero angular velocity, and  $\mathbf{a}_H = (0, 1)$  m/s<sup>2</sup>. With no applied couple,  $\ddot{\theta}_2 = -0.8$  rad/s<sup>2</sup> and  $\mathbf{F}_H = (0, 0.4)$  N. The force’s moment about the COM is  $-0.4$  N m, while its moment about the hand is zero. If  $\mathbf{v}_H = (0, 3)$  m/s, it supplies 1.2 W. Force, moment reference, acceleration and power describe different aspects of one consistent balance.

#### 7.2.5 Limiting Cases Need a Declared Family

If distal mass and rotary inertia shrink together while accelerations and external loads scale consistently, their inertial reactions can vanish. Holding a finite wrist couple while inertia vanishes is different. For a fixed hand, omit gravity and begin at rest. With  $m = \epsilon$  kg,  $I_G = \epsilon$  kg m<sup>2</sup>, a one-meter COM offset and a 1 N m couple, angular acceleration is  $1/(2\epsilon)$  rad/s<sup>2</sup> and tangential hand force remains 0.5 N. Acceleration diverges: the endpoint is not a regular massless dynamic body. The actuator’s

equal-and-opposite couple on its support also remains finite; this is not a moment supplied by an ideal free hinge.

Likewise, zero link length need not remove a retained rotary inertia or actuator. A singular mass matrix requires an appropriately reduced model, not division by zero inside the original equations. Physical limits specify which masses, dimensions, inertias, loads and states are held fixed.

An already locked wrist gives a compound rigid-body model. **Locking a moving wrist is an event:** an impulsive constraint can remove kinetic energy. For a manufactured generalized inertia  $\text{diag}(2, 1)$  and initial absolute rates  $(1, 0)$ , a perfectly inelastic wrist lock with no other generalized impulse produces rates  $(2/3, 2/3)$ ; kinetic energy falls from 1 to  $2/3$  in consistent SI units. Momentum balance includes the locking impulse. This is not an energy error in an otherwise conservative integrator.

For smooth motion, conserve total mechanical energy only when external power and dissipation are absent and every conservative storage term is included. A stationary system can accelerate under gravity or a preloaded spring; zero actuator command alone does not imply equilibrium. A rigid, gravity-free, unforced model at rest should remain at rest. State the version being tested.

### 7.2.6 Zero Command Is Not the Same as Underactuation

Setting the wrist command to zero does not freeze the wrist angle. Coupling and gravity can still move it. **Underactuation** describes insufficient independent actuator directions for the model's degrees of freedom, not an instant when an available actuator receives zero command. A shoulder-only two-link model is underactuated. A model with two independent actuators still has two actuator directions when its wrist command is temporarily zero.

A related distinction separates actuator command from net torque. For a single pendulum with fixed pivot inertia, the same **net** moment gives the same angular acceleration. The same actuator command can give different acceleration when the opposing gravity moment changes with configuration. Speed depends on the initial state and the intervening acceleration history.

### 7.2.7 A Slow-Motion Downswing Is a Conditional Story

In a chosen simulation, proximal torque may first accelerate the arm while the club lags. The hand force already moves the club through space. Later, relative rotation can increase rapidly as geometry and velocity-dependent terms change. Direct wrist torque can add work, resist release or change the proximal motion through its reaction. Those mechanisms overlap; no universal instant separates carrying from transfer.

An earlier wrist command changes both the current power and the later state. Equal command amplitudes do not imply equal work. Compare unconstrained actuator schedules and work-matched schedules as different experiments, because enforcing equal work may itself alter the controls. Finite actuator rise time, the starting state and the impact event also matter. Maximizing speed at impact is not maximizing the highest speed reached anywhere.

**Model Result:** Sharp's arm-club and shoulder-arm-club analyses produced different torque interpretations despite fitting historical swing records. His three-link optimization improved simulated impact speed mainly through changed arm-torque timing, without a distinct hold-back phase in the reported

net wrist torques ([Sharp 2009](#)). This is a conditional computational comparison, not a universal instruction to delay or drive the wrists.

**Human Evidence:** The model’s arm and club coordinates are abstractions. Putnam’s synthesis discusses motion-dependent interactions in kicking and throwing and cautions against generalizing their timing without examining the interactions ([Putnam 1993](#)). Neither that synthesis nor fitting historical golf kinematics identifies a unique human torque or muscle strategy. This chapter’s manufactured examples are not participant measurements.

**Practical Interpretation:** Delayed visible release need not mean delayed energy input. Track when work enters, how segment energy changes and which geometry produces the final velocity. The two-link model supplies no direct instruction about which muscle a golfer should activate.

### 7.2.8 Tests That Make a Toy Model Useful

For regular positive-inertia parameters, the mass matrix should be symmetric and positive definite. Equivalent coordinate formulations should predict the same physical motion, energy and power. Conservative smooth trials should maintain energy within a declared numerical tolerance; timestep refinement should converge the relevant output. Check event handling separately from smooth integration and test whether compared implementations share assumptions.

Then come causal controls. At a copied complete state, change one input and compare instantaneous acceleration. Integrate a separate forward branch to study its consequences over time. Those branches soon have different states; their later difference is not the original pointwise torque contribution. Record joint-stop modes and any actuator or spring states when they exist.

If a benchmark containing the required ingredients contradicts a proposed mechanism, investigate both assumptions and implementation. Added relevant physics may genuinely change the result. A successful simple model establishes a mechanical possibility; richer models and measurements test its applicability.

### 7.2.9 Reader Lab: Discover Coupling Yourself

Choose one parameter set and torque schedule. Record coordinates, complete initial state, input limits, gravity, joint laws, solver settings and the output event. Begin with hanging links at rest and apply a brief shoulder pulse with zero wrist command. Plot both absolute angles, relative angle, hand reaction, clubhead velocity vector and segment energies. Repeat from different relative angles; a changed distal response can arise without changing its actuator input.

Next copy a moving state and set commanded torques to zero. In the ideal conservative benchmark, motion and segment-energy exchange continue while total mechanical energy stays constant. In a separate pointwise calculation, set velocities to zero while retaining positions and other states. The difference isolates all velocity-dependent effects present in that model, including damping if added, not exclusively inertial coupling. It is an attribution calculation, not a golfer stopping instantaneously.

Finally sweep distal-torque onset and amplitude, then add finite rise time and parameter uncertainty. Show a surface rather than only the best trace. Declare whether “better” means speed, impact alignment or another task measure. Repeat with finer timesteps and perturbed initial states. A sign that changes readily belongs in a sensitivity map rather than a universal timing law.

### 7.2.10 What Carries Into Higher-Order Models

Conservation laws and coordinate-consistent accounting survive extensions. The sign, timing and benefit of a particular intervention need not. A moving base, spatial motion, separated hands and shaft flexibility add mechanisms and observables. Force allocation hidden by a net wrench, for example, becomes a separate question when two contacts replace the single wrist connection.

The useful insight is that distal motion depends on the linked state and all relevant inputs, not only distal effort. Later chapters must preserve the definitions of state, force, power and intervention while testing which conclusions survive. A reversal with added physics can be an informative model boundary; it is not automatically a coding error or an excuse to discard the transparent benchmark.

## 7.3 Where the Picture Breaks

The two-link picture encourages the “whip” metaphor. A whip is flexible and distributed, with wave propagation; a rigid double pendulum has discrete links and joints. Both can exhibit distal speed amplification, but they need not achieve it through the same mechanism.

The benchmark also collapses two arms and wrists into one connection and fixes a plane and shoulder pivot. It cannot resolve grip-force allocation, scapular muscle activity, general three-dimensional clubface control or injury loading. Those questions require additional variables, measurements and independent validation. A convincing animation cannot supply them.

**Go Deeper:** Read the monograph’s [double-pendulum mechanics](#), [numerical mechanism audit](#), and Putnam’s linked-segment synthesis ([Putnam 1993](#)).

## Chapter 8

# The Summation-of-Speed Story and Its Limits

### 8.1 A Concrete Picture

Walk forward inside a train moving at 20 m/s. If you walk at 2 m/s relative to the carriage, your ground velocity is 22 m/s along the track. Walk backward at the same speed and it is 18 m/s. Velocities add as vectors; their magnitudes add only when the vectors point in the same direction.

A clubhead likewise moves because its base moves and because the club changes orientation. The geometry is less forgiving than a straight train aisle: rotating segments continually change the directions of their contributions. A large angular speed does not, by itself, tell us how much endpoint speed it adds. Nor does an ordered series of speed peaks explain the forces that made the movement possible.

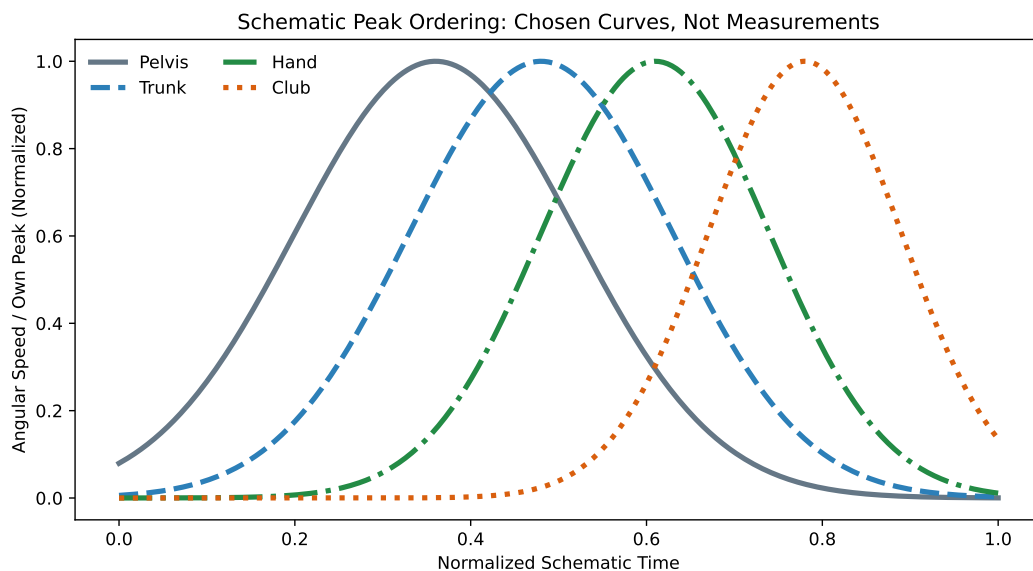


Figure 8.1: Schematic Peak Ordering With Overlapping Motion.

The figure uses chosen Gaussian curves, each scaled to its own peak. It shows what an ordering could look like. It contains no human measurements, equations of motion, calculated coupling, or contribution analysis. Equal normalized heights do not mean equal segment speeds.

## 8.2 How the Mechanism Works

### 8.2.1 A Velocity Identity With Explicit Assumptions

Choose two material points on a rigid club: a grip point  $H$  and a head point  $C$ . Express their velocities and the club's angular velocity in the same inertial frame. With  $\mathbf{r}_{C/H} = \mathbf{x}_C - \mathbf{x}_H$ ,

$$\mathbf{v}_C = \mathbf{v}_H + \omega_{club} \times \mathbf{r}_{C/H}.$$

Here  $\omega_{club}$  is the club's **absolute angular velocity**, not its angular velocity relative to the forearm. The grip point can be identified with a hand point only under a compatible no-slip contact assumption. The identity holds in three dimensions; it does not require planar motion.

For a deforming shaft, write  $\mathbf{r}_{C/H} = R(t)\rho(t)$  where  $R(t)$  rotates coordinates from a chosen moving club frame into the inertial frame and  $\rho(t)$  is the displacement in that moving frame. Differentiation adds  $R\dot{\rho}$  to the transport and chosen-frame rotation terms. A slipping hand point also needs its relative motion accounted for. A single rigid-body formula cannot silently include shaft deformation or changing grip contact.

The rotational term has magnitude  $\|\omega\|r_{\perp}$ , where  $r_{\perp}$  is the perpendicular distance from the angular-velocity axis through  $H$  to  $C$ . The familiar  $v = r\omega$  uses that radius, not an arbitrary segment length. A long club can amplify endpoint motion for a given angular rate; kinematics alone says nothing about the torque, energy, or controllability required to achieve it.

### 8.2.2 A Two-Link Calculation That Exposes a Common Error

Take two links, each one metre long, in a plane with a fixed base. Let  $q_1$  be the arm's absolute angle and  $q_2$  the club's angle relative to the arm, both positive counterclockwise. The head position is

$$\mathbf{x}_C = \begin{bmatrix} \cos q_1 + \cos(q_1 + q_2) \\ \sin q_1 + \sin(q_1 + q_2) \end{bmatrix}.$$

At  $q_1 = 0$ ,  $q_2 = \pi/2$ ,  $\dot{q}_1 = 2$  rad/s and  $\dot{q}_2 = 3$  rad/s, the hand velocity is  $(0, 2)$  m/s. The club's absolute angular rate is  $2 + 3 = 5$  rad/s, giving a rotational velocity  $(-5, 0)$  m/s and a total  $\mathbf{v}_C = (-5, 2)$  m/s. Substituting the relative rate of 3 rad/s into the rigid-body formula would incorrectly give  $(-3, 2)$  m/s. It loses the rotation of the club carried by the arm.

Differentiating the head position, or checking a small central time difference, recovers  $(-5, 2)$ . This is a geometric check on the convention, not a simulation of a golfer or a demonstration of energy transfer.

### 8.2.3 Joint Contributions Are a Declared Decomposition

For endpoint position  $\mathbf{x}_C(\mathbf{q})$ , the Jacobian gives  $\mathbf{v}_C = J(\mathbf{q})\dot{\mathbf{q}}$ . Include moving-base coordinates among  $\mathbf{q}$ , or add their transport explicitly. If position depends explicitly on time, its partial time derivative is another term. Generalized speeds that differ from  $\dot{\mathbf{q}}$  require the corresponding mapping.

In the two-link example the Jacobian columns at that posture are  $(-1, 1)$  and  $(-1, 0)$ . Their rate-weighted contributions are  $(-2, 2)$  and  $(-3, 0)$  m/s. These sum to the same head velocity as the hand-plus-club decomposition, but allocate it differently. The first joint rotates both links; hand translation alone is not its entire Jacobian contribution.

For any declared vector decomposition  $\mathbf{v}_C = \sum_i \mathbf{v}_i$  with nonzero total velocity, a useful signed speed partition is

$$c_i = \frac{\mathbf{v}_C}{\|\mathbf{v}_C\|} \cdot \mathbf{v}_i, \quad \sum_i c_i = \|\mathbf{v}_C\|.$$

These projections can be negative. Keep the posture and arm rate above, but set  $\dot{q}_2 = -5$  rad/s. The contributions become  $(-2, 2)$  and  $(5, 0)$  m/s; total velocity is  $(3, 2)$  m/s. Their projections are  $-2/\sqrt{13}$  and  $15/\sqrt{13}$  m/s, summing to  $\sqrt{13}$  m/s. Large component magnitudes would overestimate the actual speed if simply added.

The partition depends on the selected point, coordinates, reference motion and choice of terms. A common orthogonal rotation of the axes leaves these dot products unchanged; changing the decomposition is a different operation. At zero endpoint speed the unit direction, and therefore this partition, is undefined.

A negative projection does **not** establish negative joint power. Even the translational kinetic energy of a point mass contains cross terms:

$$\frac{1}{2}m \left\| \sum_i \mathbf{v}_i \right\|^2 = \frac{1}{2}m \sum_i \|\mathbf{v}_i\|^2 + m \sum_{i < j} \mathbf{v}_i \cdot \mathbf{v}_j.$$

For the opposing example with  $m = 1$  kg, total energy is 6.5 J, whereas adding the two separate squared-speed energies gives 16.5 J. The cross term is  $-10$  J. A whole golfer also has distributed mass and rotational energy. Speed projections are neither segment energy shares nor causal allocations of the work that created the state.

### 8.2.4 Peak Time Is Not Transfer Time

A smooth signed scalar velocity component has zero derivative at an interior extremum. For nonzero three-dimensional angular velocity, however,

$$\frac{d\|\omega\|}{dt} = \frac{\omega \cdot \alpha}{\|\omega\|}, \quad \alpha = \frac{d\omega}{dt}.$$

A smooth interior angular-speed peak requires  $\omega \cdot \alpha = 0$ , not  $\alpha = \mathbf{0}$ . For a dimensionless manufactured example near  $t = 0$ , take  $\omega(t) = (1 - t^2)(\cos t, \sin t, 0)$ . Its magnitude reaches 1 at  $t = 0$  while

$\alpha(0) = (0, 1, 0)$ : the rotation axis is changing direction. A stationary derivative alone does not prove a maximum; endpoints, plateaus, noise and nonsmooth signals need separate event rules.

Report whether a peak belongs to an absolute segment angular-speed norm, a relative joint rate, or an anatomical component. Merely re-expressing the same physical vector in orthonormal axes cannot change its norm or its peak time. Switching from an absolute rate to a relative joint rate, or selecting a component, can change the reported peak time. Filtering, differentiation, time normalization and event detection can also alter the reported ordering; near ties need uncertainty intervals rather than a forced ranking.

Power answers a different question. For applied force  $\mathbf{F}$  and moment  $\mathbf{M}$  at a declared rigid-body contact point,  $P = \mathbf{F} \cdot \mathbf{v} + \mathbf{M} \cdot \boldsymbol{\omega}$ , with moment and velocity referred consistently to that point. For a rotational joint, net joint power uses the joint moment and compatible relative angular velocity. Interface power, net joint power, segment energy change and external work are related ledgers, not interchangeable labels. Integrate the powers over a declared interval and compare the resulting work with changes in stored energy to test the account. Angular-speed peak times cannot close it.

### 8.2.5 What the Human and Model Evidence Establishes

**Human Evidence:** Tinmark and colleagues studied 45 golfers: 11 male tournament professionals, 21 male elite amateurs and 13 female elite amateurs. They recorded pelvis, upper-torso and hand motion at 240 Hz and calculated resultant angular-speed magnitudes. Their reported ordering and increasing peaks occurred across the partial and full-shot conditions studied (Tinmark et al. 2010). This is evidence about those measured segments and definitions. It does not supply a club-energy ledger, prove that every individual trial follows one order, or experimentally identify an optimal torque schedule. The authors' interpretation involving interaction torques goes beyond the directly reported kinematic observations.

**Mechanistic Evidence From Other Tasks:** Putnam's 1991 analysis used four subjects performing kicking, walking and running, and examined thigh-leg interaction terms (Putnam 1991). It supports investigating motion-dependent interaction, not transferring a lower-limb result unqualified to golf. The later striking-and-throwing review synthesizes descriptions and explanations (Putnam 1993). Milburn's golf paper presents a double-pendulum analysis of the downswing (Milburn 1982); its model conclusions should be stated with their assumptions rather than counted as independent proof of a universal human rule.

**Model Result:** The pinned uncertainty study uses a planar moving-base, two-hand, flexible-club model with bounded surrogate actuators. Delivery is evaluated at a fixed 0.24 s, without a ball collision; its speeds are not calibrated full-swing human performance. All eight preselected control programs are nondominated across five objectives on six synthetic held-out cases: lower-tail delivery speed, planar face/path error, upper-tail hand force, squared-torque effort proxy, and speed dispersion. That means no other member of this finite comparison set is at least as good in every objective and strictly better in one. It does not imply every possible improvement incurs a cost, or that these programs form the continuous optimum over all controls.

Speed alone selects `early_restrain`: its tenth-percentile delivery speed is highest among these candidates, about 4.706 m/s. This ranking does not settle a choice across all five objectives. The saved comparisons do not analyze segment-peak sequences, validate human strategies, or establish that

different human sequences achieve the same result.

**Hypothesis:** A useful strategy may arrange geometry and motion so that the club's rotational contribution aligns favorably with grip-point velocity near the chosen impact event, while satisfying face, path, load and robustness requirements. Whether that strategy improves performance requires trajectory and intervention evidence. Instantaneous vector alignment is not an optimization result or a prescription for when a golfer should release the club.

### 8.2.6 A Sequence as a Testable Constraint

A proximal peak can precede a distal peak because proximal drive changes, because interaction loads change, or because the measured vector changes magnitude or direction. Several mechanisms can coexist. Proximal slowing also need not mean that every associated joint has negative power. The links remain coupled while their speeds rise and fall; a peak does not disconnect one segment and hand its motion to the next.

Build the explanation around synchronized observations: segment and joint kinematics, external and grip wrenches, declared interface powers, segment energies, and uncertainty. Define transition and impact events explicitly. Then ask whether a proposed mechanism predicts the force direction, moment sign and energy changes as well as the peak order. A model with the right order but the wrong power history has not explained that history.

If the arm-dominant and wrist-dominant proposals reproduce similar motion, bilateral wrenches and motion can constrain their external mechanical allocations. Activation measurements can add physiological evidence, but they do not uniquely identify muscle forces or neural commands. More measurements are useful only insofar as they discriminate the competing predictions; see Section 21 for identifiability.

### 8.2.7 Reader Lab: Separate Motion From Its Causes

Reproduce the two-link velocity with a central finite difference of position. Plot its Jacobian contributions tip-to-tail, then use the opposing-rate example to calculate signed projections and the energy cross term. Change posture while holding rates fixed. The arrows change because geometry participates in the mapping, even before a dynamics model has been introduced.

Next, plot the manufactured angular-velocity vector above and its norm. At the norm peak, check both the acceleration vector and its dot product with velocity. This prevents the zero-acceleration interpretation from entering a subsequent sequence analysis. With measured data, repeat peak detection across justified filter settings and report ambiguous orderings instead of selecting a preferred plot.

Finally, in a forward dynamics model, copy a state and remove a finite proximal torque while retaining positions, velocities and the same constraints. The instantaneous endpoint velocity is unchanged. Acceleration can change through the constrained equations of motion, and the subsequent trajectories can diverge; it is not guaranteed that every endpoint acceleration component changes. Impulses, contact changes and prescribed-motion constraints require separate handling. An existing velocity contribution is distinct from continuing actuator power, and a finite-time response is distinct from the copied-state comparison.

### 8.2.8 Responsible Practical Translation

A sequence plot can locate a phase worth investigating and test whether a model reproduces a measured coordination feature. It becomes more informative when paired with within-person repeatability and predicted responses to perturbation. A reliably successful counterexample would refute a claim that one exact ordering is necessary under those conditions. Observing the order alone would not establish sufficiency or reveal its cause.

For coaching or equipment analysis, evaluate the whole declared objective and the uncertainty of the change. Do not turn population peak times into commands to stop one segment before starting another. State-based timing, such as an event near a particular arm-club configuration, is a candidate for testing across tempos rather than an established replacement rule.

## 8.3 Where the Picture Breaks

The train analogy explains instantaneous velocity addition. It leaves out the reaction forces, changing geometry, elastic motion and control that determine a swing trajectory. Increasing every segment's angular speed does not guarantee a faster, better-directed clubhead at the event of interest.

The chapter's central distinction is between three questions: how current motions combine, when a selected speed reaches its peak, and how forces and work created or changed the state. Jacobians answer the first, defined events answer the second, and dynamics with explicit power boundaries addresses the third. A strong swing explanation connects all three without using one as a substitute for another.

**Go Deeper:** Read the pinned [mechanics chapter](#), [empirical evidence chapter](#), and [uncertainty study record](#).

## Chapter 9

# Force Direction Changes the Job

### 9.1 A Concrete Picture

Push a shopping cart along its rolling direction and it can speed up. A sideways push may turn it or be largely balanced by wheel reactions. A downward push can change wheel loading without much forward acceleration. Direction changes the interaction, but the motion follows the net force, moments, and constraints—not one applied arrow in isolation.

At the grip, resolve a hand force along the shaft and along two transverse directions. These components can load the shaft, accelerate the club center of mass, and create a moment about that center. The moment depends on the force's line of action and chosen reference; a force applied at the hand has zero moment about that same point. An independent grip moment is a separate load. As the club rotates, a force fixed in the room changes its club-frame components. Direction is a moving relationship, not a permanent assignment to translation or rotation.

### 9.2 How the Mechanism Works

Power begins with projection. The dot product  $\mathbf{F} \cdot \mathbf{v}$  selects the component of force along point velocity. Moment begins with a cross product,  $\mathbf{r} \times \mathbf{F}$ , which selects the component perpendicular to the moment arm. The same vector can therefore do little translational work yet create a large moment, or deliver power through a moving base while its moment about a particular point is small.

For the planar illustration, let  $H$  be the hand point,  $G$  the club center of mass,  $\hat{e}_s$  the shaft direction from grip toward head, and  $\hat{n}$  a chosen unit normal to the illustration plane. Define  $\hat{e}_\perp = \hat{n} \times \hat{e}_s$ . Then  $F_s = F \cdot \hat{e}_s$  and  $F_\perp = F \cdot \hat{e}_\perp$ ; retain  $F_n = F \cdot \hat{n}$  in a spatial model. These are shaft-based components, not automatically the normal and tangent of the hand or clubhead trajectory.

The force's moment about  $G$  is

$$M_G^{(F)} = (r_H - r_G) \times F.$$

In the straight-link idealization  $r_G = r_H + \ell \hat{e}_s$ , its projected moment is  $M_G^{(F)} \cdot \hat{n} = -\ell F_\perp$ . Here  $\ell$  is the hand-to-center-of-mass distance, not necessarily the full club length. The figure labels the

## Direction and Reference Point Set the Moment

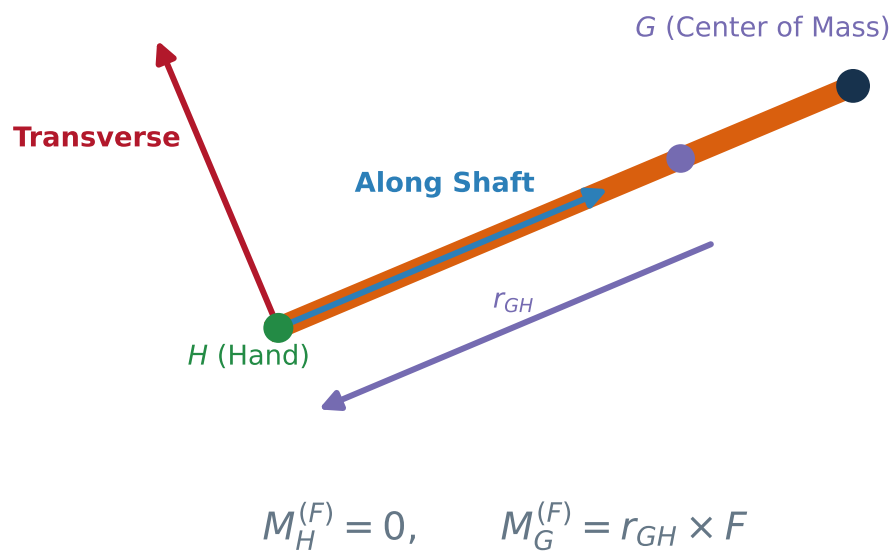


Figure 9.1: Hand-Force Components and the Moment Arm From the Club Center of Mass.

center-to-hand lever arm  $r_{GH} = r_H - r_G$ . The minus sign follows the stated basis. Its moment about  $H$  is zero. Add any intrinsic grip moment separately. A real club's center of mass may lie off the shaft line, so even an axial force can have a moment about  $G$ ; use the actual vector lever arm.

Nor does an axial force automatically equal a centripetal requirement. For a rigid offset  $r_{HG} = r_G - r_H$ , with angular velocity  $\omega$  and angular acceleration  $\dot{\omega}$ , the center acceleration is

$$a_G = a_H + \dot{\omega} \times r_{HG} + \omega \times (\omega \times r_{HG}).$$

Hand acceleration and all external loads enter the balance. Only in a declared fixed-center circular case does the inward net-force component reduce directly to  $mv^2/R$ . An axial grip force can deliver positive power when the hand moves along it, even when its moment about an on-axis center is zero.

### 9.2.1 Vectors Rotate While Labels Stay Put

Suppose the laboratory vertical force remains upward through a short interval. When the club is nearly horizontal, upward may be largely transverse to the shaft. When the club is nearly vertical, upward may be largely axial. A plot of global vertical force can remain positive while the signed shaft-transverse component  $F_{\perp}$  crosses zero. Mechanistic claims should use the frame tied to the proposed job and provide the transform back to measured coordinates.

Figure 9.1 identifies the application point and moment arm. The technical figures draw vectors on model snapshots because time traces alone hide this geometry. At each selected phase, the reader can see force origin, direction, scale, reference point, club orientation, and sign convention. A good vector figure is an argument, not decoration.

### 9.2.2 A Force Can Accelerate One Coordinate and Decelerate Another

Generalized coordinates need not be independent Cartesian directions. For a specified force on a modeled body, virtual work gives  $Q_H = J_H^T F_H$ , where  $J_H$  maps generalized velocity to its application-point velocity. Include the opposite interface load when the other body is also inside the modeled system. Generalized forces are moments only for the corresponding angular coordinates. After resolving ideal constraints, a contribution  $Q_k$  produces  $\ddot{q}^{(k)} = M^{-1}Q_k$  at the same state in a nonsingular model. For example,  $M = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$  and  $Q_k = (1, 0)^T$  give  $\ddot{q}^{(k)} = (2/3, -1/3)^T$ . Coupled response can have opposite signs without violating action–reaction; it does not identify the source of energy or muscle effort. Unreduced constrained equations require the associated reaction solve, not an unconstrained inverse mass matrix.

The mass matrix then mixes generalized forces into accelerations. Looking only at the sign of  $F_{\perp}$  does not guarantee the sign of the relative wrist acceleration  $\ddot{q}_2$ , where  $q_2$  is the wrist angle relative to the arm in the two-link example, when other terms are present. Exact decomposition solves the coupled equations at the same state and reports the contribution rather than inferring it from one arrow.

### 9.2.3 Direction Through the Downswing

Early in the downswing, the hand path and club orientation can make a large reaction mostly along the shaft. Its path-radial role depends on the declared center of curvature. Later, as relative angle

changes, a similar interaction can have a stronger tangential projection. Near the impact zone, two separated hand forces may form a negative swing-normal couple even while other pathways increase club kinetic energy.

This phase dependence is why the scientific question is not “is hand force good or bad?” It is: which component, in which frame, over which interval, about which reference point, with what power and impulse?

### 9.2.4 Three Names for Parts of One Motion Equation

The double-pendulum model gives a more precise vocabulary for the forces that appear as the arm turns and the wrist angle opens. It calls one contribution the **Coriolis Cross Term**. This part contains the arm’s rotation rate multiplied by the wrist’s relative opening rate. If either motion stops, the product is zero. It is therefore a compact way to ask how simultaneous proximal rotation and distal unfolding reshape the joint-coordinate balance.

A second contribution is the **Squared-Speed Term**: the collected terms proportional to individual coordinate-rate squares in this registered model. Curvature contributes to such terms, but this bin is not a measured inward contact force. Centripetal describes an inward acceleration requirement in a declared motion; centrifugal is an inertial term introduced in a rotating observer’s equations. Those are different balance conventions. Do not add a rotating-frame pseudo-force to an inertial free-body diagram that already accounts for the motion.

Gravity supplies a third, configuration-dependent contribution. Unlike the other two, it does not require motion. It can help or oppose the declared coordinate depending on club orientation. Damping and applied moments complete the model’s account. Adding every declared contribution reproduces the modeled acceleration; inventing a separate extra “centrifugal force” on top would count the same mechanism twice.

These categories are coordinate-dependent. In this study the arm angle is absolute and the wrist angle is relative to the arm. A different coordinate choice can move algebra between cross-speed and squared-speed bins without changing the physical trajectory. The full equations must transform together. Even the complete velocity-bias vector is not independently invariant under a nonlinear coordinate change: a free Cartesian particle  $x = q^2$  has  $\ddot{x} = 0$ , yet  $\ddot{q} = -\dot{q}^2/q$  for  $q > 0$ . Its curved coordinate description introduces a velocity term. Compare reconstructed physical accelerations and power, and carry the coordinate and decomposition convention with every named contribution.

### 9.2.5 Why an Outward Energy Story Needs More Than a Force Name

It is tempting to picture proximal energy as a parcel pushed outward by a Coriolis force. The safer picture is a state being prepared. Earlier applied moments and external forces build configuration and velocity. Once both links are moving, geometry redirects the consequences of that stored motion through the hinge. A Coriolis term can then make a large contribution even if the instantaneous distal command is small. It did not create energy from nothing; it describes how the moving system routes the effects of work already done and work still entering.

The sign also needs care. The code reports each left-hand equation term and its equal-and-opposite generalized drive. A positive equation term is not automatically positive acceleration, and a large force impulse is not automatically positive work. The drive must be paired with the appropriate velocity to

obtain power. This is why the source display shows impulse and work side by side instead of giving one arrow a permanent “speed-producing” label.

### 9.2.6 Normalization Can Hide Magnitude

Force arrows are often normalized for visibility. That communicates direction but not magnitude. Other figures use one scale across all phases so arrow length can be compared. Reports should say which convention is used. A tiny vector and a huge vector drawn at equal length can otherwise invite a false mechanism.

Likewise, unit-body-mass normalization aids participant comparison but does not restore missing contact allocation. Directional averages can cancel when participants use different frames or handedness conventions. Preprocessing is part of the interpretation.

**Model Result:** The archived planar double-pendulum audit reconstructs wrist reaction from the links’ tangential/centripetal accelerations and gravity support. That kinematic force balance is distinct from the coordinate-equation Coriolis/squared-speed split. Both require the state and declared conventions; neither alone identifies biological effort. The two-hand geometric reconstructions test the separated-force couple under frozen loads. They reverse that component under the stated transformations; they do not necessarily reverse the entire grip moment or describe a new forward trajectory.

**Human Evidence:** Net club motion alone does not uniquely identify both hand wrenches. Additional measurements or explicit allocation assumptions are needed. Koike’s single-instructor example used a strain-gauge handle with static calibration, fixed contact points, and neglected handle-piece inertia (Koike 2016). Choi and Park combined a six-axis internal-grip sensor with inverse dynamics for nine right-handed male professionals. Their estimates depend on the sensor/segment model; the heavier grip, hand overlap, and foam-ball impact limit interpretation (Choi and Park 2020). Neither study proves a passive human control strategy. Rigid-body wrench conventions provide the mathematical background (Featherstone 2008).

**Hypothesis:** A late negative projected force couple can occur without an instantaneous commanded free moment in a specified model. “Passive” must name which commands and dissipative terms were removed, the retained state and stored energy, and whether the result is a frozen-state evaluation or a new rollout. Its sign alone proves neither passive muscles nor whole-system passivity.

**Practical Interpretation:** Replace “push harder” with a more scientific question: where does the force point relative to the club, hand path, and desired turning axis during that phase?

### 9.2.7 The Same Force Can Do Different Jobs

Draw a force arrow at the grip. If its point of application is moving in the same direction, the force delivers positive power. If motion is opposite, it absorbs power. If motion is perpendicular, instantaneous translational power is zero even though the force changes momentum direction. Rotate the club and the same laboratory-frame force obtains different shaft-axis and transverse components. No arrow has a permanent mechanical job independent of state.

This is why phrases such as “pulling on the handle” need coordinates. Pulling toward the golfer, along the shaft, along the hand path, or opposite clubhead motion are different vectors. A force can be inward relative to the clubhead path but tangential relative to a joint. Resolve each vector in declared

axes while retaining the same observer and physical velocity. A common orthogonal axis rotation preserves a dot product. Changing to an observer translating with velocity  $U$  instead changes force power by  $-F \cdot U$ . Distinguish a coordinate representation from a new observer; the scalar work and energy balances must use matching conventions.

### 9.2.8 Force at a Point Becomes a Wrench

A rigid body's external loading is summarized by a force and a moment at a reference point. Move the reference point and the force remains the same while the reported moment changes by the cross product of offset and force. This does not change the physics. It changes the point about which the moment is reported.

Use net external force for center-of-mass translation. In body axes, spatial rotation about the center obeys  $\sum M_G = I_G \dot{\omega} + \omega \times (I_G \omega)$ , with constant body inertia  $I_G$ . An arbitrary reference-point moment is not simply  $I \dot{\omega}$ ; a moving-point balance needs its transport terms.

For two contact forces  $F_1, F_2$  at  $r_1 = c + d/2$  and  $r_2 = c - d/2$ , define resultant  $R = F_1 + F_2$  and differential force  $D = (F_1 - F_2)/2$ . Include intrinsic contact moments  $C_1, C_2$ . About a point with position  $r_O$ ,

$$M_O = (c - r_O) \times R + d \times D + C_1 + C_2.$$

The separated-force couple is  $d \times D$ . The common-force moment vanishes at the midpoint, but generally not at another reference point. Intrinsic moments can survive when  $d = 0$ . An axial differential load  $D$  parallel to  $d$  changes internal loading while leaving both resultant force and force couple unchanged. Thus net club loading cannot generally recover individual hand loading; Section 12 examines the additional constraints and measurements needed.

### 9.2.9 Following a Vector Through the Downswing

Early in the downswing, a large grip reaction may point largely along the club, maintaining the curved path and transmitting proximal motion. Its direct tangential acceleration contribution can be modest. As the club changes orientation, the force's shaft-axis components can change even if its laboratory-frame components change smoothly. Late in the downswing, a smaller force with a larger component perpendicular to the center-to-hand lever arm, or a longer lever arm, may contribute a larger moment about the club center.

At the same time, hand-point velocity changes. Positive grip force power can coexist with a negative swing-normal couple power because the translational and rotational pieces differ. A complete display therefore shows the force vector, moment vector, contact-point velocity, angular velocity, and the paired power terms  $F \cdot v$  and  $M \cdot \omega$  at the same reference point. One scalar cannot represent this structure.

### 9.2.10 A Geometry-Control Experiment

Take a simulated state and freeze its resultant  $R$ , differential force  $D$ , midpoint  $c$ , and intrinsic moments. Reconstruct the force couple  $d \times D$  under four controls: retain  $d$ ; replace  $d$  by  $-d$  by swapping contact positions; replace  $D$  by  $-D$ ; and shrink  $d$  to zero. The force couple respectively stays the same, reverses,

reverses, and tends to zero **if the differential force  $D$  remains bounded**. Allowing  $D$  to grow inversely with separation can keep the couple finite. Neither the common-force moment about another reference point nor intrinsic contact moments must vanish or reverse. Test the declared component, not a mislabeled total wrench.

These controls are more informative than simply changing a force magnitude. They challenge the proposed geometric pathway. If a nonzero separated-force couple survives zero separation while  $D$  remains bounded, the calculation or labeling is wrong. A surviving intrinsic moment is not such an error. If the control behaves correctly, the model has passed a necessary mechanical test but has not shown that golfers use that allocation.

### 9.2.11 Measurement Frames Are Part of the Experiment

Instrumented grips report forces in sensor axes. Motion capture reports markers in laboratory axes. The club carries a moving frame, while anatomical analysis may use segment or joint frames. A calibration must locate each sensor and its orientation relative to the club. Rotate force and moment components together, and also transport moments when the reference point changes. Correct inertial loads of sensor-side hardware when needed; a raw internal sensor wrench is not automatically either hand's complete wrench. Synchronize kinematics and loads before combining them, then propagate both rotation and position uncertainty.

Small orientation errors can mix large axial forces into smaller transverse components. Let  $\delta$  be the sensor-axis misalignment angle. For one sign convention, a true axial/transverse pair of 100 N and 1 N becomes  $1 \cos \delta - 100 \sin \delta$  N transversely: at  $\delta = 1^\circ$  it is about  $-0.745$  N. The inferred sign reverses without a physical force reversal. Such leakage can alter a moment's swing-normal projection when that moment depends on a small transverse force and its lever arm. Calibration uncertainty should therefore be propagated into couple sign and timing, and reference transforms should be tested with known loads before any biological interpretation.

### 9.2.12 What a Useful Figure Should Show

A publication-grade force-vector figure should include coordinate axes, reference point, hand locations, arrow scale, moment-arm line, phase or timestamp, and sign convention. If arrows from different trials use automatic scaling, the caption must say so. A sequence of frames should retain scale where comparison matters. Accompanying plots should show components and resultant, not just one selected direction.

For reviewers, the underlying numeric table should be linked. That makes it possible to recalculate moments, transform frames, and verify that plotted arrows match reported values. Visual intuition is valuable precisely when it is anchored to auditable quantities.

### 9.2.13 Questions That Keep the Claim Honest

Where is the force applied? About which point is moment calculated? In which frame is “negative” defined? Is the vector measured, inferred, or simulated? Does it do positive or negative power at that instant? Does reversing geometry reverse the effect? Can another hand-force allocation produce the same net club wrench? Answering these questions turns an arrow from decoration into evidence.

### 9.2.14 Reader Lab: Verify a Two-Hand Diagram

Choose two three-dimensional hand positions and force vectors. Select a club reference point. Compute net force and moment directly. Translate the reference point and recompute; then verify total wrench power is unchanged for points on the same rigid body, seen by the same observer, when the new point velocity is transformed consistently. A cut through a deforming shaft or slipping interface requires its own local velocity and energy balance. This catches the common mistake of moving the moment without moving the velocity term.

Resolve each force in laboratory, club, and swing frames. Reconstruct the original vector from every component set. Perturb sensor orientation by one or two degrees and observe how much a large axial force leaks into the smaller transverse component. Recompute the moment's swing-normal projection; the resulting change becomes part of sign uncertainty.

Animate the arrows through a simulated late downswing using one fixed scale. Overlay moment arms and hand velocities. The visual should explain the calculated couple and power rather than merely make forces look dramatic. Release the numeric array so every arrow can be audited.

### 9.2.15 Competing Force-Direction Stories

One hypothesis may say inward handle force changes club acceleration through parametric geometry. Another may say a differential hand force supplies a negative couple. A third may say shaft reaction explains the observed wrench. They can coexist, but each predicts a different relationship among force direction, hand motion, deformation, and state.

Separate algebraic checks from feasible dynamical interventions. Frozen-force separation changes test the couple formula; alternative velocities paired with the same force illustrate the power projection. Neither automatically supplies a trajectory satisfying Newton's laws and contact constraints. For a causal comparison, reintegrate a declared model after changing geometry, compliance, actuation, or controller while matching the initial state and reporting new reactions. Removing shaft flexibility changes stored energy and often other dynamics; specify what was matched. Compare those trajectories separately from reconstructions so a correct cross product is not mistaken for a demonstrated human mechanism.

## 9.3 Where the Picture Breaks

The cart and planar arrows ignore distributed contact, three-dimensional axes, and deformable tissue. The real grip force acts across an area; a six-axis sensor reduces it to an equivalent wrench at a reference point. That reduction preserves resultant loading and, for a common rigid velocity field, its power. It does not preserve local deformation work, contact pressure, discomfort or tissue load without additional information.

Force direction alone also cannot identify muscle intent. The same hand wrench can arise from many shoulder, elbow, wrist, and co-contraction patterns. Electromyography (EMG) adds activation timing but not a unique force mapping.

**Go Deeper:** Use the monograph's [force-vector mechanism chapter](#), [coordinate force-source chapter](#), [two-hand local geometry results](#), and the instrumented-grip studies (Choi and Park 2020; Koike 2016).

## Chapter 10

# Force Is Not Work Until Something Moves

### 10.1 A Concrete Picture

Hold a heavy suitcase motionless for a minute. Your muscles fatigue, but the suitcase has not moved and the external mechanical work on it is zero. Carry it horizontally at constant height and the upward hand force is nearly perpendicular to horizontal velocity, so that force again does little mechanical work on the suitcase. Lift it onto a shelf and force aligns with displacement; positive work by the hand raises the suitcase–Earth gravitational potential energy. During a slow lift, the hand and gravity do nearly equal and opposite work on the suitcase, so its kinetic energy changes little. Hand work and net work are different ledgers.

This example feels unfair because biological effort is real in all three cases. That is exactly the lesson. Mechanical work, metabolic cost, muscle activation, and perceived effort are different quantities. A large grip force may stabilize the club with little club-level work. A modest force aligned with fast hand motion may transfer large power.

### 10.2 How the Mechanism Works

Over a path, translational work is

$$W = \int \mathbf{F} \cdot d\mathbf{r}.$$

Here  $\mathbf{r}$  is the position of the material point where the force acts, measured in a declared inertial observer frame;  $\mathbf{v} = \dot{\mathbf{r}}$ . Over time,  $W = \int \mathbf{F} \cdot \mathbf{v} dt$ . For a rigid body, rotational power at the chosen reference is  $\mathbf{M} \cdot \boldsymbol{\omega}$ , so its work channel is  $\int \mathbf{M} \cdot \boldsymbol{\omega} dt$ . Only for rotation about a fixed axis does this reduce directly to  $\int M_{\text{axis}} d\theta$ . In general three-dimensional motion, Euler-angle rates are not the components of angular velocity. Generalized moments must be transformed to be power conjugate to those rates before taking their dot product.

Impulse instead integrates force over time,  $\mathbf{J} = \int \mathbf{F} dt$ . For a fixed collection of bodies, the **net**

## Power Depends on the Projection of Force Along Velocity

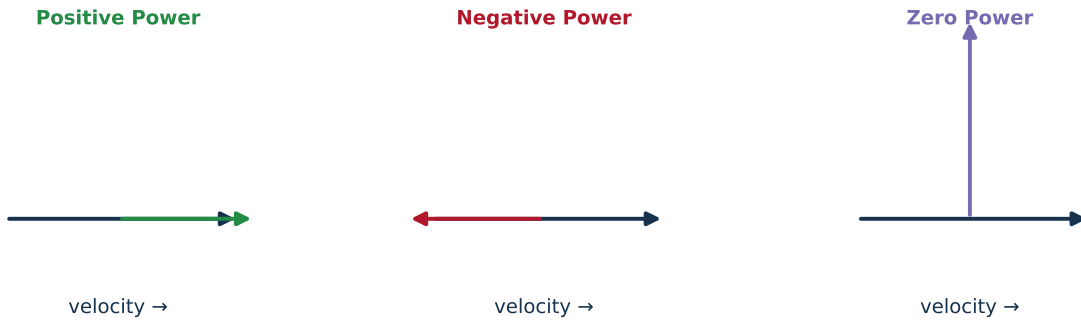


Figure 10.1: Force, Velocity, and Their Projection Distinguish Large Load From Large Mechanical Power.

**external** impulse equals the change in total linear momentum in that inertial frame. A single force contributes to that balance; other forces may cancel it. These answer different questions.

A force can have large impulse and small work when it acts mostly perpendicular to motion. It can have small impulse and meaningful work if it aligns with rapid motion over a short interval. Reporting magnitude, impulse, power, and work together exposes the difference.

### 10.2.1 Hand-Path Force Needs a Declared Path

Some golf analyses project net hand force along the hand path and integrate over path length. This can be a useful transfer observable. But a large value does not prove large biological effort. A chosen model may assign much of the instantaneous force to its drift/constraint part and an opposing contribution to its input part. This is a decomposition at the same achieved state, not a finding that muscle activity was unnecessary. Removing an input changes later states and reactions; its causal effect requires a new trajectory under a declared intervention or controller.

The pinned hand-path study evaluates this estimand in a planar double pendulum, a one-arm three-link point-mass model, and a two-arm floating-club closed loop. In two forward cases, drift supplies slightly more than 100% of signed hand-path force work because the contemporaneous control contribution is opposing. A percentage above 100 does not violate conservation; positive and negative contributions cancel to the total. In a prescribed two-arm kinematic sweep, rather than a forward rollout, drift and control nearly cancel, making fraction language unstable.

### 10.2.2 What the Coriolis-Impulse Search Actually Optimized

A separate archived pendulum example asks a deliberately narrow question: among 135 frozen torque programs, which one produces the greatest *absolute* along-hand-path impulse from the coordinate-defined Coriolis cross term before qualified impact? Absolute impulse counts both signs, so a force that reverses direction does not disappear through cancellation. Signed impulse is published beside it to retain sign relative to the changing path tangent. This scalar projected impulse is not the vector

impulse that appears in the momentum balance. The registered search uses shoulder-absolute and wrist-relative coordinates; its Coriolis label denotes their cross-speed equation terms. It is not an additional force acting on a free-body diagram.

The grid combines three proximal torques (60, 80, 100 N m), three restraint magnitudes (0, 5, 10 N m), and fifteen onset times from 0 to 0.350 s. After onset, the wrist drive is 15 N m. Ninety-one programs reach the registered event: the first upward crossing of the club angle through downward vertical, with arm angle no greater than 2 radians. The other 44 are excluded. This is a geometric pre-impact score interpolated from sampled history, not a simulated ball collision. The maximum absolute Coriolis impulse is 13.817 N s with a 100 N m proximal drive, 10 N m of early wrist restraint, and wrist-drive onset at 0.100 s. That program reaches 46.780 m/s. The highest-speed program also uses 100 N m proximally, but 5 N m of early restraint and a 0.125 s onset; it reaches 46.913 m/s with 13.701 N s of absolute Coriolis impulse. The optima are close, but they are not the same program. Maximizing a named force component is therefore not identical to maximizing clubhead speed, even inside this small deterministic model. These are discrete grid maxima, not continuous optimal-control solutions or human swing prescriptions.

The signed values deepen the lesson. The maximum-absolute-impulse program has a signed Coriolis impulse of -6.825 N s and Coriolis generalized work of -100.69 J. A large absolute impulse can coexist with negative net work because the source reverses and because force, velocity, and generalized-coordinate signs matter. Squared-speed work is -33.73 J, gravity work is +25.56 J, and applied-drive work is +358.20 J in the same impact-truncated interval. These figures are an equation-term ledger, not a claim that a Coriolis agent supplied or removed a measured amount of biological energy. Nor does their sum, by itself, equal the change in total kinetic energy: the configuration dependence of the mass matrix matters.

For example, in a smooth ideal model with time-independent constraints, let  $q$  be the generalized coordinates,  $M(q)$  the mass matrix,  $U(q)$  the potential, and  $\tau$  the applied generalized forces. Define kinetic energy  $K = \frac{1}{2}\dot{q}^T M(q)\dot{q}$  and write  $M\ddot{q} + h + \nabla U = \tau$ , with  $h$  the full velocity bias and dots denoting time derivatives. The kinetic-energy rate contains  $\frac{1}{2}\dot{q}^T \dot{M}\dot{q}$  as well as  $\dot{q}^T M\ddot{q}$ . For the usual consistent rigid-body equations,  $\dot{q}^T h = \frac{1}{2}\dot{q}^T \dot{M}\dot{q}$ ; these terms cancel to give  $d(K+U)/dt = \tau^T \dot{q}$ . Coriolis and squared-speed terms redistribute the coordinate acceleration balance without becoming extra energy sources. Include dissipation, moving constraints and explicit time dependence when they are present.

### 10.2.3 The Hand-Path Arrow Does Not Carry the Whole Wrench

The modeled wrist point moves with the arm coordinate but not independently with the relative wrist coordinate. Its task Jacobian is therefore rank-deficient for mapping a two-coordinate generalized drive into one force-only hand-path vector. The software computes the minimum-norm force that can be represented at the wrist and keeps the leftover generalized couple as a residual. It never hides that remainder inside the displayed hand force. For component  $k$ , use generalized drive  $Q_k$ , point Jacobian  $J_H = \partial x_H / \partial q$ , mapped force  $F_{H,k}$ , residual  $r_k$ , and point velocity  $v_H = J_H \dot{q}$ . Then  $Q_k = J_H^T F_{H,k} + r_k$ . Consequently,  $Q_k^T \dot{q} = F_{H,k}^T v_H + r_k^T \dot{q}$ . The generalized work totals above include the residual-power channel; they are not just force work along the hand path. A least-squares force equivalent is not automatically a measured contact force.

The archived maximum residual norm is 54.44 N m for the impulse-selected program and 52.81 N m

for the speed-selected program. Each maximum is taken over time **and all attributed components**, not solely the Coriolis component or the net physical wrist moment. This is a structural mapping diagnostic, not a measured joint load or a small numerical rounding error. It tells the reader that an along-path force plot answers a useful but incomplete question. A complete grip description needs a wrench, or an endpoint-and-couple mapping with enough independent coordinates and measurements.

At the top of the simulated swing, hand speed begins at zero, so no path tangent exists. The integration marks that interval undefined and uses only adjacent samples with a defined direction (the archived speed threshold is  $10^{-12}$  m/s). Generalized power remains defined at zero speed and its work is integrated over the full recorded interval. More than 99.6% of the impact-truncated time is valid in the highlighted runs. Reporting that fraction prevents an arbitrary initial direction from contaminating the impulse. It does not bound the missing impulse: even a short excluded interval can matter if its force is large. Report the masking threshold and convergence when interpreting a small difference between programs.

#### 10.2.4 Fractions Become Dangerous Near Cancellation

If drift contributes +10 J and control contributes −9 J, total signed work is +1 J. Calling drift “1,000% of total” is arithmetically correct and scientifically awkward. If total crosses zero, the fraction is undefined.

Better reports show signed contributions, absolute magnitudes, a cancellation index, and intervals separately. Ratios are reserved for denominators safely away from zero. This is not statistical fussiness; it prevents a small residual from turning two large opposing mechanisms into a sensational percentage.

#### 10.2.5 Power Reveals Timing Hidden by Work

Two strategies can deliver the same integrated work with different power histories. One may load the shaft early and receive energy back late. Another may deliver direct power near impact. Their club energies at one instant can match while internal loads and robustness differ.

Plotting cumulative work beside instantaneous power is useful. The power curve shows when exchange occurs; the cumulative curve shows what remains after positive and negative intervals cancel. A phase-integrated table connects both to transition, early downswing, delivery, and impact.

#### 10.2.6 Joint Power Is Not Muscle Power

A measured or inferred net joint power is a mechanical constraint on a model; it is not a muscle-energy meter. Tendon storage, biarticular transfer and opposing muscle forces can produce different biological allocations with the same net result. The segment-work section below specifies the required relative angular velocity and explains what inverse dynamics identifies.

This distinction is central to the arm-versus-wrist proposal. A net club wrench can match while direct wrist moments, two-hand force couples, internal squeeze, and proximal joint loads vary. Club-level work is an outcome constraint, not a biological allocation measurement.

**Model Result:** The hand-path attribution separates pointwise drift and control contributions, then reports impulse, power, work, force vectors, joint effects, and cancellation rather than one percentage.

Recomputing the selections from the archived 135 outcome records reproduces the stated grid winners and counts. That check does not independently reproduce their ODE trajectories or quadrature.

**Human Evidence:** Nesbit and Serrano used kinematically driven body/club models to examine four selected amateur golfers; the outputs are model-based work estimates, not direct muscle-energy measurements (Nesbit and Serrano 2005). MacKenzie and colleagues analyzed 76 right-handed golfers with handicaps below 15 using the same driver, inverse dynamics and between-golfer regressions. Their strong force/work associations do not measure the effect of instructing a golfer to pull harder (MacKenzie et al. 2020). Model boundaries, differentiation, inertial parameters and shaft treatment affect interpretation. Contact measurements and muscle-informed models can narrow the ambiguity without eliminating it.

**Practical Interpretation:** Force tells how strongly the system interacts; work tells how that force aligns with motion over distance; power tells how fast the exchange happens; effort belongs to a separate biological account.

### 10.2.7 From Force History to Mechanical Consequence

Start with a force vector through time. Its integral is that force's impulse; only the sum of external impulses fixes the system's momentum change. Pair each force sample with contact-point velocity and take the dot product to obtain power. Integrate power for work. Pair a moment with angular velocity for rotational power. These operations answer related but distinct questions; none can be safely inferred from the appearance of the force curve.

A high force applied while a point barely moves may create little work. A lower force acting through a long displacement can do more. A force perpendicular to velocity can bend a trajectory without changing kinetic energy at that instant. In a golf swing, all three patterns occur while the club follows a curved, rotating path.

### 10.2.8 The Hand Interface Has Two Power Channels

Choose a club reference point. Translational power is net force dotted with reference-point velocity. Rotational power is net moment dotted with angular velocity, all expressed in the same observer frame. Their sum is wrench power. For two material points on the same rigid body, let  $r_{AB}$  run from  $A$  to  $B$ ;  $F$  is the resultant force and  $M_A, M_B$  its equivalent moments. Then  $v_B = v_A + \omega \times r_{AB}$  and  $M_B = M_A - r_{AB} \times F$ , so  $F \cdot v_B + M_B \cdot \omega = F \cdot v_A + M_A \cdot \omega$ . The separate force and moment powers can change even though the sum does not. A change of observer is different: kinetic energy and power are frame dependent. A point across a bending shaft need not satisfy this rigid-body velocity relation. Reporting only the couple may miss positive translational power; reporting only net force may miss a large rotational exchange.

For two hands, individual contact powers can also be calculated. Their sum must match net wrench power after consistent transformations and inclusion of any contact moments, provided the contacts share the rigid-body motion used in the reduction. With slip, local deformation or distinct hand/club velocities, retain those contact powers and losses explicitly. That closure is a strong audit. Yet individual hand power does not identify which muscle supplied it, because work can cross wrist, elbow, shoulder, and trunk pathways and because co-contraction can add internal cost without changing net external power.

### 10.2.9 A Late Negative-Couple Example

Suppose the net force supplies +900 watts of translational club power while the two-hand couple supplies -150 watts of rotational power over a short interval. Net hand power remains +750 watts. The negative couple is real and locally absorptive, but the hands as a whole still add club energy. If the couple changes orientation so its power later becomes positive, its integrated work may be near zero while its angular impulse is nonzero. A change in angular momentum requires the sum of all external moments about a consistently chosen inertial origin (or a separately justified center-of-mass balance).

Now run a matched-state branch without that couple. The club may release sooner, changing geometry and reducing later positive power from another channel. The baseline could then finish faster despite the early negative work. That result would not mean negative work created energy; it would mean the intervention changed the route and timing by which other work acted.

### 10.2.10 Segment Work Is Not Muscle Work

Inverse dynamics produces net joint moments consistent with measured motion and external loads under a model. For a rotational joint, dot the net moment acting on the distal segment with its angular velocity **relative to the proximal segment**, expressed in a common frame. This gives the joint's net rotational power; moments dotted with absolute segment velocities instead describe the separate segment power flows. Include translational joint power if relative translation is allowed. Several muscles, passive tissues, contact loads, and model residuals can contribute to that moment. Biarticular muscles can absorb energy at one joint and generate it at another. Assigning net joint work to one named muscle is therefore not warranted.

Musculoskeletal optimization can propose allocations, and EMG can constrain activation timing, but redundancy remains. A strong report keeps rigid-body joint work, muscle-tendon work, and metabolic energy on separate evidence levels. This distinction is especially important for claims about wrists “resisting passively” or scapular muscles “creating” a club couple.

### 10.2.11 Numerical Closure Before Storytelling

First declare the resolved mechanical energy: kinetic energy plus the chosen gravitational and elastic potentials. A club-only boundary receives hand work; a golfer-plus-club boundary instead contains actuators or muscles that convert chemical energy into mechanical energy internally. “Closed” to mass does not mean mechanically isolated or unable to perform internal work.

A useful smooth-phase ledger is

$$\begin{aligned}\Delta E_{\text{mech}} = & W_{\text{boundary,nc}} \\ & + W_{\text{internal actuation}} \\ & - E_{\text{dissipated}}.\end{aligned}$$

where “nc” denotes the nonconservative boundary channel, and  $E_{\text{dissipated}} \geq 0$  is mechanical energy converted into unresolved heat or other dissipative modes. Boundary work excludes forces already represented by the included potentials, and internal actuation excludes elastic storage already counted in  $E_{\text{mech}}$ . This equation assumes no unlisted mass flow, time-varying potential or other energy conversion. Use a separate jump balance for impacts rather than integrating across an unresolved impulse. For a

passive full club including its shaft, hand and aerodynamic work feed kinetic, gravitational and shaft elastic energy plus dissipative losses. For the combined body and club, hand-interface work cancels internally when contact motion is compatible, but muscle/actuator conversion remains. Adding gravity work while also treating gravitational potential  $U_g$  as a destination would double-count it. Adding shaft storage to an energy that already includes it would do the same.

Calculate closure over the whole trial and important phases. Report absolute residual and residual relative to the effect being discussed. A 2 J residual may look small compared with total swing energy yet be fatal to a claimed 1 J mechanism.

Timestep refinement, alternate differentiation/filtering, and independent power calculations should preserve the conclusion. Agreement from two quantities derived from the same filtered motion is an internal consistency check; it is not an independent validation of the assumed mass properties or contact model. At contact interfaces, action-reaction and power consistency provide additional checks. A model that cannot close its ledger should not explain an unexpected sign by inventing a new transfer narrative.

#### **10.2.12 What to Compare Across Strategies**

For the arm-dominant and wrist-dominant extremes, compare more than terminal speed. Plot bilateral forces and moments; common and differential hand modes; interface power; cumulative work; club and body energy; shaft strain energy; joint load proxies; and face/path sensitivity. Align comparisons by both clock and state. Match the final club task when asking about internal allocation, and match the initial state when asking about causal input contribution.

A role reversal may preserve net club work while changing peak force, internal co-contraction, or timing sensitivity. Those are scientifically interesting differences even if clubhead speed barely changes. Conversely, a small speed gain may be unattractive if it depends on fragile timing or unmeasured internal load.

#### **10.2.13 A Reviewer Can Recompute the Claim**

The most useful release contains time, state, forces, moments, velocities, powers, cumulative works, energy terms, and residuals in machine-readable form. Units and frames live in a schema, and each figure links to its source. A reviewer can then verify dot products and integrals instead of accepting a caption. This is how a casual explanation remains connected to a scientific product.

#### **10.2.14 Reader Lab: Recompute a Power Ledger**

Download one time series of force, moment, reference-point linear velocity, and angular velocity. Check units and frames, then calculate translational power, rotational power, and their sum sample by sample. Numerically integrate each with at least two reasonable quadrature rules and compare. A difference larger than the reported mechanism needs explanation.

Next transform the wrench and velocity to another point on the same rigid club segment, keeping the observer fixed, and verify summed power is invariant. Reverse the declared swing-normal axis and confirm only the coordinate label changes, not the physical power. Compare cumulative hand work with the declared club energy and losses: count gravity and shaft storage once, include any power crossing a cut through a flexible shaft, and retain the residual with its sign.

Finally, shorten and lengthen the integration window. If a claimed positive contribution depends entirely on an unreported boundary choice, the conclusion must name that interval. This exercise is simple enough for a reviewer and powerful enough to expose many sign and synchronization errors.

### 10.2.15 The Difference Between Necessary and Sufficient

For a constant-mass particle in an inertial frame, increasing kinetic energy requires positive net force work. For an articulated system's total mechanical energy, the required positive balance may come from boundary work or internal chemical/actuator conversion, after accounting for potentials and losses. Neither statement makes one particular joint-power peak necessary: other routes can substitute. A negative couple may be sufficient to alter release in one model but not necessary for all high-speed trajectories. Counterfactuals and alternative allocations establish these logical distinctions.

Use cautious language: “contributed under this decomposition,” “improved the declared outcome in this branch,” or “was associated in this sample.” Avoid “caused all transfer” unless interventions and closure exclude competing paths. The strength of the claim should match the design.

### 10.2.16 A Compact Reporting Template

Name the system, boundary, reference point, frame, sign, and interval. Report force and moment, point and angular velocity, power, cumulative work, energy change, residual, and uncertainty. State whether each quantity is measured, inferred, or simulated. Then link the machine-readable series. This template is short enough for every mechanism result and prevents most category errors.

## 10.3 Where the Picture Breaks

The suitcase treats the hand as one contact and the object as rigid. In golf, two hands, a flexible shaft, and moving joint centers complicate the boundary. Work at one reference must be transformed consistently, and distributed contact may store energy locally.

Mechanical work also does not judge usefulness. Negative work can be essential for control, accuracy, or preparing a favorable state. Positive work at the wrong time can worsen the final task.

**Go Deeper:** Compare the [pinned hand-path attribution snapshot](#) (provider revision [69eb7e9d](#)) with the separate [coordinate force-source chapter](#) and its [135 archived outcome records](#). The latter archive is at revision [f70f4620](#); its recorded generating revision is [942c4e32](#). Archive inspection is distinct from rerunning that environment. For methodological background, Robertson and Winter examined segment power balance in a sagittal-plane walking model; that is not golf-specific validation ([Robertson and Winter 1980](#)). Winter's textbook supplies broader background ([Winter 2009](#)).

## Chapter 11

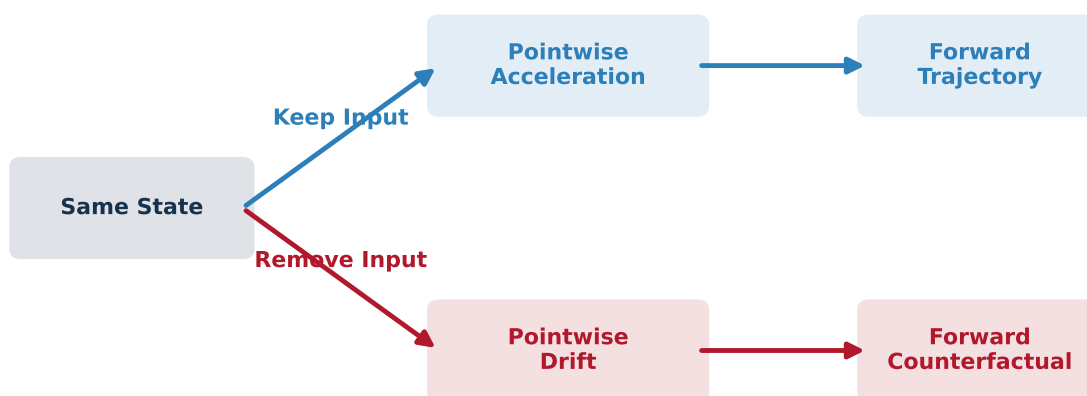
# Counterfactuals Are Scientific Scissors

### 11.1 A Concrete Picture

A baker wants to know what yeast contributed to a loaf. Removing yeast from the finished loaf is impossible. Baking a second loaf without yeast changes the rise, temperature history, moisture, and structure. Comparing one instant of the two doughs answers a different question from comparing the finished loaves.

Models let us make cleaner cuts, but each cut has a meaning. We can remove a torque at one frozen state and recompute acceleration. We can continue that altered branch forward. We can rerun the entire swing with a different schedule. Or we can remove gravity, damping, shaft flex, or a contact moment arm. These are not interchangeable “what ifs.” They are scientific scissors with different blade shapes.

#### Two Questions That Must Not Be Confused



Pointwise: what is the acceleration now? Forward: where does the changed system go?

Figure 11.1: Four Counterfactual Branches Separate Instantaneous Attribution, Short-Horizon Persistence, Full Resimulation, and Mechanism Ablation.

## 11.2 How the Mechanism Works

A pointwise counterfactual holds configuration and velocity fixed, changes one input or term, and resolves acceleration or reaction force. Because the state is identical, the difference isolates an instantaneous modeled contribution. It does not say what a complete no-input trajectory would have looked like.

A matched-state forward counterfactual begins both branches at the same state and integrates with different future inputs. At the cut, attribution is clean. As time passes, states diverge. The difference now includes direct input effect, changed geometry, changed velocity coupling, and changed impact time. This answers whether the consequence persists over a declared horizon.

A full resimulation changes an input schedule from the beginning. It answers a strategy question: what trajectory and outcome follow if the entire command program changes? It cannot by itself attribute the outcome to one instant or mechanism because every later state changes.

An ablation removes a modeled mechanism such as gravity, damping, compliance, or contact separation. If the claimed effect remains when the proposed mechanism is absent, that mechanism is not necessary within the model. If it disappears, necessity is supported only for the declared family; another unmodeled mechanism could still produce a similar observation in humans.

### 11.2.1 Cut Time and Horizon Are Part of the Experiment

Choosing a cut after seeing the largest difference can exaggerate a result. The technical work therefore registers cut times, event definitions, and horizons. It repeats tests across nearby cuts and timesteps. A result that exists for one numerical sample but vanishes under refinement is not a mechanism.

Impact creates another trap. One branch may hit the impact criterion earlier. Comparing both at the baseline impact clock time may evaluate one branch after its own impact. Comparing each at its own first valid impact changes the state being compared. Both variants can be reported, but the distinction must be visible.

### 11.2.2 Decomposition and Intervention

An algebraic decomposition divides the acceleration at one state among gravity, velocity terms, controls, and constraints. An intervention modifies the model or input and observes the outcome. Decomposition says how the equation balances now; intervention says what changes under a defined alteration. Confusing them turns a bookkeeping term into a causal claim.

The zero-torque counterfactual force is a decomposition tool when computed at a fixed state. Integrating a zero-torque branch is an intervention. The technical paper reports both so readers can see where the instantaneous contribution continues and where trajectory divergence takes over.

### 11.2.3 Negative Controls Challenge the Story

Good experiments include cases where the mechanism should vanish or reverse. Coincident hand contacts remove a force-generated couple. Reversing moment arms should reverse its sign. Setting a dead zone to zero should remove any advantage attributed specifically to crossing that dead zone. Equalizing shaft stiffness should remove differences attributed only to flexibility.

Negative controls are more persuasive than a baseline-versus-favored comparison because they ask the mechanism to predict its own failure. A model that always produces the preferred result, regardless of geometry or removed mechanism, is describing an outcome rather than explaining it.

#### 11.2.4 Counterfactuals Need Common Observables

Every branch should report the same quantities in the same frames: club speed, face/path proxy, forces, moments, powers, work, constraint residual, energy closure, and event timing. Otherwise one model can appear better merely because it reports a convenient metric.

This common-observable contract also connects model tiers. A double pendulum and a spatial contact model have different coordinates, but both can report an equivalent club wrench and interface power. Mechanisms that survive this ladder earn broader confidence.

**Model Result:** Pointwise wrist-force attribution, matched-state persistence, driver killswitches, geometry controls, zero-dead-zone controls, and stiffness ablations form a linked counterfactual program in the technical release.

**Hypothesis:** If delayed transfer depends on a particular preload or interaction mechanism, its advantage should disappear when that mechanism is removed while the task and state are appropriately matched.

**Practical Interpretation:** A fair “what if” must say what stayed identical, what changed, when the change occurred, and how long the altered branch was allowed to become a different motion.

#### 11.2.5 Four Different Meanings of “Take It Away”

The phrase “remove wrist torque” can hide several experiments. A pointwise removal changes the torque only in the acceleration calculation at one copied state. A forward killswitch removes it from that state onward while other controls follow their original schedules. A reoptimized branch removes it but lets other controls compensate. A human perturbation asks a person to change a strategy and inevitably changes feedback, intent, and perhaps the starting state. These are not interchangeable.

Pointwise removal is best for decomposition. Forward removal is best for persistence within the declared model. Reoptimization asks whether another control route can substitute. Human manipulation tests whether the mechanism appears in an adaptive biological system. A complete research program uses the first three to sharpen predictions before attempting the fourth.

#### 11.2.6 The Matched-State Rule

At the branch instant, copy every modeled state: positions, velocities, shaft modes, activation or transmission states, and controller memory. Change only the declared intervention. If one branch begins with a different club angle or force history, later divergence cannot be assigned cleanly to the removed input.

This rule is particularly important for preload. Comparing a continuously loaded backswing with a role-reversed downswing naturally produces different transition states. That comparison answers a strategy-level question but not the narrow causal question “what does the transmission state do?” A second matched-state experiment should equalize geometry and motion while manipulating only the declared gap or activation state. Both comparisons are informative when labeled correctly.

### 11.2.7 Pointwise Decomposition Does Not Add Like Percentages

At a fixed state, the acceleration equation may be decomposed into gravity, velocity, applied-control, external-load, and constraint contributions. These terms can oppose. One positive component can exceed total acceleration because another is negative. Dividing each by total and calling the result a percentage creates values below zero or above one hundred, which are mathematically possible but easy to misread.

Report signed components in physical units, cancellation indices, and residual closure instead. For nonlinear forward branches, effects generally do not add because each intervention changes the future state. A combined removal can differ from the sum of separate removals. That interaction is part of the result, not an accounting error.

### 11.2.8 A Counterfactual Sequence for the Negative-Couple Claim

Begin with baseline forward simulation. At selected late-downswing states, reconstruct bilateral contact and compute swing-normal couple. Next perform a pointwise removal of the relevant actuator or driver and observe the changed constraint reaction. Then run a short forward killswitch. Reverse hand geometry and force direction in separate negative controls. Finally repeat across model rungs and parameter samples.

The mechanism is strengthened if sign, timing, and geometric response remain coherent while magnitude varies. It is weakened if the couple persists when hand separation collapses, changes sign under a mere coordinate relabeling, or depends on one integration step. Human data must then test the predicted wrench, not merely reproduce the club trajectory.

### 11.2.9 Counterfactuals About Timing

To study onset, branch from several candidate states and apply identical command shapes shifted in state or time. Include finite rise time. Compare terminal speed, face/path proxies, peak loads, and sensitivity. The preferred command by one objective may be poor by another. Repeat under parameter uncertainty and on held-out samples.

A clock-shift experiment asks how a command schedule performs at different times. A state-trigger experiment asks whether activation at a geometric or dynamic event is more robust to tempo. These interventions can produce different answers. The study should expose both rather than translate one into a universal instruction.

### 11.2.10 Compensation Is a Separate Scientific Question

If wrist torque is removed and an optimizer increases arm torque to recover the same club motion, the result demonstrates redundancy. It does not show wrist torque was irrelevant in baseline. Compare internal loads, power routes, and robustness between baseline and compensated solutions. Multiple feasible routes are expected in a redundant biological system.

Conversely, failure to compensate within a restricted controller does not prove humans cannot. It establishes a limit of that control family. Expanding the family is a model-ladder step and should be preregistered enough to prevent the optimizer from becoming an explanation that can fit anything.

### 11.2.11 From Model Intervention to Human Falsifier

A safe human experiment cannot instantaneously remove a muscle torque. It can manipulate grip spacing, club inertia or stiffness, instruction, or perturbation while measuring the predicted mediators. The model must say which observable should change and in what direction. If a grip-spacing reduction should shrink a differential couple, that is more testable than asking a participant to “use less wrist.”

The final evidentiary bridge is therefore indirect but disciplined: model intervention, robust observable prediction, physical manipulation, synchronized measurement, and a predeclared result that counts against the mechanism.

### 11.2.12 Reader Lab: Build a Branch Manifest

For one intervention, write a small manifest containing source run, copied state hash, branch time, removed input, retained inputs, controller memory, solver, tolerance, horizon, and primary outcomes. Generate baseline and branch from that manifest. A third party should be able to confirm that no hidden state changed.

Plot zero difference at the branch instant for all states and the expected nonzero difference in acceleration or reaction. Then show divergence through time. Repeat with half timestep and a second engine. Store both successes and any failed closure. This turns a counterfactual from a narrative choice into a reproducible intervention.

Run the same design pointwise, forward, and with compensation. The pointwise effect identifies local contribution; forward effect identifies persistence; compensation identifies redundancy. If conclusions differ, report why rather than choosing the most favorable version.

### 11.2.13 Counterfactual Etiquette

Never describe an artificial zero-velocity calculation as a motion a golfer could perform. Never call a reoptimized trajectory “the effect of removal” without noting compensation. Never compare branches with different initial states as though only control changed. Never turn nonlinear component differences into percentages assumed to sum.

These rules do not make causal modeling timid. They allow strong statements within scope: “At this exact modeled state, removing this driver changed this reaction by this amount,” or “Over this forward branch, retaining the negative couple produced this later outcome.” Precision makes the result easier to challenge and reuse.

## 11.3 Where the Picture Breaks

The baking analogy suggests one ingredient can be removed without changing the meaning of the recipe. In nonlinear dynamics, interventions change the state trajectory, and mechanisms interact. Removing damping can alter phase and impact time, not just energy loss. Counterfactual design reduces ambiguity but does not make coupled systems separable.

Human counterfactuals are harder. A golfer cannot repeat an identical internal state while one muscle command is magically removed. Randomized cues, perturbations, and participant holdout approximate

interventions but include adaptation and measurement noise. Model results should guide experimental design, not substitute for it.

**Go Deeper:** Read the [matched-state counterfactual chapter](#), the [same-state spatial killswitch](#), and the registered [experimental negative controls](#).

## Chapter 12

# Two Hands Make One Visible Wrench

### 12.1 A Concrete Picture

Turn a steering wheel by pushing down with one hand and pulling up with the other. The two forces can nearly cancel as a net force on the wheel's center, yet together they create a clear turning effect. Now squeeze the wheel equally from both sides. The internal load in your hands rises while the wheel's motion may not change at all.

Two hands on a golf grip create the same distinction. The club responds to the combined **wrench**: net force plus net moment about a chosen point. The arms and wrists may experience very different individual forces while producing the same club wrench. Club motion alone cannot reveal the hidden allocation.

#### Opposing Hand Forces Can Form a Couple Without a Net Push Narrow Grip Wider Separation

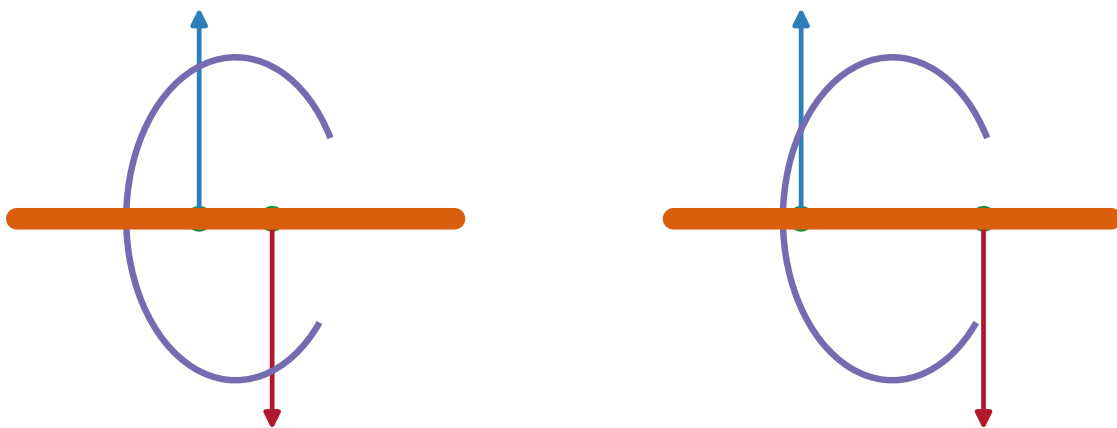


Figure 12.1: Equal and Opposite Hand Forces Produce a Turning Couple Whose Magnitude Grows With Grip Separation.

## 12.2 How the Mechanism Works

Let hand forces be  $\mathbf{F}_L$  and  $\mathbf{F}_R$  at positions  $\mathbf{r}_L$  and  $\mathbf{r}_R$  relative to club reference  $O$ . Include direct hand or wrist moments  $\mathbf{M}_L$  and  $\mathbf{M}_R$ . The equivalent club wrench is

$$\mathbf{F}_O = \mathbf{F}_L + \mathbf{F}_R,$$

$$\mathbf{M}_O = \mathbf{M}_L + \mathbf{M}_R + \mathbf{r}_L \times \mathbf{F}_L + \mathbf{r}_R \times \mathbf{F}_R.$$

All components use the same orthonormal axes at the same instant; each position vector points from  $O$  to its contact. The moments are contact moments exerted on the club, not automatically anatomical wrist joint torques. For a rigid club with known inertia and all other loads specified, the combined wrench determines the hands' contribution to acceleration. Many allocations produce the same sum. A flexible club also has deformation modes that a single resultant does not describe.

### 12.2.1 Common and Differential Modes

Define  $\mathbf{F}_c = (\mathbf{F}_L + \mathbf{F}_R)/2$  and  $\mathbf{F}_d = (\mathbf{F}_L - \mathbf{F}_R)/2$ . Let  $\mathbf{r}_c$  locate the contact midpoint relative to  $O$ , and  $\mathbf{d} = \mathbf{r}_L - \mathbf{r}_R$  point from the right contact to the left. Then, exactly,

$$\mathbf{F}_O = 2\mathbf{F}_c,$$

$$\mathbf{M}_O = \mathbf{M}_L + \mathbf{M}_R + 2\mathbf{r}_c \times \mathbf{F}_c + \mathbf{d} \times \mathbf{F}_d.$$

About the midpoint, the common force has no moment. About another point it usually does. Only the part of  $\mathbf{F}_d$  perpendicular to  $\mathbf{d}$  creates the differential force couple. Its axial part changes internal loading without changing the wrench.

If we add  $+\eta$  to one hand and  $-\eta$  to the other, resultant force stays unchanged. Depending on contact geometry, moment may change or remain unchanged along certain null-space directions. This is the mathematical source of hidden internal loading.

### 12.2.2 What Inverse Dynamics Can Identify

If full club motion and inertia are known well enough, inverse dynamics can estimate the required net external wrench. Isolating the hands requires subtracting gravity, aerodynamic loads and any ball-contact load using the same boundary and reference point. Impact may require an impulse balance rather than a smoothly differentiated acceleration. Clubhead position alone does not supply the complete rigid-body state or flexible-shaft motion. Even a perfectly known hand resultant cannot identify its split without additional measurements or assumptions.

A minimum-norm allocation is a reproducible mathematical choice. It is not a measurement of how a golfer used the arms. A muscle-effort optimization is still model-dependent. Instrumented grips with force and moment sensing provide the most direct route to individual hand wrenches.

### 12.2.3 The Invisible Axial Push–Pull Mode

For two point forces at distinct locations, with no free contact moments, the map is

$$G = \begin{bmatrix} I & I \\ [\mathbf{r}_L]_{\times} & [\mathbf{r}_R]_{\times} \end{bmatrix}, \quad [\mathbf{r}]_{\times} \mathbf{f} = \mathbf{r} \times \mathbf{f}.$$

Zero resultant force requires opposite force increments. Zero moment then requires  $\mathbf{d} \times \boldsymbol{\eta} = 0$ : the increment lies along the contact line. The null space has dimension one, so the map has rank five. Equivalently, the moment about the midpoint cannot have a component along  $\mathbf{d}$  in this point-force model. If the locations coincide, the rank drops to three and every opposite force pair is invisible.

For distinct known contacts, an added scalar measurement sensitive to the axial null direction closes the structural rank gap. A redundant measurement does not. Short separation also amplifies force errors associated with transverse moment noise, so full rank does not ensure reliable estimation. One scalar is insufficient if each hand can also apply its own three-dimensional contact moment: the full 12-to-6 bilateral-wrench map retains a six-dimensional invisible family. The calculation is a sensor-design result, not an anatomical result. It does not identify wrist action, scapular action, muscle recruitment, or intentionality, and calibrated noise, cross-talk, and moving contact centers still require separate qualification.

These dimensions concern unconstrained linear maps. Friction, unilateral contact, finite grip strength and allowable moments restrict physically possible loads. Their intersection with an algebraic solution family need not have the same dimension or contain any solution.

### 12.2.4 Internal Force Can Matter Without Moving the Club

Hidden loading may change grip stability, friction margin, joint compression, muscle activation and effective stiffness. The axial point-force null mode is not itself a measurement of normal grip pressure. At a fixed state, a zero-wrench load increment does zero power against a rigid velocity field, exactly. Its metabolic cost and tissue loading can still be nonzero. With deformation or slip, equal resultants can also conceal different mechanical power. There is no single internal metric called “effort” that settles these trades. Reducing an axial null load does not algebraically require increasing a contact moment.

This is why the technical article refuses to infer scapular recruitment from a club couple. A proximal generalized-torque subspace can be labeled and tested in the model, but the mapping to muscles, scapular motion, and hand pressure is not unique.

### 12.2.5 Action-Reaction Must Close

At a common physical interface, each force has an equal and opposite reaction. Transform moments to the same point and axes before summing both sides. An ideal rigid grasp uses closure constraints; a compliant grasp needs a constitutive law with stored-energy and dissipation terms. Opposite forces do not make contact-pair power vanish when the two sides have different velocities.

For loads on one rigid club,

$$P_{\text{hands}} = \sum_i (\mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_i \cdot \boldsymbol{\omega}) = \mathbf{F}_O \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega}.$$

Here  $\mathbf{v}_i = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_i$  is the club’s velocity field in one inertial observer. Changing the reference point transports both wrench and velocity; changing observer is a different operation. For a deforming grip or shaft, use actual local velocities and rotations. A single rigid-body power formula is insufficient. Power and energy residuals test these ledgers separately from action-reaction closure.

The [archived spatial contact test](#) uses MuJoCo and Pinocchio for independent kinematics and forward dynamics, but shares the contact law and time-stepping scheme. Agreement tests those engine implementations under shared assumptions; it cannot independently validate the shared law. The hands are finite-mass translational carriages, not anatomical arms.

### 12.2.6 Why Two Hands Can Create Negative Coupling

Near the late downswing, separated hand forces can produce a couple with a negative projection onto a declared swing-plane normal. The forces need not themselves point along that normal. Couple power is the full dot product with angular velocity: one negative component alone does not establish negative power. Even negative moment power can coexist with increasing club kinetic energy when force power and other external powers outweigh it. Inertial coordinate terms are not an extra energy source. The model asks where the couple comes from and whether the mechanism persists in a specified forward intervention.

At fixed bounded forces, collapsing hand separation eliminates the differential force couple; reversing  $\mathbf{d}$  with  $\mathbf{F}_d$  fixed reverses its sign. Free contact moments do not vanish in either control. These geometric checks are not predictions for a newly integrated swing: moving contacts can change forces, constraints, commands and motion. Keeping a nonzero force couple as separation tends to zero would require unbounded perpendicular differential force.

**Model Result:** Wrench reconstruction reveals nonunique internal modes. In the pinned reduced spatial test, switching off the declared driver at 0.18 s retains a negative swing-normal force couple for 0.0375 s, with minimum about  $-0.4093$  N m in both engines. This is a model-time interval, not a measured phase of a golfer’s downswing. Coincident and reversed geometry checks transport the same achieved forces; they are algebraic controls, distinct from the integrated driver-switch experiment. Stored energy and initial motion remain after the switch.

**Human Evidence:** Instrumented grips add information that club motion lacks, but sensor signals and inferred hand loads must be distinguished. Koike’s example used one professional instructor and statically calibrated strain gauges, with fixed hand reference points and neglected handle-piece inertia ([Koike 2016](#)). Choi and Park studied nine right-handed male professional golfers using a six-axis sensor between two grip segments and inverse dynamics. Their heavier grip, possible hand-to-hand load bypass and foam-ball impacts limit interpretation ([Choi and Park 2020](#)). Neither study identifies muscle recruitment or validates the present synthetic model’s hand allocation.

**Hypothesis:** Opposed arm-chain actions can generate a useful club couple while wrists supply resistance or impedance, but only synchronized bilateral wrench and biological measurements can test that allocation.

**Practical Interpretation:** The club can feel one net push and twist while the two hands experience a much richer internal negotiation.

### 12.2.7 From Two Contacts to One Club Load

Each hand can apply three force components and three moment components at its contact reference frame. Transform both six-dimensional wrenches to one club frame, including the moment created by each position offset, then add them. The resultant hand wrench contributes to rigid-body acceleration together with all other external loads and inertia. Shaft forces are internal to a whole-club boundary; they become boundary loads when a shaft or clubhead subsystem is cut out. Count them once at the chosen boundary. This sum is estimable in principle, but its split between hands is not determined by club motion alone.

In a planar simplification, the arithmetic is easier: two two-dimensional forces at separated points create a net force and an out-of-plane moment. Even there, multiple pairs yield the same resultant. Adding equal and opposite force to the two hands changes internal loading while leaving net force unchanged; depending on direction, it can change the couple. Distinct point-force contacts have one axial null direction in either two or three dimensions. Allowing full contact wrenches increases the nullity from three in the planar model to six in space.

### 12.2.8 Interpreting Push–Pull

The exact midpoint decomposition separates resultant force from the differential force couple and free contact moments. This coordinate change does not claim the nervous system controls those modes explicitly. It is a clean way to see which parts of bilateral loading are visible at the club and which can cancel.

The distinction also clarifies “push–pull.” Equal opposition is not required. One hand may supply a larger force, both may point partly in the same direction, and contact moments may matter. A measured push in grip-local coordinates can coexist with a laboratory-frame component that looks like a pull. The complete wrench should replace verbal labels wherever data permit.

### 12.2.9 How a Negative Couple Can Emerge

Near impact, proximal motion, club inertia, grip constraints, and contact geometry determine the reactions required at the hands. A differential force component acting through hand separation may project onto the swing-plane normal with negative sign even when neither hand is commanded with an explicit negative club torque. Calling a reaction “passive” requires a declared partition: the same contact force can depend on active proximal commands, stored elastic energy and inertia. State dependence alone does not establish muscular passivity.

That statement is narrower than “the scapulae passively torque the club.” The generalized drivers can stand for many body actions. Reconstructing a negative contact couple does not uniquely map it to shoulder-girdle motion, wrist action, or muscle activation. The model supplies an observable prediction for bilateral wrenches; anatomy remains a separate inference problem.

### 12.2.10 A Worked Wrench Example

Let the hands be separated by 0.20 metres. Suppose they apply opposite 50-newton forces perpendicular to that separation. Net force is zero and couple magnitude is 10 newton-metres. Add the same 100-newton vector to each hand: net force becomes 200 newtons and moment about the contact midpoint remains 10 newton-metres when free contact moments are zero. About an offset reference, include the additional moment of the common force. If separation drops to 0.10 metres, the same opposing forces make 5 newton-metres.

Real force vectors are not perfectly perpendicular, contact points are distributed, and hand moments may contribute. The arithmetic nonetheless gives testable dependencies: couple scales with perpendicular separation and force; reversing either reverses sign; collapsing either collapses the couple. A model or measurement pipeline should pass those controls exactly within tolerance.

### 12.2.11 Matching the Club Task, Varying the Allocation

To compare arm-dominant and wrist-dominant strategies fairly, first specify a club-level task: perhaps the same net wrench history or a set of terminal club states. Solve for distinct internal allocations that meet it. Then compare joint demands, individual hand power, co-contraction proxy, transmission state, peak load, and sensitivity. The matched task prevents one strategy from “winning” merely because it was asked to produce a larger club moment.

Next relax the task and optimize performance under common constraints. This second experiment asks whether internal freedom can improve speed or robustness. Keeping matched-task and optimized comparisons separate avoids confusing mechanical equivalence with performance preference.

### 12.2.12 What Bilateral Sensors Add

Separate six-axis grip sensors directly constrain each contact wrench. Pressure mapping can reveal distributed contact and shifts in effective application point. Shaft strain helps close the club load balance. Synchronized motion supplies velocities for contact power and transforms sensor axes to club and laboratory frames. Force plates and body motion support whole-system closure.

Calibration needs known loads, crosstalk characterization, thermal drift checks, and frame registration. Grip modification can itself change behavior, so the instrumented club should be compared with a standard club. Uncertainty in hand position and sensor orientation should propagate to couple sign and magnitude.

### 12.2.13 Why Closure Is Not Calibration

A [pinned synthetic sensor qualification](#) makes the distinction quantitative. Its net-wrench-only inverse closes the resultant to numerical tolerance while giving 11.86 N component-wise RMSE across the six hand-force components and 29.05 N RMSE for its declared axial scalar. These are errors against synthetic ground truth. Closure tests consistency with the supplied resultant; it does not establish that either the resultant or allocation is physically correct.

The same controlled study separates practical measurement errors. Correcting a declared cross-talk matrix reduces allocation RMSE from 0.94 N to 0.15 N; a 10% error in the cross-talk correction amplitude raises it to 0.18 N; this is not ten-percent device accuracy. Holding contact centers fixed

with the migration amplitude parameter set to 8 mm causes 2.02 N RMSE; this parameter scales prescribed sinusoidal coordinates, not total travel. Tracking those centers removes that modeled bias in the idealized migration case. The combined case, with noise, residual cross-talk error and untracked migration amplitude set to 4 mm, gives 1.02 N component-wise RMSE and 3.87 N at the 95th percentile of the six-component error-vector norm over 32 seeded trials with 301 samples each. The percentile and RMSE use different aggregations; they are not two statistics of one scalar hand-force error. The 8-mm and 4-mm settings are different cases.

These values come from a synthetic point-force sensor qualification, not a physical transducer, distributed-pressure experiment, or participant study. They define pre-acquisition acceptance tests: verify the augmented rank, report allocation error in addition to wrench closure, calibrate cross-talk, track effective contact centers, propagate residual calibration uncertainty, and fail closed when any required channel or registration is absent. Governed bilateral six-axis human data are still required to test the proposed hand, wrist, arm, or scapular mechanisms.

#### 12.2.14 Internal Force Is Not Wasted Force

An equal-and-opposite hand mode may cancel at the club's center-of-mass force balance while increasing grip compression, joint load, or stiffness. It may help stabilize the clubface or may be needless co-contraction. Net club motion cannot decide. Outcomes and internal-load measurements are required.

This is where the question of “slack” becomes scientific. If continuous bilateral loading reduces a measurable force-development delay or perturbation response time without excessive internal cost, preload has operational value. If the same club task occurs with no gap or robustness difference, the proposed benefit is unsupported. A feeling of connection is not yet a measured state.

#### 12.2.15 One Point per Hand Is Itself a Modeling Choice

Treating each hand as a single contact point is convenient, and the natural worry is that the convenience is doing the work. A [registered reduced-model comparison](#) examines one part of that question.

Each hand was replaced by one, three, or five tension fibers spread across a 30 mm band, while **total per-hand** stiffness and damping were held fixed at 1800 N/m and 18 N·s/m, divided equally among that hand's stations. That control matters: adding contact points without holding the total constant would silently stiffen the grip, and any change could then be blamed on stiffness rather than on distribution.

The archived design contains 288 trajectories: twelve initial states, three station counts, two velocity branches, two integration steps and two engines. The 4, 10, 25 and 50 ms outputs are nested observation horizons within each 50 ms trajectory, not integration timesteps or separate trials. The steps are 1 and 0.5 ms. Initial motion supplies the starting condition; this experiment does not use active joint drivers.

Peak single-station force reached 4.03 N. The maximum concentration of 0.77 is the largest station-force norm divided by the sum of all station-force norms at an instant. It is neither pressure nor a named finger's share. The summary also reports zero transitions of whether **any station is active**. That counter does not track every station's engagement. Indeed, the maximum fraction of samples with fewer than all stations active is 1.0. Zero recorded global on/off transitions therefore cannot establish that every contact stayed closed.

The maximum three-to-five-station comparison at the fine step is 0.00355. Its implementation compares final coordinate arrays using maximum absolute component difference divided by the larger of one and the two maximum coordinate magnitudes. This is a registered coordinate-dependent diagnostic, not a unit-invariant physical error or proof of convergence of forces, energy and contact events. The stored coordinates mix translations and rotations; units and parameterization affect the comparison. The worst normalized work-energy residual also falls from 0.03119 to 0.01560 on timestep refinement. Passing these particular checks supports a bounded comparison, not universal discretization convergence.

Holding total stiffness and damping fixed makes the comparison interpretable. It does not measure grip pressure, identify finger anatomy, or model friction or soft tissue. A number of stations is not a set of fingers. These statements describe the pinned archive, not a new solver run or current-human validation.

### **12.2.16 Minimum Evidence for a Human Claim**

The model prediction should specify phase, frame, couple sign, magnitude range, geometry dependence, and competing explanations. A human test should measure bilateral wrench, motion, shaft response, and relevant activation, with participant-held-out analysis. Only then can researchers ask whether a modeled contact mechanism is commonly used, individually variable, or absent. Until then, it remains a falsifiable mechanical hypothesis—not a prescription.

### **12.2.17 Reader Lab: Explore the Hidden Mode**

Start with one pair of hand forces that produces a target club wrench. Add a small vector to one hand and subtract it from the other. Recompute the resultant. Depending on the vector and hand separation, net force stays fixed while moment changes. Add compensating contact moments so the full target wrench is restored. You have constructed a distinct internal allocation with identical rigid club loading.

Repeat while minimizing different costs: total force squared, peak wrist moment, lead-hand load, or deviation from a measured pressure pattern. Each criterion selects a different solution. This makes the role of regularization visible. Without bilateral measurement, none should be called the observed allocation.

Now add simulated sensor noise and contact-position uncertainty. Estimate the confidence interval on swing-normal couple and on each hand's contribution. A negative mean with an interval spanning zero is not firm evidence of sign. The lab links elementary wrench geometry to realistic qualification.

### **12.2.18 How This Guides Product Design**

The software should display net wrench and the family of compatible allocations when individual forces are unavailable. When bilateral data exist, it should show measured allocation with uncertainty and closure residual. Any optimized allocation should be labeled as a model solution with its cost function.

This prevents the interface from converting hidden assumptions into anatomical facts. It also makes the tool useful for experiment design: users can see which sensor or manipulation would shrink the ambiguity most.

## 12.3 Where the Picture Breaks

The steering wheel is rigid, symmetric, and fixed to one axis. A golf grip moves in space, the shaft bends, wrist axes rotate, pressure is distributed, and both arms attach to a moving body. Two point contacts are a reduction.

The common/differential transformation also does not identify comfort or intent. The same internal wrench can arise from different muscles and co-contraction. Technique claims need participant-level evidence.

**Go Deeper:** Read the complete [two-hand wrench mechanics](#), the [forward constrained model](#), and the instrumented-grip literature ([Choi and Park 2020](#); [Koike 2016](#)). The technical chapter links follow upstream development; numerical claims above use the explicitly pinned artifacts so their definitions remain inspectable.

## Chapter 13

# Negative Torque Can Help a Positive Outcome

### 13.1 A Concrete Picture

Imagine a club receiving a force through its moving grip while a couple opposes its rotation. The couple can remove rotational power at the chosen grip point while the force supplies more power than the couple removes. The club can gain kinetic energy. No energy is created by the minus sign: two physical inputs have different effects, and their sum decides the instantaneous energy change.

A second question is harder. Could a schedule containing an opposing moment produce a better later delivery than a schedule without it? Possibly, because the moment changes the evolving state and the work subsequently done by other inputs. That outcome requires a controlled comparison. It does not follow from negative torque, apparent wrist lag, or the existence of a reaction force.

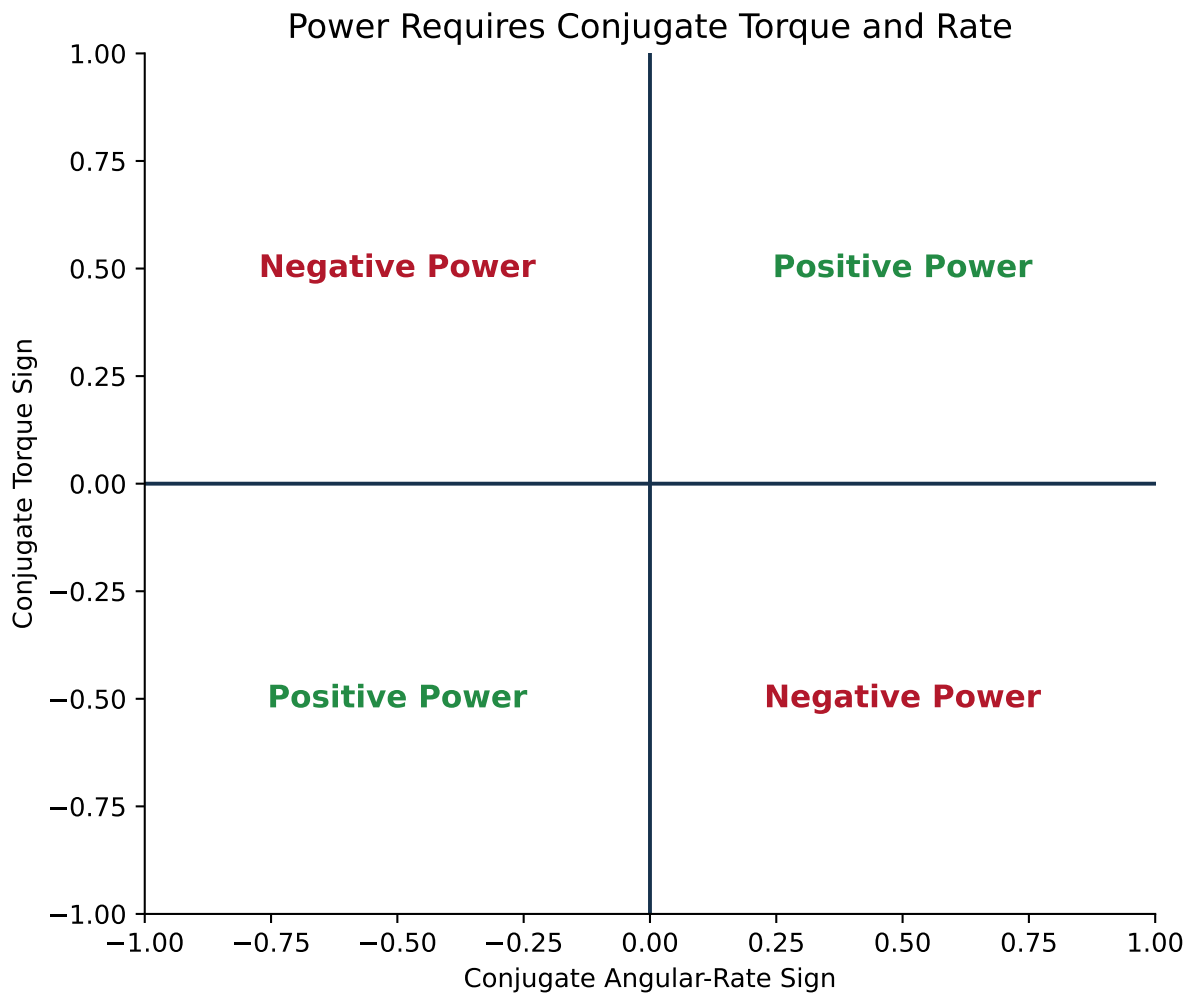
### 13.2 How the Mechanism Works

#### 13.2.1 Name the Torque and Its Conjugate Velocity

For a single rotational power channel,  $P = \tau \dot{\theta}$ . The word *conjugate* means that this torque and this rate belong to the same virtual-work pair. A club-side free moment uses the club's absolute angular velocity in the declared inertial observer. An internal wrist moment acting between club and forearm has an equal and opposite reaction on the forearm. Their combined rotational joint power uses the **relative** angular velocity; the subscript m below denotes the moment contribution:

$$P_{\text{club,m}} = \tau \omega_c, \quad P_{\text{arm,m}} = -\tau \omega_a, \quad P_{\text{joint}} = \tau(\omega_c - \omega_a).$$

For a manufactured planar example, let  $\tau = -2$  N m on the club,  $\omega_c = 3$  rad/s and  $\omega_a = 5$  rad/s. The moment supplies  $-6$  W to the club and  $+10$  W to the arm; the pair supplies  $+4$  W. Calling this a negative-power joint merely because the club moment is negative would be wrong. With coordinates  $(\theta_a, \beta)$ , where  $\beta = \theta_c - \theta_a$ , this internal motor has generalized force  $(0, \tau)$  and power  $\tau \dot{\beta}$ . With absolute coordinates  $(\theta_a, \theta_c)$  it has generalized force  $(-\tau, \tau)$ . Both descriptions give the same total power.



Club Moment: Club Rate. Joint Pair: Relative Rate.

Figure 13.1: Power Signs for One Conjugate Rotational Channel: Club-Side Moment Uses Absolute Club Rate; an Internal Joint Pair Uses Relative Rate.

In three dimensions use the full dot product  $\mathbf{M} \cdot \boldsymbol{\omega}$ , or  $\mathbf{M} \cdot (\boldsymbol{\omega}_c - \boldsymbol{\omega}_a)$  for the rotational joint pair, with vectors expressed in one common frame. A negative swing-normal component alone does not determine the full dot product. Neither Euler-angle derivatives nor anatomical labels can be substituted for angular velocity components without their coordinate mapping. Reversing a scalar axis reverses both conjugate signs and leaves power unchanged.

### 13.2.2 Follow Power Across a Declared Boundary

For a rigid club before impact, choose an inertial observer and a material grip point  $H$ . Reduce all hand contacts to a resultant force  $\mathbf{F}$  and moment  $\mathbf{M}_H$  about that point. If the only other external action is uniform gravity, the kinetic-energy balance is

$$\dot{T}_c = \mathbf{F} \cdot \mathbf{v}_H + \mathbf{M}_H \cdot \boldsymbol{\omega}_c + m\mathbf{g} \cdot \mathbf{v}_C,$$

where  $C$  is the club's center of mass. Other physical loads require their own power terms. If gravitational potential is included in stored energy instead, gravity moves to the left side; it is not counted again on the right.

For a flexible club, include shaft strain energy and distinguish dissipation from recoverable storage. The exact input is the sum of contact forces dotted with their actual contact velocities, together with any contact-moment powers. A single rigid wrench formula needs compatible rigid motion; deformation or slip can add terms. Shaft release redistributes stored club energy. It is not a second external input to a boundary already enclosing the entire shaft. Likewise, *inertial coupling* describes the equations' redistribution of motion; it is not an additional energy supply alongside the complete physical balance.

Negative instantaneous power means the selected channel removes energy at that instant. Work is its time integral. A short negative interval does not determine the sign of work over the entire downswing, total club energy, or final speed.

### 13.2.3 A Moment Can Change Sign When Its Reference Point Changes

Keep the observer fixed and let  $\mathbf{r} = \mathbf{r}_C - \mathbf{r}_H$ . Rigid velocity and moment transport give

$$\mathbf{v}_C = \mathbf{v}_H + \boldsymbol{\omega}_c \times \mathbf{r}, \quad \mathbf{M}_C = \mathbf{M}_H - \mathbf{r} \times \mathbf{F}.$$

The complete wrench power is the same about  $H$  and  $C$ , although its force and moment parts differ. Take  $\mathbf{r} = (0, 1, 0)$  m,  $\mathbf{v}_H = (2, 0, 0)$  m/s,  $\boldsymbol{\omega}_c = (0, 0, 3)$  rad/s,  $\mathbf{F} = (10, 0, 0)$  N and  $\mathbf{M}_H = (0, 0, -2)$  N m. Then  $\mathbf{v}_C = (-1, 0, 0)$  m/s and  $\mathbf{M}_C = (0, 0, 8)$  N m. At the grip, power is  $20 - 6 = 14$  W; at the center it is  $-10 + 24 = 14$  W. The reported moment changes from negative to positive without changing the physical force system. These are instantaneous manufactured values, not a golfer's measurements.

Only a pure couple, with zero resultant force, has reference-independent moment. For general hand loading, always name the moment point. Changing the inertial observer is a different operation: body power can change with observer even when reference-point transport within one observer preserves it.

### 13.2.4 What Two Hands Add

Let  $L$  and  $T$  denote the lead and trail hands. At the grip midpoint, use the archive’s convention for contact separation and differential force:  $\mathbf{d} = \mathbf{r}_T - \mathbf{r}_L$  and differential force as  $\mathbf{F}_d = (\mathbf{F}_L - \mathbf{F}_T)/2$ . The force-generated moment is  $\mathbf{M}_{F,H} = -\mathbf{d} \times \mathbf{F}_d$ . Any free moments applied by the hands must be added separately. Large opposing forces can therefore produce a substantial moment despite a small resultant.

Holding the forces fixed while reversing  $\mathbf{d}$  reverses this moment; collapsing finite-force lever arms removes it. Neither operation proves that a new forward swing will have reversed or vanished total torque. A changed grip geometry changes the constraints and their reaction forces, may require a new compatible initial state, and need not eliminate free moments. Rotating the entire force system and its measuring axis together preserves the scalar projection. Changing their relative geometry is the relevant intervention.

Ideal maintained contact can transfer energy between bodies while doing zero net work on the complete system. For stationary holonomic constraints with  $J\dot{q} = 0$ , total constraint power is  $\lambda^T J\dot{q} = 0$ , where  $q$  contains generalized coordinates,  $J$  is the constraint Jacobian and  $\lambda$  contains the multipliers producing generalized reaction  $J^T \lambda$ . One-sided club contact power need not be zero. Prescribed moving supports, compliance, friction and impacts require their own accounting; this ideal cancellation is not a universal contact law.

### 13.2.5 Three Model Results Answer Three Different Questions

The pinned [two-hand archive analysis](#) reconstructs an archived project simulation at achieved states. Its zero-command table is **pointwise**, not one torque-free forward swing. The late negative couple is a contact-force reaction with current commanded wrist moments and the designated damping removed. Earlier commands helped create the states being evaluated.

Crucially, this archive does not show a positive-power boost from that late couple. Reconstructed contact-force power, the sum  $\mathbf{F}_L \cdot \mathbf{v}_L + \mathbf{F}_T \cdot \mathbf{v}_T$  on the club, is negative at about 85% of samples inside its approximately 88 ms negative-couple interval, with about  $-149$  J of integrated contact-force work. The archive’s separate power columns have a reported pointwise discrepancy, reaching about 73 W; interval agreement does not remove that limitation. These are model records, not participant data or proof of a speed-improving strategy.

The separate [forward constrained test](#) uses two planar arms and a floating rigid club. Its zero-command branch at 0.200 s retains a negative force-generated couple for the next 50 ms. A later branch changes sign before its 50 ms window ends. This demonstrates finite, state-dependent persistence in that model. It neither reproduces the WSCG trajectory nor establishes a benefit in speed, human muscle behavior or impact delivery. Reported numerical projection corrections also remain part of its error assessment, not physical work.

The [uncertainty and control comparison](#) uses a different planar moving-base, two-hand, flexible-club model with bounded actuators. Among eight sampled schedules and six held-out parameter draws, **early\_restrain** has the highest sampled 10th-percentile endpoint speed, about 4.706 m/s. Its mean planar face/path error proxy exceeds that of **late\_drive**. All eight candidates are nondominated under five objectives: this speed quantile, mean face/path error, upper-tail hand force, squared-torque effort and speed dispersion. Nondominated means no other sampled candidate is at least as good in every objective and better in one. The endpoint is a fixed 0.24 s delivery time, with no ball collision.

These results describe a finite synthetic comparison, not a continuous optimum, population reliability bound, or calibrated human swing. The schedule name alone does not establish the sign of its work throughout the trajectory.

**Model Result:** A negative contact moment can persist without current command, and a selected restraining schedule can perform well under a separate model’s objectives. Persistence and performance are different findings.

**Human Evidence:** Morrison and colleagues studied kinematic variability with uncontrolled-manifold analysis; that citation does not establish wrist EMG or identify a passive grip couple ([Morrison et al. 2016](#)). Nesbit combined recorded motion with a full-body and flexible-club model to estimate kinetics. Its wrist motion was kinematically driven, and its hand representation was simplified; inferred interface loads are not isolated muscle measurements ([Nesbit 2005](#)).

**Hypothesis:** In some feasible states, an opposing distal contribution may improve later delivery by changing geometry and the subsequent allocation of work. The useful states, mechanisms and tradeoffs remain to be established.

**Practical Interpretation:** Ask for the body, reference point, conjugate velocity, work interval and objective before judging a torque sign.

### 13.2.6 How to Test Whether Restraint Helped

Start with a complete copied state: positions, velocities, actuator activation and any elastic or controller memory included in the model. Declare whether the intervention removes a command, a finite realized torque, damping, or a contact mode. With smooth finite forces and the same constraints, removing a torque does not instantaneously change velocity. It changes the applicable acceleration and subsequent evolution; a particular acceleration component need not change. Activation dynamics may delay the effect of command removal. An impulsive event or changed contact constraint is a different intervention.

Integrate each branch with its own evolving reactions and power balance. Compare the same declared endpoint: fixed time, a registered delivery event, or a modeled impact. If an event moves or becomes unreachable, report that instead of comparing unlike states. Subtract branch outcomes to estimate a conditional effect within the declared model. A pointwise torque decomposition is not a decomposition of separately integrated future trajectories.

For a strategy comparison, state what remains matched: initial state, proximal control policy, available actuator capacity, task requirements and uncertainty draws. Identical torque schedules do not imply equal work once velocities differ. Reoptimization asks whether another feasible route can substitute for restraint; it changes the question from removing one input in a fixed policy. Neither comparison should silently receive the other’s interpretation.

Descriptions such as arms driving while wrists regulate, or a phase-dependent reversal of those roles, are hypotheses about measured force and moment allocation. Continuous loading can change actuator and elastic history even at similar poses. Compare those proposals through matched tasks, bilateral loads and work, rather than assigning a biological role from one torque curve.

An early opposing phase might delay release and change later force directions; it might also increase proximal work, violate load limits or leave inadequate time. There need not be an interior optimum or

a useful restraint interval. Map feasible outcomes before claiming an optimum region. Keep the finite candidate comparison distinct from any newly proposed search.

### 13.2.7 Negative Work Needs a Destination

For a torsional spring with relative deflection  $\beta$ , torque  $-k\beta$  gives relative power  $-k\beta\dot{\beta} = -\dot{U}$ , where  $U = k\beta^2/2$ . Absorbed mechanical energy is stored and may later return. A viscous damper instead gives power  $-c\dot{\beta}^2 \leq 0$  for  $c \geq 0$ . The same instantaneous power sign can therefore accompany recoverable storage, transfer to another body, or dissipation. Select the actual mechanism and boundary before naming a loss.

At a human joint, a net moment and relative rate cannot uniquely separate muscle fibers, tendons, passive tissues and opposing muscle groups. EMG supplies an activation-related signal, not a unique force or work measurement. Bilateral wrench sensing, registered contact geometry, shaft measurements and a calibrated model can constrain the explanation; their uncertainty and unresolved internal loads must remain visible.

### 13.2.8 Reader Lab: Audit a “Negative Helps” Claim

Reproduce the  $-6$ ,  $+10$  and  $+4$  W joint example, then the two equivalent 14 W wrench reductions. Explain why neither calculation establishes a future speed benefit. Plot torque with its conjugate rate, instantaneous power and cumulative work for the selected interval. Add all other physical inputs and stored-energy changes, and inspect the closure residual.

Next specify a feasible copied-state intervention and the future outcome being compared. Separate evidence for moment sign, energy transfer, persistence and performance. A failure of the negative sign to persist would limit a persistence hypothesis; a worse feasible outcome would limit a strategy claim. A human measurement inconsistent with the predicted wrench would challenge transfer of that mechanism to golfers. None of these tests is replaced by an attractive sequence plot.

## 13.3 Where the Picture Breaks

The brake metaphor suggests an independent wheel. A swing has moving supports, coupled coordinates, elastic storage and several objectives. Even the sign of the reported club moment needs a point and axis; a favorable planar result can coexist with an unfavorable face orientation or load.

“Passive” must name the channel left uncommanded. Zero current command does not mean zero force, zero prior work, zero activation or biological relaxation. The useful question is how a specified action changes a feasible trajectory and its energy balance, with evidence strong enough to support that scope.

**Go Deeper:** Compare the pinned [timing and counterfactual framework](#) with the distinct archive, forward and uncertainty studies above. Their model boundaries and endpoints must accompany any claim carried into this chapter.

## Chapter 14

# Arms, Wrists, Preload, and the Meaning of Slack

### 14.1 A Concrete Picture

Turn a two-hand valve. One strategy uses the arms to create most of the opposing push-pull while the wrists remain firm. Another uses wrist moments more directly while the arms stabilize. The valve can receive the same net turning effect in both cases, but the loads inside the person differ.

Now reverse the motion. Motion reversal need not reverse the applied torque: a continuing torque can first brake one direction and then accelerate the other. A gear train that stays loaded against one tooth face avoids traversing backlash. That does not make torque tracking instantaneous; compliance and actuator dynamics can still delay the response. Reversing the load can instead require crossing a gap before the opposite face engages. Biological tissue is not a gearbox, but the picture creates a testable question: does maintaining preload and load direction reduce a measurable transmission delay compared with complete role reversal?

### 14.2 How the Mechanism Works

The first experiment matches a **scalar control contribution**, not the entire club task. At a fixed configuration and velocity, the constrained dynamics give an affine acceleration map. Subtracting the zero-control solution leaves a linear control contribution. The planar target is

$$I_C[\ddot{\alpha}(u) - \ddot{\alpha}(0)] = 8 \text{ N m},$$

where  $I_C$  is club rotational inertia and  $\alpha$  is club angle. The drift acceleration  $\ddot{\alpha}(0)$  remains part of the total motion. Proximal-only and wrist-only generalized controls can each attain this increment at the sampled states. Their convex mixtures attain it too, because this fixed-state map is linear. Feasibility under strength, contact, activation, or other constraints requires a separate check; the algebra alone does not establish human capacity.

Equal turning moment does **not** mean equal wrench. A planar wrench includes two force components

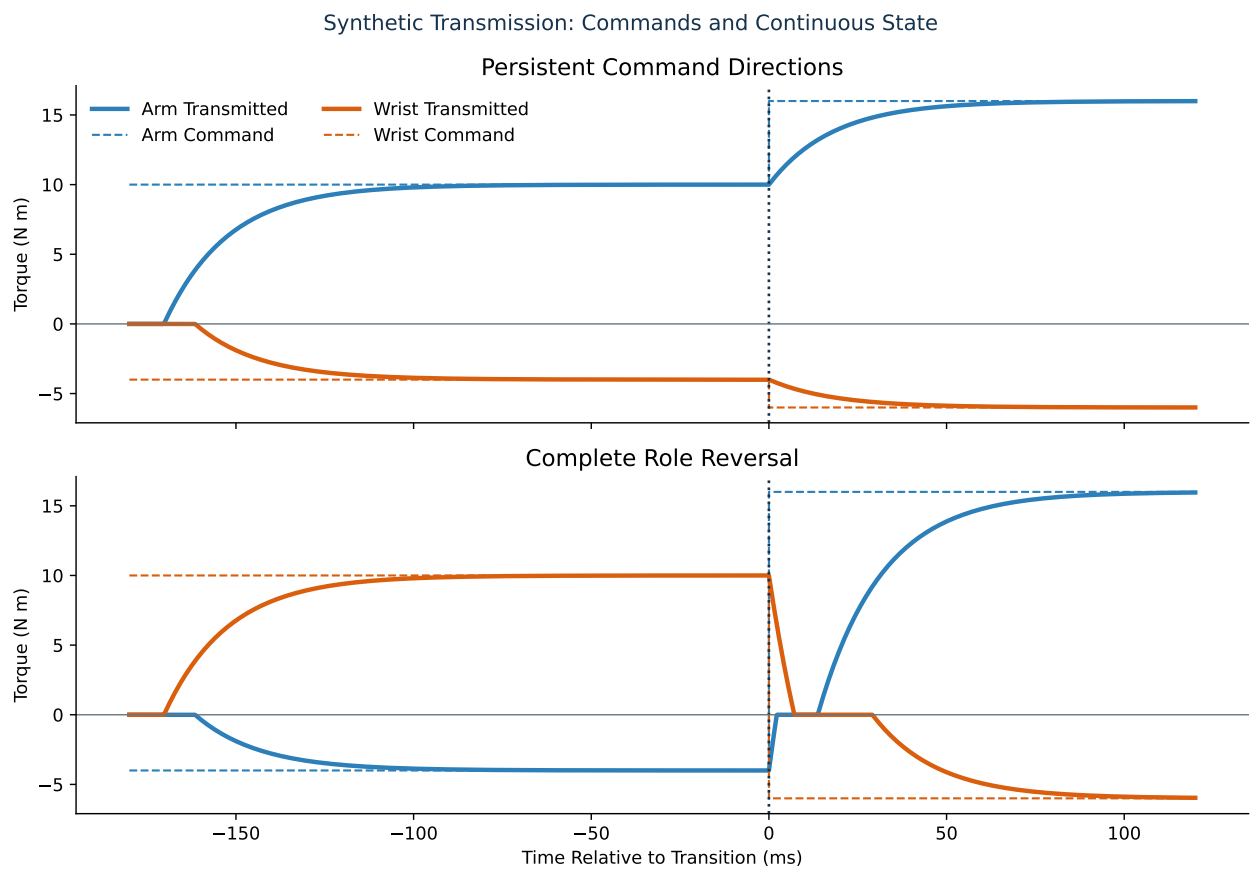


Figure 14.1: Commands and Transmitted Torques in the Declared Synthetic Preload Experiment.

as well as moment about a specified point. Equal moments can accompany different resultant forces, club translation, and work. For example, adding the same transverse force at two symmetrically placed contacts changes the resultant without changing their combined moment about the midpoint. Even matching a complete wrench would leave some internal allocations undetermined.

The study varies 19 club angles and 21 mixture fractions. Its reported hand-force RMS is calculated from the **control-induced contact-force increments**, not complete measured hand loads. Its generalized-torque norm is an unweighted Euclidean norm in the declared actuator coordinates; it is neither muscle effort nor metabolic cost. Changing coordinate scaling or cost weights changes the meaning of a minimum-norm allocation.

The labels *arm channel* and *wrist channel* identify generalized actuator subspaces. A wrist control can also induce contact forces through the closed chain; it does not isolate a pure contact moment from every other effect. Zero direct wrist actuation does not imply zero passive wrist resistance or zero constraint reaction. These results do not identify scapular retraction, muscle activation, or a preferred technique.

### 14.2.1 Two Extreme Stories

At one end, proximal arm-chain action creates the grip couple while wrists supply resistance or impedance. The same general proximal drive can continue across backswing and early downswing while signs remain loaded. At the other end, wrist moments create more of the backswing couple while arms resist; during downswing, roles reverse so arm-generated couple becomes positive and wrist action opposes release.

These are useful poles, not a binary classification of golfers. Real allocations can lie between them and vary continuously through time. The comparison asks what each pole predicts under matched tasks.

### 14.2.2 Replace “Slack” With Measured Variables

Slack can mean tendon slack length, grip gap, low muscle activation, low tangent stiffness, joint laxity, series-elastic extension, or a delay in force rise. Those are not synonyms. Having studied instantaneous allocation, we now ask a different question: how does an internal transmission state affect the response to a changing command? This second experiment uses two abstract channels, each with deflection  $x$ , commanded torque  $\tau^*$ , stiffness  $k$ , dead-zone half-width  $\delta$ , and time constant  $T$ :

$$x^* = \text{sgn}(\tau^*) \left( \delta + \frac{|\tau^*|}{k} \right),$$

$$T\dot{x} = x^* - x.$$

$$\tau = k \text{sgn}(x) \max(|x| - \delta, 0).$$

For a zero command,  $x^* = 0$ . The parameters are  $k = 600$  N m/rad,  $\delta = 0.012$  rad, and  $T = 18$  ms in **both** channels. The dead zone has total width  $2\delta$ . Deflection follows a first-order lag; torque is obtained through the dead-zone law. This is a phenomenological transmission model, not a measured activation law or a literal anatomical gearbox. It contains no calibrated muscle energetics, and its tracking score cannot establish energetic efficiency.

### 14.2.3 Continuous Preparation Matters

Starting at a loaded equilibrium would hide how preload arose. The finite-history test starts both channels at zero deflection, applies preparation commands for 180 ms, then changes commands without resetting the state. Each pair is (arm, wrist) in N m:

- **Persistent Directions:** Preparation (10, −4); after transition (16, −6).
- **Complete Role Reversal:** Preparation (−4, 10); after transition (16, −6).

Both desired sums change from 6 to 10 N m. This scalar experiment is distinct from the earlier fixed-state 8 N m allocation experiment; it does not simulate a complete anatomical backswing. Preparation lasts ten time constants, bringing the state close to equilibrium without reaching it exactly. The figure plots the archived transmitted torques alongside the commands, including the initial rise from the relaxed state.

The reported score integrates **absolute net torque tracking error** over the first 120 ms after transition:

$$J = \int_0^{0.120\text{ s}} |\tau_{\Sigma}^*(t) - \tau_{\Sigma}(t)| dt.$$

It has units N m s, but is not signed angular impulse and is not work. Finite preparation gives  $J = 0.0717151$  N m s for persistent directions and  $0.1954089$  N m s for reversal. The calculation uses a 0.1 ms forward-Euler step and a trapezoidal error integral. These are synthetic model outputs, not estimates of human delay.

Persistent directions have no post-transition zero-transmission samples. Reversal has arm and wrist zero-torque occupancies of 11.5 and 22.0 ms, with the study’s one-step brackets [11.4, 11.6] and [21.9, 22.1] ms. These sampling brackets are not total model-error or physiological confidence intervals. Occupancy counts samples with torque magnitude at most  $10^{-12}$  N m; it is distinct from time to useful opposite torque. The corresponding first attainment of 10% of the post-transition target in its new direction is 15.7 and 31.1 ms. From an identical relaxed state at transition, both programs have identical future commands and therefore identical responses ( $J = 0.2010714$  N m s).

### 14.2.4 What the Zero-Dead-Zone Control Establishes

With  $\delta = 0$  and the **same  $T$  in both channels**, the sum satisfies  $T\dot{\tau}_{\Sigma} + \tau_{\Sigma} = \tau_{\Sigma}^*$ . Equal initial sums and future net commands then give equal net responses, despite different component histories. The registered 12-case grid combines half-widths 0, 0.004, 0.012, and 0.024 rad with common time constants 8, 18, and 35 ms. Its three zero-width cases are equivalent within  $10^{-10}$  N m s; the other nine favor persistent directions on this score. This is a conditional result within the specified family, not a universal allocation optimum.

Unequal channel time constants break the sum reduction even with no dead zone. For equilibrium initial torques (10, −4) versus (−4, 10) and the common final command (16, −6), the difference in net responses is  $14[\exp(-t/T_{\text{arm}}) - \exp(-t/T_{\text{wrist}})]$  N m. It is generally nonzero when the time constants differ. Thus the negative control isolates a mechanism within this model and comparison; it cannot uniquely identify a human mechanism from a similar trace. Removing the dead zone also leaves first-order lag: no gap does not mean immediate command tracking.

### 14.2.5 Compliance Can Be Useful

Eliminating all compliance is not an implied goal. Elastic elements can store and return energy; damping dissipates it. Compliance can change peak loads, timing, and sensitivity, with benefits or costs depending on the excitation and objective. None of those outcomes is measured by the tracking score alone. The relevant question is which stiffness and preload suit the task, including internal load and physiological cost, rather than how to maximize rigidity.

The wrist-led extreme may reduce some proximal load, use elastic storage, or aid participant-specific accuracy. The current objective does not price those benefits. Its conditional continuity result therefore cannot select a universal winner.

### 14.2.6 What Must Be Measured

A serious human test needs bilateral six-axis grip wrenches and pressure, closed-chain inverse dynamics constrained by those measurements, wrist and shaft kinematics, shaft strain, transition-aligned force-development delay, and EMG from forearm, shoulder, and scapular muscles. Perturbation or system-identification methods should estimate tangent stiffness at a declared posture, activation state, perturbation amplitude, and time scale. Dynamic impedance can also include damping, inertia, and feedback. Preload is an operating load, not itself a stiffness measurement. Holding out participants tests generalization; it does not remove individual differences or justify prescribing a group average.

### 14.2.7 Define Slack Operationally

Distinguishing these mechanisms requires different measurements: grip separation can indicate contact loss, bilateral forces resolve load modes, and perturbation response estimates stiffness. EMG and tissue imaging add complementary evidence without directly measuring the net wrench. The transmission experiment alone does not tell us which mechanism generated a measured trace.

A separate synthetic constitutive audit keeps five candidate mechanisms separate: contact disengagement, transmission dead zone, structural preload, biological series compliance, and control deadband. At the selected synthetic operating points, each class has a rank-three local sensitivity matrix for its three fitted parameters under the registered excitations. This concerns parameters **within an assumed class**, not selection of the class itself, global uniqueness, or reliable estimation with noisy data. Contact and biological surrogates have a normalized output RMSE of about 0.0196 under the registered multisine, but that number is not a statistical detection threshold. Their similar traces leave single-channel class identification unestablished. This is a useful negative result: a clean-looking force trace cannot by itself tell a reviewer which kind of “slack” was present. Forward delivery and human tests therefore need independent contact, shaft, tendon, activation, and bilateral wrench measurements.

### 14.2.8 Why Continuous Loading Might Help

Force does not appear instantaneously in biological tissue or actuators. Electromechanical delay, tendon compliance, muscle short-range stiffness, grip deformation, and neural feedback all introduce history. Maintaining some load through transition might reduce the time required to develop a useful wrench in the early downswing and could improve resistance to perturbation.

Continuous load could also require more co-contraction, raise internal load, or constrain useful reorientation. These are proposed trade-offs to measure; the two-channel model does not estimate

fatigue or physiological cost. “No slack” is therefore not automatically optimal. The correct question is whether a measurable loaded state improves a declared outcome at an acceptable internal cost compared with alternatives.

### 14.2.9 Why Role Reversal Might Help

Reversal can exploit different muscle groups and mechanical advantages across phases. The backswing has more time and different geometry than the late downswing. A wrist-dominant backswing allocation might position or load the club efficiently, while proximal controls take a larger share of downswing drive and wrist controls resist. That is a proposed allocation, not a measured anatomical hierarchy established by this model. Antagonistic preactivation could also preserve stiffness despite a change in net torque sign.

Therefore role reversal need not mean mechanical disengagement. Two opposing preloaded torsional elements can exert  $+5$  and  $-5$  N m at equilibrium while each contributes 100 N m/rad of restoring stiffness: net torque is zero but restoring stiffness is 200 N m/rad. This illustrative construction is not a fitted wrist model. Net moment can reverse while individual muscles or hand contacts remain loaded. The experiment must measure force magnitude, differential modes, and stiffness rather than inferring engagement from net sign.

### 14.2.10 A Matched-Task Comparison

First state what is matched. Equal instantaneous angular moment, equal full wrench histories, equal club trajectories, and equal terminal states are increasingly different constraints; none follows automatically from the first. Select a feasible comparison and report its tolerances. Under a matched scalar task, compare the remaining wrench components as outcomes. If matching the full wrench is feasible, compare joint demands, bilateral load distribution, work, and robustness. Account for actuator limits and closed-chain consistency.

A second experiment can compare future evolution from matched configurations **and velocities**, retaining different transmission states. Match other relevant states and disturbances as well, or identify them explicitly as alternative causes. Declare whether future commands or feedback laws are held fixed: freely reoptimizing each program changes the causal question. Matching visible geometry alone leaves momentum and several possible sources of divergence unmatched.

A targeted state intervention can test dependence on preload in a model, but resetting a state may change stored energy, momentum, or constraint consistency. Those changes and any physical preparation needed to realize the reset belong in the accounting. If the entire deterministic state, future control law, and external inputs are identical, uniqueness already implies identical future motion. That equality is not independent proof of mediation. A convincing mechanism test changes the candidate state while controlling competing pathways and checks that the claimed effect survives feasible preparation and measurement uncertainty. Chapter 15 develops the related shaft-memory problem.

### 14.2.11 Predictions That Separate the Extremes

The continuous arm-dominant hypothesis predicts less force-rise delay after transition, fewer or shorter near-zero-load intervals in the relevant bilateral mode, and a particular sign continuity in arm-level

control. The role-reversal hypothesis predicts a clearer change in net allocation but may retain high co-contraction or contact load. Both can predict similar club motion.

If a verified allocation change produces no practically meaningful difference in the predicted wrench or stiffness mediator within adequately precise bounds, that particular slack explanation loses support. Instructions alone do not establish that an allocation changed. If one allocation improves speed only by increasing peak load or face/path variance, its practical value becomes conditional. If responses differ between participants, report that heterogeneity; subgroup explanations should be prespecified or clearly exploratory and independently tested.

#### **14.2.12 Safety and Coaching Boundary**

This comparison is not an instruction to apply maximal opposing force or to manipulate the wrists aggressively. The model does not estimate tissue safety, and the linked EMG evidence does not identify a preferred strategy. Human experiments require bounded tasks, clinical and ethics review where appropriate, and monitoring of discomfort and load.

For a reader or coach, the useful principle is narrower: distinguish continuous mechanical engagement from continuous net torque, and distinguish a felt cue from the physical variable it is supposed to change. A cue becomes scientifically useful only when it reliably changes the measured mediator and outcome.

#### **14.2.13 What Would Count Against Preload**

The specific dead-zone account is weakened if its predicted contrast fails when the declared model assumptions are reproduced; if an apparent advantage instead comes from unequal channel dynamics, unmatched states, or a different control law; if measured transition load or stiffness does not mediate the outcome; if continuous loading raises internal cost without a robust task benefit; or if held-out participants show no repeatable relation. These conditions prevent “connection” from becoming an explanation that fits every result.

#### **14.2.14 Reader Lab: Make the Opposite Case Strong**

Construct the strongest case for each allocation using the mechanisms above. Separate what the declared transmission model predicts from the anatomical assumptions that would need human measurements.

Write predictions that differ. Continuous loading should reduce a defined gap in a particular force mode. Role reversal may show a sign change in allocation without reduced stiffness. Both could yield equal club motion. Specify how bilateral wrench, perturbation response, EMG, and shaft state distinguish them.

Now test a middle family rather than only endpoints. Map allocation continuously from arm-dominant to wrist-dominant and allow phase-specific mixture. Plot speed, face/path, load, robustness, and transmission measures. If an interior region dominates both extremes, the spectrum has done its job by revealing that the original dichotomy was too coarse.

### 14.2.15 Interpreting a Null Result

If instructions produce no detectable allocation change, the experiment has not established the intended contrast; inadequate measurement sensitivity is also a possibility. It has not shown strategy equivalence. If allocation changes but a well-resolved preload measure does not, the proposed pathway loses support. If preload changes without a precisely estimated practical outcome, distinguish an uncertain effect from one bounded below a prespecified useful size. A nonsignificant result alone establishes neither absence nor equivalence. If the outcome changes without the predicted mediator, examine competing pathways and mediator observability before drawing a causal conclusion.

This staged interpretation prevents every outcome from supporting the same story. It also respects participant differences: one subgroup may maintain stiffness during reversal while another develops a measurable gap.

## 14.3 Where the Picture Breaks

The gearbox picture is intentionally simplified. Muscle-tendon systems can have slack-to-taut transitions and history-dependent behavior, but the chosen torque dead zone and scalar deflection lag do not identify those biological mechanisms. Surface EMG cannot measure force or stiffness directly. Scapular retraction does not map uniquely to one hand force.

The model also treats two abstract channels, not dozens of muscles and joint axes. Its numerical advantage is a conditional prediction awaiting data, not a coaching instruction.

**Model Result:** Persistent load direction outperforms complete role reversal on the chosen tracking score in the registered transmission family. Its zero-dead-zone equivalence also depends on equal channel time constants and matched initial net states and commands.

**Human Evidence:** [Kao et al. \(1995\)](#) recorded scapular-muscle EMG in 15 competitive male golfers and described phase-dependent activity. [Robinson et al. \(2023\)](#) studied 15 sub-elite male golfers and found side-dependent wrist kinematics and extensor carpi ulnaris EMG. Their reported lack of a significant relationship between downswing EMG and impact clubhead kinematics is not an equivalence test. Neither study identifies the bilateral wrench allocation proposed here. EMG is an electrical measurement, not a direct measure of muscle force or stiffness.

**Hypothesis:** Proximal push-pull during preparation may preserve useful transmission continuity while wrists resist, but the opposite allocation could win on load, accuracy, or participant-specific objectives.

**Practical Interpretation:** “No slack” should become a measurement question: was there a force gap, low tangent stiffness, delayed transmission, or merely a different internal allocation?

**Go Deeper:** Read the full [arm-wrist allocation and preload chapter](#), the [dynamic constitutive audit](#), the administrative [epic record](#), and the linked scapular and wrist activation studies above.

## Chapter 15

# The Shaft Is a Spring With a Memory

### 15.1 A Concrete Picture

Bend a flexible ruler and release it. The ruler does not respond only to the force you apply now. Its present shape and motion carry a short history of earlier loading. It can return energy after your hand force has fallen, overshoot, and vibrate in several patterns.

A golf shaft behaves far more subtly than the ruler, but the central point survives: flexibility adds state. Treating the club as perfectly rigid may be adequate for one question and misleading for another, especially when late timing, grip moment, and impact-window energy are under study.

The illustration prescribes  $x = A \sin^2(\pi t/T)$  with  $A = 0.025$  m,  $T = 0.6$  s,  $m = 0.2$  kg,  $k = 80$  N/m and  $c = 0.8$  N s/m. The required drive force is  $m\ddot{x} + c\dot{x} + kx$ , so input power minus  $c\dot{x}^2$  equals the rate of change of kinetic plus elastic energy. Negative drive power means energy returns to the actuator; it does not label energy delivered to a clubhead.

### 15.2 How the Mechanism Works

The simplest flexible model adds one coordinate  $s$  for shaft deflection. Its stored elastic energy might be  $U_s = \frac{1}{2}ks^2$ , and damping force might be proportional to  $\dot{s}$ . The state now includes  $s$  and  $\dot{s}$ . Grip work can increase  $U_s$ ; later shaft motion can return energy to rigid-body coordinates.

#### 15.2.1 Timing, Not Free Speed

An ideal passive spring stores and releases energy without creating it. Its strain energy is not constant while it exchanges work with the rest of the system. For a fixed linear stiffness  $k > 0$ ,  $\dot{U}_s = ks\dot{s}$ : the same nonzero deformation can be storing or releasing energy. Mechanical energy is conserved only for the complete conservative system with no net boundary work. Damping converts mechanical energy to heat; contact can transfer energy to the ball or other bodies.

The potentially useful feature is timing: accepting work and returning it when the geometry and task make the transfer useful. That usefulness is an outcome to test, not a property guaranteed by elasticity.

In a rigid-flexible comparison, the work-energy ledger must track boundary input, gravity, damping and all retained kinetic and strain energy so that any change in the specified delivery outcome has an

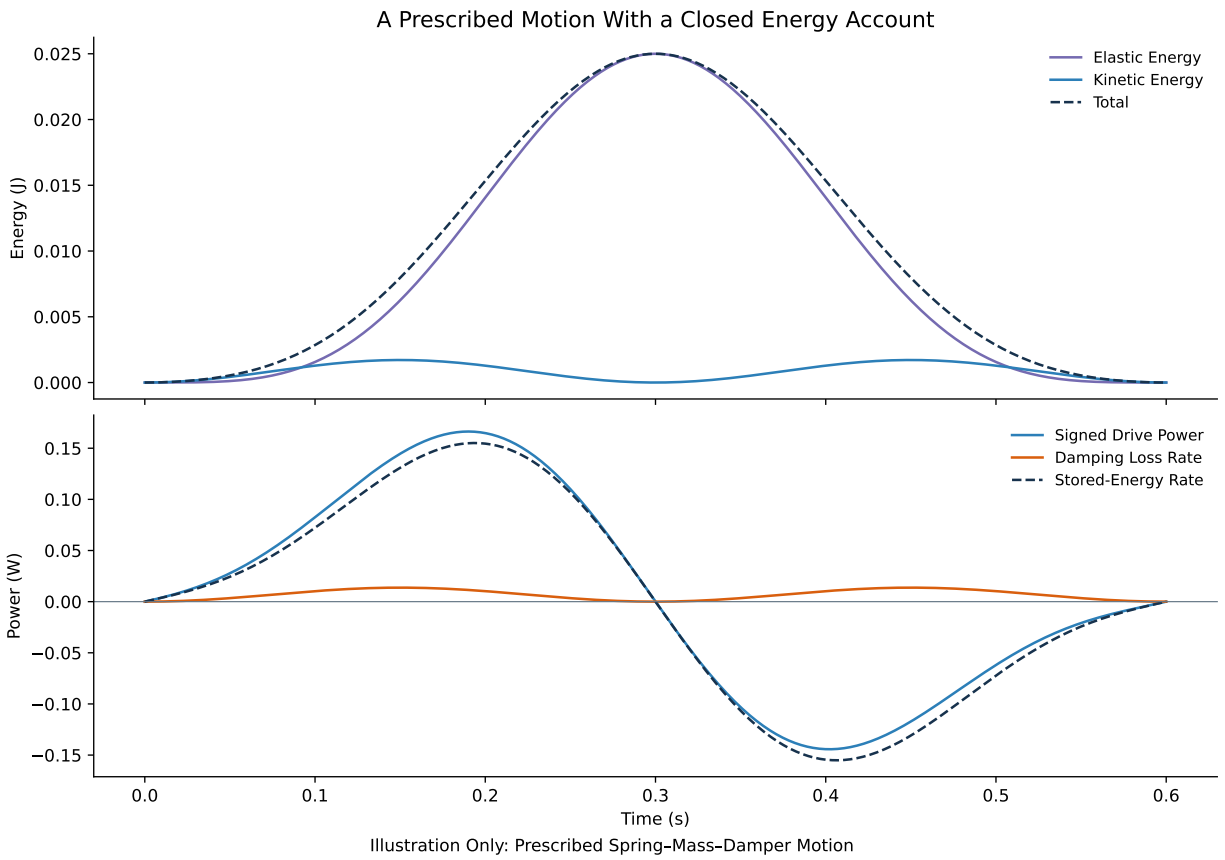


Figure 15.1: A Prescribed Spring–Mass–Damper Motion Separates Stored Energy From Input Power and Damping Loss. This Is an Illustrative Energy Ledger, Not a Shaft Simulation.

accounted source.

### 15.2.2 One Mode Versus a Distributed Shaft

A real shaft bends continuously. Modal analysis represents its shape as a sum of vibration patterns. The first mode may capture most low-frequency deformation; higher modes matter when excitation is rapid or spatially complex. A one-mode model is efficient but should be compared with a higher-order beam reference.

An isolated structural study assembles an Euler–Bernoulli beam with tip mass and rotary inertia, computes modal frequencies, identifies synthetic stiffness and mass, and compares reduced responses. A separate study transports bending modes into forward two-hand dynamics. Its baseline fails its declared small-deflection screen, as discussed below; numerical closure does not remove that failure. Mode truncation should be judged by specified load, power, energy and trajectory errors, not by appearance or frequency agreement alone.

### 15.2.3 Excitation Decides Required Complexity

Slow loading may barely excite high modes, making a rigid or one-mode model adequate. A sharp command, impact, or rapid contact change can excite them. The right model therefore depends on the frequency content of the question.

This is analogous to describing ocean swell with one broad wave versus describing a splash. More modes are not inherently more truthful; they are needed when the input and observable are sensitive to them.

### 15.2.4 Interface Power Connects Hand and Shaft

For an ideal point load  $F$  at a deforming shaft station, write its material-point velocity as  $v = V + \omega \times r + B\dot{\eta}$ . Here  $V$  and  $\omega$  describe the reference motion,  $r$  is the current offset, and  $B\dot{\eta}$  is the relative deformation velocity. In one observer frame the same power can be written as

$$\begin{aligned} P &= F \cdot v \\ &= F \cdot V + (r \times F) \cdot \omega \\ &\quad + (B^T F) \cdot \dot{\eta}. \end{aligned}$$

These are two representations of one port. Adding the last modal term to  $F \cdot v$  would count deformation power twice. Applied contact moments require their own compatible angular-velocity decomposition. A resultant wrench alone represents all contact power only when the interface motion is rigid, or the unrepresented deformation power vanishes. Distributed tractions can do work on deformation even when their net wrench is zero.

The flexible model separates grip moment from internal shaft moment. A negative grip couple can coexist with shaft energy release and positive club kinetic-energy growth. This makes the shaft relevant to the negative-torque discussion without turning it into the sole explanation.

### 15.2.5 Impact Is a Severe Boundary

Near ball contact, high-frequency response and contact duration matter. A model that stops at a pre-impact criterion can discuss delivery but not claim a full impact prediction. A distributed shaft needs measured stiffness, damping, geometry, clubhead inertia, grip conditions, and timestep resolution before it becomes equipment calibration.

**Model Result:** The isolated beam and articulated first-mode atlas report bounded synthetic checks. The separate planar forward-modal baseline retains numerical coupling evidence but fails its small-deflection screen. These are different models and acceptance results, not interchangeable validation claims.

**Human Evidence:** Betzler and colleagues tested 20 golfers with two clubs designed to differ in shaft bending stiffness. The less stiff club produced a small average clubhead-speed increase, alongside changes in shaft recovery and grip motion; the authors judged the overall effects small for the shafts tested (Betzler et al. 2012). This does not identify a universal fitting rule or calibrate the synthetic models here. Strain or optical measurements need their own calibration and uncertainty treatment.

**Hypothesis:** Shaft state may mediate how a late negative grip couple and proximal power coexist, returning stored energy while the grip channel opposes rotation.

**Practical Interpretation:** Flex behaves like short-term mechanical memory: what the shaft does now depends on how it was loaded and how fast that loading changed.

### 15.2.6 What the Shaft Adds to the State

A rigid club needs pose, including orientation, and compatible linear and angular velocity. A flexible shaft needs additional deformation coordinates and their rates. A one-mode model might use one bending amplitude; a distributed beam uses many modes or elements. These states carry memory because current shaft force depends not only on grip and clubhead position but also on how the shaft is bent and moving.

That memory complicates matched-state comparisons. Two swings with identical grip and clubhead pose can have different shaft curvature or modal velocity. Removing a command from those apparently identical frames can produce different responses. Within the same flexible model, a same-state intervention must match the complete state, including activation or contact-memory variables if present. Between rigid and flexible models, the state spaces differ: declare a mapping of common pose and velocity, the elastic initial state, mass properties, contact geometry and energy. Equal rigid coordinates alone do not establish equal total energy or a compatible constrained state.

### 15.2.7 Loading, Storage, and Return

When relative motion bends or twists the shaft, load power can increase strain energy. Inertial coupling redistributes energy; it is not a separate external energy supply. Damping turns some mechanical energy into heat. As deformation changes, the shaft can return energy to the rigid motion of grip and head. Return is not guaranteed to occur in the useful direction or at the useful time; phase matters.

It is tempting to say the shaft “kicks” the clubhead. A better account plots strain energy, damping loss, power at grip and head interfaces, and modal phase. If shaft energy decreases while clubhead

kinetic energy increases, return may be part of the explanation, but whole-system closure and other power ports still matter.

### 15.2.8 A Simple Spring Thought Experiment

Take a spring with stiffness  $k$  and displacement  $x$ . Stored energy is  $\frac{1}{2}kx^2$ . Doubling deformation quadruples energy, but a real shaft's distributed bending and torsion do not reduce exactly to one scalar spring. Add damping and a changing boundary, and release timing depends on natural frequencies, mode shapes, and excitation history.

If the downswing duration aligns poorly with a mode, extra bending can return too late or increase presentation variability. A linear spring in static equilibrium stores  $F^2/(2k)$  under a prescribed force, so increasing  $k$  reduces storage in that comparison. At fixed displacement it stores  $kx^2/2$ , so increasing  $k$  increases storage instead. Dynamic loading also depends on inertia, damping and drive frequency: resonance can reverse a simple stiffness trend. Increasing stiffness raises  $\sqrt{k/m}$  only with the same effective mass and boundary model; it does not guarantee a faster useful response. There is no universal “more flex equals more speed” rule.

### 15.2.9 Interaction With the Negative-Couple Mechanism

A two-hand couple loads the handle and can excite shaft bending or torsion. Shaft reaction then changes the wrench at the hands and the clubhead's future state. A negative contact couple may partly regulate deformation, or it may be a reaction to deformation generated elsewhere. Causal separation requires killswitches and energy accounting in the flexible model.

Compare rigid, one-mode, and distributed-shaft rungs with an explicit initial-state mapping and the same endpoint definition. Does the late couple retain sign? Does its timing shift with modal frequency? Does a speed benefit remain after shaft strain energy and damping are accounted for? If a mechanism vanishes in the flexible rung, that is important evidence about the rigid approximation.

### 15.2.10 Interaction With Preload and Slack

Continuous hand loading can maintain shaft deformation through transition, while a role reversal can unload and reload one mode. Yet “shaft preload” is not the same as muscle or grip preload. The shaft may remain bent while a hand-force component crosses zero, or vice versa. Each state requires its own measurement.

The phenomenological dead zone should not absorb shaft dynamics implicitly. If shaft deformation is modeled, transmission parameters and modal states must be identifiable enough to avoid two mechanisms explaining the same delay. Ablation tests—rigid shaft, zero dead zone, altered damping—show which declared model mechanism changes the result. They do not by themselves identify the mechanism in a golfer; distinct models can still fit the same observations.

### 15.2.11 Model Validation Ladder

Begin with static deflection under known loads. Continue with free-vibration frequency and damping tests, then dynamic boundary excitation. Verify modal convergence and timestep stability. Compare strain, tip displacement, and energy balance with an independent beam or finite-element reference.

These are verification steps. After coupling to the swing, repeat the power, convergence and deformation-domain checks: a sound isolated beam can be driven outside its valid range by the coupled model. Experimental validation is a separate step.

Human validation uses instrumented clubs with calibrated strain or optical shape measures. Club properties should be measured, not selected merely from a label such as “stiff.” Repeated trials reveal whether deformation phase and contact wrench are consistent within a golfer. Equipment changes should be bounded and randomized to reduce learning and expectation effects.

#### **15.2.12 What a Shaft Result Can Support**

A validated model can say that, under declared properties and motion, certain modal forces, powers, and energy exchanges occurred. It can predict how a controlled property change shifts them. It cannot infer a golfer’s preferred feel, injury risk, or universal fitting recommendation without additional data.

For the proximal-distal framework, the shaft is most valuable as a test of mechanism survival. If the interaction and allocation conclusions remain under realistic flexibility and uncertainty, confidence grows. If not, the model ladder has found the boundary where the simpler explanation must be revised.

#### **15.2.13 Reader Lab: Watch Phase Change the Answer**

Excite a declared one-mode shaft model with the same peak handle force at three different times. Plot deformation, modal velocity, strain energy, grip power, head power, and damping loss. The same force magnitude can store or return different energy because phase and boundary motion differ. Equal peaks do not match force history, impulse or work; report those differences too.

Copy two states with identical rigid pose and velocity but opposite modal velocity. Apply the same future hand wrench and test whether the chosen clubhead observable differs. Symmetry, mode coupling or observation timing can hide a state difference; a different internal state does not guarantee a different measured endpoint.

Next, reset shaft deformation and modal velocity to zero while retaining the rigid coordinates and rates. This is a model intervention with a new initial condition. Record its energy change, momentum change and any constraint or contact projection. Removing strain energy does not necessarily reduce total energy: rigid-modal cross terms in kinetic energy can change either way. The reset requires explicit work and impulse bookkeeping before interpreting the subsequent trajectory. It is not an available human command or a measured causal contribution in a golfer.

Repeat the negative-couple branch across stiffness and damping. Report whether couple sign, onset, work, and terminal speed remain. A result that survives realistic property uncertainty deserves more confidence than one tuned to a single resonance.

#### **15.2.14 Avoiding Double Explanations**

If both a dead-zone transmission and flexible shaft can produce delay, fit or vary them jointly. Look for observables that separate them: grip force may change before shaft strain under one mechanism, while modal phase dominates another. Without such information, their parameters can compensate and neither is identifiable.

The release should therefore include ablations and joint sensitivity, not one best-fit flexible trace. When two explanations remain equivalent, the honest output is an experiment recommendation—usually synchronized wrench and shaft state—rather than a preferred story.

### 15.2.15 A Minimum Flexible-Club Result

At minimum, publish the measured or declared club properties, modal frequencies and damping, deformation and modal velocity, grip and head power, strain-energy history, damping loss, and work-energy residual. Show timestep and modal convergence. Declare the rigid-to-flexible state mapping and distinguish fixed-input comparisons from comparisons selected to match an outcome.

Then qualify the conclusion. A synthetic property sweep shows mechanical possibility and sensitivity. A calibrated participant club permits predictions for that equipment; held-out observations test their accuracy. Replication can extend support only across the equipment, golfers and conditions actually tested. Skipping these levels turns shaft “memory” into a metaphor too broad to fail.

### 15.2.16 The Registered Test Says the Benefit Is Not Universal

The [articulated shaft atlas](#) tests a passive synthetic shaft with the first bending shape in each of two orthogonal planes, plus torsion. These are not the first and second bending frequencies in one plane. Bending is anchored to a 24-element reference at 5.2399 Hz; torsion is declared at 70.1260 Hz. Twelve initial states, four activations (rigid, bending, torsion, coupled), two velocity branches, two timesteps and two engines give **384 trajectories**. The motion starts from initial conditions, with no active driver or joint torque. Zero modal coordinates and rates give a common initial energy in this declared comparison. The velocity branches add  $\pm 0.05$  m/s to club-frame  $x$  translation; the timesteps are 0.25 and 0.125 ms, using MuJoCo and Pinocchio. Five tension-only spring-damper stations per hand supply grip loads and grip dissipation. These are synthetic contact elements, not measured fingers or pressure distributions.

The observed maxima are 1.696 mm of tip bend and 0.00186 rad of twist; these are results, not the acceptance thresholds. The configured limits are 5% of the 1.08 m shaft length (54 mm) and 10 degrees of twist (about 0.175 rad). Two additional coarse-timestep probes that left the linear domain were excluded and reported separately; they do not reduce the accepted 384-trajectory grid.

The coupled-versus-rigid comparison uses four horizons: 4, 10, 25 and 50 ms. Its **384 comparison cells** count twelve states, two velocity branches, two timesteps, two engines and four horizons. This happens to equal the trajectory count, but counts something different. Repeated horizons, engines and timesteps are not independent golfer trials.

Only **126 of 384** comparison cells pass the post-registration load-and-work matching rule: 28 at 4 ms, zero at 10 ms, 40 at 25 ms and 58 at 50 ms. For each quantity  $a, b$ , the rule is

$$\frac{|a - b|}{\max\{f, (|a| + |b|)/2\}} \leq 0.05.$$

The floor  $f$  is 1 N for the largest station-force norm over the horizon and  $10^{-6}$  J for grip-plus-shaft damping loss. When the mean magnitude falls below its floor, the absolute tolerances are 0.05 N and  $5 \times 10^{-8}$  J, respectively. These compare dissipation, not total grip boundary work. Selecting cells by

these two outcomes is a descriptive comparison; it does not match every load history or identify a causal shaft effect in a golfer.

Across those selected cells, the reported coupled-minus-rigid speed difference is  $-0.0285$  to  $+0.0212$  m/s. Both signs appear. Crucially, the [recording code](#) uses the norm of the club-frame translation rates, with origin at the modeled grip mass. This omits the rotational and deformation contributions to clubhead-point velocity. It is measured at fixed horizons, not at impact. Sometimes the flexible shaft is slower **in this translation-speed observable**; the quoted interval is not an impact clubhead-speed interval. At 50 ms alone, the matched range is  $-0.0208$  to  $+0.00634$  m/s.

A positive effect on this observable is therefore **rejected as a universal rule** within the registered synthetic comparison. A human clubhead-speed or equipment-fitting conclusion requires a different endpoint and calibrated data. The atlas also omits ground reaction and free moment, so it cannot establish a contribution from the feet.

Two other studies answer different questions. In the isolated beam reference, one-mode versus six-mode tip-displacement RMS disagreement rises from about 0.69 micrometres under an 80 ms pulse to 41 micrometres under a 4 ms pulse with a different force-and-moment input. That comparison motivates checking excitation and truncation; it is not a universal frequency cutoff. In the separate planar forward-modal baseline, maximum absolute modal tip displacement is **13.48% of shaft length**, above its **5%** screen. Its small contact-power residual verifies a coupling identity but cannot establish valid small-deflection dynamics outside the screen. The articulated atlas passing its own checks does not repair this distinct baseline.

### 15.2.17 Chapter Checkpoint

Before crediting shaft return, check the complete state, energy-transfer direction, damping and ledger closure. Test the specified endpoint across property uncertainty and independent models. Distinguish verification, domain screens and held-out validation. The current translation-speed result supports both signs of change; clubhead-speed and individual control/load tradeoffs still require calibrated human data.

## 15.3 Where the Picture Breaks

The ruler bends in one obvious plane. Shafts bend and twist in space, taper, vary in material, and interact with a distributed grip and clubhead. Classical Euler–Bernoulli bending neglects shear deformation and distributed beam rotary inertia; a tip body’s rotary inertia can still be included, as in this reference. Torsion requires its own model. Synthetic parameter recovery checks an inverse pipeline under its generating assumptions, not equipment validity.

The phrase “kick” can also imply a free late boost. Any returned energy was previously supplied and survived damping. A fair comparison counts the complete cycle.

**Go Deeper:** Read the [gravity, damping, and shaft contribution chapter](#), the [beam reference](#), and the [forward modal coupling](#).

## Chapter 16

# The Ground Is Part of the Conversation

### 16.1 A Concrete Picture

Jump from a small boat and the boat slides backward. Push against a wall and the wall returns a force through your body. Walking, throwing, and swinging all rely on external contact with the environment. The ground is not merely a stage; it is one boundary through which force and sometimes power enter the body.

Force plates make this interaction visible, but their curves are easy to overinterpret. A vertical-force peak does not identify which muscle created it. A center-of-pressure shift does not prove a unique segment sequence. The ground reaction is a measured boundary load. A model must reproduce it consistently with the measured motion, inertias, gravity, and other external loads; that consistency does not identify the muscles that produced the motion.

### 16.2 How the Mechanism Works

In an ideal constrained-contact model, ground reactions appear as multipliers enforcing the selected contact constraints. Solving the equations can decompose reaction at a fixed state into configuration/gravity, velocity-dependent, control, and external-load terms. The technical study defines ground-reaction versions of zero-torque and zero-velocity counterfactual forces.

#### 16.2.1 The Reaction Reflects the Whole System

Accelerating the club changes body reactions. Body motion changes required foot forces. Gravity contributes. If the feet are constrained, the solver finds a reaction satisfying both motion and contact. A large ground force is not a separate motor delivering energy directly to the club; it is part of the coupled external interaction.

Ground power depends on motion at the physical boundary. For a rigid contact wrench referred to point  $O$ , in one observer frame,

$$P_g = \mathbf{F}_g \cdot \mathbf{v}_O + \mathbf{M}_{g,O} \cdot \boldsymbol{\omega}.$$

## Modeled Net Reaction Does Not Uniquely Specify Foot Allocation

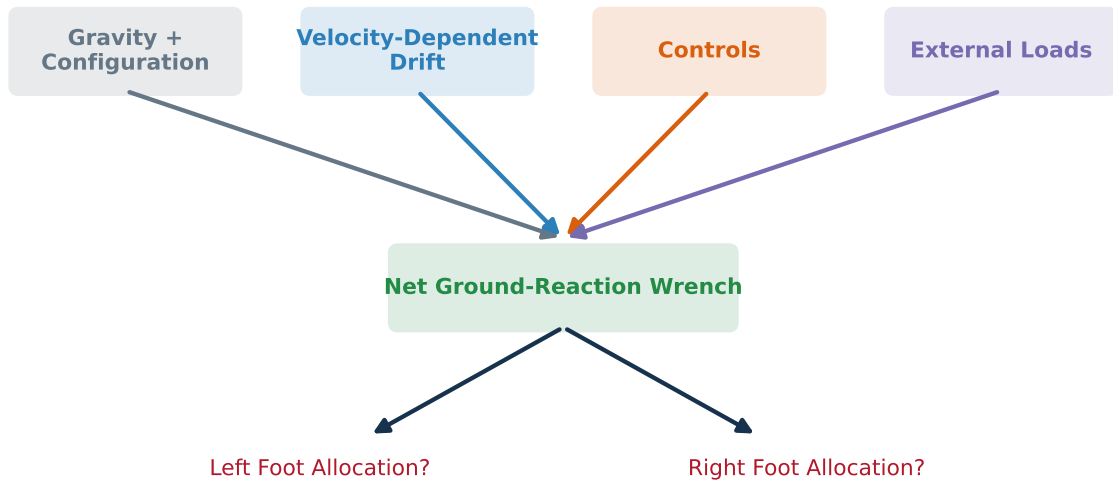


Figure 16.1: A Model Decomposes Net Reaction; a Summed Wrench Alone Leaves Foot Allocation Unresolved.

The moment and velocity must refer to the same point on the same rigid body. An ideal nonslipping rigid contact at rest in the chosen observer frame has zero boundary power despite large reactions. For a deforming foot, use traction dotted with local material-point velocity over the contact surface, including any applied contact couples in the chosen continuum model. A center-of-pressure path is a changing force-resultant location, not a material trajectory: its migration alone is not ground work. Internal muscle, joint, and tissue work can occur while the external ground port does zero work. A force plate measures wrench, not metabolic source.

### 16.2.2 ZTCF Family and ZVCF Ask Narrow Questions

At one achieved state, the ground-reaction ZTCF sets declared controls to zero and recomputes reaction. In the pinned technical study, ZVCF sets **both velocity and declared control to zero**, retaining configuration and declared other loads. Keeping control while zeroing velocity is the separate **zero-velocity control-preserved reaction**. These pointwise calculations do not advance a new trajectory or describe a person instantaneously stopping all motion.

In a fixed-support double-pendulum benchmark, pointwise ZTCF predicts the general reaction waveform but retains substantial amplitude error. Opposing control can make one ZTCF impulse exceed the total because contributions cancel. This is a useful warning against treating a component as a measured fraction.

### 16.2.3 Two Feet Add Allocation Ambiguity

A single net wrench under the body may be distributed across left and right feet in many ways, especially if both contact regions can support moments. Separate plates reduce ambiguity. Pressure insoles add spatial information. Yet foot models, friction, and coordinate transforms remain.

The same common/differential logic used for hands applies to a summed wrench: its net value does not specify its split. Separate six-axis plates directly resolve each foot’s aggregate wrench within measurement uncertainty, though neither plate uniquely resolves within-foot pressure or muscle co-contraction. Two feet do not automatically make every forward dynamics problem indeterminate; constraint rank and the contact constitutive law determine that question.

### 16.2.4 What Ground Data Can Test

Synchronized whole-body kinematics, separate force plates, club motion, and bilateral grip wrenches can test the predicted **total** reaction in timing, direction, and amplitude. Individual counterfactual components are not separately measured by a plate. Their attribution requires additional model assumptions and perturbation predictions. Residual forces and moments can expose inconsistencies in segment parameters or filtering. Participant holdout tests whether a fitted pattern generalizes.

**Model Result:** The fixed-support benchmark establishes a reproducible reaction decomposition with partial waveform agreement and explicit amplitude limitations.

**Human Evidence:** Han and colleagues studied 63 skilled men, with handicap at most 3, using optical motion capture and two force plates for driver, 5-iron, and pitching-wedge shots. Their abstract reports associations between selected force/moment peaks and maximum clubhead speed, not a muscle-level causal decomposition (Han et al. 2019). Ball and Best identified two major direction-of-hit COP clusters in 62 golfers; this does not establish two exhaustive patterns of two-dimensional foot pressure (Ball and Best 2007). A 2026 systematic review retained 24 studies, mostly cross-sectional, and identified inconsistent methods and remaining reliability and intervention questions (Watson et al. 2026). These studies measure different quantities and populations; none validates the present counterfactual decomposition in humans.

**Hypothesis:** State-dependent reaction components may explain part of observed ground-force timing without requiring each peak to be directly commanded, but full-body human data must test the prediction.

**Practical Interpretation:** The ground tells us what net external conversation occurred at the feet. It does not provide a transcript assigning every sentence to a muscle.

### 16.2.5 Force Plates Measure a Net External Wrench

A six-axis force plate reports three force components and three moments at its declared origin. Transform these to the support surface before deriving COP. For a planar surface  $z = 0$ , reaction on the golfer, and  $F_z \neq 0$ ,  $x_{\text{COP}} = -M_y/F_z$  and  $y_{\text{COP}} = M_x/F_z$ ; the remaining  $M_z - (x_{\text{COP}}F_y - y_{\text{COP}}F_x)$  is the surface-normal free moment. Declare axes and signs, and mask low-load intervals where the ratios are unstable. Separate plates preserve left–right aggregate allocation. Pressure insoles usually measure normal pressure; shear and a full wrench require additional sensing or a calibrated model.

The plate does not report muscle force, joint torque, or energy “from the ground.” It reports the reaction required at the contact boundary. To understand how that reaction relates to the club, synchronize it with body and club motion and solve the dynamics with uncertainty.

### 16.2.6 Large Force Does Not Necessarily Mean Large Power

At an ideal stationary rigid support, both terms in the wrench-power expression vanish. Body kinetic energy can still increase through active muscle work, gravitational potential decrease, or release of elastic storage. Which changes count as new mechanical input depends on the declared energy boundary. Force magnitude is not an energy-flow rate, and a kinetic-energy increase alone does not establish new active work.

This is similar to pushing a locked wall during a squat. The wall or ground provides a reaction that permits body acceleration, while chemical energy in muscle supplies work. Saying the golfer “gets energy from the ground” can be a useful cue, but it is not a precise mechanical ledger unless contact work is actually calculated.

### 16.2.7 A Whole-Body Reaction Walkthrough

Early in the downswing, changes in pelvis and trunk motion require horizontal, vertical, and rotational ground reactions. As arms and club accelerate, their momentum changes alter the reaction needed by the rest of the system. Late club acceleration can therefore appear in foot-force features even when no new foot command specifically targets the club.

The influence is reciprocal. Ground reaction enables changes in whole-body momentum that shape proximal motion, and distal acceleration changes the reaction required for balance and constraint. Cross-correlation of peaks cannot determine direction of causation. A forward model and perturbations are needed.

### 16.2.8 Pointwise Ground-Reaction Decomposition

For an autonomous holonomic contact model, write

$$\begin{aligned} M\ddot{q} + h_0 + h_v &= Bu + Q_e + J^\top \lambda, \\ J\ddot{q} + \gamma &= 0. \end{aligned}$$

Here  $h_0$  is the declared velocity-independent bias,  $h_v$  the velocity bias,  $Bu$  the selected control, and  $Q_e$  other loads. With positive-definite  $M$  and  $J$  of full row rank, let  $A = JM^{-1}J^\top$ . A well-conditioned linear solve yields

$$A\lambda = JM^{-1}(h_0 + h_v - Bu - Q_e) - \gamma.$$

Holding configuration, contact rows, and the declared load conventions fixed gives an additive split  $\lambda = \lambda_0 + \lambda_v + \lambda_u + \lambda_e$ . ZTCF retains  $\lambda_0 + \lambda_v + \lambda_e$ ; ZVCF retains  $\lambda_0 + \lambda_e$  when  $h_v(q, 0) = \gamma(q, 0) = 0$ . Thus total minus ZTCF isolates the declared control term, and ZTCF minus ZVCF isolates the velocity term. **ZTCF and ZVCF overlap and are not additive parts of the total.**

Time-dependent constraints, velocity-dependent controllers or other loads, slip, and contact opening require recomputation under the actual intervention. They cannot inherit this fixed-operator shortcut without qualification. Compute COP from each reconstructed wrench; dividing each component separately does not produce an additive COP decomposition.

Components can cancel, so a counterfactual reaction may exceed total in one direction. That does not mean it contributed more than one hundred percent of a measured force. It means another component opposed it. Report signed waveforms, impulses, cancellation, and residual rather than forcing an additive percentage story.

### 16.2.9 Two Feet Create an Allocation Problem

The same net body wrench may be supported by different left and right contact wrenches within friction and balance constraints. Separate plates observe much of this allocation. If both feet occupy one plate, a model or pressure data must supply the split, and nonuniqueness should be shown.

Foot models also need contact assumptions. A fixed point, rolling contact, distributed pressure, and compliant shoe produce different reaction paths. The choice should match the measurement question. A mechanism that depends on an unrealistic fixed foot may not survive a richer contact rung.

### 16.2.10 Connecting Ground and Hand Ledgers

For the **body-plus-club** system before ball impact, ground, gravity, and any other external contacts or aerodynamic loads determine total momentum change. Hand-club interface loads are internal: include both sides with consistent points and signs, so their complete wrench cancels. For the body alone, the club-on-hand wrench is external; for the club alone, the opposite wrench is external. Do not add hand loads again to the combined-system external ledger.

About a fixed inertial origin  $O$ ,  $\dot{\mathbf{H}}_O$  equals the total external moment about  $O$ . If the reporting origin moves, include the transport term appropriate to the chosen angular-momentum definition. Segment powers and internal elastic/dissipative storage then connect the system ledger to the hand interface; action-reaction alone does not make internal work vanish.

This creates powerful error checks. If measured hand and ground loads combined with motion leave large residuals, segment inertias, filtering, synchronization, or unmeasured contacts may be wrong. A polished transfer story should wait until those residuals are bounded.

### 16.2.11 What Ground Manipulations Could Test

Researchers can compare stance width, lead/trail pressure constraints, surface friction within safe limits, or externally paced tempo while measuring the predicted hand and club mediators. The mechanism should specify whether changing a ground variable shifts proximal state, hand wrench, negative-couple interval, or only balance strategy.

A finding that ground-force peaks change without the predicted hand or club change would weaken a simple ground-to-club transfer narrative. A finding that hand-wrench timing mediates the relation would strengthen the chain. Participant holdout is essential because people may adopt different compensations.

### 16.2.12 Practical Boundary

Force-plate patterns can characterize an individual and expose constraints, but they should not become universal target traces. Similar club outcomes can arise from different foot allocations, and larger peaks can reflect braking or balance as readily as useful propulsion. The valuable question is how the measured external wrench fits the complete system ledger and whether a proposed change improves robust outcomes without unacceptable load.

### 16.2.13 Reader Lab: Close the External Wrench

Take synchronized body and club kinematics, segment parameters, separate foot wrenches, gravity, and any other external loads. Compute body-plus-club linear and angular momentum change and compare with integrated external wrench about a fixed inertial origin. Use hand measurements to close the body and club subsystems separately, without double-counting them in the combined system. Plot residual by direction and phase. A large residual near peak club acceleration may reflect synchronization, filtering, inertia, or omitted contact—not a new force.

Next sum the two foot plates and verify the result against a single combined wrench calculation. Shift reference point consistently. Compute center of pressure only when vertical force and moment assumptions make it well defined. Flag low-force intervals rather than allowing unstable ratios.

Finally, validate the total modeled reaction against the plates and run the declared pointwise counterfactuals. Show component timing as model attribution. A component-to-total difference is not itself a validation error: the other components are expected to contribute. Test counterfactual predictions through separately specified perturbations and held-out trials.

### 16.2.14 Competing Explanations for a Force Peak

A vertical peak can reflect accelerating the body center of mass, resisting downward motion, club reaction, balance, or a combination. A horizontal peak can turn or translate the system. A free moment can relate to rotational balance but does not map uniquely to pelvis or club torque.

Test proposed pathways with mediation and intervention. If a stance manipulation changes ground force but neither proximal state nor hand wrench as predicted, the simple transfer chain is unsupported. If all links change in order and a forward model predicts them on held-out trials, the explanation strengthens.

### 16.2.15 From an Observation to a Causal Claim

Suppose higher club speed is associated with a larger lead-side vertical impulse. That is an observation. A causal hypothesis might say the impulse changes pelvis and trunk state, which changes hand wrench and later club acceleration. The hypothesis requires each mediator and its timing. A model can predict the chain; a manipulation and synchronized data can test it.

Alternative explanations include stronger golfers producing both variables, different stance or club, balance response to the faster club, or processing choices. Randomized safe manipulations, within-person contrasts, and held-out prediction reduce these alternatives. The ground trace alone cannot.

### 16.2.16 What the Product Should Display

Show separate and summed foot wrenches, center of pressure only in valid intervals, whole-system momentum residual, predicted reaction components, and uncertainty. Link peaks to concurrent body, hand, and club state. Let readers toggle pointwise ZTCF and ZVCF components while explaining that they are model counterfactuals.

Do not rank a golfer from force magnitude alone. Normalize only when the scientific comparison justifies it and retain dimensional values. A transparent display should make cancellation, allocation ambiguity, and residuals as visible as the headline pattern.

### 16.2.17 A Final Ground Check

For a closed accounting of kinetic, gravitational, and elastic mechanical energy, any increase must be supplied by declared active work or external ports, with dissipation also recorded. Release of already-counted elastic storage redistributes that energy; it does not create it. If the support is compliant, specify whether its stored energy belongs inside the system or enters through an external port. This keeps everyday “use the ground” language tied to an explicit ledger.

### 16.2.18 What Happened When the Ground Was Actually Added

The archived finite-base experiment adds horizontal/vertical translation, pitch about world  $y$ , or both to an articulated model. Each pathway is paired with the fixed-base version at the same registered initial state. Its intrinsic **pitch moment** differs from a force plate’s usual surface-normal free moment. The support law is linear spring-damper mechanics, without two feet, unilateral ground contact, Coulomb friction, or measured plantar pressure.

The design comprises 12 states, four primary pathways, two signed initial velocity perturbations, two timesteps, and two native dynamics backends:  $12 \times 4 \times 2 \times 2 \times 2 = 384$  primary trajectories. Two control families add 192 trajectories. Each is summarized at 4, 10, 25, and 50 ms, giving 1536 primary pathway summaries. The coupled-versus-fixed subset contains 96 trajectory pairs, each compared at four horizons. Consequently, it has **384 summary comparisons**,  $12 \times 2 \times 2 \times 2 \times 4$ , rather than 384 independent human trials or 384 distinct trajectory pairs. Engine and timestep variants are numerical replications of the modeled conditions.

MuJoCo and Pinocchio supply independent native dynamics operators but use a **shared contact law** and semi-implicit integration in this experiment. The maximum archived trajectory relative discrepancy is about  $1.77 \times 10^{-10}$ . The worst primary normalized energy residual decreases from about 0.01986 to 0.00995 when the timestep halves. These results support implementation agreement and refinement over the sampled domain; they are not independent validation of all contact physics. All primary and control numerical gates pass, while the separate load/work matching criterion fails.

### 16.2.19 What the Matching Criterion Actually Measures

The archive’s **declared primary criterion** requires both peak grip load and total dissipated work to match within 5%. Here load is the maximum single-station force norm over the interval, not the summed hand force. A separate timestamped prospective registration was not verified here. The work comparison is

$$\delta_W = \frac{|W_c - W_f|}{\max(10^{-6} \text{ J}, (|W_c| + |W_f|)/2)},$$

where  $c$  and  $f$  denote coupled and fixed support and  $W$  is nonnegative dissipated work. Peak load uses the same symmetric normalization with a 1 N floor. **Zero of 384** comparisons meet both criteria; 202 meet the load criterion alone. The work discrepancy ranges from 1.723 to 1.999. This does **not** mean twice as much energy. With positive work and no active floor, the symmetric discrepancy approaches 2 as the ratio diverges. The actual coupled/fixed work ratios range from **13.44 to 3519.40**. Small fixed-base dissipation makes a large ratio possible; the archived fixed and coupled ranges are approximately  $4.46 \times 10^{-7}$ –0.00290 J and 0.00157–1.872 J.

The sampled tolerance sweep finds no matches through 100% and all 384 at 200%. For nonnegative quantities this symmetric discrepancy cannot exceed 200%, so that last threshold imposes no meaningful matching requirement. It is not an estimated minimum tolerance, nor proof that unequal work is unavoidable in every possible ground model. Adding support damping changes dissipation directly and also changes motion, hence grip and shaft dissipation.

The supported conclusion is narrow: **no eligible comparison exists to estimate a speed benefit under the joint load-and-total-work matching criterion**. An empty matched cohort does not estimate a zero effect. That criterion must not be replaced silently after seeing the result.

### 16.2.20 A Total Simulated Contrast Is a Different Question

At the finer timestep and 50 ms, the coupled pathway has higher terminal club translation speed in all 48 summaries, by approximately 0.028–0.177 m/s. These are 12 states, two velocity signs, and two backends. The atlas has no active driver or joint torque; motion begins from the assigned state and perturbation, with initial mechanical energy matched across pathways.

Changing support alters the equations, energy redistribution, and losses. The resulting paired trajectory difference is a **total effect of that declared model intervention**. Dissipation is partly downstream of the support change, so equalizing it would ask a different question. Failing the matching screen does not erase the total simulated contrast; conversely, the contrast does not establish an equal-load/equal-work advantage, optimal coordination, energy supplied by stationary ground, or a causal benefit in golfers. The modeled endpoint is 50 ms translation speed, not impact speed or whole-swing performance.

A post-hoc sensitivity excludes ground damping from the matched-work quantity while retaining grip and shaft dissipation. It finds 60 matches: **48, 8, 4, and 0** at 4, 10, 25, and 50 ms respectively, across both steps and backends. Their speed differences range from approximately  $-0.00075$  to  $+0.01394$  m/s, with 20 positive and 40 negative. These are not 60 successful 50 ms comparisons. This alternative estimand remains exploratory and does not replace the primary result.

### 16.2.21 Initialization Is Part of the Experiment

A separate 4 ms diagnostic uses seven branches, three steps, and two backends: 42 trajectories. At the 0.125 ms step, three coupled-support initializations give the following peak ground forces and terminal club translation speeds:

| Initialization   | Force (N) | Speed (m/s) |
|------------------|-----------|-------------|
| Natural-Zero     | 32.8      | 0.264       |
| Gravity-Only     | 565.5     | 1.908       |
| Conditional Base | 510.3     | 0.946       |

Natural-zero matches the registered initial coordinates and energy but is not static equilibrium. Gravity-only balance omits the grip preload contribution. Conditional equilibrium balances the three base coordinates while holding the remaining posture and club fixed; it is not equilibrium of every coordinate. The resulting initial storage and transients differ, so these numbers demonstrate initialization sensitivity, not three human standing strategies.

This archive also uses distributed central tension-only grip fibers. Their equal-and-opposite endpoint forces have zero combined pair moment. The noncentral vector-dashpot limitation discussed in Section 20 belongs to a different archived pathway and must not be attributed to these fibers.

### 16.2.22 Chapter Checkpoint

Ground reaction is external force evidence, not a muscle transcript or automatic energy source. Use it to close momentum, test contact models, and constrain the causal chain through body and hands. Report left/right allocation and model components without converting them into additive percentages.

And when someone shows you a speed difference between a moving base and a fixed one, ask what was held equal before the comparison was made.

A practical pattern becomes mechanistically meaningful only when its predicted mediators and outcome survive within-person manipulation and participant holdout.

## 16.3 Where the Picture Breaks

The wall and boat pictures use simple rigid contacts. Feet roll, deform, slip microscopically, and distribute pressure. Fixed-support pendulums omit balance, pelvis translation, and bilateral contact.

Correlation between force peaks and segment motion cannot establish causation. Timing can reflect shared state, feedback, or filtering. Reaction decomposition must be validated against synchronized data before becoming a human claim.

**Go Deeper:** Read the pinned monograph's [ground-reaction attribution](#), [finite-ground experiment](#), and [human falsification protocol](#).

## Chapter 17

# When the Base Moves, Every Shortcut Changes

### 17.1 A Concrete Picture

Swing a pendulum while standing on a rolling board. Your hand accelerates the pendulum, and its reaction can move you and the board. Now imagine a powerful actuator holding your hand to a recorded trajectory. The hand can still move, but the pendulum cannot change its assigned path. The actuator supplies the reaction needed to maintain that motion.

These experiments answer different questions. A prescribed hand path lets us calculate a club response or infer loads compatible with measured motion. A finite-mass base lets the body-side motion respond to the loading. Comparing them is useful only when we say which inputs change, which motions are enforced, and where the enforcing force and its work enter the calculation.

### 17.2 How the Mechanism Works

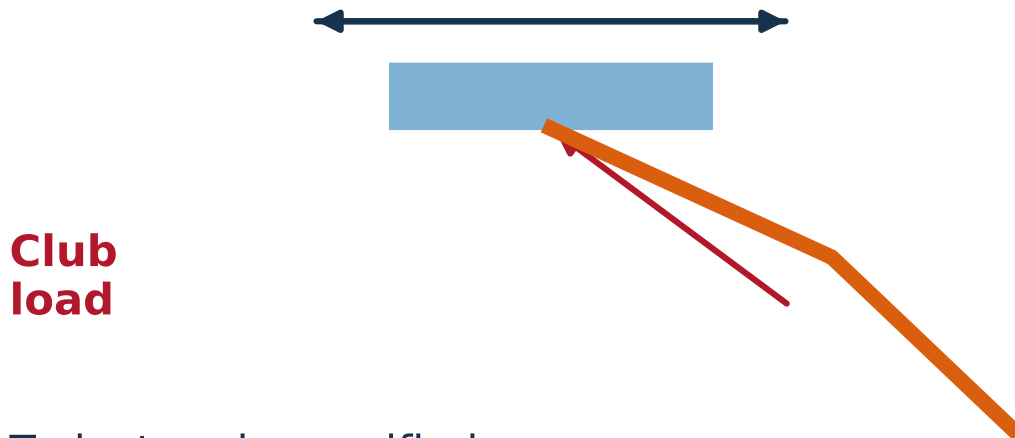
#### 17.2.1 The Boundary Condition Comes First

A **fixed base** is constrained to remain stationary. A **prescribed base** has a specified position and orientation as functions of time; stationary support is one special case. A **dynamic base** has equations of motion in which applied loads, inertia, gravity, contacts, and constraints determine its acceleration. None of these descriptions alone identifies a golfer's muscles or controller.

For prescribed motion, the enforcing reaction is the residual needed after the other loads have been declared. It may be large even when the trajectory looks smooth. Multiple contacts can leave the allocation of that reaction nonunique. Calling the trajectory measured improves its connection to an observed trial; it does not make the motion responsive to a new control strategy.

Assigning an applied force as well as a prescribed trajectory is not automatically an error. It is a well-defined experiment if an additional constraint reaction absorbs the difference and its work is reported. The error is to treat the assigned force as the sole cause of the motion while silently allowing the trajectory actuator to supply the remaining force and energy.

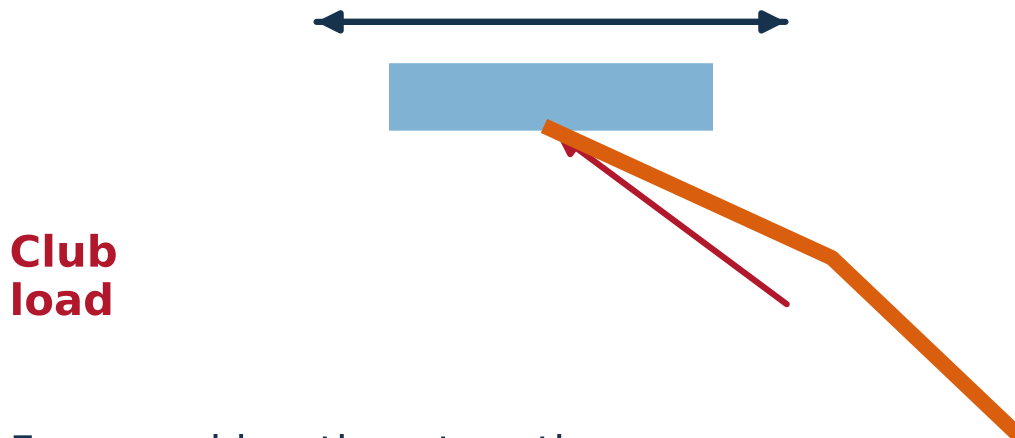
## Prescribed Path



**Club  
load**

Trajectory is specified  
Report reaction and power

## Finite-Mass Base



**Club  
load**

Force and inertia set motion  
Club load can change the path

Figure 17.1: Both Bases Can Move: One Follows an Assigned Trajectory, While the Other Responds to Force and Inertia.

### 17.2.1.1 Finite Mass Is Not a Trajectory Actuator

A translating body’s acceleration decreases as its mass increases under a bounded resultant force. In a simple translating-base limit with a fixed time interval and initial velocity, increasing mass does not produce an arbitrary accelerating hand path. Maintaining such a path instead requires an appropriately increasing force. Mass, stiffness, feedback gain, and actuator force capacity are different model assumptions.

A reduced moving hub also remains a reduced hub. Adding finite mass does not supply pelvis and trunk dynamics, shoulder anatomy, balance, or realistic ground contact. Its value is that one formerly imposed motion can now respond to load.

### 17.2.2 A Small Model That Makes the Difference Visible

Consider a cart of mass  $M$  on a frictionless horizontal track. A point bob of mass  $m$  is attached by a massless rigid link of length  $l$ . Let  $x$  be cart position and  $\theta$  the link angle from downward vertical, so the bob is at  $(x + l \sin \theta, -l \cos \theta)$ . Apply horizontal cart force  $F$  and pivot torque  $\tau$ , positive with increasing  $\theta$ . The cart does not rotate; the support moment that maintains this orientation has zero angular power. Take  $M, m, l > 0$  and let  $g$  be the magnitude of gravitational acceleration.

In an inertial frame, the equations are

$$\begin{bmatrix} M + m & ml \cos \theta \\ ml \cos \theta & ml^2 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \ddot{\theta} \end{bmatrix} + \begin{bmatrix} -ml \sin \theta \dot{\theta}^2 \\ mgl \sin \theta \end{bmatrix} = \begin{bmatrix} F \\ \tau \end{bmatrix}.$$

The first row is total horizontal momentum balance. The second is the bob’s tangential balance about its moving pivot. Both rows are needed: the pivot acceleration is part of the bob’s equation, while the bob contributes to the force needed to accelerate the cart.

Use the **manufactured** values  $M = 2$  kg,  $m = 1$  kg,  $l = 1$  m, and  $\theta = \dot{\theta} = 0$ . Keep  $\tau = 1$  N m. The following are simultaneous accelerations at that state, not measured golf results or finite-duration performance predictions. Force entries are in newtons; the acceleration pair is  $(\ddot{x}, \ddot{\theta})$  in  $\text{m s}^{-2}$  and  $\text{rad s}^{-2}$ , respectively.

| Experiment                                            | Applied $F$ | Extra Enforcing Force $\lambda$ | Acceleration Pair |
|-------------------------------------------------------|-------------|---------------------------------|-------------------|
| Free cart, baseline                                   | 3           | 0                               | (1, 0)            |
| Free cart, remove $F$                                 | 0           | 0                               | (−0.5, 1.5)       |
| Enforce baseline cart acceleration after removing $F$ | 0           | 3                               | (1, 0)            |

The free-cart intervention changes both accelerations. The prescribed replay instead replaces the removed force with  $\lambda$ . Its unchanged pendulum response does not show that the original cart force was unimportant. It shows that another driver is now enforcing the same base motion.

### 17.2.2.1 Coordinate Changes Do Not Remove Physical Coupling

Off-diagonal inertia terms are informative in a declared coordinate system, but they are not a coordinate-independent definition of mechanical interaction. For this example, put  $a = ml/(M + m)$  and use the horizontal center-of-mass coordinate  $r = x + a \sin \theta$ . The kinetic energy becomes diagonal:

$$T = \frac{1}{2}(M + m)\dot{r}^2 + \frac{1}{2}\left(ml^2 - \frac{m^2 l^2}{M + m} \cos^2 \theta\right)\dot{\theta}^2.$$

However, the same applied loads have virtual work

$$F \delta x + \tau \delta \theta = F \delta r + (\tau - aF \cos \theta) \delta \theta.$$

The cart force still contributes to the angular equation. A coordinate change moves terms between the inertia description and the generalized loads; it does not alter the physical accelerations. Convective terms, gravity, constraints, and the way controls enter the equations must travel with the coordinate change.

### 17.2.3 What a Counterfactual Can Establish

Start two free-base simulations from the same compatible state. Keep the distal input history fixed and remove a declared base input from one branch. Compare instantaneous accelerations before either trajectory has moved. Then integrate them and compare the accumulated changes in base motion, contact loads, club speed, and work. This identifies an effect of that intervention **within the specified model**, not a physiological strategy effect by itself.

Now prescribe the baseline base trajectory. If the removed input enters only the base equations, while distal equations, inputs, and initial state remain identical, the baseline distal trajectory solves the replay problem too. Under uniqueness it is the same solution; the enforcing reaction supplies the missing base load. **Baseline replay is a consistency check, not evidence that dynamic feedback was unimportant.** The acceleration table is an instantaneous example; a trajectory comparison additionally needs compatible position and velocity.

This conclusion has limits. Removing an actuator that also supplies a distal generalized load need not leave the distal equations unchanged. Discontinuous contact changes, incompatible constraints, or a different initial state can also invalidate the comparison. State those changes instead of attributing every difference to feedback.

Keeping a torque **history** fixed differs from keeping a feedback **policy** fixed. A policy evaluated along diverging trajectories can issue different commands. Both experiments can be useful: the first isolates a prescribed input comparison, while the second evaluates the chosen controller interacting with the altered mechanics. They estimate different effects and should not share one unqualified label such as “proximal contribution.”

### 17.2.4 Follow Work Across the Same Boundary

For a declared collection of bodies in an inertial frame, a useful mechanical ledger is

$$\frac{d}{dt} (T + U_g + U_{\text{shaft}} + U_{\text{contact}}) = P_{\text{act}} + P_{\text{other,ext}} - D, \quad D \geq 0.$$

Here  $T$  includes the kinetic energy of every included body and any represented flexible modes. The  $U$  terms are the conservative gravitational, shaft, and contact energies actually present in that model; absent elements contribute nothing.  $P_{\text{act}}$  is the power of the declared actuators, including any trajectory-enforcing actuator;  $P_{\text{other,ext}}$  is other external power not already represented by the conservative potentials. Dissipation  $D$  includes the specified damping or friction losses. Account for gravity through  $U_g$  or through gravitational power, not both. Likewise, shaft work entering stored energy is not an additional unexplained whole-system energy input.

If the prescribed base coordinate is  $x$ , its enforcing actuator contributes  $\lambda \dot{x}$ . For several prescribed coordinates the corresponding term is  $\lambda^\top \dot{\mathbf{q}}_b$ , where  $\mathbf{q}_b$  contains the prescribed coordinates and  $\lambda$  their power-conjugate enforcing loads. In the cart example,  $E = T + mgl(1 - \cos \theta)$  and

$$\dot{E} = (F + \lambda)\dot{x} + \tau\dot{\theta}.$$

The force needed to enforce motion is not free merely because the solver calls it a constraint reaction. At an instant with zero base velocity, that reaction can change acceleration while doing zero instantaneous translational work.

When the system boundary includes both sides of an ideal passive internal joint, their constraint powers cancel under its compatible relative motion. A compliant interface instead exchanges energy with its storage and dissipative elements. Do not add club-side contact power as a new external input to an enlarged system that already contains the hand, club, and interface. Relative joint motion is power-conjugate to an internal actuator torque; an absolute segment angular velocity generally is not. These distinctions connect the moving-base question to the ledgers in Section 5 and Section 10.

Numerical convergence tests whether the computed result stabilizes under a specified refinement. It does not establish that the contact law, controller, or anatomy is valid. Check force and moment balance, power signs, bookkeeping, and constraint-projection work separately; report any residual with its scale and refinement behavior rather than treating a small energy residual as a universal validation certificate.

#### 17.2.4.1 The Moving-Handle Contribution

For a rigid club, let  $H$  be a handle point,  $C$  a head point,  $\mathbf{r}_{C/H}$  the vector from  $H$  to  $C$ , and  $\omega$  the club's angular velocity. Their inertial-frame velocities satisfy  $\mathbf{v}_C = \mathbf{v}_H + \omega \times \mathbf{r}_{C/H}$ . A flexible club needs its additional deformation velocity under a declared reference description. Changing handle motion can therefore change head speed, club loading, and relative release together. An inward handle motion is not a universal speed instruction: its effect depends on timing, force, orientation, shaft state, and the rest of the trajectory.

For a locally rigid grip region, let  $\mathbf{F}_H$  be the resultant force on the club and  $\mathbf{M}_H$  its moment about  $H$ . With grip velocity  $\mathbf{v}_H$  and angular velocity  $\omega_H$ , their power is

$$P_H = \mathbf{F}_H \cdot \mathbf{v}_H + \mathbf{M}_H \cdot \omega_H.$$

A negative projected moment is neither this total power nor its time integral. Even negative rotational power can coexist with positive translational power. Moving the wrench reduction point changes its moment and the separate power terms; consistent transport preserves their total for the same rigid-body loading and twist. It does not permit treating a flexible assembly as one rigid body without accounting for deformation work.

## 17.3 Where the Picture Breaks

### 17.3.1 What the Provider Benchmarks Establish

The pinned coupled-base/flex study supplies a reduced planar example with explicit base dynamics, grip and shaft moments, and a constrained solve. Its registered driver-removal branch follows a short, 50 ms continuation from the selected state. The spatial study instead uses finite-mass hand carriages, compliant contact forces, and a rigid club. Its archived driver-removal traces retain the provider-defined negative projected contact moment for a bounded 37.5 ms interval. These are model results with declared inputs and observables, not estimates of a typical golfer’s passive release duration.

MuJoCo and Pinocchio provide separate native inertia and bias operators in the spatial comparison. The experiment shares its contact law and integration procedure, so agreement is narrower than two independently developed physical models reproducing a human experiment. A reproducible sign within that shared construction can motivate a further test without establishing physiological passivity, net energy transfer, or robustness over untested parameters.

#### 17.3.1.1 Force Closure Is Not the Whole Contact Contract

The spatial force-pair limitation already discussed in Section 20 matters here too. With club-point position  $\mathbf{p}$ , hand-point position  $\mathbf{h}$ , displacement  $\delta = \mathbf{h} - \mathbf{p}$ , and the supplied law  $\mathbf{F} = k\delta + c\dot{\delta}$ , where  $k$  is stiffness and  $c$  is damping coefficient, applying  $+\mathbf{F}$  at the club point and  $-\mathbf{F}$  at the hand point gives zero resultant force but pair moment

$$(\mathbf{p} - \mathbf{h}) \times \mathbf{F} = -c\delta \times \dot{\delta}.$$

The central spring contribution vanishes; the general vector damping contribution need not. Calling this a closed internal interface requires the corresponding connector moment or angular-momentum account, or an explicit external torque pathway. The existing power identity and native-engine agreement do not supply that missing interpretation. This qualification is not proof that the pair moment causes the reported negative club moment. Changing the contact law or adding a couple changes the dynamics and power ledger, so one must rerun the registered trajectories before claiming the sign survives. The central-fiber model discussed in Section 16 has a different contact law; the diagnosis cannot be transferred to it by name alone.

#### 17.3.1.2 Declare the Moment Point, Axis, and Geometry Control

The spatial record reduces its contact wrench at the club reference point. Its signed projection uses a normalized cross product of the club longitudinal axis and that point’s linear velocity, with a declared world-axis fallback and sign convention. This is a chosen observable, not a universal anatomical

swing-plane normal. Near a degenerate cross product, the fallback and sign convention need to be exposed in the record.

Expressing the same physical moment and axis in another orthonormal frame preserves their dot product. Changing the moment reference point, changing the axis, or moving a contact point is a different operation. The spatial coincident-point and reversed-arm controls recompute moments from the same force traces; they test wrench geometry, not newly simulated contact trajectories. The planar coincident-grip control does run a new continuation. A one-arm benchmark also needs explicit second-hand contacts and controls before it can answer a two-hand allocation question.

### 17.3.2 Human Evidence and Strategy

Nesbit and McGinnis analyze one selected swing from each of four right-handed amateurs. Their three-dimensional marker measurements are mapped into a rigid planar club model to infer loads and examine hub-path effects. This supports a mechanical reason to include handle motion; it is not a population intervention showing that an imposed path improves performance ([Nesbit and McGinnis 2009](#)).

Miura combines measured examples with simplified moving-pivot models to examine inward pulling and parametric acceleration. A prescribed-pivot model explains a possible route for work to enter the club under its assumptions. It does not establish the required whole-body control, energy cost, or benefit for every golfer ([Miura 2001](#)). The relation among measured motion, inferred loads, and biological explanations is developed further in [Section 24](#).

Testing a body-side strategy needs synchronized whole-body and club motion, external and interface load estimates, inertial parameters, and uncertainty in joint centers and soft-tissue motion. A force plate constrains an external wrench; it does not identify individual muscle forces. A handle driver can be a useful reduced input without acquiring anatomical precision from its name.

Declare the model, controller, comparison, and failure criterion before testing. If their combination fails, that result counts against the tested combination. Blaming the controller rather than the proposed mechanism requires additional comparisons or evidence. A state-triggered command may transfer better across base tempo than a clock-triggered one, but that remains a hypothesis to assess on held-out conditions with observable triggers and appropriate load limits.

## 17.4 Reader Lab: Prescribe, Then Release

Begin with the manufactured cart example. Verify the acceleration table from both Newton balances and the coupled equations. Compare the energy rate with applied and enforcing power. Then simulate a baseline, remove  $F$  with the cart free, and replay the baseline cart trajectory under the same distal torque. Report which reaction replaces the removed force and the numerical tolerance of the replay comparison.

Only then enlarge the model: add a club link, shaft compliance, declared contact points, and body-side degrees of freedom as the question requires. Preserve wrench points, projection axes, energy boundaries, and event definitions across comparisons. Physical parameter changes need newly consistent states and trajectories; relabeling coordinates does not create a new experiment.

The useful golf question is how body-side controls create a handle state and club interaction while remaining responsive to loading. The hand path, club reaction, shaft storage, and distal control are intertwined through the same equations and energy boundaries. A moving-base model makes those assumptions available for testing; it does not settle the golfer's best strategy by itself.

**Go Deeper:** See the pinned [coupled base and compliance chapter](#) and [spatial contact study](#). These immutable records specify the experiments; they are not a new simulation or an endorsement of every physiological interpretation attached to them.

## Chapter 18

# How a Constrained Forward Model Solves the Next Instant

### 18.1 A Concrete Picture

Imagine a computer model of two hinged arms holding a bar. You specify its configuration, motion and applied joint torques. The program must find how the arms and bar accelerate together. It cannot move each arm independently and then declare that both hands still hold the same bar. Their shared geometry helps determine the interaction forces.

A forward swing model performs this calculation repeatedly. Within a declared contact mode, it balances inertia, gravity, velocity effects, applied loads and constraint reactions. Flexible deformation and actuator memory add state variables that must also evolve. If a hand slips or a contact opens, the equations governing that mode must change; keeping every contact closed is not a universal success criterion.

### 18.2 How the Mechanism Works

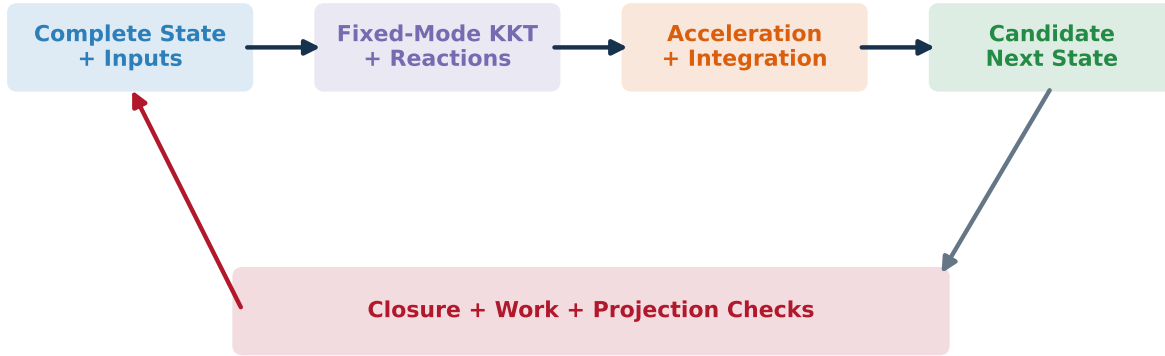
For local generalized coordinates  $\mathbf{q}$  with  $\mathbf{v} = \dot{\mathbf{q}}$ , write

$$\mathbf{M}\dot{\mathbf{v}} + \mathbf{h} = \mathbf{Q} + \mathbf{J}^T\lambda, \quad \mathbf{J}\dot{\mathbf{v}} = \mathbf{b}.$$

The vector  $\lambda$  contains reaction multipliers. The mass matrix  $\mathbf{M}$  expresses inertia and coupling. The constraint Jacobian  $\mathbf{J}$  maps coordinate velocities to rates of change of the constraint equations. Here  $\mathbf{h}$  contains the selected velocity, gravitational, elastic and dissipative terms;  $\mathbf{Q}$  contains the remaining applied loads. The allocation can differ between implementations, but each term belongs in the balance once. An elastic force already represented by a potential must not be added again as an independent energy source. Representations with generalized velocities other than  $\dot{\mathbf{q}}$  require their own configuration-rate map.

For a fixed set of ideal bilateral constraints, the simultaneous solve has the Karush–Kuhn–Tucker (KKT) saddle-point form

## A Forward Model Advances State and Audits Its Approximation



Schematic: Fixed Bilateral Mode; Refinement Compares Separate Runs

Figure 18.1: A Schematic Forward-Dynamics Loop Connects State, Constrained Acceleration, Integration and Separate Numerical Audits.

$$\begin{bmatrix} \mathbf{M} & -\mathbf{J}^\top \\ \mathbf{J} & \mathbf{0} \end{bmatrix} \begin{bmatrix} \dot{\mathbf{v}} \\ \lambda \end{bmatrix} = \begin{bmatrix} \mathbf{Q} - \mathbf{h} \\ \mathbf{b} \end{bmatrix}.$$

The right-hand side  $\mathbf{b}$  follows from differentiating the constraints, including any prescribed boundary motion. Symmetric positive-definite  $\mathbf{M}$  and full row rank of  $\mathbf{J}$  are sufficient for a unique solve in this form. Redundant constraints, singular configurations and different contact laws need separate treatment. Reaction multipliers are solved alongside acceleration; they are not arbitrary controls chosen to create a preferred swing.

### 18.2.1 What a Reaction Multiplier Means

The physical generalized reaction is  $\mathbf{J}^\top \lambda$ . A multiplier's units and sign depend on the constraint equation. Multiplying one constraint row and its right-hand side by seven divides its multiplier by seven while leaving the reaction and acceleration unchanged. A raw multiplier is therefore not automatically a Cartesian hand-force component.

The same caution applies to residual norms. Translational and rotational force equations, acceleration constraints and geometric closure have different units and scales. Report them separately or normalize them with declared reference scales. A small unscaled norm alone does not tell the reader which physical quantity is accurate.

### 18.2.2 Integration Turns Local Rules Into a Trajectory

An integrator approximates the coupled evolution of configuration, velocity and internal states. It may evaluate dynamics several times per accepted step. Large steps can miss events or distort fast flexible motion even when the animation looks smooth. Stability, accuracy and constraint consistency are related but distinct requirements.

Compare declared observables over a sequence of step sizes, solver tolerances and, where relevant, mode-transition settings. Look for convergence and report the range over which it is observed. Projection tolerances, event-location error, roundoff and competing errors can prevent each residual from decreasing monotonically. The name of an integration scheme does not establish its order after velocity-dependent forces, projections or contact switches are added.

Define each event explicitly. In the revision-bound two-degree-of-freedom double-pendulum study, “impact” is a specified club-angle crossing located by interpolation; it does not simulate ball contact. The planar forward two-hand example instead runs to a fixed time horizon. Neither should be described as a resolved club-ball collision. An actual collision requires an impact law or a resolved compliant-contact model, with its own verification.

### 18.2.3 Initial Conditions and Projection

For a holonomic constraint  $\Phi(\mathbf{q}, t) = \mathbf{0}$ , compatible velocities satisfy the following condition, where  $\Phi_t$  is the explicit partial derivative with respect to time:

$$\mathbf{J}\mathbf{v} = -\Phi_t.$$

For a stationary boundary,  $\Phi_t = \mathbf{0}$  and the condition reduces to  $\mathbf{J}\mathbf{v} = \mathbf{0}$ : admissible motion does not open or stretch the constraints. Give flexible modes, activation states and controller memory consistent initial values too. Otherwise the solver may spend its first steps repairing the initialization rather than predicting the intended experiment.

Projection changes a state to satisfy the chosen constraints. Its correction and energy change are numerical bookkeeping, unless a separately justified physical reset law gives them a different interpretation. Record whether the energy ledger starts before or after initial projection. A genuine impact can change velocity discontinuously; silently correcting an inconsistent measured state is a different operation.

For a manufactured example at fixed configuration, take two translational coordinates with masses 2 and 1 kg, velocity  $(3, -1)$  m/s and the stationary condition  $v_1 + v_2 = 0$ . Mass-metric velocity projection gives  $(7/3, -7/3)$  m/s. Kinetic energy changes from 9.5 to 49/6 J: a loss of 4/3 J introduced by the correction. Calling that loss muscle work or physical damping would tell the wrong story. Large corrections should trigger investigation of initialization, discretization or model consistency.

### 18.2.4 Inputs Need a Declared Physical Meaning

A finite step in directly applied torque produces a finite acceleration jump in a regular fixed-mode model; it is not an impulse and does not itself jump velocity. For example, a 3 N m step on an ideal rotor with inertia 2 kg m<sup>2</sup> changes acceleration to 1.5 rad/s<sup>2</sup>. Starting at 0.4 rad/s, speed is still 0.4 at

the step and reaches 0.55 rad/s after 0.1 s. The increase in kinetic energy equals the torque’s work in this example.

Realistic input models may add activation or transmission dynamics. A first-order filter smooths its output but does not impose a universal hard torque-rate bound. A delay shifts a signal without necessarily smoothing it. Preload and actuator memory can retain applied load after the command is set to zero. Rise-time and delay studies therefore compare different input models and initializations; they do not directly identify human activation.

The pinned two-hand example maps its six command values directly to joint torques. Its zero-command branch removes those torques. That interpretation must be revised if an actuator state is inserted between command and load.

### 18.2.5 Building a Counterfactual Correctly

At a forward intervention, copy the complete model state and preserve the boundary prescription. Change only the declared future input, contact law or other intervention variable. Keep solver settings comparable and record mode changes. Do not silently reset velocities, controller integrators or shaft modes. The trajectories can then diverge through both state evolution and changed reactions.

For an instantaneous input comparison, retain the state and solve the dynamics and reactions again under each input. If the applied generalized load has the affine form  $\mathbf{Q} = \mathbf{Q}_0 + \mathbf{B}\mathbf{u}$ , then  $\mathbf{Q}_0$  is the input-independent load and  $\mathbf{B}$  maps inputs to generalized loads. With all coefficients fixed at that state, the same nonsingular KKT system maps input differences linearly to acceleration differences. Here  $\mathbf{u}$  must be the declared input level; it is not automatically the actuator command. Contact-mode changes or nonlinear actuator maps can invalidate that simple comparison. A zero-velocity construction changes the state and requires a separate feasibility check; it is not another input change at the original state. See Section 11 for the different counterfactual questions.

### 18.2.6 Physical Work and Numerical Error Need Separate Ledgers

Choose the modeled system and its mechanical energy before assigning power. Count applied work, prescribed-boundary power and dissipation consistently with the conservative terms already included in that energy. An ideal stationary constraint has zero net power across the complete constrained system, but a hand force can still transfer energy into the club viewed alone.

An ideal moving constraint may supply or remove energy. In a second manufactured example, two masses of 2 and 3 kg satisfy  $q_1 - q_2 = s(t)$ , where  $q_1, q_2$  are their positions and  $s(t)$  is a prescribed relative displacement. With  $\dot{s} = 0.5 \text{ m/s}$ ,  $\ddot{s} = 0.2 \text{ m/s}^2$ , velocities (2, 1.5) m/s and no other loads, the reaction is (0.24, −0.24) N. Its net power is 0.12 W, equal to the kinetic-energy rate. Resetting both velocities to zero would violate the retained moving-boundary condition. Compliant contact can store energy and may dissipate it; neither effect should be confused with an ideal stationary reaction.

Record numerical projection energy separately from the physical work ledger. A small net energy residual can conceal cancellation among larger numerical errors. Momentum checks add useful information when the external wrench and the reference point/frame are specified; angular momentum about a moving point requires the appropriate transport terms.

### 18.2.7 What the Pinned Forward Example Actually Shows

The saved planar two-hand example has seven coordinates, four independent bilateral constraints and no prescribed swing trajectory. Its report records closure, interface-power identities, work-energy residuals, geometry controls and selected step/tolerance comparisons. These records support bounded mechanical feasibility under the stated model, not a guarantee for every forward tier or controller.

The saved baseline reports about 0.096 J of absolute work-energy residual alongside about  $-0.778$  J of accumulated projection-energy change. The latter is not physical work to insert merely to improve the former. It shows why the separate error terms matter. In the saved step comparison, the raw work-energy residual falls from about 0.695 to 0.373 to 0.191 J as the step falls from 2 to 1 to 0.5 ms; position closure does not decrease strictly at every refinement. These are inspected stored results, not a new simulation performed for this chapter.

The model's direct tests also examine a short driven rollout, a zero-torque step comparison, a branch from an achieved moving state, coincident grip offsets and selected invalid requests. Their finite cases do not prove general convergence, symplecticity or global positive definiteness. Checking a release's file hashes establishes correspondence to recorded artifacts; rerunning its dynamics is a separate act that must be reported separately.

### 18.2.8 Inverse and Forward Models Complement Each Other

Inverse dynamics uses motion, inertial assumptions and external-load information to infer net generalized loads. It does not uniquely recover individual muscle forces or redundant internal contact forces without additional assumptions. Forward dynamics predicts motion from a complete initial state and a declared input model. It enables interventions within that model, but its parameter, controller and contact assumptions remain consequential.

Agreement between inverse and forward calculations can expose implementation errors. It can also follow from using the same equations and fitted data in both directions. Independent observations and held-out interventions provide stronger evidence for predictive validity than replay agreement alone.

**Model Result:** The revision-bound planar study reports selected numerical checks and a feasible negative sum of grip-force moments about the club center, using its declared planar angular sign convention and excluding direct wrist torques. Those checks neither establish biological passivity nor identify a human control strategy.

**Human Evidence:** MacKenzie and Sprigings fitted a selected swing of one golfer using single-camera measurements; their quantitative comparison did not cover the model's axial arm rotation. A separate optimization then targeted clubhead speed (MacKenzie and Sprigings 2009). Balzerson and colleagues combined a fitted golfer model with shaft, impact and flight components having different comparison datasets (Balzerson et al. 2016). These studies illustrate useful modeling and bounded empirical comparisons, rather than independent validation of every prediction of an optimized whole-system model.

**Hypothesis:** Mechanisms that survive declared forward, geometry and numerical checks are stronger candidates for targeted experiments. That survival does not establish that golfers exploit them.

**Practical Interpretation:** Trust grows when the reader can distinguish the equations, their numerical solution, the fitted data and genuinely predictive tests of the physical system.

### 18.2.9 Coordinates, Contact and Independent Implementations

Minimal coordinates can eliminate simple holonomic joint constraints locally. Maximal coordinates often make body poses and complex connectivity easier to describe, at the cost of explicit constraints. Spatial-vector algorithms organize dynamics calculations and can be used with different coordinate choices; they are not a competing coordinate representation.

Compare mapped positions, velocities, wrenches and powers for the same physical model, inputs and initial state. Components in different bases or at different wrench reference points need the corresponding transformations. Coordinate invariance is not a claim that kinetic energy has the same numerical value for observers moving relative to one another. Agreement between independent engines is strongest when they do not share the same derivative implementation or unexamined assumptions.

An ideal bilateral grasp places no friction or strength limit on its reaction. Human hands and feet do. A no-slip model must be checked against the specified contact capacity before interpreting its loads physically. Unilateral contact requires separation and admissibility conditions; an active-set KKT solve can describe a regular fixed mode, while switching, complementarity, compliant contact or impacts require additional equations and numerical choices. Test sensitivity to those choices when the proposed mechanism depends on contact.

### 18.2.10 Reader Lab: Separate a Broken Calculation From a Different Model

Introduce a frame mismatch deliberately: reverse one measured axis while leaving the model transformation unchanged. A frame-consistency check should detect that error. Transforming all quantities consistently describes the same physics and should pass. Likewise, a documented damping removal or input delay can define a legitimate new physical scenario. It should change predictions, not automatically fail the solver. Energy may rise when applied work is positive.

For a geometry control, setting both grip offsets to zero removes their force-generated moment about the coincident point. By contrast, reversing an offset and solving a new trajectory also changes the reactions; the resultant couple is not guaranteed to reverse sign. State the exact limiting prediction before using it as a pass/fail test.

Next perturb the initial state or enlarge the timestep and compare correction sizes and converged observables. Preserve failed cases and classify them: infeasible contact, singular geometry, a numerical failure and a different physical model are not interchangeable explanations. Do not loosen tolerances until a desired animation appears and then call it verified.

### 18.2.11 Reproducibility and Extending the Model

Record the exact code and model revision, parameters, complete initial state, input schedule or feedback law, solver settings, event definitions and relevant random seed. State units and frames in machine-readable outputs. A stateful controller needs its own state evolution; repeatedly calling a hidden mutable callback can change the experiment and make reconstructed diagnostics disagree with the inputs actually used.

Retain earlier benchmarks when adding a shaft, moving base or spatial joint. Add new limiting cases and explain which earlier assumptions no longer apply. Version changed output meanings and

regenerate figures from recorded results. Publish a small runnable benchmark, its expected tolerances and failed-case examples so review does not require the entire optimization stack.

The same reproducibility requirement extends to the reader’s visualization. An educational interface should consume the same versioned model or verified outputs used for the scientific evidence. That is a design recommendation, not proof that every existing front end already shares one engine. Inspect actual dependencies before making that architectural claim.

## 18.3 Where the Picture Breaks

The two-arm picture assumes known geometry, constitutive laws and boundaries. Human anatomy, contact and control are uncertain. Verification checks how well the implementation solves the declared equations; validation compares relevant predictions with physical observations over a specified use range. Good numerics can integrate an inadequate model, and a good fit can hide compensating errors. Neither an optimized swing nor a small residual proves that a person used the modeled controls, that a particular muscle supplied a net torque, or that a proposed intervention is beneficial.

**Go Deeper:** Read the revision-bound [forward constrained two-hand model](#), the [solver and platform methods](#), and the [qualification instructions](#).

## Chapter 19

# A Ladder of Models, Not One Final Model

### 19.1 A Concrete Picture

A map can show a continent, a city, a building, or the pipes inside one room. The city map is not false because it omits plumbing. The plumbing drawing is not better for planning a cross-country trip. Scale and detail should match the question.

The proximal-distal project uses a model ladder for the same reason, but its models form branches rather than one sequence of increasingly complete replicas. A double pendulum isolates coupling. A two-hand model represents separate contacts. A flexible shaft adds storage. A dynamic base adds back-reaction. Spatial and anatomical models address other omissions; neither their detail nor their software name establishes an independently verified result. Each model must answer a declared question through observables it can actually represent.

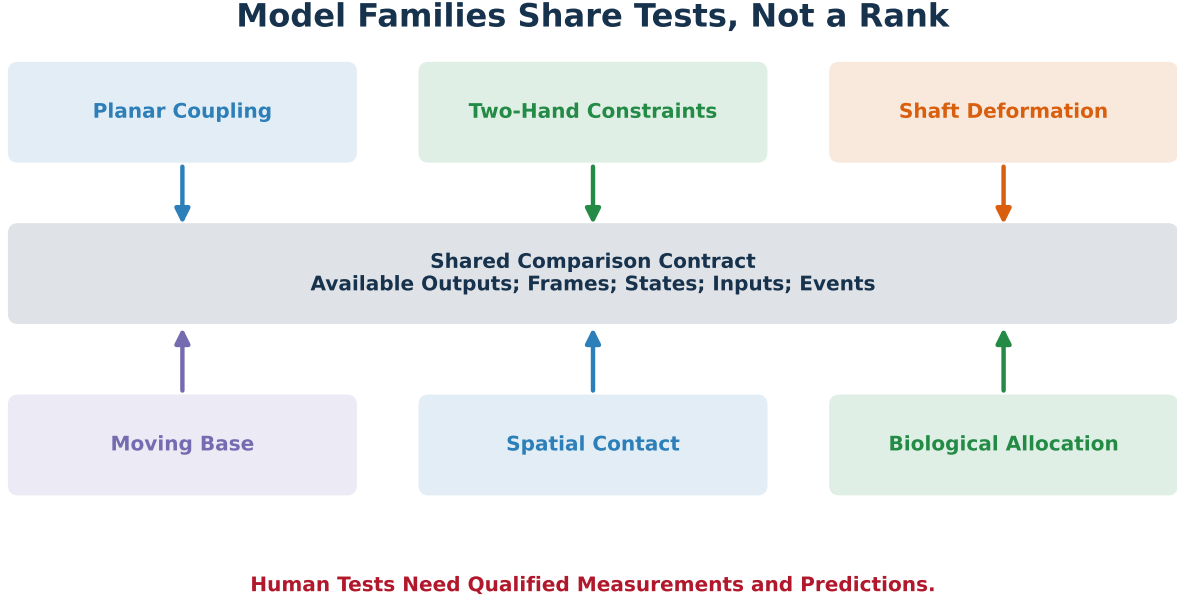


Figure 19.1: Model Families Meet Through a Declared Comparison Contract; More Detail Does Not Establish Human Validity.

## 19.2 How the Mechanism Works

The bottom rung should be analytically transparent. Exact equations, limiting cases, and pointwise decompositions reveal why a mechanism appears. Higher rungs ask whether it survives added realism. If the sign reverses or effect vanishes, the model boundary becomes a scientific result rather than an embarrassment.

### 19.2.1 Common Observables Are the Handrail

Coordinates differ across tiers. A planar wrist angle has no one-to-one match in a spatial model with two wrists and a flexible shaft. Comparing raw coordinates would confuse model architecture with mechanism.

Instead, specify a comparison map for each model  $m$ :

$$\mathbf{y}_m = h_m(\mathbf{x}_m, \mathbf{u}_m, \theta_m).$$

Here  $\mathbf{x}_m$  is the full state,  $\mathbf{u}_m$  the declared input, and  $\theta_m$  the model parameters. The common output  $\mathbf{y}_m$  might contain club motion, a wrench at a named reference point, interface power, work, and closure residuals. Every component needs units, signs, frame, event window, system boundary, and an uncertainty or numerical-error allowance. A planar result is a projection with restrictions; storing it in a three-component array does not add out-of-plane dynamics.

Use the intersection of available outputs. A single-wrist pendulum does not measure two individual hand wrenches; a rigid club has no shaft strain state. Mark an unsupported quantity **unavailable** or

**not applicable**, with the reason. A zero imposed by a model assumption is not an observed zero. A shared record format ensures compatible labels, not common physics or equal trajectories.

The power comparison needs particular care when one branch deforms. At separate contact ports, retain

$$P_{\text{ports}} = \sum_i (\mathbf{F}_i \cdot \mathbf{v}_i + \mathbf{M}_i \cdot \boldsymbol{\omega}_i).$$

The moments here are local contact moments, paired with their work-conjugate angular velocities; positive power enters the chosen system. Reduction to one resultant wrench and one rigid-body twist requires compatible rigid motion at all those ports. For a counterexample, apply outward axial forces of 10 N at two ends moving apart, each with outward velocity 0.2 m/s in the same inertial frame. The resultant force and moment are zero, yet the input power is 4 W. A rigid center-of-mass motion alone misses that deformation power. This connects the system boundary in Chapter 2, power accounting in Chapter 5, and shaft storage in Chapter 15; it does not violate reference-point invariance for a rigid body.

### 19.2.2 Matching Outputs Does Not Match the State

Two full states can have the same comparison output. In a manufactured two-mass spring example, initially set the observed mass's position and velocity to zero. Let  $q_1$  and  $q_2$  be displacements from the two masses' equilibrium positions along the same axis, with spring extension  $z = q_2 - q_1$  and force  $+kz$  on mass 1. With mass 1 kg and stiffness 100 N/m, an omitted extension of +0.01 m or −0.01 m gives that mass acceleration +1 or −1 m/s<sup>2</sup>. The reduced snapshot is identical; the stored elastic state is not. Activation, contact history, and modal velocities can create analogous ambiguities.

Therefore register how a reduced initial condition is lifted into each full state, including preload and stored energy, and whether the lift is unique. Holding the same motion while reconstructing required loads is an inverse comparison. Applying matched input histories or matched feedback policies and letting motion evolve is a forward comparison. Deleting a command at one state does not erase its earlier impulse or reproduce the trajectory obtained by removing it throughout the swing. Chapters 3 and 11 supply those distinctions.

### 19.2.3 Complexity Must Pay Rent

Adding states can expose uncertain parameters and alternative explanations, but does not necessarily increase every uncertainty or destroy identifiability. Known parameters, informative measurements, or a justified constraint can make a richer model more predictive. Conversely, a smaller model can be precise about the wrong quantity because it excludes important physics. Distinguish approximation bias, numerical error, parameter uncertainty, and ambiguity in the unobserved internal variables. A full-body animation does not resolve any of these by itself.

The distributed shaft earns its extra states when omitted modes affect the target beyond the declared tolerance, or when shaft behavior is itself the target. A moving base matters when prescribed support hides relevant work or back-reaction. Spatial contact matters when out-of-plane geometry or wrench components matter. Compare predictions on data or conditions not used to tune the model; improved training fit alone does not establish the improvement.

### 19.2.4 Surviving and Changing Claims

Some statements are structural within their stated mechanics. Consistent action-reaction, wrench transformation, and complete work-energy accounting should survive a change of representation. Closure must include moving-support work, storage, dissipation, and numerical residuals where relevant; zero energy change is not the universal test. Some statements are model-specific: the exact duration of a negative couple, optimal torque onset, or numerical force magnitude can change with parameters and degrees of freedom.

The release matrix labels what each tier supports, omits, and must not claim. This prevents a result from climbing the ladder merely because a later model exists elsewhere in the repository.

### 19.2.5 Cross-Formulation Agreement

Inverse, forward, reduced, and spatial formulations can share a common state or observable. Agreement under independently implemented equations reduces the chance that one code path created the effect. Disagreement helps locate hidden assumptions.

The reviewed upstream release makes a narrower distinction than “two engines agree.” Its nonplanar common-state inverse comparison uses MuJoCo and a separate Lagrange–Christoffel formulation with prescribed hand loads. Its reduced spatial forward-contact comparison uses native MuJoCo and Pinocchio inertia-and-bias operators, but a **shared contact law and integrator**: projected compliant contacts and the same semi-implicit update. That checks transport through those rigid-body operators; it is not agreement between independently authored native contact solvers or integration schemes. The separate native-formulation control reports a discrepancy rather than proving their equivalence.

Report which equations, code, calibration, contact law, and time integration are independent. Compare the physical generalized reaction or a declared contact wrench, not raw redundant multipliers whose values depend on constraint coordinates (Chapter 6). Check converged observables and residuals under refinement, with event alignment stated. Identical bugs in a shared layer can survive perfect cross-engine agreement. Pinocchio is a rigid-body dynamics library; using it in a simulation does not imply that it independently supplied every component of that simulation.

### 19.2.6 The Ladder Ends at an Open Gate

No computational branch becomes a human trial. Select measurements from the claim and its competing explanations: bilateral hand wrenches for hand-load allocation, shaft measurements for deformation, or activation observations for an activation hypothesis. Not every question requires the same sensor list; EMG does not directly measure muscle force. The full project protocol requests multiple synchronized modalities because its joint claims cross these boundaries. Chapters 25 and 26 connect measurement sufficiency to identifiability and design.

**Model Result:** The pinned release reports bounded interaction and force-generated-couple findings in specified planar and reduced spatial branches. It also includes qualified short-horizon articulated cases, while broader anatomical transport remains unresolved. “Articulated” is not one universal pass/fail label: capability, duration, contact law, and calibration must accompany it. In its finite-base work-matching experiment, numerical gates pass for 576 trajectories but no fixed-versus-coupled comparison cell meets the registered total-work match. Consequently, that experiment identifies no delivery benefit; numerical success cannot supply a missing matched comparison.

**Human Evidence:** The reviewed release has no qualifying governed participant deposit for its complete bilateral-wrench protocol. That is an evidence boundary for this project and revision, not a claim that no relevant measurements exist anywhere. Published kinematics or net kinetics support narrower questions.

For example, McNally and McPhee’s six-degree-of-freedom model combines pelvis motion, a second shoulder coordinate, flexible-club dynamics, and an impact/flight objective (McNally and McPhee 2018). Its optimization constrains some joint timings together and assumes candidate center-face impacts. The authors report a torso-before-pelvis timing result and grip-kinematic discrepancies, discussing rigid-torso limitations. This is a useful example of testing a richer model against selected observations, not evidence that its optimum identifies a golfer’s muscle control.

**Practical Interpretation:** Ask for the smallest model that can answer today’s question and a comparison that can expose a consequential simplification. One successful extension supplies bounded support, not proof against every omitted mechanism.

### 19.2.7 Why No Single Model Is “the Model”

Every model answers a question by omitting something. A double pendulum isolates linked inertia. A two-hand constraint model exposes force allocation. A flexible shaft adds stored energy and phase. A whole-body spatial model adds moving base and ground interaction. A musculoskeletal model asks about candidate muscle allocations. None supersedes all earlier models; each changes the balance between interpretability, fidelity, and identifiability.

The ladder works when observables and interventions remain aligned across rungs. If every model uses a different definition of transfer, comparison is impossible. This project carries forward available hand-wrench, swing-normal-couple, interface-power, work, club-state, closure, and intervention records with their capability limits. None may be silently filled from a different model.

The following families organize modeling questions. They are not a claim that every listed capability has been executed or validated together in one release.

### 19.2.8 Rung Zero: Algebra and Geometry

Before integrating motion, test wrench arithmetic, sign conventions, moment-arm dependencies, and coordinate transforms. Symbolic or static examples should recover known limits exactly. This rung catches errors that a complex animation can conceal.

It also states propositions clearly: collapsing hand separation collapses a pure differential couple; a consistent reference-point transform preserves wrench power; equal-and-opposite contacts can have zero net force and nonzero moment. These are mechanics identities, not human findings.

### 19.2.9 Rung One: The Planar Double Pendulum

The fixed-base two-link model tests interaction dynamics, carried motion, relative release, and torque timing. Its small state permits dense parameter sweeps, analytic checks, and transparent pointwise decompositions. It cannot represent bilateral hands, spatial axes, ground contact, or shaft deformation.

A result promoted from this rung should be qualitative and mechanism-focused. Exact speed gains or optimal milliseconds are provisional. Negative controls and sensitivity maps determine which signs or

orderings are stable enough to carry upward.

#### **19.2.10 Rung Two: Two Hands and a Moving Base**

Add separated contact points and declare either prescribed or dynamic proximal motion. An algebraic two-hand representation can display common and differential force modes, but geometry alone does not determine their forces. A forward contact model needs the constitutive law, state, drivers, and constraints that close its dynamics. A finite-mass dynamic base allows club-to-base reaction to change base motion; a prescribed base instead requires whatever external actuation enforces its trace. Contact remains idealized and anatomy generalized.

This rung asks whether the mechanism survives when the very feature it concerns —two hands—is represented explicitly. It also exposes a possible inverse ambiguity: the same club wrench can admit different hand allocations. Additional measurements or constitutive restrictions can reduce that set; its dimension and feasible extent are not fixed solely by having two contacts.

#### **19.2.11 Rung Three: Flexible Club**

Add one or more shaft modes, damping, and calibrated boundary coupling. Repeat the same counterfactuals while including modal state and energy. This rung asks whether rigid-club conclusions were artifacts of omitting deformation and whether negative coupling interacts with shaft return.

Modal convergence and independent beam checks matter. A single arbitrary spring can generate an attractive delay without representing a club. Parameters should come from measured or explicitly synthetic sources and remain qualified.

#### **19.2.12 Rung Four: Spatial Whole-Body Dynamics**

Add three-dimensional pelvis, trunk, shoulder/arm, hand, and club motion with ground contact. Spatial frames permit out-of-plane moments and full wrenches. The number of parameters and controls grows sharply, so identifiability becomes a central result rather than an inconvenience.

Independent implementations are valuable here, provided the physical problem and compared outputs are matched. Similar trajectories alone can hide incorrect load or power bookkeeping. Conversely, different contact laws need not predict identical reactions: first decide whether the test concerns implementation agreement or sensitivity to physical assumptions. One implementation is not automatically wrong, but it leaves that replication question open.

#### **19.2.13 Rung Five: Biological Allocation**

A musculoskeletal or activation-aware model proposes how generalized loads might be shared among muscles and passive tissues. It can study co-contraction, stiffness, and internal load, but it needs EMG, strength, anatomy, and optimization assumptions. Multiple allocations usually fit the same motion.

This rung should not retroactively turn earlier generalized arm and wrist drivers into named muscles. Instead it tests candidate mappings and reports their nonuniqueness. Scapular action, in particular, requires adequate kinematics and measurement-aware shoulder-girdle representation.

### 19.2.14 Promotion and Demotion Rules

A claim earns support in an additional model only for the registered outputs, interventions, and domain that were tested. Keep four different outcomes apart: a failed closure check prevents numerical qualification; a resolved adverse prediction can contradict the registered claim; insufficient measurements leave it inconclusive; an unrun comparison remains untested. Changing a constitutive law and reversing an effect demonstrates sensitivity to that assumption, not automatically a bug or a contradiction of the original conditional statement.

Demotion is not failure of the project. It locates a boundary: “this effect is specific to a fixed-base rigid planar model” is scientifically useful. The reader can then see what remains established, what is model-conditional, and what human evidence still owes.

### 19.2.15 Cross-Rung Tables and Figures

For each mechanism, a comparison table should list degrees of freedom, contact model, shaft model, controls, identified parameters, pass/fail gates, and common outcomes. Figures should overlay normalized time histories with consistent signs, not juxtapose unrelated axes. Differences in magnitude should remain visible rather than normalized away when magnitude is the question.

Machine-readable manifests make this comparison reproducible. Each row links to input, output, code commit, and generated figure. Reviewers can rerun one rung or replace it with an independent implementation without adopting the whole software ecosystem.

### 19.2.16 Deciding When Complexity Is Worth It

Add a degree of freedom or state when it changes a target observable, explains a structured residual, enables a discriminating experiment, or removes an unacceptable assumption. Do not add complexity merely for visual realism. Weakly constrained flexibility can absorb residuals and weaken discrimination among explanations. It does not literally allow every possible observation; the question is which alternatives the chosen data and test can distinguish.

The final modeling product should therefore remain modular: a transparent core, optional higher-fidelity rungs, shared contracts, and explicit qualification. That architecture invites criticism and extension as an open resource rather than presenting one branded engine as final authority.

### 19.2.17 Reader Lab: Write a Promotion Matrix

Choose one claim, such as a late negative swing-normal contact couple. Across rows list algebraic geometry, planar double pendulum, two-hand moving base, flexible shaft, spatial whole-body, and biological allocation. Across columns list sign, onset, duration, magnitude, work, outcome effect, negative controls, uncertainty, and human evidence. Populate every cell with result, not applicable, or unresolved.

The matrix makes a common error obvious: a claim may be well verified in early rungs but unresolved in humans. It also reveals where complexity changes the result. Link each cell to machine-readable evidence and tests.

Define comparison criteria before looking at outcomes. A chosen rule might require the sign in 95 percent of a declared ensemble, with geometry controls passing, while permitting bounded changes in

magnitude. That percentage is a decision rule about a specified sample and weighting, not a universal probability or confidence guarantee. Three positive results among four scenarios give 75% with equal weights, but 30% with weights (0.1, 0.1, 0.1, 0.7) when the last result is negative. Neither weighting becomes a golfer population without evidence.

Even under independent sampling from a specified Bernoulli population, 95 successes in 100 trials give an approximate 95% Wilson interval of [0.888, 0.978] for its success probability, not proof that it is at least 0.95. A deterministic parameter grid or correlated trajectories do not automatically justify that binomial calculation. Declare scenario selection, dependence, weights, adverse cases, and held-out checks. Keep statistical uncertainty separate from untested model discrepancy.

### **19.2.18 Independent Extension**

Invite a second implementation to reproduce one rung using the published contracts. If it disagrees, preserve both outputs and reconcile specification, frames, and numerics. The goal is not to defend an engine. It is to discover which physical conclusion survives independent expression.

A modular open resource makes such extension inexpensive. Shared schemas and benchmarks let a contributor add a contact or anatomical rung without rewriting the manuscript's conceptual core.

### **19.2.19 A Claim Can Live at More Than One Rung**

The identity that rigid-body wrench power is reference-point invariant requires paired transport of moment and velocity for the same physical port motion. The finding that a particular negative-couple interval appears belongs to specific model rungs. The hypothesis that golfers use it belongs to the human-evidence rung. Separating them prevents a human null result from invalidating an identity or a verified identity from being offered as human validation.

Each public statement should therefore carry rung and status. Readers can see which parts are durable foundations and which remain contingent. New evidence updates the appropriate layer without rewriting the entire framework.

### **19.2.20 Completion Is a Maintained Ladder**

A release can be complete as an audited record while leaving proposed branches untested. Each included result needs a specification, executed verification, available observables, limitations, and the next discriminating test; proposed extensions need explicit untested status. Documentation completeness is neither completion of every experiment nor validation of maximal anatomical detail.

Maintaining simple benchmarks alongside rich models prevents regression and keeps the project teachable. The lay book, technical monograph, simulator, and evidence register should all reference the same ladder identifiers.

## **19.3 Where the Picture Breaks**

The map analogy suggests clean nested scales. Model families are not always nested; a spatial engine may use different contact or damping laws rather than simply adding detail. Common observables and negative controls are needed to compare them fairly.

More agreement also does not guarantee truth if all tiers share a wrong premise. Independent measurement remains decisive.

**Go Deeper:** Explore the monograph’s [common-observable ladder](#), [model completion matrix](#), and [reviewer workbench](#) at reviewed revision [85cce4d](#). The [MuJoCo computation documentation](#) and [Pinocchio overview](#) describe package algorithms; their capabilities alone do not certify a project comparison. This chapter audits published evidence and manufactured examples, without rerunning those provider simulations or collecting human measurements.

## Chapter 20

# From a Plane to Three Dimensions and Independent Engines

### 20.1 A Concrete Picture

A shadow puppet can reveal a hinge mechanism while hiding motion toward the wall. A planar golf model is a carefully chosen shadow. It can expose in-plane forces and swing-normal moments, but it omits motion out of the plane, moments about axes within the plane, changing swing plane, and spatial joint axes.

Moving to three dimensions is not merely drawing the same links with perspective. Finite rotations generally do not commute, moments require declared frames and reference points, and constraints can contain new internal modes. A six-component wrench record does not mean that every contact can exert six independent loads.

### 20.2 How the Mechanism Works

In a plane, one angle and one signed moment describe each hinge. In space, body orientation belongs to the rotation group  $SO(3)$ . Angular velocity is a vector, and a moment's components depend on frame. Applying rotations in different orders can yield different orientations.

#### 20.2.1 Spatial Wrenches Preserve the Physical Contract

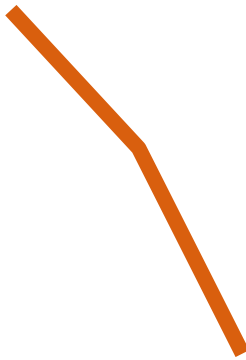
A three-dimensional wrench records force  $\mathbf{F}$  and moment  $\mathbf{M}_O$  about a specified point  $O$ . For  $\mathbf{r} = \overrightarrow{OQ}$ , expressing the same loading at another point  $Q$  of a rigid body gives

$$\begin{aligned}\mathbf{M}_Q &= \mathbf{M}_O - \mathbf{r} \times \mathbf{F}, \\ \mathbf{v}_Q &= \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}, \\ P &= \mathbf{F} \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega} \\ &= \mathbf{F} \cdot \mathbf{v}_Q + \mathbf{M}_Q \cdot \boldsymbol{\omega}.\end{aligned}\tag{20.1}$$

All velocities here describe the same rigid motion for the **same observer**. Rotating the components of

## A Plane Reveals One Projection and Hides Others

### Planar Projection



### Spatial Wrench and Changing Axes

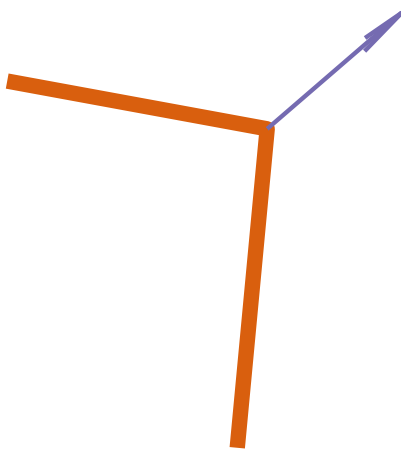


Figure 20.1: A Planar Projection Hides Directions That a Spatial Model Must Resolve.

every vector into another orthonormal basis preserves these dot products. The wrench components themselves change. Likewise, shifting the moment reference requires the matching change in rigid-body velocity; using the old velocity with the new moment generally changes the reported power.

Changing the observer is a different operation. An observer translating with constant velocity  $\mathbf{U}$  measures  $\mathbf{v}'_O = \mathbf{v}_O - \mathbf{U}$  and  $P' = P - \mathbf{F} \cdot \mathbf{U}$ . External power and kinetic energy need not have the same value for different observers. Rotating observers require additional relative-motion terms in dynamics. Expressing laboratory-observed velocities in club-aligned axes does not by itself make them club-relative velocities.

A point-force contact has no independent contact moment: its moment about  $O$  is fixed by its position and force. Friction, unilateral separation, and contact area further restrict admissible loads. A distributed grip may transmit a resultant moment, but its six recorded components still obey the contact model. For a deforming shaft, one rigid wrench–velocity pairing does not capture all deformation power; sum the actual contact powers and account for internal energy.

This frame-explicit contract is the bridge from the planar negative couple to a spatial test. The swing-normal component can be compared while additional components reveal additional twisting or bending loads.

### 20.2.2 Full Body Adds Redundancy

A spatial full-body model has many coordinates capable of producing similar club motion. Inverse dynamics may estimate a net generalized-force vector for a prescribed trajectory, but two-hand contact and muscle redundancy remain. More coordinates do not automatically identify more biology.

The common-state spatial model therefore emphasizes action-reaction, wrench closure, and matched

interventions rather than claiming a unique human strategy. It reports what survives and what remains open.

### 20.2.3 Contact Closure Comes Before Contact Dynamics

A constraint Jacobian can have full row rank even when the points it would constrain are far apart. The project tested that distinction with six deterministic subject-scaled engineering profiles, three grip spans, and 61 prescribed states per case. The anatomical hand points miss the declared grip points by 0.171–0.616 m, and none meets the registered 5 mm tolerance.

The same open configurations still produce a rank-six bilateral contact Jacobian. That is not a contradiction. Rank asks whether six local correction directions are locally available without coordinate bounds; closure asks whether a correction is still required. Rank alone does not guarantee a finite correction within joint limits or without collision. Calling a rank test a contact-feasibility result would therefore hide the most important geometric failure.

This adverse result narrowed the next model rather than weakening the scientific program. A bounded subject-scaled solver then closed both hands on the grip while respecting its declared joint and collision guards. It held the club at 13 prescribed samples over 0–0.24 s, while allowing pelvis, trunk, and arm coordinates to change. All **234 of 234** profile–span–sample cases passed the **0.5 mm** closure gate, with no stored contact error above  $1.16 \times 10^{-10}$  m. That numerical residual is not an anatomical measurement accuracy. A deliberately unreachable 2.0 m grip span failed, preserving a negative control.

That success is a reduced-tree necessary condition, not proof that a person can adopt the poses. The limits are broad engineering guards, collision clearance uses coarse spheres, and the model omits scapular motion, forearm rotation, multi-axis wrists, fingers, and distributed grip contact. The spatial tier still cannot answer whether a passive mechanism reduces timing demand, creates useful slack, improves delivery, or specifies a human technique. Subject-specific anatomy and calibrated compliant forward contact are the next gates.

### 20.2.4 Scapular Mobility Changes Reachability Without Identifying Strategy

The next screen makes a paired comparison within a different feasible set. It holds the trunk and club pose fixed, then compares fixed shoulder centers with shoulder centers that can move on a declared thoracic ellipsoid. Four screening coordinates per side represent protraction, elevation, upward rotation, and winging. In this surrogate, the first two move the shoulder center; the latter two change joint orientation. This is a reduced geometry experiment, not a validated anatomical shoulder. Its arm-only solve is not a rerun of the 234-case solve that allowed trunk motion.

The fixed branch closes **0 of 54** stored states. The mobile branch reaches the 0.5 mm residual tolerance in **31 of 54**. The optimizer reports successful termination in **39 of 54**, but only **16 of 54** satisfy both criteria: fifteen closed cases lack successful termination, and twenty-three successfully terminated cases remain open. Twenty-eight states touch a declared bound, and an intentionally impossible 2.0 m grip span remains open. Mobility clearly changes modeled reach, but unresolved cases may reflect range choice, omitted coordinates, or numerical termination rather than human infeasibility.

There is a second warning. Both contact Jacobians retain rank six, while local coordinate nullity rises from two to ten. Many scapular and glenohumeral coordinate changes therefore leave the contact residual unchanged to first order in the unconstrained local linearization. At an open pose this is not

yet a closed-contact manifold, and active joint bounds restrict admissible directions. Contact geometry alone cannot tell a reviewer which internal allocation a person used or should use.

The scientific next step is not to rename these coordinates as muscles or coaching actions. It is to repeat the comparison with a validated articulated shoulder, subject-specific bone geometry where governed, calibrated distributed grip contact, and paired forward dynamics. Those additions support tests of force, work, passivity, and delivery with increasing anatomical relevance. Contact geometry alone supplies none of those results or a human strategy.

### 20.2.5 Independent Engines Test Implementation Dependence

Running the same conceptual experiment in two engines tests the implementation parts that differ, provided observables use a common convention. Report which model construction, contact laws, derivatives, and numerical updates are shared. Matching screenshots prove little. Matching trajectories, wrenches, powers, constraint residuals, and negative controls is stronger.

The archived finite-mass hand-carriage experiment calls native MuJoCo `mj_forward` and Pinocchio articulated-body forward dynamics, both version 3.8.0. It uses a shared Kelvin–Voigt point-interface law and a shared semi-implicit update after native acceleration evaluation. It therefore compares rigid-body dynamics and force mapping; it does not independently test the engines’ native contact solvers or integrators. Neither branch commands the club directly.

The driver killswitch is a genuine forward intervention: starting from the same state at 0.180 s, it removes the grounded hand-carriage driver while retaining gravity, inertia, and interface storage and damping. In contrast, the coincident and reversed moment-arm controls reuse achieved forces **without advancing a new trajectory**. They test the moment calculation, not the later motion of a physically altered grip.

### 20.2.6 Angular Momentum Is a Separate Check

Equal and opposite forces can still produce a net moment when their application points differ. In the archived vector spring–damper law, let  $\mathbf{h}$  be the hand position,  $\mathbf{p}$  the club contact point, and  $\mathbf{d} = \mathbf{h} - \mathbf{p}$ . The club receives  $\mathbf{F} = k\mathbf{d} + c\dot{\mathbf{d}}$  and the hand receives  $-\mathbf{F}$ . With both positions measured from one origin, the pair contributes

$$\begin{aligned}\tau_{\text{pair}} &= \mathbf{p} \times \mathbf{F} + \mathbf{h} \times (-\mathbf{F}) \\ &= -c\mathbf{d} \times \dot{\mathbf{d}}.\end{aligned}\tag{20.2}$$

The spring term cancels, but transverse relative velocity makes the damping forces noncentral. The power ledger can still show nonnegative dissipation. Thus force balance and passivity alone do not establish angular-momentum balance for a closed internal interface. A declared support reaction or contact-couple pathway is needed to account for this moment, or the constitutive model must change.

Direct calculation from the archived traces gives a peak norm of the summed two-interface pair moment of about 0.083 N m. The native adapters apply no additional contact couples, and the recorded ground-pathway proxy accounts for the driver. This is a shared modeling limitation, not a discrepancy between the two engines. Its effect on the proposed club-moment mechanism requires a new, governed comparison; the existing agreement does not quantify that effect. The [provider follow-up](#) tracks this

unresolved moment accounting. The archived sign and power results remain results of the declared engineering model, not validation of an isolated biological grip.

### 20.2.7 Agreement Has a Boundary

If two engines agree on a reduced spatial mechanism, code-specific error becomes less likely. Shared parameter errors, oversimplified contact, and missing anatomy remain possible. Cross-engine agreement is verification evidence, not human validation.

### 20.2.8 Three-Dimensional Measurement Is Hard

Marker-based motion capture suffers soft-tissue artifact, especially around the scapula and forearm. Joint centers and axes are estimated. Clubface orientation requires careful rigid-body tracking. Bilateral grip sensors must be registered to the club frame. These uncertainties should enter the model-data comparison.

**Model Result:** The archived carriage model retains a negative swing-normal projection of its point-force moment after the driver is removed in both native dynamics realizations. The moment-arm identities also pass their algebraic controls. This projection is about the declared club reference; it is not an origin-independent pure couple when net contact force is nonzero.

**Human Evidence:** Measured trunk and pelvic motion has three-dimensional structure ([Meister et al. 2011](#); [Joyce et al. 2013](#)). Those kinematic studies do not test this specific bilateral compliant-contact mechanism. McPhee reviews the broader modeling and measurement methods ([McPhee 2022](#)); neither that review nor agreement between reduced models supplies participant validation for this mechanism.

**Hypothesis:** The planar swing-normal couple is one projection of a spatial wrench pattern that should be measurable without requiring a unique muscle allocation.

**Practical Interpretation:** A planar model can explain a shadow mechanism; a spatial model must show which parts remain when the lights move.

### 20.2.9 What the Plane Captures

A planar model captures rotation in one chosen swing plane and forces projected into it. It can represent link inertia, relative release, radial/tangential reaction, and an out-of-plane moment. Because signs and geometry are easy to see, it is an excellent place to derive and falsify the basic negative-couple mechanism.

The plane is not a claim that a golf swing is flat. It is a controlled slice. Projecting a spatial trajectory is not a closed planar dynamics model: omitted motion can alter inertia, reactions, and forces within the displayed plane. Imposing planar constraints may itself introduce reactions absent in free space. The key question is whether a spatial extension retains the corresponding physical projection when out-of-plane motion and extra axes are admitted.

### 20.2.10 What Three Dimensions Add

Each rigid body now has spatial orientation and angular velocity. The club's moment is a vector. Hand loading can be recorded as a six-component wrench at each declared reference, subject to the contact restrictions above. Swing-plane normal can vary with time and must be defined operationally. Forearm

rotation, wrist deviation, shaft droop, torso side bend, and shoulder-girdle motion introduce pathways absent from the plane.

These freedoms can redistribute a planar effect rather than merely add to it. A moment that looked entirely negative in one axis may split across changing axes. Preserve the complete wrench, its reference point, and the observer used for power. A passive coordinate rotation changes components while preserving the physical loading and scalar power; choosing a different physical projection axis changes the reported component. The swing-normal projection remains useful if its construction is reported and tested for sensitivity.

### **20.2.11 Frames Must Be Registered**

Laboratory, pelvis, trunk, arm, hand, club, sensor, and swing frames all move differently. A general spatial wrench transform includes rotation and a moment shift; either part can be trivial when axes or reference points already coincide. Angular velocities require consistent representation; Euler-angle derivatives are not the angular-velocity vector. These are common sources of plausible but wrong signs.

Frame tests use known loads and orientations, action–reaction at contacts, reference-point invariance of power, and round-trip transforms. Coordinate conventions should be machine-readable. A reviewer should be able to reproduce the swing-normal axis from released kinematics.

### **20.2.12 Defining the Late Impact Zone**

“Near impact” can be a clock window, a range of club position, a state-space region, or a phase normalized between events. A spatial study should declare the primary definition and test alternatives. If the negative-couple interval appears only under one post hoc window, confidence is low.

State-based regions may include clubhead height, hand path, shaft orientation, and arm–club geometry. They allow comparisons across tempo but can introduce selection bias if tuned on outcomes. Freeze them before participant holdout.

### **20.2.13 Independent Engines**

Implement the same spatial mechanism in two dynamics engines or formulations. Share model parameters, initial state, control histories, and output schema. Compare trajectories, hand wrenches, constraints, energy, and the projected couple within tolerances. Differences should be investigated, not averaged.

Independence is strongest when implementations do not share derivative code or hidden convention helpers. Agreement then reduces the chance that one algebraic or frame error created the result. The released comparison should include both successful agreement and known deviations.

### **20.2.14 A Spatial Geometry-Control Suite**

These are proposed extension and measurement tests, beyond the two archived same-force moment-arm identities. Reverse the defined swing-normal axis and verify only the reported projection changes sign, not physical power. Swap hand positions in the model and verify the differential couple responds. Collapse separation, align forces with the separation vector, and rotate the swing plane. Perturb sensor-frame orientation to quantify sign confidence.

These tests distinguish coordinate behavior from physical mechanism. If a purported negative couple disappears whenever the reporting plane shifts a few degrees, its interpretation should include that uncertainty rather than a binary label. At a fixed reference point, if two unit normals differ by angle  $\theta$ ,

$$|\mathbf{M} \cdot (\mathbf{n}' - \mathbf{n})| \leq 2\|\mathbf{M}\| \sin(\theta/2). \quad (20.3)$$

A projection close to zero can reverse sign under a small angular perturbation even when the moment vector is accurately known. For example, a moment  $(1, 0, -0.01)$  N m is negative along  $(0, 0, 1)$  but positive after that normal tilts one degree toward the positive first axis. A sign probability additionally needs a specified uncertainty distribution; a collection of plausible axes alone gives a sensitivity range, not a probability.

### 20.2.15 Moving Anatomy Without Overclaiming It

Spatial arm segments permit richer kinematics, but a ball joint at the shoulder does not represent scapulothoracic motion or muscle recruitment automatically. Marker-based shoulder centers and scapular orientation have substantial measurement uncertainty. A generalized shoulder driver remains an abstraction.

The model can predict a required hand wrench or joint-level generalized load. Mapping that to scapular retraction, protraction, or named muscle action requires additional kinematics, physiological measurements, and a biological model. EMG constrains activation hypotheses but does not uniquely identify muscle force. Keeping that boundary explicit protects the very hypothesis the work aims to test.

### 20.2.16 Quantities That Should Survive the Extension

Whole-system energy balance for a declared observer, contact action–reaction, consistent wrench–velocity power under passive coordinate changes, and geometric control responses should survive exactly or within numerical tolerance. Qualitative timing and sign of the proposed contact contribution may survive over a declared region. Exact optimal controls and magnitudes are expected to change.

The spatial model earns its complexity if it makes new falsifiable predictions: out-of-plane force components, axis-dependent couple timing, or conditions under which the planar mechanism fails. If it merely draws a more realistic golfer while hiding the causal tests, it has moved away from science.

### 20.2.17 Human Validation Path

Use synchronized full-body and club motion, bilateral grip wrenches, separate force plates, and shaft deformation. Calibrate every frame and report soft- tissue and sensor uncertainty. Evaluate participants held out from parameter tuning. Compare predicted three-dimensional wrench trajectories, not only clubhead speed.

Residuals should be inspected by phase and direction. A systematic out-of-plane residual may identify missing wrist, shaft, or shoulder-girdle physics. Updating the model in response should be declared as a new rung and retested on untouched participants.

### **20.2.18 Reader Lab: Rotate the World**

Take one spatial state and apply one constant proper rotation to every position, orientation, velocity, force, moment, gravity vector, and boundary/input definition. Express inertias and constraints consistently, then recompute dynamics. This is a passive relabeling for the same observer. The rotated trajectory and the same scalar energy and wrench power should match. A quantity tied to the declared swing frame should transform predictably. Failure reveals a frame-dependent implementation error.

Next project the same contact moment onto several plausible swing-plane normals. Report sign confidence and angular sensitivity. If all definitions agree over the late interval, the negative-couple claim is robust to plane construction. If they disagree, publish the dependence and prefer the complete vector result.

Finally compare two independent spatial engines on geometry-reversal and hand- separation controls. Match initial state and input at high precision. Differences in constraint stabilization or quaternion handling should remain below declared observable tolerances.

### **20.2.19 Spatial Visualization Without Illusion**

A three-dimensional rendering can make a model look biologically authoritative. Include visible joint definitions, hand contact frames, force-vector scale, uncertainty, and model-rung label. Let readers rotate the scene and inspect the numeric state. Do not use a detailed human mesh to imply anatomical fidelity not present in the equations.

The best spatial figure pairs an intuitive view with declared observables: wrench power for a named observer, projected moment with axis sensitivity, and closure residual. Beauty should make the mechanism easier to audit, not harder to question.

### **20.2.20 A Minimal Spatial Report**

Publish body and joint frames, orientation representation, state definition, inertias, constraints, controls, solver, and swing-plane construction. Report complete three-dimensional hand wrenches before selected projections. Include energy, constraint, and engine-agreement residuals. Provide geometry-control animations with fixed vector scales and downloadable data.

For the negative couple, show the moment vector in club, laboratory, and swing frames, the angle between vector and plane normal, and sign probability under frame uncertainty. Report the interval by both time and state. This makes a spatial claim reviewable rather than visually persuasive only.

### **20.2.21 When the Planar Result Fails in Space**

Failure can occur because out-of-plane motion redirects the contact wrench, because a different axis dominates, because hand geometry changes, or because the planar constraint created an artificial reaction. Diagnose which term changes. A spatial failure is not proof that coupling is irrelevant; it narrows the specific projection or validity region.

The revised explanation might say that a negative total swing-normal couple is not robust, while one component or state-dependent regulation effect remains. Such refinement is exactly why the model ladder exists.

### 20.2.22 Accessibility of a Three-Dimensional Result

Pair an interactive scene with two-dimensional explanatory slices and textual alt descriptions. A phone reader should understand hand positions, arrows, and sign without rotating a model. A specialist can open the interactive view and data. Providing both prevents visual sophistication from becoming an access barrier.

### 20.2.23 Chapter Checkpoint

The spatial extension succeeds when it preserves invariant balances, survives frame and engine tests, and either confirms or precisely narrows the planar mechanism. More degrees of freedom alone are not progress. New, measurable predictions and exposed failure regions are.

For a reviewer, the full wrench vector remains primary. A convenient negative projection should never conceal uncertainty in the axis used to define it.

## 20.3 Where the Picture Breaks

The shadow analogy makes spatial extension look visually obvious. In fact, coordinate conventions, rotation representations, and contact frames can create subtle sign errors. Every comparison needs automated transformation and power checks.

Engine agreement can also become false reassurance if one result is tuned to the other. Parameters and tolerances should be registered before comparison, with held-out cases and negative controls.

**Go Deeper:** Read the [spatial common-state formulation](#), the [native forward-dynamics comparison](#), and the project's [reproduction boundary](#). These links identify the reviewed provider revision; proposed validation steps above are not claims that those studies have already been completed.

## Chapter 21

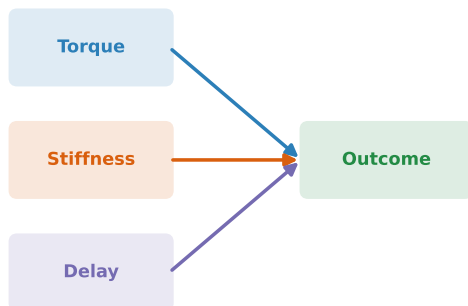
# Sensitivity and Identifiability

### 21.1 A Concrete Picture

Taste a soup and notice that it is salty. Its taste changes when salt is added, but tasting it may not tell whether the cook added salt, reduced the broth, or used a salty ingredient. **Sensitivity** asks how an observation changes when an input changes. **Identifiability** asks whether the observations distinguish the unknown inputs. A strong effect does not guarantee a recoverable cause.

The same distinction matters in a swing. A model's clubhead speed can respond to both a torque schedule and a shaft parameter. Matching that speed does not show which quantity was responsible. Conversely, uncertainty about individual parameters need not prevent a useful prediction: several compatible parameter sets may predict the same club outcome but different hand loads.

**Sensitivity: What Moves the Outcome?**



**Identifiability: Can We Recover the Recipe?**

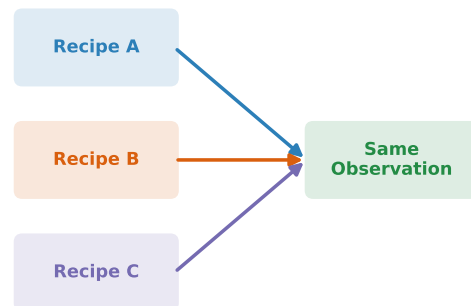


Figure 21.1: A Schematic Contrast Between Changing Inputs and Distinguishing Explanations.

## 21.2 How the Mechanism Works

Write a forward model as  $y = h(p; u)$ , where  $p$  contains the unknown parameters,  $u$  contains the declared inputs and experimental conditions, and  $y$  contains the observations being compared. Initial states, sensor offsets, time alignment and unknown actuator histories must either be known or included among the unknowns. Calling them fixed does not establish that they were measured.

An observation might be one terminal speed, six summaries, or a collection of waveforms from several trials. These are different inverse problems. A richer waveform can contain information discarded by a summary, although thousands of correlated samples are not thousands of independent measurements. The parameter domain, contact assumptions and sensor model also belong to the problem definition.

For this declared model and experiment, **global structural identifiability** means that ideal noise-free observations distinguish the parameter vector over the allowed domain; **local** identifiability concerns nearby alternatives. Usual structural claims concern generic parameter values, with exceptional cases stated separately. **Practical identifiability** asks whether the actual noisy, finite data constrain the quantities sufficiently for the intended question. State the required precision; a numerical optimizer returning one answer is not a demonstration of either property.

These distinctions also separate calibration, optimization and explanation. Calibration seeks parameter values consistent with data. Optimization selects an input or design for an objective. Neither makes the selected solution a unique account of a person's control strategy.

### 21.2.1 A Worked Ambiguity and a Useful New Measurement

Consider a manufactured, dimensionless model with two unknowns. Its first observation is  $y_1 = p_1 + p_2$ . Both partial derivatives equal one, so both parameters influence the observation. If  $y_1 = 10$ , however,  $(6, 4)$ ,  $(5, 5)$  and  $(8, 2)$  fit equally well. The **sum** is identified; its allocation is not. Requiring nonnegative parameters leaves a finite line segment of possibilities, not a unique solution. Among exact fits, minimizing the parameter norm selects  $(5, 5)$  by assumption.

Now add an independently measured channel  $y_2 = p_1 - p_2$ . The two observations permit the inverse

$$\hat{p}_1 = \frac{y_1 + y_2}{2}, \quad \hat{p}_2 = \frac{y_1 - y_2}{2}.$$

With  $y_1 = 10$  and  $y_2 = 2$ , the estimates are  $(6, 4)$ . If the two measurement errors have zero mean, are independent and have variances  $\sigma_1^2$  and  $\sigma_2^2$ , the covariance of this inverse estimator is

$$\text{Cov}(\hat{p}) = \frac{1}{4} \begin{pmatrix} \sigma_1^2 + \sigma_2^2 & \sigma_1^2 - \sigma_2^2 \\ \sigma_1^2 - \sigma_2^2 & \sigma_1^2 + \sigma_2^2 \end{pmatrix}.$$

For variances one and four, the diagonal entries are 1.25 and the off-diagonal entries are  $-0.75$ . The model is invertible, yet recovery becomes imprecise as the second channel becomes noisy. This covariance calculation does not require Gaussian errors; distributional confidence statements require additional assumptions. Correlated sensor errors require their full covariance matrix.

The lesson for golf is conditional. A shaft-strain or bilateral-wrench channel is useful if its predicted response separates explanations beyond measurement and modeling uncertainty. Its name alone does

not guarantee that it identifies stiffness, timing or muscle action. Demonstrate the separation under a declared measurement model, as the second channel does in this example.

### 21.2.2 Local Sensitivity and What Rank Can Establish

For fixed declared inputs  $u$ , let  $\Delta p = p - p_0$  and  $\Delta y = h(p; u) - h(p_0; u)$ . Near the reference  $p_0$ , differentiable observations satisfy

$$\Delta y \approx J(p_0)\Delta p, \quad J_{ij} = \left. \frac{\partial h_i}{\partial p_j} \right|_{p_0}.$$

Here  $h_i$  is observation component  $i$  and  $p_j$  is parameter component  $j$ . The approximation is local. Analytic derivatives, automatic differentiation or finite differences can estimate  $J$ , but check perturbation size, numerical convergence and the active contact or event regime. A derivative across a switch or changing impact event needs particular care. A derivative per kilogram cannot be compared directly with one per millisecond. Declare the parameter scales and observation scales; noise weighting serves a different purpose from merely standardizing units.

For a continuously differentiable map, full column rank is a sufficient local recovery condition. Rank deficiency describes first-order directions that the observations cannot distinguish; it is not a universal proof of nonlinear nonuniqueness. For example,  $y = p^3$  is one-to-one even though its derivative is zero at  $p = 0$ , where recovery is poorly conditioned. Conversely,  $y = p^2$  has nonzero derivative away from zero but cannot distinguish  $p$  from  $-p$  if both are allowed. The domain and the distinction between local and global claims matter.

For an inverse problem with 12 freely varying continuous parameters and six scalar summaries, as in the study considered below, a Jacobian has rank at most six. At a regular rank-six point, the model's local constant-observation family has six dimensions. This prevents generic recovery of the full vector from that experiment alone. It does not identify which six parameters could be recovered, exclude identifiable combinations, or rule out an isolated exceptional solution at a singular point. More experiments, known parameters or independently justified restrictions change the problem.

### 21.2.3 Global Screening Is a Different Calculation

Global sensitivity studies variation across a specified range or joint input distribution. Variance decomposition is one approach; regression, rank correlation and elementary-effect screens answer other questions. A ranking belongs to its outcome, input distribution and estimator. It is not an intrinsic percentage of a muscle's or parameter's causal importance.

Partial rank correlation removes linear associations with the other ranked inputs before correlating the residuals. It is a screen for monotonic association after that adjustment, not a general allocation of output variance. Inspect nonmonotonic response, dependent inputs and estimator stability before interpreting it (Marino et al. 2008). A symmetric sample of  $y = p^2$  can have zero linear slope despite strong variation. For samples  $p = (-2, -1, 0, 1, 2)$ , the linear regression slope of  $p^3$  on  $p$  is 3.4, while the local derivative at zero is zero. A fitted slope and a local derivative are not interchangeable.

Report sensitivity separately for speed, face/path, hand loading and effort. A parameter can affect one strongly and another weakly. Combining the outcomes into a weighted score is a decision rule whose

weights need explanation; it does not remove the tradeoff.

### 21.2.3.1 What the Coupled Study Actually Computes

The pinned provider study screens 12 inputs in 24 engineering-envelope cases and records six outputs. Its so-called standardized sensitivity matrix contains linear least-squares coefficients across those cases after input/output standardization. It is a **finite global regression screen**, not a local Jacobian, noise-weighted information matrix or likelihood profile.

Reconstructing the saved arrays reproduces the six singular values of the 6-by-12 coefficient matrix; the input design itself has 24 rows and 12 columns. Relative cutoffs of 1%, 5% and 10% retain all six; a 20% cutoff retains five. Those ranks describe the fitted matrix under the chosen threshold and scaling. They do not constitute a structural-identifiability proof for the nonlinear simulator or a practical-identifiability analysis with measured sensor noise. High in-sample regression fit is also not held-out parameter recovery.

The stored leave-one-out audit changes the leading hand-force parameter among limb-mass scale, actuator time constant and delay. This makes the screen useful for generating questions, while limiting claims of a stable ranking. The separate six-case training and six-case held-out designs assess selected model programs across synthetic cases. They are not participant-held-out experiments. The reconstruction here checks saved results and selected implementation; it does not rerun the provider dynamics.

### 21.2.4 Force Allocation Is Another Observation Map

Club motion, a net club wrench and two individual hand wrenches are different observations. Even a net wrench inferred from motion depends on inertial, external-load and contact assumptions. Consider known contact positions and express all forces and moments in one frame about one point  $O$ . Let  $f_i$  be each hand force and  $r_i$  its position vector from  $O$ ;  $F$  and  $M$  are the net force and moment about  $O$ :

$$F = f_1 + f_2, \quad M = r_1 \times f_1 + r_2 \times f_2 + m_1 + m_2.$$

Here  $m_i$  is a direct contact couple, included only when the contact model allows it. For two point forces, set both  $m_i$  to zero. The following are ranks of the unconstrained instantaneous linear maps, before friction or other contact restrictions:

| Contact Description                 | Unknown Scalars | Net Wrench Map Rank | Nullity |
|-------------------------------------|-----------------|---------------------|---------|
| Two distinct planar point forces    | 4               | 3                   | 1       |
| Two distinct spatial point forces   | 6               | 5                   | 1       |
| Two coincident spatial point forces | 6               | 3                   | 3       |
| Two complete spatial hand wrenches  | 12              | 6                   | 6       |

For separated spatial point forces, adding a force along the line joining the hands to one hand and subtracting it from the other leaves the net wrench unchanged. Complete hand wrenches permit further compensating force/couple changes. Thus “one hidden hand-force mode” is correct for particular point-force models, not for every two-hand representation.

Friction, anatomy, actuator dynamics and constitutive laws may restrict feasible allocations. A justified additional law can narrow the solution set; a regularizer selects according to a preference. Neither should be presented as a new observation. Kinematic constraint rank, instantaneous force-allocation nullity and parameter identifiability over time are distinct questions.

Calibrated bilateral wrench sensors can directly constrain the hand resultants they measure. They do not by themselves resolve the pressure distribution within a hand, individual muscle forces or a unique neural strategy. Likewise, activation timing, shaft strain and body motion have their own forward models, uncertainties and nuisance variables. Evaluate which ambiguity each channel actually reduces.

### Different Coordination Histories

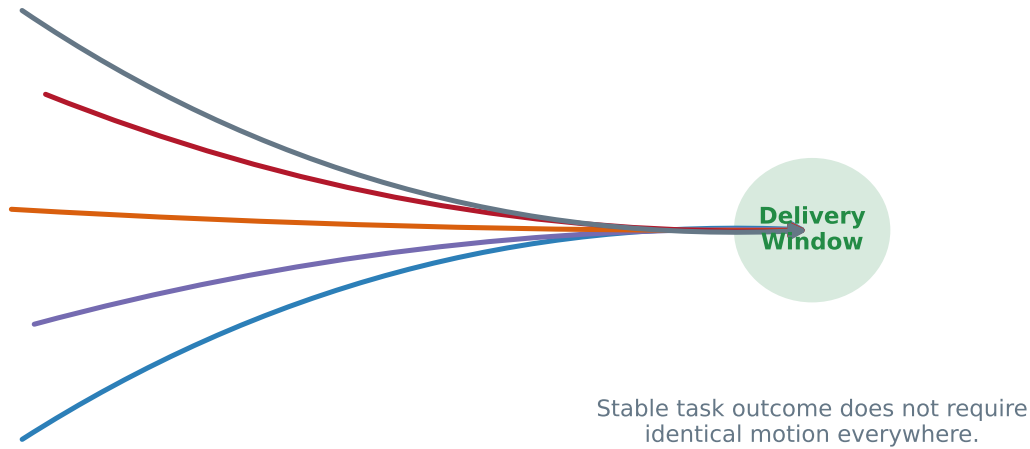


Figure 21.2: A Schematic of Different Histories Sharing a Task Outcome.

#### 21.2.5 Let Other Parameters Compensate During the Test

A likelihood profile varies one parameter while reoptimizing the remaining parameters. Holding the others fixed instead produces a slice that can hide compensation and exaggerate precision. Raue and colleagues use profiles to expose such tradeoffs and guide additional measurements; their particular criterion for practical nonidentifiability is an unbounded likelihood-based interval (Raue et al. 2009). That criterion is narrower than declaring an estimate too imprecise for an application. State which meaning is intended.

In the dimensionless example with unrestricted real parameters, the first-observation cost is  $(10 - p_1 - p_2)^2$ . For every  $p_1$ , choosing  $p_2 = 10 - p_1$  gives zero cost. Holding  $p_2 = 4$  would instead give  $(p_1 - 6)^2$ , falsely suggesting that the first observation alone isolates  $p_1$ . With the second observation and the stated variances one and four, divide each squared residual by its measurement variance. Minimizing

$$(10 - p_1 - p_2)^2 + \frac{(2 - p_1 + p_2)^2}{4}$$

over  $p_2$  gives  $p_2 = 7.6 - 0.6p_1$  and profile cost  $0.8(p_1 - 6)^2$ . An independent numerical minimization checks these values. They illustrate how an informative observation changes the inverse problem; they are not a fitted swing or a calibrated confidence interval.

Profiles require reliable optimization and a justified likelihood. Local minima, finite search bounds and a chosen threshold can alter the result. A frequentist confidence procedure concerns repeated-sampling coverage; a Bayesian credible region depends on the likelihood and prior. A tight posterior or bounded search interval can owe its width to assumptions. Show which combinations the data constrain, what the prior or penalty supplies, and how predictions vary along the remaining ambiguity.

### 21.2.6 Carry Uncertainty Into the Mechanical Claim

Consider a mechanical hypothesis such as a late negative hand couple. Define its reference point, axis, time/event window and sign before counting cases. Report the fraction of the **declared ensemble** exhibiting that sign, its magnitude and duration, and which assumptions change it. Engineering ranges and synthetic design points are not automatically a golfer-population probability distribution. Nor is 50% a universal chance baseline for a mechanical sign.

A sign stable across many cases can support a conditional prediction without identifying the underlying mechanism. Different force allocations or constitutive models may predict the same sign. Geometry controls need their own derivation: making hand separation zero removes the moment of an equal-and-opposite force pair, but does not remove an independently applied couple. A geometry reversal must specify what else is held fixed and use compatible sign/frame conventions.

Compare decision-relevant predictions across all adequately supported models and parameter sets. A well-constrained speed prediction may coexist with uncertain hand loading. Robustness of one output does not justify an unqualified performance or physiological claim about another.

### 21.2.7 Turn Ambiguity Into an Experiment

Choose two or more explanations consistent with current observations. Simulate the proposed sensor channels under each, including noise, calibration, synchronization, filtering and plausible model discrepancy. Locate an event or manipulation where their predictions differ reliably. A more precise version of the same insensitive observation may help less than a new channel with a different parameter dependence.

Freeze preprocessing choices, parameter tuning, sensor selection and decision thresholds before the final evaluation. Split repeated swings by participant when the claim concerns new people; reserve other appropriate groups when the claim concerns equipment or conditions. Synthetic holdouts test the stated model sampling process, not transport to golfers. Report individual errors, uncertainty calibration and failure cases, rather than only an average fit.

**Provider Model Finding:** The coupled study supplies a finite sensitivity screen and synthetic control comparison. Its point-force null mode follows from its particular wrench map. These do not uniquely identify human parameters or control policies.

**Human Evidence Boundary:** No participant parameter-recovery experiment is established by the manufactured examples or saved-array reconstruction here. A claim that additional sensors resolve an actual golfer's allocation needs its own measured-data evaluation.

**Hypothesis:** Additional wrench and shaft observations can distinguish some competing explanations when their predicted channel differences exceed the combined measurement and model uncertainty.

#### 21.2.7.1 Reader Lab: Design One Discriminating Measurement

Start with the sum-only example. Plot its family of exact solutions, then add the difference channel. Increase its noise variance and plot how estimator uncertainty grows. Compare the profiled cost with the misleading fixed-parameter slice. These are three separate questions: uniqueness, precision and whether nuisance parameters were allowed to change.

Next choose two golf-model explanations that fit the same current data. Compare candidate measurements with an explicit noise model. State what would reject each explanation, what could remain ambiguous, and which prediction matters for the decision. If both fail, investigate model discrepancy rather than forcing one to win. If neither is separated, report that result without retuning on the test data and declaring success.

#### 21.2.7.2 A Minimum Report

Publish the unknowns and their units, known inputs, observation definitions, parameter domain, ranges/distributions and correlations. Name the sensitivity estimator, scaling, numerical checks and any rank threshold. Separate ideal identifiability arguments from finite-data fits. Report the noise model, priors/penalties, nuisance treatment, calibration/holdout split, parameter tradeoffs and predictive checks. Link numerical summaries to their inputs and implementation revision.

### 21.3 Where the Picture Breaks

The soup analogy suggests independent ingredients; mechanical parameters are coupled and can have different meanings in different models. Even perfect identification within an inadequate model can produce a precise parameter that compensates for missing physics. Residual structure, independent observations and comparison across model assumptions remain necessary.

The chapter's central connection is between **what affects the swing**, **what our measurements distinguish**, and **what a decision actually requires**. Explain the remaining family of possibilities, identify which conclusions hold across it, and specify the observation that could change the conclusion.

**Go Deeper:** Read the pinned provider's [uncertainty and identifiability chapter](#) and [machine-readable study record](#).

## Chapter 22

# Timing Is a State Question, Not Just a Clock Question

### 22.1 A Concrete Picture

A cook watches both the timer and the bread. Twelve minutes specifies elapsed time; temperature and structure describe what has happened during it. Neither one thermometer reading nor a timer reveals the entire loaf's condition.

The same distinction matters in a swing. A wrist command applied 100 ms after transition can encounter different club angles, velocities, shaft deflections and existing loads. Triggering at one club angle removes that particular angle difference, but the other quantities can still differ. The useful question is which information predicts a feasible opportunity to improve the chosen outcome.

### 22.2 How the Mechanism Works

#### 22.2.1 Specify What Reaches the Trigger

Let  $x$  contain every state needed to predict the declared model: positions, velocities, actuator states, elastic memory and any controller memory. A measured quantity is generally only a function of that state. A club-angle threshold, for example, does not specify bilateral hand force, shaft strain or activation. Even identical poses can have different futures.

Define an event by a scalar function  $g(x, t) = 0$ , then specify which root counts. One useful rule is the first upward crossing after a registered transition, inside a declared admissible region. An unsigned equality alone also admits backward crossings, repeated crossings and tangential touches. Define what happens if no admissible event occurs before the horizon. Noise suppression, hysteresis and a one-shot latch change the policy and must be included in it.

An event detector also has numerical limits. A solver that searches for sign changes between integration steps can miss multiple crossings inside one step; a tangential touch need not change sign. Event localization and step-size sensitivity therefore need their own checks. A small differential-equation residual is not proof that the intended event was found.

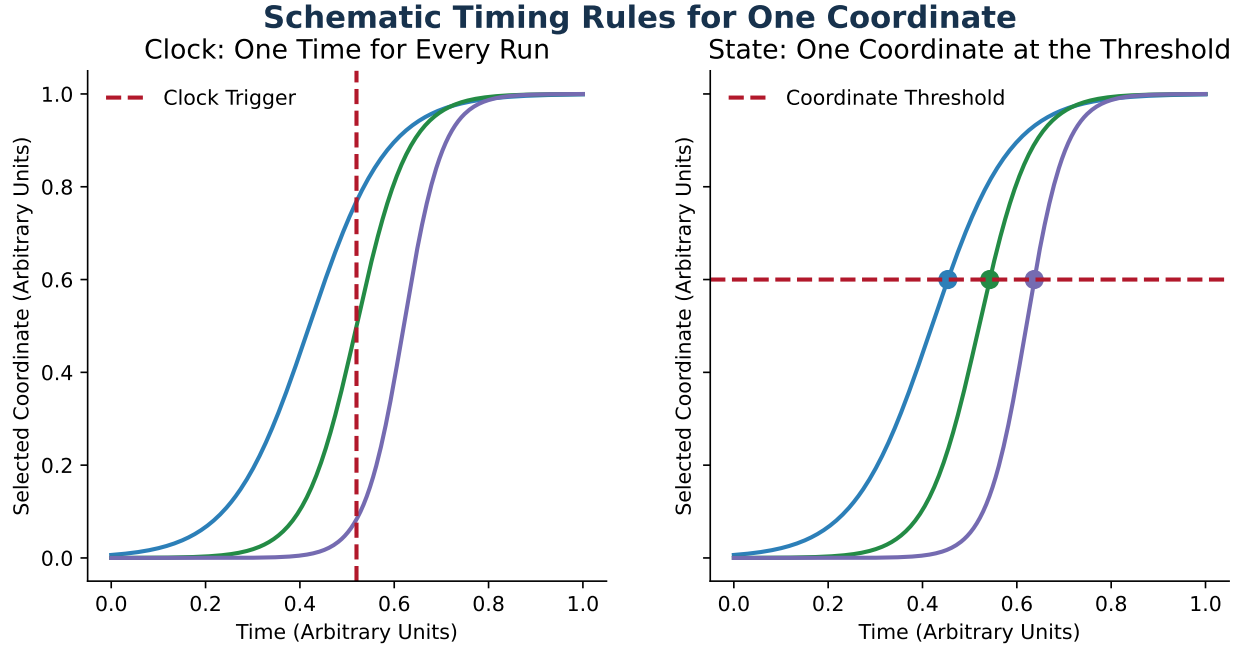


Figure 22.1: Schematic Clock and Coordinate Triggers for Three Artificial Trajectories: Equal Coordinate Values Can Occur at Different Times and Rates.

The figure uses three invented scalar curves. It illustrates the two timing rules, not an advantage of one controller. Their threshold values agree while their slopes differ; no claim of full-state matching follows.

### 22.2.2 A Slow Crossing Magnifies Timing Error

For smooth pre-event motion  $\dot{x} = f(x, t)$ , the rate of change of the event function is

$$\dot{g} = g_t + \nabla_x g \cdot f.$$

A *transverse* crossing has  $\dot{g} \neq 0$ . If a small perturbation changes the event-function value at the nominal event time by  $\delta g$ , first-order implicit differentiation gives

$$\delta t_e \approx -\frac{\delta g}{g_t + \nabla_x g \cdot f}.$$

Evaluate the denominator on the approach side if detection switches the control law. The numerator can include a threshold error or the changed trajectory's state error at the original event time. This is a local approximation for the same identifiable crossing, not a rule for arbitrary event changes.

For a manufactured angle crossing, a positive 0.002 rad error shifts detection about 0.5 ms earlier at 4 rad/s, but 50 ms earlier at 0.04 rad/s. The latter calculation is exact for a linear angle trace; a curved trace over that longer interval needs explicit root evaluation. At a grazing event the denominator vanishes. In dimensionless time  $s$ , the example  $g = (s - 0.2)^2 + \epsilon$  has no real event for positive  $\epsilon$ , two for negative  $\epsilon$ , and a double root at  $s = 0.2$  when  $\epsilon = 0$ . A confidence interval around one assumed time cannot describe all three possibilities.

Report missed and multiple events as outcomes. Add synchronization error, processing delay and uncertainty in the observable proxy. A timestamp reported to a millisecond need not be accurate to a millisecond.

### 22.2.3 Separate the Command From the Torque Response

Command onset, transmitted torque, contact wrench, acceleration, a velocity peak and delivery are distinct events. An ideal command step produces an instantaneous torque step only in a model that directly equates command and applied torque. A finite step in torque still does not instantaneously change velocity in a smooth mechanical system.

For an illustrative first-order torque response  $a$  to a constant target  $u$ , with time constant  $T_a > 0$ ,

$$\dot{a} = \frac{u - a}{T_a}, \quad a(t) = u + (a_0 - u)e^{-t/T_a}.$$

Time is measured from the target change and  $a_0$  is the existing torque state. With zero initial output and a unit target, the response reaches about 63.2% after one time constant. Its 10–90% rise time is  $T_a \ln 9$ . Thus a **35 ms time constant means approximately 76.9 ms of 10–90% rise**, not 35 ms of rise. A pure delay postpones the response; it is a different parameter.

Initial load matters as well. For a target change from  $-5$  to  $+15$  N m and  $T_a = 35$  ms, the existing output remains  $-5$  N m at the change, crosses zero after  $T_a \ln(4/3) \approx 10.1$  ms, and approaches  $+15$  N m gradually. These are manufactured actuator examples, not measured muscle timings. Rate limits, saturation, contact changes and elastic transmission can change this response. A preloaded channel can already carry force without responding instantaneously to a new target.

Compare step, ramp and filtered commands with declared initial actuator states and capacities. Equal targets and nominal onset do not guarantee equal torque impulse, mechanical work or load. Work requires the applied torque paired with its conjugate rate, as in Section 13. To test how preparation established a load, include that history and its cost. To test a subsequent intervention, copy the complete prepared state before branching. Those are different questions.

### 22.2.4 What the Recorded Timing Studies Establish

The pinned [two-link methods](#) use a fixed hub, rigid club, projected gravity and retained joint damping. They compare constant shoulder commands of 60 or 100 N m with zero wrist drive, drive-only, or restrain-then-drive programs. The positive wrist target is 15 N m, with early restraint magnitudes of 5 or 10 N m where included. Fifteen onset values from 0 to 0.35 s, separated by 25 ms, form 92 attempted programs across the two shoulder levels. The zero-wrist-command condition retains the model's damping; it is not necessarily a torque-free anatomical wrist. Constant shoulder command also does not mean equal shoulder work when the angular trajectories differ.

The endpoint is the first upward crossing of the club's absolute angle through zero, scored only if the arm angle is at most 2 rad. It is a registered geometric delivery proxy, **not a simulated ball collision**. In the unfiltered sweep, 63 of 92 programs satisfy that rule. Excluded crossings remain recorded. A schedule that fails the rule cannot quietly disappear from the denominator or be compared at a later, more favorable crossing.

The [command-filter audit](#) repeats the grid at time constants of 0, 20, 35 and 50 ms. Its implementation copies the initial command directly into the filter state and smooths subsequent changes. This is a declared preloaded initialization; it does not simulate the process that established that initial torque.

Across these four settings and both shoulder levels, the four comparison speeds increase from early drive (onset at zero) through zero wrist command and grid-selected drive-only to grid-selected restrain-then-drive. The selected onset is allowed to change with the filter. At 60 N m, the selected drive-only onset moves from 0.200 s without filtering to 0.175 s at 35 ms. Accepted counts change too: 63, 59, 59 and 56 out of 92 as the time constant increases. These are stored finite-grid results, not a continuous optimum, one fixed policy succeeding everywhere, or human timing prescriptions.

A [separate uncertainty comparison](#) uses moving-base, two-hand, flexible-club dynamics, delayed bounded actuators, eight schedules and six held-out parameter draws. Its nominal wrist targets differ: `early_restrain` requests -2 N m before 0.10 s and +4 N m afterward; `late_drive` requests zero before 0.14 s and +5 N m afterward. Delivered drive remains subject to delay, lag and bounds. This comparison changes early torque, switching time and late amplitude; it does not isolate early restraint alone. The named `early_restrain` schedule improves the sampled 10th-percentile speed relative to `late_drive` while worsening mean planar face/path error. All eight schedules are nondominated across five objectives: that speed quantile, mean face/path error, upper-tail hand force, squared-torque effort and speed dispersion. The last four are minimized; the speed quantile is maximized. This describes tradeoffs among the sampled schedules, not a proven boundary of every feasible policy. Its endpoint is the fixed time 0.24 s, not the preceding study's crossing. Neither comparison tests a general advantage of observable state feedback over clock schedules, and the engineering draws are not a golfer population sample.

Springs and MacKenzie separately compared three optimized conditions in a planar torso-arm-club model with constrained torque generators. They reported a small advantage from resisted delay and an advantage from active wrist torque after release ([Springs and MacKenzie 2002](#)). That published simulation supports a bounded model question, not an identified human command history or the parameters of our separate filter audit.

**Model Result:** Selected timing and speed rankings depend on the model, candidate family, initialization and endpoint. The recorded filter comparisons retain their stated ordering within those boundaries.

**Human Evidence:** The simulations above do not measure human command onset, preactivation or a physiological delay. Kinematics and EMG are different observations; neither alone identifies the complete transmitted wrench history.

**Hypothesis:** Preparation can arrange geometry, motion and loading so that a later distal action becomes useful. An observable event might select that opportunity more reliably than a fixed clock, but this requires a prospective comparison with delays, failures and competing objectives included.

**Practical Interpretation:** Define what “late” is relative to, which signal is available, and what outcome improves before turning it into a cue.

### 22.2.5 Alignment Changes the View, and Sometimes the Units

Subtracting impact time from every timestamp translates a trial’s time axis; it does not stretch the trial. Scaling each downswing to a common percentage is an additional operation. With duration  $D = t_{\text{end}} - t_0 > 0$  and phase  $\phi = (t - t_0)/D$ , corresponding phase values can still contain different states. For an angular coordinate  $q$ , the derivative  $dq/d\phi = D\dot{q}$  has units of radians per unit phase, not rad/s.

For physical power  $P$ , preserve the time Jacobian when computing work:

$$W = \int_{t_0}^{t_{\text{end}}} P(t) dt = D \int_0^1 P(t_0 + D\phi) d\phi.$$

Two manufactured traces with the same displayed curve  $P = 10 + 5\phi$  W but durations 0.2 and 0.4 s deliver 2.5 and 5 J. Their identical phase plots do not imply identical work. A nonlinear time warp needs its local  $dt/d\phi$ , and averaging before integration need not reproduce the mean of trial work.

An event used to align curves will be sharp at the alignment origin by construction. Greater sharpness after state alignment does not, by itself, show a causal mechanism. Publish physical time, absolute duration, the alignment rule and uncertainty alongside phase plots. Compare proposed explanations on independent outcomes rather than rewarding the feature used to align them.

### 22.2.6 A Real-Time Policy Cannot Know Future Impact

Retrospective “100 ms before actual impact” is available after the trial. A controller cannot use that future timestamp directly. It can use a prediction from present and past observations, with an explicitly evaluated prediction error. The same restriction applies to normalized phase if its denominator uses the future downswing duration.

An open-loop schedule chooses commands from elapsed time and declared initial conditions. A state-triggered rule uses available state estimates and detector memory. Elapsed time can itself be included in an augmented mathematical state; that formal conversion does not give a clock schedule information about an unobserved deviation.

For a fair comparison, declare the sensor set, sampling, latency, estimator, actuator bounds, training data and available preparation. A trigger based on an unmeasured shaft mode or an engine’s internal force decomposition is initially a simulation diagnostic. Its experimental proxy needs validation. Robustness of an ideal full-state controller does not establish robustness of a delayed measurement-based implementation, much less a complete neural controller.

### 22.2.7 Test Preparation, Causation and a Usable Timing Region

A moment’s sign crossing is not necessarily the beginning of the physical interaction: the same channel may have acted earlier with the other sign. A later action cannot alter the physical past, but measured event order alone does not isolate the cause of a state change. Sensor delays, filtering, shared state dependence and opposing contributions complicate that inference. Specify the body, moment point and projection axis before comparing couple onsets; pair power channels with their own conjugate velocities.

Compare forward branches from a complete copied state to ask what removing or changing a declared input does next. Each branch must evolve its own reactions and reach the registered endpoint, or report failure. A separate preparation experiment should account for how different states were reached. Reoptimizing other controls after removal asks whether another strategy can substitute; it does not estimate the same fixed-policy effect.

Continuous arm loading and phase-dependent redistribution between arm and wrist channels remain candidate explanations. A change of allocation need not produce a net torque sign reversal, and an unchanged net sign can hide large internal changes. Relate the predictions to bilateral loads, stored energy, work and estimated stiffness, rather than treating one onset as a biological role label. A changed mediator consistent with a cue is informative but does not by itself establish mediation: competing pathways and confounding still need a design.

There may be a broad useful region, a narrow peak, a boundary solution, several separated regions or no feasible benefit. Search and report the actual response surface under declared speed, direction and load requirements. A plateau is a result to establish, not a consequence of calling timing robust. Include onset jitter, measurement error, event failure and held-out uncertainty. If the proposed region is narrower than the experiment can resolve, do not claim that the experiment has located it precisely.

### 22.2.8 Reader Lab: Audit One Proposed Timing Rule

First reproduce the filter's 63.2% response, 76.9 ms rise interval and 10.1 ms sign-reversal example. Then compare the crossing rates of 4 and 0.04 rad/s and identify where the local timing approximation breaks. Finally integrate the phase power traces with their actual durations. These exercises separate actuator history, event uncertainty and energy accounting.

For a proposed swing rule, freeze the event definition and training choices before testing. Compare clock and observable-event policies on the same declared task and information budget. Report all attempted trials, event ambiguity, failures, latency, complete outcome distributions and uncertainty, rather than only the successful mean. Include raw-time and phase views with reproducible detection code. A prospective performance comparison is stronger evidence than a sharper aligned plot.

## 22.3 Where the Picture Breaks

An oven's few visible indicators make the analogy attractive; a golfer's relevant internal state is much less accessible. One angle can be a useful predictor without being either a complete state or the cause of a favorable outcome. A simulated trigger can also exploit measurements and response times that an experiment cannot provide.

Preparation determines the state available for a later action. Event detection determines when the policy recognizes it. Actuator dynamics determine what load actually changes before the task ends. Their interaction, together with the objective and feasible constraints, is the timing question.

**Go Deeper:** Compare the pinned [timing and counterfactual framework](#) with the distinct methods and uncertainty studies above. Read Section 13 for the power boundaries needed to decide what a timed torque actually does.

## Chapter 23

# Fast Once Is Not the Same as Robustly Fast

### 23.1 A Concrete Picture

A racing setup that wins one qualifying lap may lose its advantage in wind, tire wear, or traffic. Comparing setups therefore requires a task and a declared range of conditions. A probability distribution helps when the question concerns how often an outcome occurs; a worst-case comparison can instead use a specified set of disturbances without assigning probabilities.

The same distinction matters in a swing model. A command that gives high nominal club speed may become sensitive to timing, initial state, shaft properties, or the permitted face error. Robustness asks which conclusions survive specified changes. It does not mean immunity to every disturbance, and a repeatable slow swing is not automatically a useful one.

The points in Figure 23.1 are recomputed from the archived outputs discussed below. They are two projections of a five-objective comparison. A point that looks inferior in one panel can remain nondominated when another objective is included. Lines between programs would not establish that intermediate commands achieve intermediate outcomes.

### 23.2 How the Mechanism Works

#### 23.2.1 State the Question Before Choosing a Score

Let a policy  $\pi$  specify a command program or a feedback rule, and let  $\theta$  describe uncertain parameters, initial state, and disturbances. Their modeled outcome can be written  $Y(\pi, \theta)$ . Name the endpoint as carefully as the policy: speed at a fixed time, speed at a specified crossing, and ball-impact delivery answer different questions. If the event never occurs, the outcome needs an explicit missing-event or failure rule.

A useful comparison fixes the model, admissible policy family, actuator limits, initial-state convention, and endpoint. It then states what varies and what counts as acceptable. The policy itself may react to measurements, but the feedback rule and available information must be fixed before evaluation.

### Eight Preselected Programs: Six Held-Out Cases

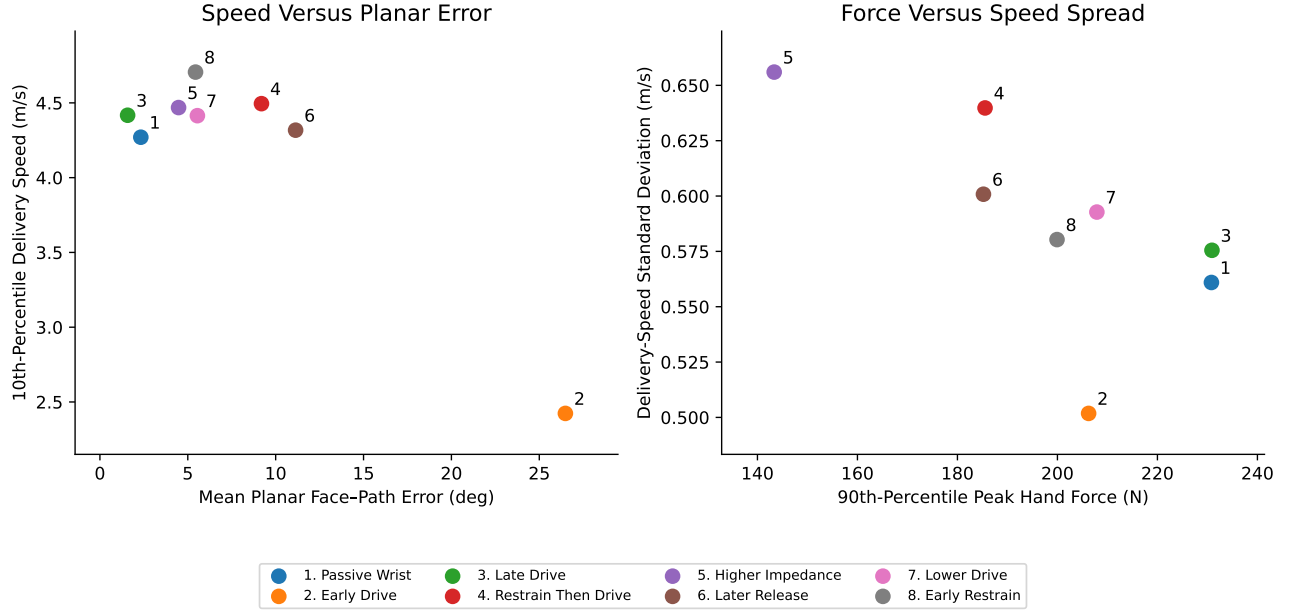


Figure 23.1: Two Projections of the Same Eight Programs on Six Held-Out Cases.

Allowing one program perfect knowledge of a future impact time changes its information advantage, not just its mechanics.

Mean speed, lower-tail speed, accuracy, load, and energy are different objectives. A weighted score requires units or normalization and explicit weights. Changing metres per second to another unit must not silently change the selected policy. A force minimum also does not imply minimum joint torque or muscle effort; those relationships require the relevant mechanical and physiological model.

#### 23.2.2 The Tail Depends on Whether Larger Is Better

For speed treated as a reward, a low percentile summarizes poor-speed cases. For a nonnegative face error treated as a loss, large values are adverse, so an upper percentile describes the difficult cases. Reducing *lower-tail error* could merely improve the already accurate swings while leaving large misses unchanged. A hypothetical speed/error trade must identify the adverse tail and must not be presented as a measured result.

For a scalar outcome  $Z$ , define its population quantile at level  $0 < \alpha < 1$  by

$$q_\alpha(Z) = \inf\{z : \Pr(Z \leq z) \geq \alpha\}.$$

A quantile is a threshold, not the average severity beyond it. An empirical quantile also needs an interpolation convention. With six speed observations, the linear 10th-percentile estimator used in the provider code is halfway between the smallest and second-smallest values. It is not six observations of a 10-percent tail, nor a demonstrated guarantee for 90 percent of future golfers.

Use the same sampled cases for paired policy comparisons where the experiment allows it. This reveals

whether one policy helps the same difficult cases. The difference between two policy quantiles generally differs from the quantile of their paired differences. In the archived early-restrain/late-drive comparison, the former is about 0.289 m/s and the latter about 0.271 m/s. Neither says that every case improves by that amount.

### 23.2.3 Severity, Probability, and Worst Case Are Different

For an integrable loss  $L$  calculated from the modeled outcomes, conditional value at risk can be defined by the Rockafellar–Uryasev formula (Rockafellar and Uryasev 2002):

$$\begin{aligned}\text{CVaR}_\alpha(L) &= \min_{\eta \in \mathbb{R}} F_\alpha(\eta), \\ F_\alpha(\eta) &= \eta + \frac{\mathbb{E}[(L - \eta)_+]}{1 - \alpha}, \\ (z)_+ &= \max(z, 0).\end{aligned}$$

It measures the average of the worst  $1 - \alpha$  fraction of loss probability. At a probability atom, use the appropriate fraction of mass at the threshold; simply averaging observations strictly above the quantile can give another quantity. For four equally weighted losses 0, 0, 0, 10, the 50-percent CVaR is 5, whereas the mean strictly above the median is 10. This example explains the definition, not a golf measurement.

A chance constraint asks for a probability of acceptable outcomes. If all delivery conditions matter together, write a joint condition, such as

$$\Pr(E \leq e_{\max}, F \leq f_{\max}) \geq 1 - \varepsilon,$$

where  $E$  is declared delivery error,  $F$  is a declared force metric, and the thresholds and distribution are specified. Two separate 90-percent success rates need not give 90-percent joint success: disjoint 10-percent failure sets leave only 80 percent satisfying both. A chosen hand-force threshold is a model constraint, not an established injury threshold.

A set-based worst-case objective instead examines  $\sup_{\theta \in \Theta} L(\pi, \theta)$  over a declared uncertainty set  $\Theta$ . The largest loss among a finite set of sampled cases is generally only a lower bound on that supremum. Dense sampling, local derivatives, and a successful optimizer do not alone certify the unsampled set. Conversely, an implausible range edge can make a worst-case design unnecessarily conservative. Quantiles, CVaR, probability constraints, and worst-case bounds answer different questions; none is the automatic definition of a good swing.

### 23.2.4 What the Registered Coupled Study Actually Compares

The provider study uses a planar moving-base model with two articulated arms, two solved hand reactions, and one flexible shaft mode. It varies 12 engineering inputs spanning geometry, inertias, base stiffness, damping, grip separation, shaft properties, command delay, lag, and an impedance coefficient. A deterministic Latin-hypercube design supplies 24 global screening cases, six training cases, and six independently generated held-out cases. The seeds are 8426, 8449, and 8450, respectively. Every candidate uses the same cases within a split.

These ranges are not fitted golfer-population distributions. The implementation permutes strata and jitters within the middle portion of each stratum; it is a declared space-filling design, not a collection of independent identically distributed human trials. Delivery is evaluated at 0.24 s with a 4 ms integration step, not at a ball collision. Small constraint and contact-power residuals check particular numerical identities; they do not by themselves bound trajectory integration error or establish convergence of tail estimates.

Eight programs were preselected: passive wrist, early drive, late drive, restrain then drive, higher impedance, later release, lower drive, and early restrain. Their names identify parameter choices, not eight observed human techniques. In particular, early restrain blends its right-wrist command from  $-2$  to  $+4$  N m beginning at 0.10 s, while late drive blends from 0 to  $+5$  N m beginning at 0.14 s. The command transition lasts 35 ms; the paired left-wrist target is  $-0.65$  times the right-wrist target in the model's signed coordinates. These signs do not alone identify a net club couple or an anatomical muscle action.

The actuator applies pure delay, a discretized first-order command lag with a rate-limited state increment, velocity-dependent resistance, and asymmetric torque–velocity clipping. Its so-called impedance term is proportional to actuator velocity in this implementation; it is not a measured muscle stiffness or a full reflex model. Thus the command schedules are preselected, while delivered torque also depends on velocity. The rate limit on the lag state does not by itself bound every change after velocity-dependent clipping. Even the program named passive wrist can retain modeled velocity resistance.

The five objectives are:

1. Maximize 10th-percentile delivery speed.
2. Minimize mean absolute planar face–path error.
3. Minimize the 90th percentile of peak individual-hand force.
4. Minimize mean integrated squared delivered joint torque.
5. Minimize the sample standard deviation of delivery speed.

The face–path proxy is the absolute wrapped difference between the modeled planar face direction and terminal velocity direction. It is not three-dimensional clubface delivery or ball dispersion. The force statistic first takes the largest force norm over both hands and all recorded times in each rollout, then takes a percentile across cases. The squared-torque integral has units  $\text{N}^2 \text{m}^2 \text{s}$ ; it is not mechanical work, metabolic energy, or measured muscular effort. Standard deviation measures spread, not tracking stability, recovery from disturbance, or probability of meeting a target.

### 23.2.5 What the Archived Results Establish

Independent recomputation from the stored arrays gives seven nondominated programs on the training cases and eight on the held-out cases. Early restrain has held-out 10th-percentile speed of approximately 4.706 m/s versus 4.417 m/s for late drive. Its mean planar error is approximately  $5.44^\circ$ , compared with  $1.58^\circ$  for late drive. This is a speed/error tradeoff for two specified schedules in six cases. It does not isolate the early opposing interval: the compared schedules also have different onset and late-command values.

The implementation selects programs using **training** objectives. Its balanced, equal-range Chebyshev rule chooses later release, the speed-priority rule chooses early restrain, and the load-priority rule

chooses higher impedance. The balanced rule minimizes the largest objective disadvantage after scaling each objective by its range across the training candidates. Those data-dependent scales encode a choice and can change when candidates change. The held-out outputs evaluate the selected programs and other candidates; they are not the data used by those selection rules.

Omitting one held-out case at a time changes Pareto membership. Passive wrist, early drive, late drive, and higher impedance remain nondominated in all six omissions; restrain then drive, lower drive, and early restrain do so in five; later release does so in four. This is a perturbation check on six cases, not a confidence interval for a population frontier. The complete recomputation and original arrays are linked in the review evidence.

### 23.2.6 A Pareto Set Is Relative to Its Feasible Comparisons

A feasible policy is dominated if another feasible policy is no worse on every declared objective and strictly better on at least one. Nondominated policies form the Pareto set; their objective vectors form the corresponding front. Restricting the comparison to eight candidates gives a *candidate* Pareto set. Another admissible policy not evaluated here might improve every objective.

For example, the loss vectors  $(0, 2)$ ,  $(2, 0)$ , and  $(1, 1)$  are mutually nondominated. An untested feasible vector  $(-1, -1)$  would dominate all three. The archived study therefore leaves a universal optimum unsupported; it does not prove that none exists over every admissible control program. Likewise, nondominance in five sample summaries is neither stochastic dominance nor superiority in every individual case.

A tradeoff plot makes the decision visible but does not assign its values. Speed, accuracy, force exposure, and repeatability can be weighted, constrained, or considered separately. Normalization and weights must be chosen and reported. A weighted sum can miss parts of a nonconvex front, and a continuous line drawn between discrete policies is not evidence of realizable intermediate performance.

### 23.2.7 Construct Uncertainty Without Hiding the Experiment

Separate execution variability, participant variation, uncertain calibration, unknown physical parameters, and model discrepancy where possible. Declare ranges, distributions, dependence, temporal structure, and their evidence. Independent draws of geometrically related parameters can create impossible bodies or grips; constraint-consistent initial states and admissibility checks belong inside the design. Merely increasing the number of simulations does not repair an omitted mechanism or an unjustified distribution.

Timing, amplitude, direction, and allocation perturbations can expose different weaknesses. Choose temporal correlations and smoothness to match the mechanism under investigation; white or independent noise is not automatically wrong, but a numerical sampling convention is not an execution model. A local derivative describes a neighborhood and can miss distant failures, contact-mode changes, or a missing event. Broad timing plateaus are useful only relative to a performance tolerance, the other disturbances, and the resolution of the search.

Paired simulations should share exogenous cases while retaining the state changes caused by each policy. Resetting transmission state to make two trials look matched changes the intervention. Chapter 3 explains that distinction; Chapter 14 explains finite preparation and role reversal. A comparison of policy families should declare their admissible controls, resource constraints, tuning procedure, and

computational budget. Equal budgets reduce one confound but do not prove equal optimization quality or equal physiological feasibility.

### **23.2.8 Preserve Failures and the Denominator**

Record how many cases were attempted, solved, physically admissible, and evaluated at the required endpoint. A numerical failure, an impossible contact, a missing impact event, and a valid but poor delivery are different outcomes. Investigate numerical failures with refinement or an independent method; do not silently delete them or automatically relabel them as physical failures.

A distribution conditional on successful solves must carry that condition. If a case's loss is unknown, report missingness and bounds or an explicit penalty rule rather than inventing its value. Discarding the worst observed case can improve a reported tail without improving any policy. A failure rate among completed trials also has a different denominator from a failure rate among all attempted trials. Publish both the accounting and the metric rules.

### **23.2.9 Keep Selection Separate From Evaluation**

Use development cases to explore policies, validation cases for tuning choices, and an untouched evaluation set for the final claim. The eight-program archive contains training and held-out splits, not a separate three-way validation exercise. It demonstrates a bounded comparison, not a completed robust-control optimization pipeline. Repeatedly modifying the program after viewing held-out results consumes that independence.

Choose splits for the deployment question. Unseen-person prediction requires participant separation; personalization for a known golfer requires evaluation on genuinely new sessions or conditions and separation from that person's calibration data. Randomly mixing correlated swings between training and test sets can exaggerate transfer. Calibration also has scope: nominal 80-percent coverage should be assessed across enough relevant unseen cases, with sampling uncertainty and subgroup checks. Good average coverage can conceal poor coverage for one participant or regime.

Personalization does not require every parameter to be uniquely identifiable if the target prediction is stable across all plausible parameter sets. Conversely, a precise parameter estimate under the wrong model is not reliable prediction. Show the uncertainty that changes the decision, and withhold confident technique prescriptions when predictive qualification is missing.

### **23.2.10 Compare Model Tiers and Human Evidence Explicitly**

Parameter sensitivity, numerical convergence, structural model changes, and transfer to people are separate tests. A planar fixed-base ranking may change when base motion, shaft memory, spatial contact, or delay is introduced. Define how policies, observations, initial states, and constraints correspond between tiers. A ranking reversal identifies a dependency to investigate; it does not automatically identify the added physical mechanism as its cause.

Likewise, a negative torque component must name an axis, frame, and time window. Persistence of its sign is different from persistence of a performance benefit. A bimodal outcome distribution can arise from nonlinear transformations or sampling choices as well as subgroups or competing mechanisms. A histogram alone cannot decide among those explanations.

Glazier’s discussion of movement variability and McPhee’s review of golf models provide methodological context ([Glazier 2011](#); [McPhee 2022](#)). They are not validation of the eight-program ranking or a certificate of individual robustness. Measured responses to appropriately designed perturbations would test specified predictions, including timing, recovery, and uncertainty. Human adaptation absent from the model can require narrowing the claim or changing the controller model; it cannot be inferred from the synthetic ensemble.

### 23.2.11 Reader Lab: Make the Robustness Claim Auditable

Start with two policies, a common set of admissible cases, and fixed outcome definitions. Plot their individual results as well as means, adverse tails, and paired differences. State whether uncertainty is sampled probabilistically or bounded as a set. Keep the original sample counts, unsuccessful cases, and the quantile convention alongside the figure.

Then vary one declared assumption: the error tolerance, a parameter dependence, the sampling design, or the model tier. Recompute the comparison without retuning on the evaluation set. Add a simple baseline and report where the ordering reverses. A ranking that survives these tests is evidence within their scope, not proof against every future change.

An open modeling tool should preserve these qualifications in exported tables and figures. It can flag violations of known input bounds but cannot guarantee detection of unknown model inadequacy. Separate an exploratory prediction from a qualified recommendation. A useful report identifies which measurement or experiment would most reduce the uncertainty affecting the choice.

## 23.3 Where the Picture Breaks

The racing analogy motivates a comparison across conditions; it does not imply that a golf downswing has the same feedback opportunities as a lap. A constructed ensemble also differs from repeated measurements of people. Denser sampling of known variables cannot make an omitted disturbance disappear.

**Model Result:** The archived eight-program comparison reproduces a conditional speed/error tradeoff and sample-dependent nondominance. It supports no universally best policy and does not exclude better untested policies.

**Measurement Boundary:** The endpoint, force statistic, and effort metric are declared model quantities. Participant-held-out measurements, appropriate outcome definitions, and calibrated predictive uncertainty are still needed for human robustness claims.

**Hypothesis:** Preload or an opposing command interval may improve selected outcomes under some delays and uncertainties. Isolating that mechanism requires matched comparisons beyond two schedules that differ in several parameters.

**Practical Interpretation:** Ask how often the intended delivery succeeds, under which changes, with which error and load consequences, and how much of that answer comes from measurements rather than modeling assumptions.

**Go Deeper:** Read the monograph’s [coupled uncertainty and control chapter](#), the [archived experiment record](#), and the [small-sample stability audit](#).

## Chapter 24

# What Has Actually Been Measured in Golfers

### 24.1 A Concrete Picture

A courtroom separates what a witness saw from the explanation offered for it. Golf science needs the same separation. A marker trajectory, a calibrated grip load, and an electrical signal from a muscle can constrain an explanation of the swing. They do not carry the same information, and collecting them together does not automatically reveal the internal mechanism.

The useful question is specific: what observation would distinguish two plausible explanations? If both predict the same club motion but different hand loads, another camera may add less information than a carefully calibrated grip measurement. If both predict the same hand loads through different muscle combinations, even a perfect grip sensor leaves a biological question open.

### 24.2 How the Mechanism Works

#### 24.2.1 Separate the Signal From the Explanation

A camera records image information from which marker positions and segment motion are reconstructed. A force plate converts transducer signals into a calibrated external wrench. A grip sensor records a load at its sensing section; the force attributed to a palm can require a further mechanical calculation. EMG records electrical activity associated with selected muscles, not muscle force directly. Every measurement has a location, reference frame, bandwidth, calibration, and uncertainty.

Inverse dynamics combines motion, external loads, and body-segment inertias to estimate net joint loads. Its answer depends on filtering and differentiation, joint definitions, and the assumed contact model. In the closed chain formed by two arms and one club, the net club wrench alone generally does not identify how the hands share the load. Additional sensing or assumptions are required.

Even uniquely determined net joint moments would not uniquely determine muscle forces. Antagonists can increase their opposing forces while leaving a similar net moment. EMG can constrain activation timing and a muscle model, but the conversion to force also depends on muscle length, contraction velocity, activation dynamics, and tendon behavior. Mechanical identification and anatomical identification

# Observations, Inferences, and Tests

## **Calibrated Observations**

Motion, external loads,  
electrical activity

## **Mechanical Inference**

Frames, inertias,  
contact assumptions

## **Anatomical Inference**

Multiple muscle allocations  
can remain

## **Prospective Tests**

Frozen predictions;  
relevant new data

Measurements constrain explanations.  
A unique mechanism is not guaranteed.

Figure 24.1: Observations Constrain Inferences; Each Explanation Still Needs a Test.

are different problems.

### 24.2.2 A Bounded Register of Human Evidence

The following studies illustrate what the literature can constrain. They are a selected register, not a new systematic search or a pooled dataset. Participant counts refer to people, not independent replications of every recorded swing. The linked sources identify the studies; unspecified trial counts or data access should be checked in the full methods before reanalysis.

- **Kinematics Across Studies.** Bourgain and colleagues’ systematic review included 92 studies and documented heterogeneous definitions and methods ([Bourgain et al. 2022](#)). It supports careful comparison of coordinate systems, clubs, samples, and events; it does not establish one universal sequence. This is distinct from McPhee’s narrative model-and-measurement review ([McPhee 2022](#)).
- **Professional and Amateur Motion.** Zheng and colleagues compared 72 men, with 18 in each of four professional/amateur skill groups, using motion capture at 240 Hz ([Zheng et al. 2008](#)). Group differences describe the sampled swings. They do not establish that copying a professional’s movement causes improvement or prevents injury.
- **Scapular Muscle Activity.** Kao, Pink, Jobe, and Perry studied 15 competitive male golfers with bilateral EMG of four scapular muscles and cinematography ([Kao et al. 1995](#)). Phase- and side-dependent activity supports shoulder-girdle involvement. It does not identify a unique route from a scapular muscle to a handle moment.
- **A Specific Wrist Muscle.** Robinson and colleagues studied bilateral extensor carpi ulnaris activity in 15 subelite, right-handed men, with five shots each using a pitching wedge, 7-iron, and driver ([Robinson et al. 2023](#)). Surface EMG and forearm/hand inertial sensors characterized selected activation and movement variables. These are not measurements of all wrist muscles. The reported nonsignificant downswing-EMG association with impact clubhead kinematics does not establish no effect or equivalence.
- **A Strain-Gauge Handle.** Koike demonstrated reconstructed palm forces and moments for one right-handed professional instructor using a driver ([Koike 2016](#)). Their calibration and mechanical calculation use fixed palm reference points and neglect handle-piece inertia. This is a useful instrument demonstration, not population validation or direct muscle measurement.
- **A Split Grip and Inverse Dynamics.** Choi and Park studied nine right-handed male professionals with five regular-grip driver swings followed by five instrumented-grip swings ([Choi and Park 2020](#)). One six-axis sensor between two grip sections supplied an internal wrench; a nine-segment rigid-body model inferred hand and upper-limb joint loads. Separate hand loads are therefore estimates supported by sensing and a model, not two independent raw sensor outputs.
- **Ground Interaction.** Han and colleagues studied 63 skilled male golfers with handicaps at most three, using driver, 5-iron, and pitching-wedge shots, optical motion capture, and two force plates ([Han et al. 2019](#)). They computed ground-interaction moments and found associations with clubhead speed. Those associations do not identify a unique ground-to-club energy pathway.
- **Shaft Stiffness.** Betzler and colleagues studied 20 golfers with two clubs designed to differ only in shaft bending stiffness ([Betzler et al. 2012](#)). Their examination of shaft strain, wrist motion, club presentation, and ball launch found small speed differences for the tested shafts. This bounds a specific equipment comparison; it does not establish that shaft flexibility is always negligible or that one stiffness suits everyone.

### 24.2.3 Why Instrument Design Changes the Claim

The Choi–Park instrumented grip weighed about 350 g versus about 100 g for a regular grip; the instrumented club weighed 551 g. Foam/sponge balls, possible hand-to-hand load bypass, and regular-club-first ordering further limit the comparison. Similar average trajectories do not prove unchanged muscle control or mechanical equivalence. The reported roughly threefold right-hand torque relative to the left is a model-derived result in this setting, not a universal hand contribution or muscle-power ratio (Choi and Park 2020).

The central lesson is constructive: record what the instrument closes in the mechanical accounting and what remains assumed. A six-component signal at one section is not automatically a six-component contact wrench at each hand. Loads must be expressed about declared points in compatible frames. Acceleration of material between sensor and contact introduces inertial terms; a path that bypasses the sensor can carry an unrecorded internal load. Static calibration alone does not establish dynamic accuracy over the swing’s frequency range.

### 24.2.4 Motion, Ground, Hands, and Shaft Belong in One Test

Kinematic peak order is a description of motion. It does not uniquely reveal intersegmental energy transfer, and changing the angular-velocity component or event definition can change the reported sequence (Bourgain et al. 2022; Marsan et al. 2019). Likewise, a large ground force is not itself a large supply of mechanical work: an ideal stationary contact can exchange momentum while its contact power is zero. The body’s muscular work and stored energy must still be accounted for.

At the hands, power pairs a wrench with the corresponding linear and angular velocity; work integrates that power over time. Force magnitude alone is insufficient. At the shaft, deformation can store and return energy and alter clubhead presentation. A head-speed change can therefore accompany changes in body motion, bilateral loading, shaft recovery, or several together. Synchronized measurements let a model predict their joint pattern instead of explaining one curve after the fact. They also require synchronized uncertainty: a timing offset between force and velocity can corrupt a computed power curve.

The EMG result supplies another constraint, not a shortcut around this accounting. Different activation patterns can produce similar net wrenches; similar EMG amplitudes can accompany different forces under different muscle states. Crosstalk, electrode placement, normalization, and limited sampling matter. A nonsignificant association leaves an uncertainty interval to examine; an equivalence claim needs a justified equivalence margin and an appropriate test.

### 24.2.5 Repeated Swings and Independent Golfers

Extra swings help estimate a person’s repeatability. Extra participants address variation between people. These contributions should not be confused when estimating a population mean or comparing groups. Nested observations require an analysis that accounts for their dependence, such as an appropriate mixed model or participant-level summaries (Lazic 2010).

Consider a **manufactured example**, not measured golfer data. Let  $y_{ij}$  be a scalar outcome on swing  $j$  from person  $i$ . Write

$$y_{ij} = \mu + b_i + e_{ij}.$$

Here  $\mu$  is the population mean,  $b_i$  is a zero-mean person-specific offset with variance  $\sigma_b^2$ , and  $e_{ij}$  is zero-mean within-person variation with variance  $\sigma_e^2$ . Assume independent participants, errors independent of the offsets and of each other, equal trial counts  $m$ , and no additional session or time dependence. For  $N$  people and their grand mean  $\bar{y}$ ,

$$\text{Var}(\bar{y}) = \frac{\sigma_b^2}{N} + \frac{\sigma_e^2}{Nm}.$$

The person offset repeats in every one of that person’s swings, so its contribution shrinks with the number of people, not the total number of swings. With  $N = 15$ ,  $m = 10$ , and both variances set to one in squared outcome units, the variance of the grand mean is  $11/150$ . Falsely treating all 150 swings as independent gives  $2/150$ : uncertainty variance is understated by a factor of 5.5. The corresponding variance-equivalent independent-observation count is  $300/11$ , about 27.3. That number is neither 27 independent people nor a universal degrees-of-freedom rule.

More repeats reduce the second variance term but cannot eliminate the first at fixed  $N$ . If between-person variance is zero, the formula reduces to the independent-observation case; if each person contributes one swing, there is no repeated-measure inflation. Real analyses may need condition effects, random slopes, unequal trial counts, session effects, or residual autocorrelation. Sample-size calculations must target the actual contrast and dependence model; these invented variances are not a golf-study power calculation.

#### 24.2.6 Prediction, Estimation, and Causation Need Different Designs

An analysis estimating a population association does not automatically need a prediction holdout, but it does need an appropriate sampling and uncertainty model. A claim about predicting a new golfer requires evaluation on participants excluded from fitting and model selection, with preprocessing learned only from training data. Randomly splitting swings can leak a person’s characteristics into both sets. A different target, such as predicting a new session for a known golfer, calls for a different split.

Neither a hierarchical model nor successful prediction alone identifies a causal mechanism. A causal claim must state the intervention or contrast, potential confounders, and assumptions that distinguish rival explanations. A model fitted to a mean curve has not passed a held-out test merely because it reproduces that curve. Multiple tested outcomes and exploratory choices also require transparent reporting; select the primary outcome and analysis before inspecting results.

#### 24.2.7 The Specific Gap in This Project

The evidence reviewed here does not provide this project with a single synchronized human dataset that validates the complete proposed passive negative-couple and preload pathway. That is a statement about the evidence assembled for this mechanism, not a claim that no relevant human study exists anywhere. Existing reviews provide context (Hume et al. 2005; McPhee 2022), but combining results from different participants and instruments does not create a joint validation experiment.

**Human Evidence:** Motion patterns, selected muscle activity, ground reactions, grip measurements, and shaft behavior constrain parts of the explanation within their study designs. **Model Result:** Simulations predict coupled changes in hand loads, club motion, and timing under specified assumptions. **Hypothesis:** The proposed complete biological pathway remains to be tested with measurements and contrasts capable of separating it from alternatives.

For the full pathway, seek synchronized whole-body and club motion, separate foot reactions, sufficiently resolved bilateral hand loading, shaft deformation, selected forearm/shoulder/scapular EMG, and launch outcomes. Contact pressure can help characterize grip location and distribution but does not replace a full wrench. Not every smaller scientific question requires every channel; select channels according to what distinguishes the competing predictions.

### 24.2.8 Turn Disagreement Into a Diagnostic

Suppose a model predicts a negative hand couple in a specified interval, but the estimated experimental couple is positive. First check sign conventions, reference points, synchronization, and uncertainty. If the disagreement persists beyond the prespecified tolerance under comparable conditions, that prediction has failed. Retuning the model can produce a new hypothesis, but does not erase the failed prospective test.

If hand loads agree while EMG timing differs, examine activation measurement and the proposed anatomical allocation. Agreement in net mechanics need not validate the muscle explanation. If motion agrees while hand or ground loads disagree, the model fails that additional constraint; the discrepancy may come from parameters, missing dynamics, or measurement error. It does not uniquely diagnose one cause. Failure in one observable should remain visible alongside success in another.

### 24.2.9 A Reproducible Human-Evidence Release

Before testing predictions, specify calibration acceptance, synchronization tolerance, coordinate registration, sensor range, residual checks, exclusions, and missing-data treatment. Relate thresholds to the intended measurement and claim; there is no universal residual limit that validates every model. Freeze the model version and primary comparisons, and disclose later deviations.

Release the protocol, data dictionary, processing code, participant and trial counts, uncertainty, and consent-compatible data. Show individual traces and residuals alongside group summaries. Report raw-data availability explicitly; open access to a paper does not imply that participant-level data are available. Synthetic pipeline examples and digitized published curves must remain labeled as such. Neither substitutes for a new human validation sample.

### 24.2.10 Reader Lab: Follow One Claim Through the Chain

Take the sentence “elite golfers sequence segments proximally to distally.” Check which people, club, coordinate components, filters, and peak definitions support it. Determine whether the pattern occurs within individuals or only in an average. A kinematic description needs reliable motion measurements; a claim that this sequence transfers energy needs additional kinetic and power evidence.

Now examine “scapular muscles create the handle torque.” Phase-dependent EMG supports selected muscle involvement. Attribution of a particular handle torque requires a closed-chain mechanical and anatomical model constrained by more measurements. Even then, check whether competing muscle

allocations remain possible. Write separate sentences for the observation, the model inference, and the untested causal claim. This makes the next experiment’s purpose concrete.

## 24.3 Where the Picture Breaks

The courtroom analogy makes observations sound neutral. Markers move relative to bone, electrical signals have artifacts, and force estimates depend on calibration and modeling choices. A richer instrument package improves the constraints only when its uncertainties and load paths are understood.

Small or specialized samples also limit generalization across sex, age, skill, injury history, anatomy, and equipment. Evidence can strengthen a bounded mechanism without becoming a universal technique. The goal is a connected test of motion, contact, energy, and biological allocation whose successes and failures are both available to the reader.

**Go Deeper:** Read the monograph’s [empirical evidence chapter](#), the kinematics review ([Bourgain et al. 2022](#)), and the distinct model-and-measurement review ([McPhee 2022](#)). The primary grip methods ([Koike 2016](#); [Choi and Park 2020](#)) make the distinction between sensing and inverse dynamics explicit.

## Chapter 25

# Muscles, Scapulae, Wrists, and Biological Ambiguity

### 25.1 A Concrete Picture

Several teams can move the same piano through a doorway. One pushes, another pulls with straps, and a third combines both. The piano can follow the same path while the people share the load differently. Watching the piano does not reveal each person's force or establish equal effort.

Club motion poses a similar inverse problem. A net grip wrench describes the combined force and moment acting on the club. It does not usually identify the separate hand loads, anatomical joint moments, muscle forces, or neural commands behind that result. These are successive inference problems with different unknowns. Resolving one does not automatically resolve the next.

The anatomical boxes are coupled parts of a body, not five interchangeable actuators. Different allocations must still satisfy anatomy, contact constraints, force limits, and state history.

### 25.2 How the Mechanism Works

#### 25.2.1 Three Different Inverse Problems

First, a rigid-club model can infer a resultant grip wrench from motion and inertia, after accounting for gravity and other external loads. Second, body inverse dynamics can estimate net joint moments if segment motion, inertial parameters, and the necessary external loads are known. A net club wrench alone generally leaves the separate hand loads underdetermined in the two-arm closed chain. Third, a musculoskeletal model allocates joint demand among muscles and passive tissues.

#### 25.2.2 Two Allocations With the Same Full Club Wrench

Separate contact ambiguity from muscle ambiguity. Fix a club reference  $O$  and contact points  $\mathbf{r}_L = (-0.1, 0, 0)$  and  $\mathbf{r}_R = (0.1, 0, 0)$  m. All components below use one right-handed frame. Here  $\mathbf{F}_i$  is hand  $i$ 's force and  $\mathbf{C}_i$  is its contact couple; both act **on the club**:

## Club Motion Does Not Uniquely Identify the Biological Allocation

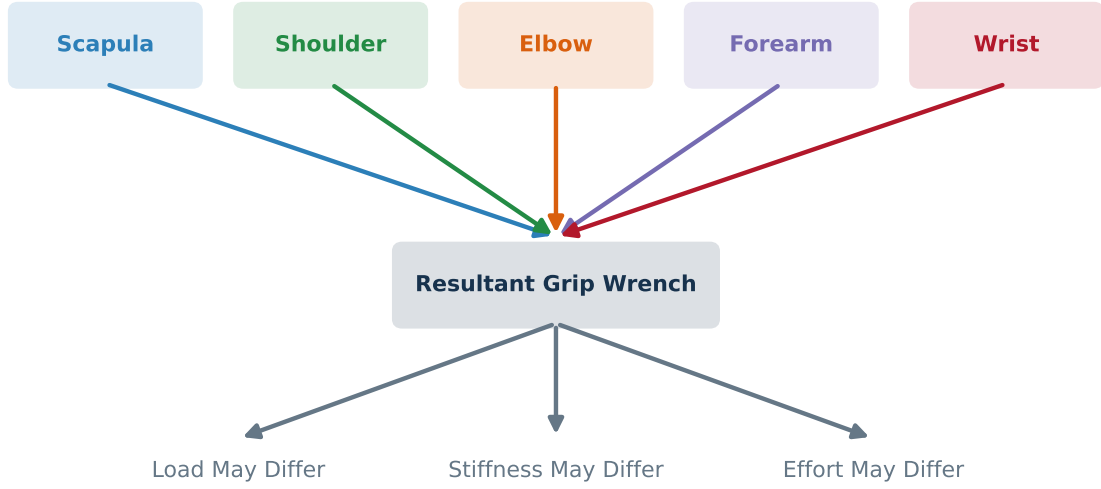


Figure 25.1: Different Feasible Biological Allocations Can Share a Resultant Grip Wrench.

| Allocation | $F_{Ly}$ (N) | $F_{Ry}$ (N) | $C_{Lz}$ (N m) | $C_{Rz}$ (N m) |
|------------|--------------|--------------|----------------|----------------|
| A          | -10          | 110          | 0              | 0              |
| B          | 30           | 70           | 4              | 4              |

All unlisted components vanish. Both allocations satisfy

$$\begin{aligned}\mathbf{F} &= \mathbf{F}_L + \mathbf{F}_R = (0, 100, 0) \text{ N}, \\ \mathbf{M}_O &= \sum_{i=L,R} (\mathbf{r}_i \times \mathbf{F}_i + \mathbf{C}_i) = (0, 0, 12) \text{ N m}.\end{aligned}\tag{25.1}$$

A obtains all 12 N m from the force distribution. B obtains 4 N m that way and 8 N m from contact moments. These are full resultant-wrench matches within an ideal contact model. Grip pressure, friction, and tissue limits determine actual feasibility.

The equal force at each hand,  $\mathbf{F}_c = \mathbf{F}/2$ , contributes  $(\mathbf{r}_L + \mathbf{r}_R) \times \mathbf{F}_c = 0$  at this midpoint. Changing arm posture while keeping the contact points, force components, and reference fixed cannot create a new common-force moment arm. Physically shifting both contacts 0.02 m along  $+x$  relative to the same  $O$  raises the moment to 14 N m if the loads are retained. That is changed geometry and a changed wrench, not a third allocation matching the original task. Changing only the reporting origin is a separate wrench-transport operation.

At the same rigid-club twist,  $\mathbf{v}_O = (0, 2, 0) \text{ m s}^{-1}$  and  $\boldsymbol{\omega} = (0, 0, 3) \text{ rad s}^{-1}$ , both allocations deliver  $\mathbf{F} \cdot \mathbf{v}_O + \mathbf{M}_O \cdot \boldsymbol{\omega} = 236 \text{ W}$ . Summing contact powers with  $\mathbf{v}_i = \mathbf{v}_O + \boldsymbol{\omega} \times \mathbf{r}_i$  gives the same answer:

left/right contributions are  $-17/253$  W for A and  $63/173$  W for B. Equal total club power does not identify each hand's contribution or each muscle's expenditure. Bilateral wrench sensing addresses the first ambiguity; anatomical and physiological evidence addresses the second.

### 25.2.3 Matching a Scalar Task Is Weaker Than Matching a Wrench

The companion benchmark in UpstreamDrift, linked below at its reviewed revision, matches a same-state control-induced angular-acceleration target expressed in moment units:  $I_C \ddot{\alpha}_{C,u} = 8$  N m. Here  $I_C$  is the model club's planar rotational inertia, and  $\ddot{\alpha}_{C,u}$  is the control-induced part of its angular acceleration. It subtracts the zero-control acceleration at the same state and allocates controls within proximal and wrist generalized-torque subspaces. This scalar constraint does not by itself match all resultant-force components, the full grip wrench, or a future trajectory. The table above demonstrates a different, explicitly full-wrench construction.

For a rigid club with identical initial state and other external loads, identical resultant-wrench histories expressed consistently imply identical rigid-club motion. Matching one instantaneous moment, one speed, or one scalar acceleration does not supply those conditions. Shaft deformation adds states if the club is not modeled as rigid. Name precisely which task was held equal before interpreting a successful simulation.

### 25.2.4 From Joint Demand to Muscle Forces

Let  $\tau_{\text{ID}}$  denote the joint demand remaining after the chosen inverse-dynamics formulation accounts for external loads and constraint reactions. One declared allocation model is

$$\tau_{\text{ID}} = R(q) \mathbf{F}_{\text{mt}} + \tau_{\text{passive}}, \quad R_{ji} = -\frac{\partial \ell_i}{\partial q_j}. \quad (25.2)$$

Here  $q$  collects joint coordinates,  $\mathbf{F}_{\text{mt}}$  collects musculotendon tensions,  $\ell_i$  is musculotendon path length,  $F_i \geq 0$  is its tensile force, and  $R_{ji}$  is its signed moment arm about rotational coordinate  $q_j$ . A positive entry means tension produces positive generalized torque. Separate passive joint loads exclude passive muscle contributions already included in  $F_i$ . Grip forces and joint reactions must not be added again if the inverse-dynamics demand already accounts for them.

More muscles than independent joint equations permits a force-map null space, but does not guarantee an unlimited feasible family. Nonnegative tension, strength limits, tendon state, activation history, and measurements constrain it. An allocation can be infeasible, unique, or nonunique. Optimization selects a solution conditional on its cost and constraints; it does not identify a neural strategy merely by producing a smooth answer (Zajac and Gordon 1989).

### 25.2.5 A Bounded Muscle Example

Consider a manufactured one-joint example with opposing moment arms of  $+0.03$  and  $-0.03$  m, zero separate passive torque, and a required moment of  $3$  N m. Suppose  $0 \leq F_A \leq 300$  N and  $0 \leq F_B \leq 200$  N. In SI units,

$$0.03(F_A - F_B) = 3, \quad F_A = F_B + 100, \quad 0 \leq F_B \leq 200. \quad (25.3)$$

The pairs (100, 0), (200, 100), and (300, 200) N produce the same net moment with different individual forces. Pairs outside the bounds are inadmissible. Pairs inside them are only instantaneously feasible; generating their histories also requires compatible muscle and tendon dynamics. Specifying those internal states can remove ambiguity that remains when only net torque is known.

At  $\dot{q} = 2 \text{ rad s}^{-1}$ , the two path-length rates are  $-0.06$  and  $+0.06 \text{ m s}^{-1}$ . Every listed pair gives

$$P_{\text{paths}} = - \sum_i F_i \dot{\ell}_i = \tau_{\text{muscle}} \dot{q} = 6 \text{ W}. \quad (25.4)$$

Here  $\tau_{\text{muscle}} = \sum_i r_i F_i$  is the net torque from the musculotendon paths, including any passive force they contain. This virtual-work equality gives net musculotendon power delivered to the modeled coordinate, not each muscle fibre's power or metabolic expenditure. Tendon storage and pennation separate fibre mechanics from path mechanics. Equal net joint power does not make the allocations biologically interchangeable. These numbers illustrate accounting, not fitted golfer parameters.

### 25.2.6 Electrical Excitation, Activation, and Force

Surface EMG measures electrical potentials associated with muscle-fibre action potentials in an electrode's detection region. With suitable processing it informs sampled excitation, not a direct force or activation-state measurement. Activation dynamics link excitation to contractile state; contraction mechanics depend on length, velocity, history, architecture, and tendon loading. Normalization does not remove those dependencies. Electrode position, tissue conduction, cancellation, and cross-talk also limit comparisons (Vigotsky et al. 2018).

A detected burst alone establishes neither joint-torque magnitude nor the sign of club power. An apparently silent channel cannot exclude force: passive tension, retained internal state, unmeasured muscles, and detection limits remain possibilities. EMG helps discriminate candidate timings and constrain models when its measurement assumptions are explicit. It does not replace the mechanical measurements needed to test a proposed force or power pathway.

### 25.2.7 Co-Contraction, Stiffness, and Impedance

Opposing muscle forces can increase without changing net torque. Whether this improves resistance to disturbance requires a different measurement: the response to a perturbation about a declared state. Stiffness, damping, inertia, and delayed feedback contribute differently to mechanical impedance. A moment-versus-angle slope traced during a swing also contains changes in activation and history; it is not automatically the local stiffness measured by perturbing a specified state.

For a one-coordinate elastic illustration, hold activation and other internal variables fixed, set  $k_i = \partial F_i / \partial \ell_i$ , and use  $r_i = -\partial \ell_i / \partial q$ . The restoring stiffness contribution is

$$K = - \frac{\partial \tau}{\partial q} = \sum_i \left( k_i r_i^2 - F_i \frac{\partial r_i}{\partial q} \right). \quad (25.5)$$

The first term maps material stiffness through the moment arm. The second comes from changing geometry. At fixed moment arms, larger positive local muscle stiffness increases the first term. Geometry, other tissues, and feedback prevent that observation from becoming a universal claim about whole-joint

or whole-swing stability. Joint compression likewise requires force directions and joint equilibrium, not just a scalar net moment.

A manufactured counterexample makes the geometric term visible. Near  $q = 0$ , use path changes  $\Delta\ell_1 = -0.03q - 0.01q^2$  and  $\Delta\ell_2 = 0.03q$  in metres, with  $F_i = 100 + 1000\Delta\ell_i$  in newtons. Here radians are dimensionless, and the coefficients carry the corresponding SI units. Both tensile paths have positive  $1000 \text{ N m}^{-1}$  stiffness. At zero angle their net torque is zero, but the restoring stiffness is  $1.8 - 2.0 = -0.2 \text{ N m rad}^{-1}$ . Holding the moment arms constant instead gives  $+1.8 \text{ N m rad}^{-1}$ . Positive element stiffness alone does not determine generalized restoring stiffness. This is a counterexample for two ideal elastic paths, not a stability assessment of a human joint.

Hu, Murray, and Perreault combined a short-range-stiffness model with an upper-limb model and compared it with previously collected small-perturbation endpoint-stiffness measurements from five people. Their formulation includes moment-arm geometry; the tested conditions involved specified postures and voluntary forces. It supports studying this mechanism, not a claim that rapid golf transition dynamics or an optimal co-contraction strategy has been validated (Hu et al. 2011).

### 25.2.8 Tendons Store Energy

An elastic tendon stores strain energy, not force. For an ideal linear element  $F = kx$ , with extension  $x$  from its unloaded reference,

$$\Delta E = \int_{x_0}^{x_1} F(x) dx = \frac{1}{2}k(x_1^2 - x_0^2). \quad (25.6)$$

With  $k = 10,000 \text{ N m}^{-1}$ , extension from 5 to 10 mm raises force from 50 to 100 N and stores an additional 0.375 J. Force is an instantaneous load; stored energy accounts for work. A static force-extension law contains no time delay. Force-rise timing requires system dynamics, including excitation, contraction, movement, and any dissipative behavior. Compliance can affect that timing but is not itself a fixed delay.

For two positive linear stiffnesses in series,  $k_{\text{eq}}^{-1} = k_1^{-1} + k_2^{-1}$ ; a compliant tendon can limit the effective stiffness of the combined path. Tendon slack length in a model defines a force-length reference. It is not evidence for a backlash-like interval at a golfer's wrist.

The UpstreamDrift benchmark also studies a separate synthetic law: a rotational dead zone, linear transmitted torque beyond it, and first-order deflection dynamics. Its persistent-direction advantage disappears in its declared zero-dead-zone comparison. That result is conditional on the chosen transmission family. Neither the calculation nor a sensation of firmness establishes a human transmission gap. Muscle short-range stiffness, tendon elasticity, grip contact, and shaft deformation need distinct state descriptions and tests.

### 25.2.9 Scapula, Forearm, and Wrist Are Distinct Parts

Scapulothoracic motion describes the scapula moving relative to the thorax; it is not a fixed hinge. Scapular orientation changes the orientation of the glenoid and the geometry available to the arm. Distinguish clavicle, scapula, and humerus when reporting shoulder motion. The International Society

of Biomechanics (ISB) recommendations provide a reporting framework, not a shortcut from one reduced shoulder torque to an identified scapular action (Wu et al., 2005).

Forearm pronation and supination involve the radioulnar mechanism; they are distinct from wrist flexion-extension and radial-ulnar deviation. They interact in orienting a gripping hand. A planar model's wrist coordinate represents its declared relative rotation. Projecting a three-dimensional moment onto an axis gives an axis moment, but it is a work-conjugate generalized torque only when the coordinate-to-angular-velocity mapping justifies it.

A grip contact moment is not automatically an anatomical wrist joint moment. Segment inertia, forces, lever arms, and muscle loads intervene. An opposing club-moment contribution can coexist with positive translational contact power or active muscles. Pair the correct wrench and velocity before classifying power; an opposing moment alone does not identify a passive tissue mechanism.

Skin-mounted measurements need a validated relationship to bone motion. Ludewig and colleagues used bone-fixed tracking to study shoulder-complex motion during arm elevation in twelve people; this describes their measurement scope, not a golf-specific scapular force pathway (Ludewig et al., 2009). Testing a proposed scapular contribution to bilateral hand loads needs compatible kinematics, load measurements, and an anatomical model with uncertainty. Renaming an abstract arm driver cannot provide those observations.

#### **25.2.10 What the Golf EMG Studies Establish**

Kao, Pink, Jobe, and Perry studied fifteen competitive male golfers with bilateral EMG of four scapular muscles and cinematography (Kao et al. 1995). Their phase-dependent observations concern those sampled muscles and participants, not a uniquely identified two-hand wrench allocation.

Robinson and colleagues studied bilateral extensor carpi ulnaris activity and wrist kinematics in fifteen subelite, right-handed men (Robinson et al. 2023). This is one selected muscle on each side, not a complete wrist-muscle inventory. The reported absence of a significant downswing EMG association with impact clubhead kinematics does not establish equivalence or mechanical irrelevance. Both studies motivate specific anatomical questions; neither resolves muscle forces from club motion or proves the proposed preload benefit.

#### **25.2.11 A Biological Comparison That Can Fail**

The working hypothesis remains worth testing: directionally persistent loading might preserve useful force transmission for some golfers, while another allocation might reduce effort or suit different anatomy. Neither mechanics example chooses a universal strategy.

Select measurements for the hypothesis. To distinguish contact allocations, measure bilateral wrenches at declared reference points and check resultant closure. To investigate stiffness, define the perturbation or identified dynamic model, amplitude, time window, and feedback assumptions. To test an anatomical pathway, add suitable segment kinematics and muscle observations. Ultrasound or other internal-state measurements may help particular tendon questions; not every study requires every sensor. The linked protocol specifies a particular combined test, not a universal minimum for biological claims.

Randomizing instructions can estimate the effect of assigned instructions under a sound experimental design. It does not randomize the muscle allocation actually adopted. Check whether the intended

load pattern changed. Association among a cue, stiffness, and outcome does not alone establish causal mediation; confounding and other pathways remain. Prespecify contrasts, uncertainty, alternative explanations, and participant-level holdout.

A failed wrench-closure check challenges the mechanical reconstruction. A well-measured mismatch between predicted and observed scapular motion challenges that anatomical mapping. Ambiguous EMG or unresolved alternatives can leave the biological question inconclusive even when contact mechanics remains supported. Separating these outcomes helps avoid attributing failures to an unspecified hidden muscle strategy.

### 25.2.12 Reader Lab: Follow the Evidence in Both Directions

Start with allocations A and B. Compute force, moment, and power, then identify the bilateral measurements that distinguish them. Next hold a net joint moment fixed and construct two candidates from the bounded antagonist example. State which internal measurements or model assumptions could reject one. Do not call a modeled joint reaction a directly measured force.

Reverse the exercise from an EMG burst. Identify the sampled region, processing, normalization, and timing uncertainty. List the additional state and mechanics needed to infer force, joint torque, and club power. A statement of sampled excitation cannot skip those steps. Contact mechanics and biology inform one another without becoming interchangeable.

## 25.3 Where the Picture Breaks

Piano teams choose explicit roles. Human control also involves distributed coordination, feedback, and history. Coaching words such as push, pull, firm, and passive do not specify a muscle allocation or a state equation.

More anatomical detail adds parameters as well as explanatory possibilities. Uncertain paths or tendon properties can dominate an optimized answer. Report sensitivity and feasible alternatives instead of equating detail with identification. Similar club speed does not establish similar internal loading or safety; these examples do not evaluate clinical outcomes.

A useful biological claim states the observation, mechanical demand, anatomical mapping, alternatives, and uncertainty. It may show consistency with evidence, contradict a pathway, or identify a discriminating measurement. Ambiguity becomes productive when it specifies the next test.

**Go Deeper:** Read the [allocation and transmission assumptions](#) and [experimental protocol](#) at the reviewed provider revision. Its human predictions remain untested; these mathematical examples are verification fixtures, not new human measurements or a rerun of that provider.

## Chapter 26

# Design an Experiment That Can Say No

### 26.1 A Concrete Picture

A smoke alarm that remains silent may have encountered clean air, a flat battery, or a blocked sensor. Those outcomes mean different things. Likewise, a golf experiment can contradict a prediction, lack the precision to distinguish it, or fail to measure the quantity that the prediction concerns. A credible test makes those possibilities distinguishable before looking at the result.

The question is not simply whether a force, an EMG burst, or a speed increase appears. It is whether a specified mechanical explanation predicts an observable under conditions that could expose its failure. A negative club moment does not identify muscular inactivity; an instruction that changes speed does not by itself identify how that change occurred. The experiment must connect its measurements, model assumptions, intervention and decision rule.

#### A Test Must Distinguish Its Possible Outcomes



An Unexecuted Test Remains Untested.

Figure 26.1: A Registered Prediction Leads to Qualified Measurements and Distinct Supported, Contradicted, or Inconclusive Outcomes.

## 26.2 How the Mechanism Works

### 26.2.1 Start With the Quantity the Claim Predicts

Define the **estimand**: the quantity the study intends to estimate, for a stated population, task, phase window and comparison. For a late negative-couple claim, that might be a participant-level mean of one declared moment projection over a registered preimpact interval. It is not interchangeable with peak force, total club work, or the fraction of trials with any negative sample.

Separate the measured channels from their interpretation. A six-axis sensor measures a wrench at its calibrated reference. Transporting that wrench requires geometry; calculating its power also requires the matching velocity and angular velocity. Individual muscle force, effective stiffness, model-attributed acceleration and a zero-command counterfactual require further assumptions or experiments. More channels can constrain an inference without uniquely solving it. Chapter 25 follows this distinction from hand loads into biological demand.

For each proposed mechanism, register the model, intervention, primary observable, expected direction or interval, competing explanation, and result that would narrow the claim. Four examples organize the program:

- **Interaction forces:** Test a specified force–velocity projection or work prediction. An acceleration decomposition is computed within a declared model; it is not another force sensor reading. A human instruction is not the same intervention as an instantaneous model torque deletion at a fixed full state.
- **Two-hand moments:** Define the reference, axis, separated-force contribution and directly applied hand moments. Test the predicted component and its geometry dependence before interpreting a negative total moment.
- **Preload and transmission:** Name the mechanical channel: hand–grip contact, musculotendinous tissue or shaft. Distinguish contact loss, low differential load, force-development delay and tangent stiffness. Test whether a manipulation changes the proposed variable, and separately whether that change explains an outcome under defensible causal assumptions.
- **Robust control:** Declare the disturbance distribution, task and load limits, outcome distribution and comparison policy. A favorable synthetic perturbation result is not automatically a prediction about variation across golfers.

### 26.2.2 Register the Plan and Preserve Its Changes

Specify outcomes, windows, exclusions, sample-size rationale, stopping rules, filters, coordinate conventions, model versions and statistical analyses before relevant outcome inspection. Record what data were already known. A timestamped plan makes later changes visible; it cannot prevent them or guarantee a sound experiment.

Useful amendments and exploratory analyses remain possible. Preserve the original plan, dated reasons, data-access state and revised analysis; distinguish predeclared contingencies from decisions informed by results. Report planned outcomes even when they are unfavorable. These are the transparency functions of [preregistration](#), not a prohibition on learning. A pilot can inform feasibility and variance assumptions, but reusing its inspected outcomes as untouched confirmation requires a different argument.

### 26.2.3 Qualify the Measurement Chain

Choose instruments from the estimand, rather than treating the longest sensor list as the best experiment. The UpstreamDrift computational research protocol (the provider linked below) proposes synchronized body and club kinematics, separate ground-reaction and bilateral hand wrenches, with launch, EMG, pressure or shaft measurements as particular hypotheses require. A split-grip transducer’s internal wrench does not by itself directly measure every component of each hand’s distributed contact load. State the instrument boundary and the balances needed to infer the remaining quantities.

Before interpretation, characterize calibration, axis orientation, crosstalk, range, drift, hysteresis and dynamic response. Register reference points and frames, sensor and club mass, motion reconstruction, segment inertias and their uncertainty. A calibration load checks particular errors; it does not establish accuracy throughout an untested dynamic range. Instrument mass or altered grip geometry can change the swing being measured.

Sampling interval, clock offset, clock drift, device latency, resampling and event localization are different sources of timing error. A high sampling rate does not prove synchronization. Combine their uncertainties according to their dependencies; a root-sum-of-squares rule needs a justified covariance model, whereas bounded errors may need a conservative combined bound. Report an event ordering as unresolved when its uncertainty permits reversal. The provider’s rates and millisecond tolerances are specifications for its frozen protocol, not universal evidence that every proposed golf observable is resolvable.

### 26.2.4 Filtering Can Break Closure Without Shifting Phase

Differentiation amplifies high-frequency position noise. Filtering can alter both amplitude and timing, but unequal cutoffs do not necessarily produce a relative phase shift. Two zero-phase filters can retain different amplitudes and thereby make force and acceleration inconsistent.

For a periodic illustrative signal, let  $F = ma$  and filter force and acceleration with linear operators  $H_F$  and  $H_a$ . The residual becomes

$$r = H_F F - m H_a a = m(H_F - H_a)a.$$

It can be nonzero even when both filters have zero phase at every retained frequency. Applying the same linear filter closes this simple constant-mass example; it does not automatically close nonlinear multibody inverse dynamics, where products, rotations, lever arms and differentiation also matter. Acausal forward–backward filtering additionally needs care near events and record boundaries: future samples contribute to the estimate.

Studies of [sidestep cutting](#) and [drop-jump landing](#) illustrate sensitivity of inferred joint moments to filtering. Those tasks and their findings do not determine a universal golf cutoff. Qualify bandwidth and temporal response for the intended observable, freeze a primary method, and report registered sensitivity analyses. Do not tune a filter until a preferred sign appears.

Force, moment and energy residuals test consistency of the selected model and measurement chain. A failed gate may indicate calibration error, missing loads, timing problems or structural model discrepancy. Keep a disposition for every trial, report failure rates by condition, and investigate

differential loss. Invalid measurements cannot support a force claim, but silently deleting them can also bias the comparison. Missing or unqualified data are neither a measured zero nor a successful falsification.

### 26.2.5 Separate Physical Controls, Invariance Checks and Deliberate Errors

A consistent change of coordinates must preserve the physical prediction. Rotating a complete wrench and its conjugate twist together preserves power. Transporting both to another reference does too. Merely changing the sign of one channel is a corrupted-input test, not a physical reversal. Specify which of these operations each control performs and its expected result.

There is a similarly important limit to collapsing hand separation. For equal-and-opposite forces  $\mathbf{f}$  and  $-\mathbf{f}$ , the net moment is

$$\mathbf{M} = (\mathbf{r}_L - \mathbf{r}_R) \times \mathbf{f} + \mathbf{C}_L + \mathbf{C}_R,$$

where  $\mathbf{r}_L$  and  $\mathbf{r}_R$  locate the left and right force application points relative to a common reference, with  $+\mathbf{f}$  on the left and  $-\mathbf{f}$  on the right. The vectors  $\mathbf{C}_L$  and  $\mathbf{C}_R$  are directly applied contact moments, all expressed in one frame. Coincident application points eliminate the separated-force term. They do not eliminate those other moments. In a manufactured example, two coincident hands with zero resultant force and direct moment projections  $-0.3$  and  $-0.2$  N m along a common  $z$  axis retain a net projection of  $-0.5$  N m. A zero-separation null is therefore valid for the named force-couple contribution, or for a point-force-only model, not for every negative club moment.

Changing grip geometry in a person also changes attainable forces, comfort and control. Do not assume that only the lever arm changed. A model intervention that holds forces fixed answers a narrower question.

Temporal shifts can probe an alleged event relation, provided the surrogate destroys that relation while retaining the relevant noise and dependence. Repeated, autocorrelated samples are not exchangeable independent rows. Likewise, label permutations must respect the actual randomization and clustering. Document what a control rules out; survival of one control does not identify a unique mechanism.

### 26.2.6 Randomizing an Instruction Does Not Randomize Its Mediator

Let  $T$  be an assigned instruction,  $M$  a measured preload variable and  $Y$  a delivery outcome. Randomization can support an effect of the assigned instruction under the study's design and adherence assumptions. It does not make differences in  $M$  random: strength, initial state, comfort or other variables may affect both  $M$  and  $Y$ .

A manufactured example makes the problem explicit. Take independent, centered, unit-variance variables  $T, U, E$ , where  $U$  is a pretreatment common cause and  $E$  supplies independent variation in the mediator. Define

$$M = T + U + E, \quad Y = U.$$

Changing  $M$  alone has no effect on  $Y$  in this structural model. Yet regressing  $M$  on  $T$  gives coefficient

1, and regressing  $Y$  on  $T$  and  $M$  gives a mediator coefficient  $1/2$ . Their product suggests an indirect effect of  $1/2$  where the true effect through  $M$  is zero. The omitted common cause  $U$  creates the association. Adjusting for  $U$  when it is observed removes it in this example; a hierarchical fit or a large sample alone does not.

**Causal mediation analysis** therefore needs an explicit causal model, temporal ordering, measurement assumptions, and defensible treatment–outcome and mediator–outcome identification. Variables changed by treatment can introduce further confounding that ordinary covariate adjustment does not solve. Report sensitivity to unmeasured causes; an analysis designed only for pretreatment confounding does not cover every post-treatment problem.

A failed manipulation check weakens the proposed explanation through preload. It does not make a randomized instruction’s total-effect comparison meaningless: the instruction could act through another pathway. Nor should researchers select only participants whose measured mediator changed and treat that selected group as if randomization were preserved. Force-gap, stiffness and EMG measurements remain distinct; EMG alone does not supply a calibrated stiffness measurement.

### 26.2.7 Repeated Swings Improve Precision Within a Defined Design

For prediction in new golfers, keep participants separate between fitting and evaluation. Freeze feature selection, normalization, thresholds and hyperparameter choices using the training data. A held-out participant’s prescribed baseline calibration may be legitimate if it is part of the intended use; accessing their test outcomes to tune the prediction is different. Predicting a later swing of a known golfer is also a different validation target from predicting a new golfer. Name the target instead of applying one split rule to every question.

Repeated swings are useful, but they do not create new independent people. For the balanced random-intercept illustration  $Y_{ij} = \mu + u_i + e_{ij}$ , let  $\mu$  be the population mean, with independent participant effects of variance  $\sigma_u^2$  and independent within-participant errors of variance  $\sigma_e^2$ , the variance of the mean  $\bar{Y}$  across all  $nm$  observations is

$$\text{Var}(\bar{Y}) = \frac{\sigma_u^2}{n} + \frac{\sigma_e^2}{nm},$$

for  $n$  participants and  $m$  swings each. Twelve participants with ten swings,  $\sigma_u^2 = 4$  and  $\sigma_e^2 = 1$  give variance 0.3417, rather than the independent-row calculation  $5/120 = 0.0417$ : an 8.2-fold difference. These are illustrative squared units. This grand-mean formula is not a universal penalty for paired treatment contrasts, where a stable participant intercept can cancel.

Plan power or interval precision for the actual estimand, covariance, condition order, expected missingness and multiplicity. More trials cannot necessarily substitute for more participants; neither is a fixed participant count or holdout fraction universally adequate. Use justified ranges of plausible effects and variances rather than treating a noisy pilot estimate as certain. Randomize or counterbalance conditions, allow familiarization, and model carryover or fatigue when the design requires it. Report participant variation and uncertainty in subgroup claims rather than promoting an exploratory reversal into a new rule.

## 26.2.8 Distinguish Contradiction From Insufficient Precision

Predeclare a scientifically justified negligible region and a decision rule. A nonsignificant test against zero is not evidence that the effect is negligible. Conversely, a precisely estimated nonzero effect can still be too small to matter.

For illustration only, choose an equivalence interval  $[-0.5, 0.5]$  N m for a mean moment contrast. Under a normal model with known standard error  $s$ , a 90% interval for the estimated mean contrast  $\hat{\theta}$  is  $\hat{\theta} \pm 1.645s$ . At estimate zero and  $s = 0.2$  N m, the interval is about  $[-0.329, 0.329]$  N m and fits inside the equivalence bounds. At the same estimate with  $s = 0.5$  N m, it is about  $[-0.822, 0.822]$  N m and is inconclusive about equivalence. Neither case establishes exact zero.

The 90% interval here corresponds to two one-sided equivalence tests at  $\alpha = 0.05$  under the stated model. This is not a generic license to use 90% intervals for every claim or ignore multiplicity. Estimated variance, small samples and clustered data require the appropriate inference procedure. The [equivalence-testing framework](#) clarifies the distinction; the numerical margin above is a teaching choice, not an established golf threshold.

Classify qualified results against the registered prediction as supported, contradicted or inconclusive. Keep an unexecuted test untested. A residual-gate failure is an acquisition/model problem to resolve, not automatically biological contradiction. Repeated precise failures should narrow the claimed mechanism; repeated imprecision should change the study design. Neither licenses moving the primary outcome after inspection.

## 26.2.9 A Staged Study With an Honest Completion Record

Begin with bench tests and complete-pipeline synthetic cases that include known signs, nulls, timing errors, saturation and missing channels. Then use a pilot to establish instrument usability, attainable manipulations and uncertainty. A first human study could address one carefully defined bilateral moment prediction; another could add a qualified stiffness perturbation and activation observations. No single sensor list automatically identifies a scapular or neural strategy.

Before confirmatory collection, preserve the protocol, analysis version, qualification records, oversight and consent scope, data authority, and rules for discomfort, fatigue and equipment failure. Public availability of an article does not imply access to its participant records or permission to reuse them. Publish governed derived results without inferring identities from filenames, session order or performance. Hashes identify bytes; they do not anonymize data.

After collection, report the primary results, uncertainty, exclusions, quality failures and deviations regardless of direction. For model-to-data comparison, distinguish a fitted reconstruction from a held-out prediction and examine signed errors, event timing, work residuals and interval coverage. A residual pattern can suggest a missing mechanism but need not identify it uniquely. Freeze a revised model before testing it on fresh data. Independent laboratories can then test transport across instruments, populations and tasks rather than demanding numerically identical curves.

**Model Result:** At the pinned provider revision, the experimental protocol reports a synthetic intake dry run and declared validation gates. Its model matrix distinguishes bounded mechanical support from open anatomical and human questions. This chapter does not rerun those validators or qualify instruments.

**Human Evidence:** The pinned protocol reports no governed dataset meeting its full bilateral-wrench contract; its human predictions remain untested. That is a statement about the declared evidence boundary, not proof that no relevant human study or dataset exists anywhere.

**Hypothesis:** The registered force, moment and transmission predictions may survive qualified participant-level testing. Their success, failure and practical size must remain open outcomes.

### 26.2.10 Reader Lab: Specify Both the Result and Its Interpretation

Write a failure sentence naming the observable, population, task, window, uncertainty and competing explanations. Then state which instrument failure would make the result uninterpretable rather than contradictory. Run the collapsed-separation, filtering, confounded-mediation and interval examples above as synthetic checks of that distinction.

Before accepting a diagram or a statistical label, ask what was held fixed and what was allowed to change. A consistent frame rotation, a physical grip change, a model torque deletion and a randomized instruction are four different tests. Their connection is useful precisely because their conclusions are not identical.

## 26.3 Where the Picture Breaks

Human strategies adapt to instrumentation and instructions. Changing grip, preload or timing can also alter path, effort and comfort. A good manipulation check documents that problem; it does not eliminate confounding by itself. Likewise, a well-registered analysis cannot create information absent from the measurements or establish a model's anatomical adequacy.

The useful standard is a traceable claim with a test capable of distinguishing its alternatives. Progress may support one mechanical component while narrowing a biological explanation. That is a reason to update the model and the writing, not to preserve an attractive story by redefining its success.

**Go Deeper:** Read the pinned [experimental falsification protocol](#), [model-falsification matrix](#), and [open qualification boundary](#). Chapters 6, 24 and 25 connect the protocol to constraint mechanics, existing measurements and biological identifiability; Chapter 27 separates reproduction from validation.

## Chapter 27

# Open Evidence and the Reviewer Path

### 27.1 A Concrete Picture

A museum label can say a vase is ancient. A serious catalog adds provenance, material analysis, restoration history, photographs, and access rules. Each record answers a different question. An unchanged photograph does not establish age; a reliable date does not establish who owned the object.

A golf-swing calculation needs similarly distinct records. A speed curve can be copied accurately, regenerated from code, recovered by an independent equation, and compared with measured motion. Those are different accomplishments. The reviewer's task is to establish which have actually occurred and what each allows us to infer about the swing.

#### The Publication Is an Entry Point, Not the Evidence Boundary



A reviewer can walk backward from prose to reproducible evidence.

Figure 27.1: A Schematic Reviewer Path Connects Claim, Figure, Data, Script, Test, and Manifest.

The diagram is a navigation aid. Following its arrows should expose assumptions and checks, but the drawing itself supplies no numerical dataset or completed reproduction. Some arguments lead directly to an analytic derivation; some figures are schematics. Neither needs an invented table merely to resemble a simulation result.

## 27.2 How the Mechanism Works

### 27.2.1 What Must Be Reproduced?

Terminology differs across disciplines. Here, *computational reproduction* means recomputing a result from the specified inputs, methods, and code, with a stated criterion for agreement. *Replication with new observations* asks a different question using newly collected evidence. This follows the convention of the National Academies report, while keeping the actual operations explicit ([National Academies of Sciences, Engineering, and Medicine 2019](#), ch. 3).

For this book, distinguish four questions:

- **Artifact Identity:** Is this the file associated with the reviewed release, under its declared byte-comparison or normalization rule?
- **Computational Reproduction:** Does executing the recorded calculation recover the specified outputs within justified tolerances?
- **Independent Verification:** Does a separately derived solution or suitably independent implementation agree under matched assumptions?
- **Empirical Validation:** Does the model predict relevant observations, with uncertainty and applicability assessed for the intended use?

Agreement at one level does not automatically establish the next. Replotting a saved speed array checks the plotting path; rerunning dynamics tests a larger calculation. Two programs calling the same force routine may share its mistake. A correct solution of a simplified arm model remains conditional on whether its geometry, actuation, contact, and material assumptions describe the golfer.

#### 27.2.1.1 Numerical Agreement and Artifact Identity

Suppose two JSON files contain the same values but use different indentation. Their parsed records can agree while their raw SHA-256 hashes differ. Conversely, a file with an erroneous speed can have a perfectly consistent checksum. Updating that checksum records the changed artifact; it supplies no reason to accept the speed. Preserve the earlier expected result and explain the change before accepting a new baseline.

A manifest therefore needs a defined comparison contract. The pinned provider’s release-bundle code normalizes CRLF to LF for a named set of text formats before hashing; it leaves binary artifacts unchanged. That rule differs from raw-byte identity. A reviewer must use the declared rule, rather than silently changing normalization until a comparison passes.

For a deployed PDF intended to be an exact copy of a release asset, byte identity is an appropriate requirement. An independently rebuilt PDF may differ because of metadata or rendering details even when the underlying numbers agree. Compare its scientific payload and inspect its presentation separately. A changed hash is a reason to investigate, not by itself a diagnosis of changed physics.

A bare hash also does not authenticate an author. A cryptographic digital signature involves signing and verification keys; a PDF’s identifying file header is only a format marker ([National Institute of Standards and Technology, n.d.](#)). Neither a recognizable PDF header nor a checksum establishes that the model is valid.

### 27.2.2 Match the Physical Quantity Before Comparing Numbers

A clubhead speed of  $43 \text{ m s}^{-1}$  equals  $154.8 \text{ km h}^{-1}$ . Comparing those unconverted scalars would report a large, meaningless discrepancy. Equal scalars are insufficient too: speed relative to the laboratory at impact is not the same observable as speed relative to the hand at a different event.

A comparison record should specify body and point, units, axes and observer, event definition, time interval, and filtering or differentiation method. For power, add the force application point and its compatible velocity; for work, add the integration interval. Chapter 2’s system boundary and Chapter 17’s moving-base conventions are prerequisites for interpreting a reproduced curve. Without them, software can agree on a column name while answering different mechanical questions.

For matched scalar observables, one possible numerical agreement rule is

$$|y_{\text{new}} - y_{\text{ref}}| \leq a + r|y_{\text{ref}}|,$$

where  $y_{\text{new}}$  and  $y_{\text{ref}}$  are new and reference values of the matched observable. The absolute tolerance  $a \geq 0$  has its units; the relative tolerance  $r \geq 0$  is dimensionless. State both values and justify them before examining the discrepancy. Near zero, the absolute term matters; for a trajectory, specify the sampling alignment and whether agreement means a maximum error, an integrated norm, or an event quantity. Do not combine angles, velocities, forces, and energies in an unscaled norm.

Numerical tolerance is not measurement uncertainty. A solver can reproduce a trajectory to many digits while uncertain segment inertias or contact parameters shift the prediction substantially. Conversely, a difference smaller than a measurement’s uncertainty does not excuse an unexplained implementation error. Keep numerical error, uncertain inputs, model discrepancy, and measurement error visible as different contributors.

### 27.2.3 Seeds, Arithmetic, and Reproducible Runs

Recording a seed is useful, but it does not specify a whole computation. With the same pseudorandom generator and seed, consuming an extra draw before assigning a model parameter changes which sample that parameter receives. Preserve the sampled inputs and their assignment to cases, as well as the generator, seed, and call order. A repeated-seed test demonstrates replay for the tested setup, not a guarantee across unspecified software versions.

Arithmetic order matters independently of randomness. In ordinary binary64 arithmetic, evaluating  $(10^{16} + 1) - 10^{16}$  gives zero, while evaluating  $(10^{16} - 10^{16}) + 1$  gives one. The exact mathematical sum is one. Parallel reductions may change addition order; this example explains a possible source of difference without asserting that every parallel run is nondeterministic. The National Academies discusses why agreement criteria sometimes require numerical analysis rather than universal bitwise equality ([National Academies of Sciences, Engineering, and Medicine 2019](#), ch. 4).

Record source revision, dependencies, solver options, precision, relevant hardware and execution settings, and actual inputs. Containers or environment locks make this easier but do not establish agreement without running and comparing outputs. Keep the command, exit status, logs, and comparison result. A command printed in a document is an instruction; it becomes a reproduction record only after a specified execution has been examined.

### 27.2.4 Why Energy Closure Cannot Validate the Swing

Consider a manufactured rotational oscillator with constant inertia  $I = 1 \text{ kg m}^2$ , angular displacement  $q(0) = 1 \text{ rad}$ , and initial angular velocity zero. Its equations and mechanical energy are

$$\ddot{q} + \omega_0^2 q = 0, \quad q(t) = q_0 \cos(\omega_0 t), \quad E = \frac{1}{2} I \dot{q}^2 + \frac{1}{2} I \omega_0^2 q^2.$$

Here  $q_0 = 1 \text{ rad}$  is the initial amplitude,  $\omega_0$  is the natural angular frequency, and the torsional stiffness is  $\kappa = I\omega_0^2$  (with radians treated as dimensionless in the SI energy relation). The initial energy is  $0.5 \text{ J}$  for  $\omega_0 = 1 \text{ s}^{-1}$  and  $2 \text{ J}$  for  $\omega_0 = 2 \text{ s}^{-1}$ . Each exact solution conserves its own energy. At one second, however, the angles are approximately  $0.5403$  and  $-0.4161 \text{ rad}$ . Energy conservation cannot determine which stiffness represents a physical apparatus.

A small explicit-midpoint integration of this example gives approximately fourfold reduction in final-angle error for integration from zero to one second with steps of  $1/20$ ,  $1/40$ , and  $1/80 \text{ s}$ , at both stated frequencies. The check compares against the analytic solution for the *same* frequency. A convergent computation using the wrong stiffness approaches the wrong physical prediction more accurately. This example is a verification exercise, not a measured golf-swing model or a general claim that this integrator conserves energy.

The same distinction applies to the book’s more complex arguments. Algebraic contact-power closure checks that force and compatible velocity have been paired consistently at a state. Integrated work-energy closure additionally depends on the trajectory, quadrature, constraint handling, and integration step. Neither alone establishes realistic anatomy, friction, muscle recruitment, or tissue loading. The pinned reviewer workbench explicitly distinguishes contact-power closure from its finite-step projected-integrator energy residual.

Independent checks should therefore attack different failure modes: analytic limits, action-reaction, momentum under appropriate isolation assumptions, constraint residuals, timestep refinement, and separately formulated dynamics. For a proposed proximal-to-distal transfer, retain signed interface powers, storage and loss terms, and the same event window. Then test whether the hypothesized explanation survives changed boundary definitions and plausible parameter uncertainty. A small residual is useful evidence about a declared balance; it is not a universal certificate for the causal story.

### 27.2.5 What the Pinned Release Checks Actually Establish

The technical provider examined here is [UpstreamDrift revision 85cce4d](#). Its [qualification entry point](#) provides separate `write`, `validate`, and `list-presets` actions. The validation path checks the release manifest, invokes numerical-claim and claim-evidence checks, validates external-source review records, and inspects the PDF against the selected publication profile. It does more than compare file hashes.

However, `list-presets` prints command strings from the manifest; it does not execute those commands. Passing the validation path must be interpreted in terms of its actual checks, rather than as a claim that every listed simulation has been rerun. This chapter’s inspection of the pinned documents and entry-point code is not a fresh execution of the provider’s scientific ladder.

The pinned qualification document reports a **computational profile** publication candidate and a failed **archival profile**: the inspected PDF remained untagged and had font-related failures. These

are reported results at that revision, not a new inspection of the current website. A computational-profile pass cannot be relabeled as archival accessibility. Counts of links and outlines can reveal structural omissions; they do not establish reading order, figure readability, scientific relevance, or future availability of the destinations.

The same provider keeps held-out human evaluation, equipment calibration, tissue identification, full-contact, and external-archive requirements unresolved. Its held-out synthetic cases are not held-out participants. Its coordinate/pose adapters do not establish independent forward-dynamics agreement across engines. Qualification must name what was tested and retain those limits beside the associated claim.

**Model Result:** A revision-bound bundle and named checks make particular calculations inspectable. Any claim of completed reproduction still needs the corresponding execution and comparison record.

**Human Evidence:** Publishing or validating a release adds no new participant observations. Calibration, study design, and comparison with suitable measured outcomes remain separate work.

**Practical Interpretation:** Follow the mechanism from the prose to its observable, assumptions, calculation, and evidence level before deciding what it says about a golfer.

### 27.2.6 Keep Different Kinds of Evidence Distinct

A useful bundle separates raw observations, processed measurements, model inputs, generated numerical outputs, and conceptual illustrations. Preserve processing steps from observations to fitted parameters and from parameters to predictions. JSON and NPZ can be useful formats; neither is a requirement of scientific validity. Units, shapes, conventions, access conditions, and interpretation need to be documented whatever format is chosen.

A quantitative plot should expose its plotted values and the calculation or measurement behind them. A schematic needs an accurate caption, accessible explanation, and editable source. A screenshot documents a view; camera settings and a run identifier may make it traceable without turning it into a measurement. Do not manufacture synthetic observations to fill a missing empirical link.

A valid evidence record may say *untested*, *unsupported*, or *provisional*. Such a status should block promotion to a stronger claim, while remaining representable and visible. Missing required fields, invalid schema values, and scientifically unresolved questions are different situations. The pinned release metadata itself retains unresolved claim statuses and open gates.

Preserve failed runs, negative controls, and contradictory results with their conditions. Replacing a failed comparison by newly generated expected hashes can hide the very disagreement the evidence path is meant to expose. An independent implementation that disagrees deserves diagnosis: mismatched definitions, numerical error, shared assumptions, and a genuinely different physical model lead to different conclusions.

### 27.2.7 A Reviewer Exercise and a Durable Record

Choose one numerical figure and write down its claim in physical terms. Identify the plotted quantity, boundary, frame, event, units, and evidence level. Follow the release-pinned links to inputs and generating code. On a suitable machine, run the documented calculation in a recorded environment and retain its output and comparison, including any failure.

Next change one assumption deliberately, keeping the baseline intact. For example, change a modeled inertia while holding the remaining declared inputs fixed. State whether the resulting motion is a prescribed replay or a new forward-dynamics solution: those answer different questions. Compare signed work and outcome measures over the same interval. A changed figure is an exploratory result until its provenance and limitations are recorded.

The following are maintenance requirements for a useful resource, not claims that every current article already satisfies them: provide claim-level links, complete citations, figure alternatives, accessible navigation, and concise execution instructions; distinguish verified checks from recommendations; retain corrections and earlier reviewed versions. Issues and pull requests can preserve these decisions, but their number is not a measure of scientific validity.

Immutable commit links identify the inspected source; an archive and an access plan address preservation. A tag, anchor, or hash alone does not guarantee that a host will keep serving a file. Archive appropriately licensed releases, retain citation metadata, and document access restrictions for participant or licensed data. Assess public URL reachability and deployed-copy identity separately from local build success.

## 27.3 Where the Picture Breaks

The museum catalog preserves a record, but scientific models remain open to revision. Reproducibility can preserve a systematic error just as faithfully as a sound calculation. Its value is that a disagreement can be localized and examined. Independent reasoning and new observations determine whether the mechanical explanation deserves broader trust.

**Go Deeper:** Read the pinned [reviewer workbench](#), [open release qualification](#), [data dictionary](#), and [release-bundle comparison contract](#).

## Chapter 28

# A Practical Synthesis Without a Swing Prescription

### 28.1 A Concrete Picture

A weather map can explain pressure, wind, and uncertainty without telling every person whether to carry an umbrella. The decision depends on destination, tolerance for rain, and changing local conditions. A mechanics model can clarify the swing in the same way without declaring one motion correct for every golfer.

We can now assemble the picture: proximal work moves a linked system; constraints and inertia transmit and redirect effects; geometry changes leverage; two hands create visible and hidden wrench modes; negative torque can retain or stabilize; the shaft stores and returns energy; timing depends on state; robustness exposes tradeoffs; human evidence remains the final gate.

### 28.2 How the Mechanism Works

The useful framework is a sequence of questions.

First, choose the system and outcome. Are we explaining club kinetic energy, clubhead velocity, face/path, joint load, repeatability, or some combination?

Second, reconstruct the state and geometry. Where are the segments, how are they moving, and what elastic or activation states are loaded?

Third, close the instantaneous mechanics. Which forces, moments, constraints, gravity, and velocity terms balance acceleration? Are frames and reference points consistent?

Fourth, close work and energy over time. Which interfaces supply, transfer, store, return, or dissipate energy? Do positive and negative contributions cancel?

Fifth, intervene fairly. Hold state for pointwise attribution, integrate matched branches for persistence, and resimulate full schedules for strategy. Include negative controls.

## The Framework Connects Mechanisms Without Selecting One Technique

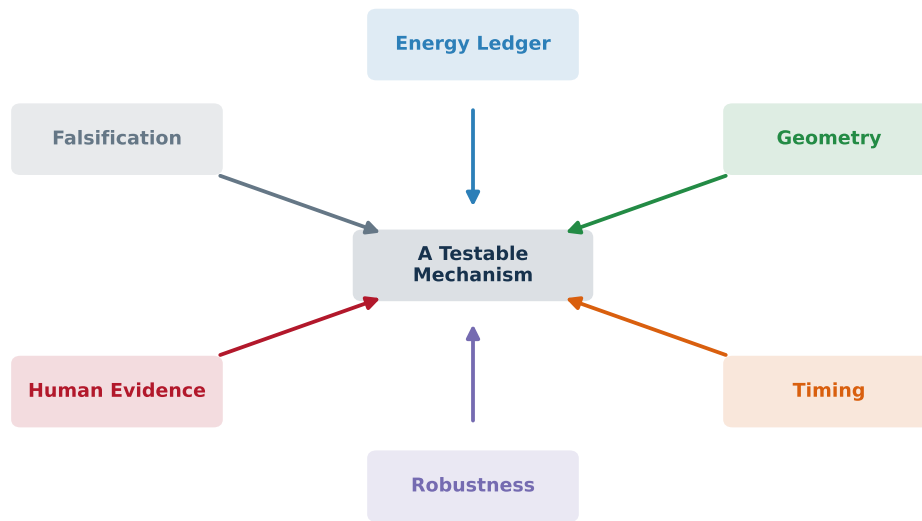


Figure 28.1: A Final Map Connects Energy Supply, Transfer, Storage, Geometry, Timing, Robustness, and Falsification Without Selecting One Technique.

Sixth, climb the model ladder. Does the mechanism survive two hands, compliance, a moving base, spatial motion, independent engines, and uncertainty?

Seventh, design the human test. Measure the variables that distinguish competing stories and preregister what would count against each.

### 28.2.1 What the Current Evidence Supports

The reduced models support a real mechanical possibility: club acceleration and hand-path work can be strongly influenced by interaction and state-dependent terms even with selected distal torque removed. Two-hand geometry can create a late negative club couple. Flexibility can modify timing while obeying energy closure. Arm- and wrist-dominant allocations can produce the same club task with different internal demands. Preload continuity helps only under a declared transmission family.

### 28.2.2 What Remains Open

The work does not establish a universal human technique, scapular mechanism, muscle preactivation strategy, injury benefit, or calibrated equipment effect. It does not identify one best allocation. The held-out model ensemble explicitly rejects a universal optimum across its outcomes.

### 28.2.3 Better Practical Questions

Instead of “should I hold lag?” ask which state variable and outcome the cue is supposed to change. Instead of “are the wrists passive?” ask whether direct wrist moment, impedance, and force-generated couple were measured separately. Instead of “is slack good?” ask which gap, stiffness, compliance, or

activation variable was observed. Instead of “did the hips transfer speed?” ask which boundary power and energy balance supports that statement.

These questions do not make instruction impossible. They make claims testable and participant-specific.

#### 28.2.4 A Reader’s Next Experiment

Use the interactive workbench to compare clock and state triggers, remove a selected torque at one state, reverse a two-hand moment arm, or change shaft stiffness. Predict the result before running it. Then inspect forces, powers, work, energy closure, and falsifiers rather than only final speed.

**Model Result:** The project delivers a connected, openly reviewable set of mechanism tests across a common-observable ladder.

**Human Evidence:** The literature motivates and bounds the questions but does not yet validate the complete strategy mechanism.

**Hypothesis:** Some golfers may benefit from strategies that maintain useful proximal drive, manage distal resistance and preload, and permit state-dependent late transfer without force-transmission gaps. The opposite allocation remains a serious competing hypothesis.

**Practical Interpretation:** The framework is not a recipe. It is a way to ask better questions of a motion, a model, a sensor, or a coaching claim.

#### 28.2.5 What the Framework Now Says

The swing is a coupled mechanical system. Proximal work, moving geometry, constraint reactions, direct distal control, bilateral hand allocation, shaft deformation, and ground interaction overlap. Clubhead speed is an outcome of their integrated trajectory. No single peak, torque sign, or body-part label contains the whole explanation.

The proximal-distal description remains useful when it means changing emphasis and energy pathways in a continuously linked system. It becomes misleading when turned into a strict relay or a universal timing script. The framework replaces that script with state, force, wrench, power, work, and counterfactual tests.

#### 28.2.6 What the Models Establish

Verified reduced models demonstrate that interaction forces can carry and accelerate a distal link without direct distal drive; geometry controls the sign and leverage of that contribution; negative torque or contact couple can coexist with a positive later outcome; and multiple hand allocations can yield the same net club task. Flexible, moving-base, and spatial rungs test whether those mechanisms survive added physics.

These are model results under declared assumptions. Their strongest value is the observable predictions they generate. They do not establish a unique muscle strategy or prove that golfers exploit every simulated pathway.

### **28.2.7 The Two Strategy Extremes**

One family emphasizes continuous arm/shoulder-girdle drive with wrist resistance from backswing through early downswing. Its proposed advantage is a loaded transition state with less transmission delay. The opposite family emphasizes greater wrist-created backswing club moment, arm resistance, then arm drive with wrist opposition in downswing. Its possible advantage is phase-specific use of different mechanical and biological resources.

Both can maintain contact and stiffness; both can create or avoid a measured force gap depending on implementation. Real strategies can lie between them. The meaningful comparison matches club task and measures bilateral wrench, stiffness, activation, work, load, and robustness.

### **28.2.8 A Decision Tree for the Researcher**

If the question concerns linked inertia, begin with the double pendulum. If it concerns push–pull allocation, add two hands. If delay or “kick” is central, add shaft and transmission states. If ground or handle path is central, release the base. If out-of-plane couple matters, use spatial dynamics. If muscle claims are made, add biological measurements and a qualified musculoskeletal rung.

At each step, retain the smallest model that answers the question, common observables, negative controls, and closure tests. Add complexity only when a residual or hypothesis requires it.

### **28.2.9 A Decision Tree for the Evidence**

If only motion is measured, report kinematics. With external wrenches and inertial assumptions, report net dynamics. With bilateral grip sensing, test hand allocation and contact couple. With shaft strain, test elastic pathways. With EMG and perturbation, constrain activation timing and effective stiffness. Do not climb the claim ladder without the corresponding evidence.

When evidence disagrees, localize the disagreement. A failed anatomical mapping does not automatically falsify the contact mechanics; a failed wrench prediction does. A model fitting speed but failing reactions needs revision.

### **28.2.10 What This Means for Creating Clubhead Speed**

Within the mechanical framework, favorable conditions include sufficient whole- system work, a hand path and geometry that permit useful late club acceleration, control timing that avoids premature or incomplete release, and allocation that remains robust to execution error. An opposing moment can be useful if it shapes state and later power rather than simply wasting energy.

Those conditions are evaluation criteria, not body instructions. A specific golfer may meet them through different coordination. Speed must be considered with face/path, impact, internal load, and repeatability.

### **28.2.11 What “Manage the Negative Couple” Can Mean**

Management is not necessarily conscious application of a negative wrist torque. It may mean arranging geometry and proximal motion so contact reactions have a favorable sign, maintaining enough engagement to avoid unwanted delay, and allowing direct control to regulate rather than fight the coupled motion. It may also mean avoiding excessive retention that leaves too little time.

The phrase should remain tied to measurable couple, power, timing, and outcomes. Without those, it is only a cue. The project does not recommend a cue from the current evidence.

### **28.2.12 A Practical Analysis Workflow**

Begin with synchronized, qualified data. Reconstruct state and uncertainty. Close whole-system and interface ledgers. Plot force vectors and bilateral wrenches in declared frames. Run pointwise decompositions and matched-state forward branches. Compare model rungs and independent engines. Propagate parameter uncertainty and evaluate held-out participants.

Then classify each conclusion: mechanics identity, verified model result, human observation, hypothesis, or practical interpretation. Link it to the evidence graph. This workflow makes a sophisticated modeling product usable without pretending it is already a coaching authority.

### **28.2.13 Near-Term Completion Priorities**

The highest-value next work is a qualified synchronized bilateral dataset, a spatial whole-body model with measured base and shaft state, participant-held-out prediction, and explicit arm/wrist allocation experiments. The workbench should expose counterfactuals, uncertainty, and failure regions with shared canonical engines. Documentation should update automatically from evidence manifests.

These priorities deepen falsifiability more than adding another attractive nominal animation. They also invite independent replication because the observables and contracts are software-neutral.

### **28.2.14 Questions a Reader Should Keep**

What system and boundary are used? Which quantities were measured? Which were inferred or simulated? Does the ledger close? Is the comparison matched by state or task? Does geometry control the predicted sign? Can another allocation explain the same motion? Does the result survive uncertainty, higher model rungs, and held-out people? What finding would make the authors change their minds?

If those questions remain visible, the framework can grow without becoming a closed doctrine. It becomes an open map of mechanisms, evidence, and remaining uncertainty.

### **28.2.15 Final Boundary**

Nothing in this synthesis establishes a universally superior swing, a prescribed scapular action, or a safe wrist-loading pattern. The research supports a more precise way to ask how transfer can occur and how competing explanations can be tested. Practical use requires qualified individual evidence and professional judgment appropriate to performance and health.

### **28.2.16 Reader Lab: Evaluate a New Claim**

Suppose a new analysis says, “Trail-hand pull creates passive wrist lag and more speed.” Translate it into testable pieces. Define trail-hand force in a frame; define lag as state; define passive channel; identify power and work; specify the comparison and outcome. Ask whether arm, wrist, shaft, and geometry alternatives can reproduce it.

Then choose a model rung and negative controls. Predict bilateral wrench and shaft response. Match state for causal removal and club task for allocation comparison. Propagate uncertainty. Finally define a human measurement and a failure condition. Only after these steps should practical interpretation be considered.

This workflow does not drain intuition from the question. It converts intuition into an argument that another group can test.

### 28.2.17 The Desired End State of the Project

The complete product is a linked set of readable chapters, technical derivations, canonical simulation libraries, interactive experiments, qualified evidence, and reproducible publication. A newcomer can understand why a force vector matters; a specialist can audit equations; an experimentalist can see the next discriminating measurement; and an independent developer can implement the contracts elsewhere.

Completion does not mean the biological hypothesis is proven. It means the framework is sufficiently explicit, broad, and falsifiable that future data can change it cleanly. That is a more durable scientific objective than presenting one software package or swing strategy as finished truth.

## 28.3 Where the Picture Breaks

A synthesis map can create the illusion that every arrow has equal evidence. They do not. Structural mechanics, simulated model results, and human observations occupy different evidentiary levels. The claim labels must travel with any excerpt from this book.

The final danger is using scientific caution as vague neutrality. The models do make conditional predictions. They should be tested decisively, updated when wrong, and retained when they survive—not promoted prematurely and not dismissed because they are reduced.

**Go Deeper:** Read the monograph’s [discussion and conclusions](#), open the [interactive workbench](#), and use the [reviewer path](#) to challenge any claim that matters to you.

## Chapter 29

# Walk Through a Whole Swing Without Losing the Ledger

### 29.1 A Concrete Picture

Imagine standing beside a complicated relay race. The runners do not merely pass a baton from one hand to the next. They are connected by springs, they can push on the track, several of them can hold the baton at once, and the finish line is moving. If we watch only the baton, the race can look beautifully simple: one runner speeds up, then another, and finally the baton crosses the line. If we want to explain *why* it crossed quickly, however, we need a ledger. Who did work? Which connection redirected a force? Where was energy stored? Which apparent contribution was just a change of coordinates? Which conclusion survives when we rerun the race with one action removed?

A golf swing is not literally that relay, but the picture is useful because it forces us to keep several questions separate. The clubhead can gain speed while the hands slow. A wrist moment can be negative while the clubhead continues to accelerate. A force can be large while doing little work because it is nearly perpendicular to the point velocity. A shaft can store energy during one interval and return it during another. Two hands can create the same net club wrench with very different internal loads. None of those statements is a contradiction. These are mechanical possibilities under specified states and loads, not measurements of the imagined golfer. They become confusing when the ledger columns are mixed.

This chapter follows one imagined swing from preparation to impact. It does not claim that this is the unique swing of a skilled player. Instead, it shows how to carry the book's concepts through one continuous story without quietly changing the system boundary, reference point, model, or meaning of a word.

### 29.2 How the Mechanism Works

#### 29.2.1 Before Motion: Write Down the Accounting Rules

Our imagined observer begins before the club moves. The chosen system is the golfer plus club when asking about external work from the ground. It becomes the club alone when asking what the hands

## The Framework Connects Mechanisms Without Selecting One Technique

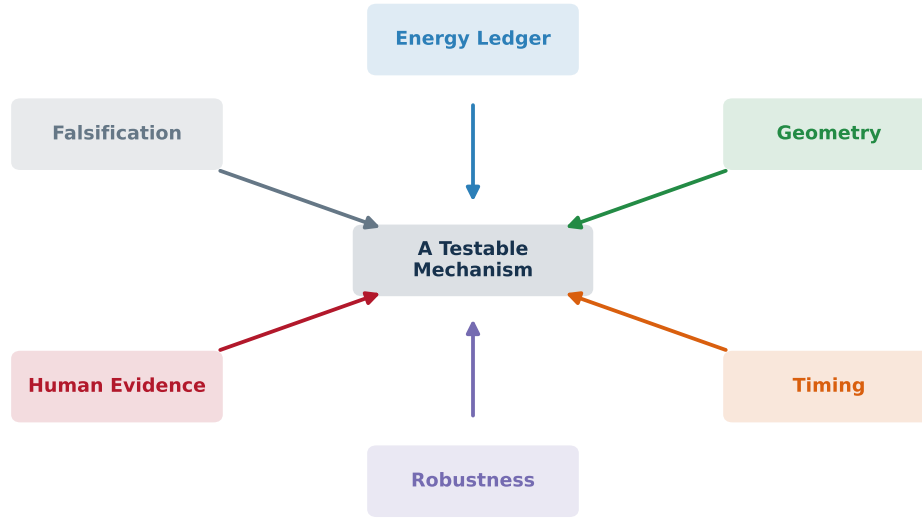


Figure 29.1: A Synthesis Map Connects State, Geometry, Forces, Power, Counterfactuals, Evidence, and Falsifiers Without Collapsing Them Into One Claim.

do to the club. It becomes the complete flexible club when asking about external grip work, but it may be divided into proximal and distal club segments when asking where shaft strain energy lives. Those are different ledgers. Changing between them is allowed; changing without saying so is not.

The observer also declares a laboratory frame, a club-attached frame, the grip center as the main wrench reference point, and positive angular directions. This sounds fussy until a force is transported from one hand to the grip center. The force remains the same physical interaction, but its reported moment changes with the reference point. For a rigid grip region in the same observer frame, the paired reduction is  $P = \mathbf{F} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \omega$ . For  $\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A$ , use

$$\begin{aligned}\mathbf{v}_B &= \mathbf{v}_A + \omega \times \mathbf{r}, \\ \mathbf{M}_B &= \mathbf{M}_A - \mathbf{r} \times \mathbf{F}.\end{aligned}$$

The two evaluations give the same total power, although their force and moment parts change. This transports the same physical velocity field; changing to a moving observer is a different operation and generally changes power. A deforming grip cannot be reduced to one rigid twist without retaining the missing deformation power. Its individual contact forces need their actual material-point velocities. A persuasive diagram cannot repair inconsistent bookkeeping.

Finally, the observer lists the state variables available in the chosen model: segment angles and angular velocities for the simplest pendulum; pelvis and trunk motion for a moving-base model; shaft deflection and deflection rate for a flexible club; activation and series-force histories for a muscle-driven model. With specified future inputs or input laws, the retained state determines the model evolution. If an internal state is omitted, the same visible pose can hide different futures. Activation is a state, not an

energy quantity; an activation history does not itself measure elastic storage or metabolic cost. The state list determines which timing questions the model can answer.

**Practical Interpretation:** Before asking which body part “caused” speed, ask what objects are inside the accounting boundary, what instant or interval is being studied, and which stored states are being carried forward.

### 29.2.2 Preparation: Motion Creates a State, Not Merely a Position

During the imagined preparation, the golfer moves the club away from the ball. The visible result is a backswing pose, but the mechanical result is richer. Segments have velocities, tissues and the shaft may carry strain, muscles have activation histories, the two hand contacts carry some distribution of force, and the ground has exchanged impulse with the body. Two golfers can arrive at nearly the same photograph with different values in those hidden columns.

This is why “the top” is not a complete initial condition. A model initialized from a still pose and zero velocity has erased the route into that pose. Such a model can still answer a deliberately posed question, but it cannot be used as evidence that preparation history is irrelevant. The continuous-preparation experiments in the technical work exist precisely to avoid resetting an internal transmission state at the command transition.

Consider the two allocation extremes developed earlier. In one, a proximal push–pull contribution remains directionally loaded while the wrists provide an opposing contribution. In the other, the modeled roles reverse across the transition. Both command programs can match the same net desired moment before and after the change while taking different routes through internal transmission states. That does not establish the same transmitted moment or club trajectory throughout transition. The [archived allocation study](#) reports an advantage for persistent direction in nine positive-dead-zone cases, and equivalence of the chosen net error-impulse metric in three zero-width cases. The equivalence threshold is  $10^{-10}$  N m s within that numerical study.

Removing a dead zone does not remove finite force-development time or make every internal trajectory equal. Two linear channels with the same time constant and matching summed commands can have the same summed output while their individual forces reverse at different times. The result is conditional on the transmission, initialization and metric. It neither identifies a human wrist dead zone nor selects a universal wrist or scapular strategy.

### 29.2.3 Transition: Replace the Magic Instant With a Neighborhood

The swing reverses direction over an interval, not at a mathematical point that contains the whole explanation. Near transition, some segment angular velocities approach zero while accelerations and internal forces need not. The club may still be moving one way as proximal segments begin accelerating the other way. A graph aligned to a named event can make this overlap visible, but the event label does not create the mechanism.

Our observer therefore records a transition neighborhood. The ledger contains configuration, velocity, applied generalized moments, constraint reactions, hand wrenches, shaft state, ground-reaction wrench, and the power of each declared interaction. If the observer wants to discuss preactivation, activation must be estimated from declared measurements or represented by a biological model; an EMG signal is not a direct measurement of every muscle force. If the observer wants to discuss slack, the word is

replaced by a quantity such as contact gap, tangent stiffness, low force, series strain, or force-development delay.

This disciplined language also makes opposing hypotheses stronger. Continuous loading might improve immediate transmission and perturbation resistance. A controlled unloading or role reversal might reduce internal loading, permit useful compliance, reshape the club path, or improve a participant's accuracy. The experiment must price both benefits and costs rather than declaring one side “connected” and the other “slack.”

#### 29.2.4 Early Downswing: The Ground and Moving Base Join the Story

As the downswing begins, the golfer pushes on the ground and the ground pushes back. This ground wrench contributes to the golfer-plus-club momentum balance, together with gravity and any other external loads. Integrate all external forces for linear momentum and all external moments about a fixed inertial point for angular momentum. A moving angular-momentum reference requires the appropriate transport terms.

Impulse and work answer different questions. At an ideal stationary no-slip contact, material-point velocity is zero, so the contact force does zero work despite supplying impulse. A synthetic 100 N force maintained for 0.1 s has impulse 10 N s; at zero contact velocity its work is zero. Multiplying that force by center-of-mass velocity gives a different quantity used in center-of-mass kinetic accounting, not the contact's boundary power. The moving center of pressure is also not a material point whose migration speed can simply be multiplied by the ground force.

Real shoe/turf deformation, slip or a moving support can change this idealization. Their power requires the actual distributed tractions and contact velocities or a justified equivalent model. The ground can make a swing mechanically possible without serving as its source of mechanical energy. Muscle conversion of chemical energy and changes in gravitational or elastic storage remain separate entries.

Pelvis and trunk motion make the arm base move. In a prescribed-base pendulum, that motion is supplied as an input trajectory; the model does not pay the energetic or dynamic cost of producing it. In a finite-mass moving-base model, the base responds to applied forces, restoring terms, damping, and connected segment reactions. The second model can test whether a mechanism survives when the base is allowed to react. It is not automatically more truthful in every parameter, but it closes a question that the prescribed model leaves open.

The familiar proximal-to-distal pattern may now appear in segment speeds. Pelvis speed peaks, then trunk, arms, and club. This is a useful observation, not yet a causal decomposition. Overlapping accelerations, interaction forces, joint moments, and changing geometry can produce the ordered peaks. A peak-speed sequence alone cannot tell us which term did work or whether deliberately slowing one segment would improve the final outcome.

**Human Evidence:** Tinmark and colleagues studied 45 elite golfers and reported proximal-to-distal timing of peak angular-speed magnitudes for the pelvis, upper torso and hands across their partial/full-shot conditions (Tinmark et al. 2010). That result does not establish a universal pelvis–trunk–arm–club sequence or its cause. Marsan and colleagues compared seven angular-velocity definitions in 13 golfers' driver swings and found that the identified sequence depended strongly on the selected component (Marsan et al. 2019). Define the segments, measured quantity, processing and event rule before

comparing peak times; observations motivate mechanism tests rather than prescribe one control strategy.

### 29.2.5 Mid-Downswing: Geometry Changes What the Same Force Can Do

The arms and club now form a changing geometric machine. A force at the hands has a moment about a joint according to its moment arm. The same force vector can create a large turning effect at one configuration and almost none at another. Likewise, the same joint moment can produce different accelerations because the mass matrix and dynamic coupling terms change with configuration.

Suppose a hand force points mostly along the instantaneous hand path. Its power at that point may be large because force and velocity align. Suppose another force is almost perpendicular to the path. Its direct power is smaller, yet it may redirect the path, maintain a grip constraint, or create a couple with the other hand. Calling the first force “useful” and the second “wasted” would miss the constraint and geometry columns of the ledger.

The double pendulum makes the changing leverage visible. When the arm–club angle is large, proximal motion can create a different induced acceleration of the club than when the segments nearly align. The model’s off-diagonal inertial terms are not a mysterious extra force; they express how coordinates in a linked system share kinetic energy and acceleration. A counterfactual can then ask how the club acceleration changes at the same state when one declared control moment is set to zero. That pointwise result is not the same as rerunning the rest of the swing without the moment.

### 29.2.6 Delivery: Two Hands Can Produce Force and Couple Simultaneously

Near delivery, reducing the hands to a single point hides an important degree of freedom. The resultant hand force contributes to center-of-mass acceleration together with the other external forces. Its moment about the club center of mass also matters. For hand separation  $\mathbf{d} = \mathbf{r}_R - \mathbf{r}_L$  and the midpoint  $G$  of the two hand contacts (not the club center of mass), the force-generated moment is

$$\mathbf{M}_G^{\text{force}} = \frac{1}{2} \mathbf{d} \times (\mathbf{F}_R - \mathbf{F}_L).$$

Equal and opposite forces form a pure couple, independent of reduction point. For a nonzero resultant, the reduced moment is point-dependent. To obtain the moment about the club center of mass  $C$ , add  $(\mathbf{r}_G - \mathbf{r}_C) \times (\mathbf{F}_R + \mathbf{F}_L)$  to  $\mathbf{M}_G^{\text{force}}$ . Direct applied moments add to this wrench; they are not hidden inside the force moment. Many hand-load combinations can yield the same rigid-club wrench, while internal loads and compliant-contact power can differ.

This is the setting in which a negative grip-moment component may coexist with increasing clubhead speed. “Negative” refers to an oriented axis, and spatial moment power is the dot product  $\mathbf{M} \cdot \boldsymbol{\omega}$ , not one component’s sign. On a rigid club, hand power is the sum of the force and moment terms at one consistent point. At an anatomical rotational joint, the paired moment contribution instead uses the relative angular velocity of its two bodies. Neither net quantity uniquely identifies individual muscle work.

Gravity and other external loads can supply power while one component absorbs it. Elastic release can convert stored energy into kinetic energy within a flexible club; already acquired kinetic energy is part

of the state, not an additional power source. These distinctions explain coexistence without showing that an opposing moment improves a particular task. That requires an intervention and an objective.

**Model Result:** In the [archived forward two-hand study](#), a branch inherits the baseline state at 0.20 s and sets its applied ideal joint torques to zero. The force-generated moment about the study’s declared club reference remains negative at all stored samples over the next 0.05 s. Gravity, fixed-shoulder constraints and ideal bilateral hand contacts remain. This is archived model evidence, not a new run, a finding that human muscles are inactive, or identification of scapular retraction.

The archive separately reports near-machine-level wrench-power identities and nonzero integration work-energy residuals that decrease under step refinement. An algebraic identity closing tightly does not make the trajectory exact. Changing geometry, contact laws or actuator dynamics requires another scoped test; removing neural excitation with activation retained is a different intervention from removing ideal applied torque.

### 29.2.7 The Shaft: A Short Memory With Direction and Timing

The flexible shaft adds internal state. When it bends, it stores strain energy; when it straightens, it can return some of that energy. Damping dissipates a portion, and the deformation can alter clubhead orientation as well as speed. It is therefore misleading to describe the shaft only as a catapult that adds free energy. Its return comes from energy previously supplied to the club-body system, and its practical effect depends on timing, mode shape, damping, geometry, and impact conditions.

In our ledger, strain energy remains stored energy even when the shaft is partitioned. Choose material subdomains and a named cut section: each part has its own kinetic and strain energy, while traction power across the cut is an exchange term. Storage is measured in joules and power in watts; one does not turn into the other merely because the boundary moves on a diagram. A spatially moving cut additionally needs transport accounting.

For the whole flexible club, one useful convention is

$$\frac{d}{dt}(K + U_{\text{shaft}} + V_g) = P_{\text{hands}} - P_{\text{diss}},$$

with no other external power, gravity included once through  $V_g$ , and all retained translational, rotational and deformation kinetic energy in  $K$ . If  $U_{\text{shaft}}$  decreases, kinetic energy may rise without an extra external shaft-return term. The distribution of that rise among head, shaft and proximal parts depends on their interface powers; release need not all help the head. The shaft also makes two visually similar delivery poses dynamically different because bend and bend-rate states can differ.

This gives “late release” a more careful meaning. A delayed visible angle does not, by itself, reveal whether the shaft is storing energy, whether wrists are doing positive or negative work, or whether a constraint reaction is creating the observed acceleration. Those mechanisms need their own measurements and model variables.

### 29.2.8 Impact: Do Not Let One Number Swallow the Task

Clubhead speed immediately before impact is important, but it is not the only task variable. Face orientation, path, strike location, dynamic loft, shaft state, and variability all affect the ball outcome.

A simulated strategy that adds a small amount of peak speed while greatly increasing face or strike variability may not be better for the actual task.

The observer therefore reports a task vector rather than a victory number. That vector can include speed, path, face, strike, peak hand force, joint-load proxies, energy cost, and robustness to timing or parameter perturbations. The choice of weights is an objective, not a fact of mechanics. Different players and tasks can rationally occupy different points on the tradeoff surface.

At impact, the ledger also changes because the ball enters the interaction. The very short collision requires contact mechanics beyond the pre-impact swing model. The pre-impact state supplies initial conditions for that event; it does not by itself determine every launch outcome without a validated club–ball contact model.

### 29.2.9 Rewind the Swing With Counterfactual Scissors

Now the observer rewinds. First comes a pointwise decomposition: at selected states, set one declared control input to zero and recompute acceleration or constraint force without advancing time. This is a finite same-state comparison, not necessarily a derivative. For the illustrative nonlinear input map  $a(u) = u^2$ , changing  $u = 2$  to zero changes  $a$  by  $-4$ , while the local derivative predicts  $2u\Delta u = -8$ . In a fixed-mode control-affine model, the input derivative gives the finite change exactly; contact switches, saturation and actuator dynamics can invalidate that simplification. Recompute constraint reactions consistently.

Next comes a forward intervention: declare which command or policy changes, retain compatible configuration, velocity and internal state, and integrate. Input removal at each baseline state does not recreate this divergent path. Compare a declared diagnostic time or each branch’s own impact event, and report a branch that misses the ball. These answer different questions about consequences over time.

The observer distinguishes model controls from numerical verification. With bounded hand forces, collapsing separation removes the force moment about the common contact point; it does not remove a direct free moment or the resultant’s moment about a different point. Demanding a fixed nonzero couple as separation shrinks instead requires unbounded forces. Any geometry intervention must initialize a compatible state and check constraint rank.

Consistent coordinate representation and rigid-point transport should preserve the same observer’s total power. Timestep refinement checks numerical stability and convergence of the particular outcome; track constraint, equation and work-energy residuals with units and declared tolerances. A change under a real intervention can be the predicted effect, whereas failure of a representation identity indicates an implementation or convention problem. None of these checks alone validates the mechanism in a golfer.

Finally, the observer moves up the model ladder. Does the double-pendulum result survive a two-hand constrained model? Does it survive a moving base and flexible shaft? Does it survive spatial motion and parameter uncertainty? Do measured bilateral wrenches, kinematics, shaft strain, ground reactions, and activation timing show the predicted relationships in held-out participants? Adding complexity does not guarantee reduced uncertainty; it can introduce new unidentified parameters. Advance when a new tier answers the declared question and survives its verification and measurement tests. No earlier rung licenses skipping evidence needed for the eventual claim.

### 29.2.10 Read the Finished Ledger in Plain Language

At the end, the observer can say something careful and still useful. The linked geometry allows proximal motion, interaction forces, hand-force couples, direct moments, and shaft state to influence distal motion in overlapping ways. Ordered speed peaks are compatible with transfer but do not uniquely identify it. Negative torque can coexist with a positive distal outcome because sign, power, system boundary, and time interval are different questions. Continuous loading and role reversal are separable hypotheses whose predicted differences depend on stiffness, gaps, activation history, and the task being optimized. The present models make those statements testable; they do not select a universal human technique.

That conclusion is less dramatic than “energy flowed down the chain,” but it is far more informative. It tells us which quantities to measure, which model states not to reset, which counterfactuals to run, and what evidence could make the explanation fail.

### 29.2.11 Reader Lab: Build a One-Page Ledger

Choose one interval—transition, mid-downswing, delivery, or impact approach. Draw six columns labeled state, external wrench, internal exchange, stored energy, task outcome, and falsifier. Fill each column with quantities rather than body-part stories. If you write “hips transferred energy,” replace it with a measurable term: pelvis-to-trunk joint power, interaction work over the interval, or a counterfactual change in club kinetic energy. If you write “wrists resisted,” specify joint moment, angular velocity, and the sign of their product. If you write “no slack,” name the gap, stiffness, force, or activation variable.

Then ask what your one-page ledger cannot establish. That final blank space is not a weakness. It is the beginning of the next experiment.

### 29.2.12 Chapter Checkpoint

The whole-swing story remains coherent only when the accounting rules stay visible. Preparation creates state. Transition is an interval. The ground contributes impulse; its boundary work depends on actual contact motion. Geometry changes leverage. Two hands create both resultant force and couple. The shaft stores and returns energy. Impact adds a new contact problem. Counterfactuals separate local contributions from forward consequences. Human evidence is required before a model allocation is called a biological strategy.

## 29.3 Where the Picture Breaks

The relay picture suggests discrete handoffs, whereas a real swing contains overlapping power, simultaneous constraints, distributed tissues, and feedback. The walkthrough also combines lessons from several model tiers; no single executed model currently contains every element in one participant-calibrated simulation. A coherent ledger is not proof that every entry is experimentally accurate. Parameters, soft-tissue motion, muscle forces, contact behavior, and individual objectives remain uncertain. The synthesis earns a research program, not a swing prescription.

**Go Deeper:** Compare the complete [scientific monograph](#), the machine-readable [transmission robustness evidence](#), and the [reviewer workbench](#).

## Chapter 30

# A Curious Golfer and a Skeptical Reviewer Ask the Hard Questions

### 30.1 A Concrete Picture

Picture two readers at the same table. One is a curious golfer. This reader wants the mechanics to connect with sensations and visible motion: Why does the club seem to accelerate when the hands are not trying to fling it? Why might a negative torque help? Is “keeping tension” meaningful? The other is a skeptical reviewer. This reader keeps asking what was measured, what was assumed, which model produced the number, and what result would make the authors change their minds.

They are not opponents. The curious golfer prevents the explanation from hiding behind notation. The reviewer prevents the explanation from becoming a confident story that outruns the evidence. A mature open research resource needs both voices.

The conversation below is deliberately sustained. Each question is followed through mechanics, evidence, and limitations rather than answered with a slogan. The point is not to memorize a preferred response. It is to learn how to move from an intuitive question to a claim that can be checked.

The figure describes evidence provenance. The dialogue adds another obligation: definitions, assumptions, uncertainty, and rival explanations must be checked at every step. A reproducible result can still answer the wrong question.

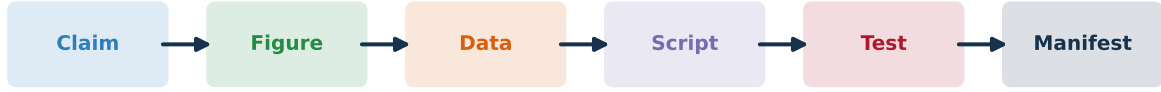
### 30.2 How the Mechanism Works

#### 30.2.1 “Is Proximal-to-Distal Transfer Just a Fancy Name for Sequencing?”

The golfer begins with the visible pattern: pelvis, trunk, arms, and club often reach peak speed in that order. The reviewer agrees that this is a useful kinematic description, then separates it from the mechanism. A sequence tells us *when* selected maxima occur. Transfer asks how forces, moments, constraints, work, and stored energy change the motion of a declared subsystem.

Two swings can show similar peak ordering while differing in joint power, ground-reaction impulse, hand forces, shaft deformation, and robustness. Two other swings can differ in peak timing yet reach

## The Publication Is an Entry Point, Not the Evidence Boundary



A reviewer can walk backward from prose to reproducible evidence.

Figure 30.1: A Reviewer Follows the Claim Through Its Figure, Data, Script, Test, and Manifest.

similar club states at impact. The sequence is therefore one observable among many. It becomes mechanically explanatory only when connected to a model that predicts the measured forces and energy changes and survives counterfactual tests.

The golfer’s useful takeaway is not that sequencing is unimportant. It is that “good sequence” should be unpacked. Does it mean a reproducible ordering of segment speeds, a particular phase relation, favorable hand-path geometry, low peak loading, or robust impact conditions? Each meaning suggests different measurements.

### 30.2.2 “Does Energy Literally Travel From the Hips Into the Club?”

The everyday phrase sounds like a parcel moving through a delivery network. Mechanical energy is a scalar accounting quantity, not a material substance. When we isolate the club, hand forces and moments can do work on it. When we isolate the trunk, joint interactions and external forces appear at its boundary. With a flexible shaft, strain energy can be stored internally and later returned. The numerical transfers depend on where the system boundary is drawn.

The reviewer asks for a power balance. For a rigid club, reduce all hand forces and directly applied moments to a resultant force  $\mathbf{F}$  and moment  $\mathbf{M}_A$  about a declared body point  $A$ . Their power on the club is

$$P_{h \rightarrow c} = \mathbf{F} \cdot \mathbf{v}_A + \mathbf{M}_A \cdot \boldsymbol{\omega}. \quad (30.1)$$

All vectors use the same axes and velocities are measured relative to the declared laboratory frame. The moment includes force moment arms about  $A$ ; one must not add those moments again as a second interaction. This is the wrench–velocity pairing explained in [Lynch and Park’s treatment of wrenches](#).

For a club subsystem with gravitational potential  $V_g$  and elastic storage  $U_s$ , a useful ledger is

$$\frac{d}{dt}(T + V_g + U_s) = P_{h \rightarrow c} + P_{\text{other}} - D, \quad D \geq 0. \quad (30.2)$$

Here  $T$  includes the modeled motion of the club's mass,  $D$  is dissipated mechanical power, and  $P_{\text{other}}$  includes external interactions other than the hands and gravity. Counting gravity through  $V_g$  means not adding gravitational power again on the right. A flexible club generally requires power at each moving contact and deformation degrees of freedom; one rigid-body twist cannot represent every contact velocity. Shaft energy return is a decrease in internal storage, not another external energy supply. Integrate this ledger over an interval and report its residual separately. For an impact, use the corresponding impulse/work or resolved-contact balance rather than assume that smooth-time power captures an instantaneous reset.

The golfer can still use a simple image: energy changes hands through boundary work and internal storage. The image stays honest only if we remember that several exchanges overlap and that energy is not tagged with the name of the body part that first supplied it.

### 30.2.3 “If the Hands Slow Down, How Can the Clubhead Speed Up?”

Speed belongs to a point or segment, not to the whole chain at once. The hands can reduce their tangential speed while the club rotates faster about them. Changing geometry can convert the motion of the proximal endpoint into angular motion of the distal segment. Constraint forces and joint moments determine how that conversion unfolds.

A skater analogy helps but is incomplete. Pulling mass inward changes rotational inertia in a nearly isolated system; a golf swing is externally forced, linked, and actively controlled. The full explanation requires the arm-club geometry, mass matrix, applied moments, constraint reactions, and hand-path motion.

The reviewer therefore rejects the statement “slowing the hands causes clubhead speed” unless a counterfactual defines the intervention. Were the hands slowed by removing force, applying an opposing force, changing the path, or changing the entire upstream motion? Those interventions produce different futures. Observed correlation between hand deceleration and club acceleration does not select one of them.

### 30.2.4 “How Can Negative Torque Possibly Help?”

Torque sign is defined about an oriented axis. In a planar model, a negative moment acting on a positively rotating club contributes negative power  $\tau\omega$ . In three dimensions, use  $\mathbf{M} \cdot \boldsymbol{\omega}$ ; one negative component does not determine the full dot product. A joint's net moment instead pairs with its relative angular velocity, or with the matching generalized speed. These are different boundaries and different power accounts.

Momentum can sustain motion; it does not by itself cause acceleration. Acceleration follows from the complete force and moment balance, including the linked system's geometry and current velocity. For a constructed planar rotor with fixed inertia  $I = 0.1 \text{ kg m}^2$  and  $\omega = 4 \text{ rad/s}$ , simultaneous moments of  $-2$  and  $+5 \text{ N m}$  give  $\dot{\omega} = (-2 + 5)/0.1 = 30 \text{ rad/s}^2$ . The opposing moment removes  $8 \text{ W}$ , the other supplies  $20 \text{ W}$ , and rotational kinetic energy increases at  $12 \text{ W}$ . The acceleration is caused by the net

moment. This example explains how an opposing contribution can coexist with speeding up, not why it is advantageous.

For a translating, rotating, deforming club, increasing total kinetic energy does not guarantee increasing clubhead speed or increasing angular speed. Those quantities require their own kinematics. Negative handle-moment power can coexist with positive hand-force power or release of previously stored strain energy, but the complete subsystem ledger decides the net result.

Negative torque may regulate release, alter face or path, manage shaft state, or reduce an overshoot. None of those possibilities makes negative torque automatically desirable. A negative moment applied too early, too strongly, or in the wrong configuration can reduce speed or disrupt the task.

The reviewer requests a signed plot of moment, angular velocity, moment power, resultant-force power, and net energy change. A plot of torque alone cannot support an energy claim. The golfer gains a better question: “What job is this negative moment doing during this interval, and what changes when it is removed in a matched counterfactual?”

### 30.2.5 “Is the Late Negative Couple Passive?”

The [archived forward two-arm study](#) reports a specific test: at 0.20 s, all six applied joint torques are set to zero and a new trajectory starts from exactly the achieved position and velocity. The force-generated moment about the declared club center remains negative for the tested 0.05 s. This is a forward branch, not a set of isolated zero-command calculations along the original motion.

It demonstrates feasibility for that planar rigid model, with fixed shoulders, gravity, and ideal bilateral hand constraints. The constraints enforce grip without a human contact-capacity model. The archived timestep and projection checks support a numerical model result; they do not establish a physiological strategy. Persistence also depends on the branch state: the 0.18 s cut starts with a nonnegative moment and becomes negative later, so the archive reports zero *immediate* negative persistence for that cut.

That does not mean the golfer’s body is passive. Muscles can remain active to maintain posture, hand contact, joint stability, and the state from which the constraint reaction emerges. Nor does it identify which muscles created the earlier state. “Passive mechanism” and “inactive person” are different claims.

The reviewer also demands a precise geometric control. The archive sets both grip offsets to zero at the club center, recomputes a compatible state, and reports zero force-generated moment there. More generally, shrinking the separation of two bounded forces eliminates their differential couple. It does not eliminate a directly applied free moment, nor the moment of a resultant force about a different reference point. Maintaining a fixed force-generated couple while its lever arm tends to zero would require unbounded forces.

### 30.2.6 “Do the Scapulae Create the Push–Pull Couple?”

Scapular protraction, retraction, rotation, and tilt change the shoulder socket and the geometry from which the arms act. They are plausible contributors to a bilateral push–pull pattern. Establishing that contribution requires a prediction linking measured scapular motion to the bilateral hand wrench; an anatomical possibility alone does not establish its importance in a swing.

The mapping is not unique. The same net club wrench can result from different combinations of trunk motion, scapular motion, shoulder and elbow moments, wrist moments, grip pressure, and muscle

co-contraction. Surface EMG reports electrical activation timing imperfectly; it does not directly report muscle force, hand-force direction, joint work, or a unique causal pathway.

The reviewer’s minimum evidence package includes bilateral grip wrenches, full-body and scapular kinematics of adequate quality, closed-chain inverse dynamics constrained by the internal wrench, forearm and shoulder-girdle EMG, and participant-level replication. Until then, “scapular action is a candidate mechanism” is supported more strongly than “scapular retraction creates the club couple.”

### 30.2.7 “If the Wrists Resist, Are They Passive?”

No. A joint can show little visible angular motion while muscles actively co-contrast, while passive tissues bear load, or while opposing moments nearly cancel. A negative wrist moment can involve active eccentric work, isometric activation at small angular velocity, elastic tissue loading, or a combination. The word “resist” describes a net mechanical relation only after the axis, moment, motion, and interval are defined.

This distinction matters for the two allocation extremes. The proximal-drive case in a reduced model is not equivalent to “arms active, wrists off.” The wrist contribution can remain negative and actively regulated. The wrist-led preparation case is not equivalent to “arms doing nothing.” The arms must still maintain geometry and contact. Biological redundancy prevents net joint moments from revealing a unique activation pattern.

The golfer may feel firmness, pressure, or stretch. Those sensations are useful reports, but the reviewer will not equate them directly with stiffness or torque without calibration. A sensation can motivate an experimental condition; it cannot substitute for the measurement.

### 30.2.8 “Is Slack Bad?”

The reviewer first removes the word from the question. Does slack mean a literal loss of grip contact, a low-force interval, tendon slack length, low tangent stiffness, a transmission dead zone, delayed activation, or simply a relaxed sensation? These quantities behave differently.

A contact gap prevents that contact from carrying compression until it closes; the effect on task precision depends on what other contacts can transmit. Elastic compliance can store energy, whereas damping dissipates it. Their effects on peak load, tracking, and disturbance response depend on direction, frequency, geometry, and control. Neither softness nor stiffness guarantees stability. A tendon has a slack-length parameter without implying that the entire movement feels loose.

There is direct evidence that force alone need not identify stiffness: [Höppner and colleagues](#) measured grip stiffness during perturbations in ten seated participants and found that voluntary co-contraction could change stiffness at similar grip force. Their posture-maintenance experiment does not establish a dynamic golf swing strategy. It supports measuring stiffness rather than inferring it from grip force or a cue to tense the hand.

The [archived transmission study](#) uses a phenomenological dead zone in two actuator channels. Its parameter sweep favors persistent loading direction in nine cases with positive dead-zone width; the three zero-width cases have equivalent net torque-error impulses within the declared  $10^{-10}$  N m s tolerance. This comparison concerns that metric, matched commands, and initialization. It does not mean that every channel history becomes identical or that all real actuators lose their delay when a

gap is removed. The next experiment should estimate the actual gap, stiffness, or force-rise behavior instead of arguing over the word.

### 30.2.9 “Which Is Better: Arm-Led Preparation or Wrist-Led Preparation?”

The honest answer is that the current evidence does not establish a universal winner. In the matched-task model, proximal-dominant and wrist-dominant controls can produce the same club-level moment with different hand-force and generalized- torque demands. The allocation that minimizes one demand changes with geometry and need not minimize another.

Persistent proximal drive with wrist resistance may preserve a loaded transmission state through transition. Wrist-led preparation followed by role reversal may exploit compliance, reduce a proximal load, or suit another coordination objective. Intermediate allocations may dominate both extremes for a particular participant. Speed, accuracy, tissue loading, repeatability, and adaptability are not the same objective.

The reviewer asks for a preregistered tradeoff surface and held-out participants, not a group-average declaration. The golfer is invited to understand the options, not to treat the chapter as medical or coaching advice.

**Hypothesis:** Allocation strategies will differ most when the transmission has measurable state dependence and when the task includes perturbation or accuracy constraints, not merely an unperturbed peak-speed target.

### 30.2.10 “Does Preactivation Eliminate Slack?”

Preactivation can increase force readiness and effective stiffness, but it does not erase every kind of slack. It may take up a contact gap or develop series force before a rapid command. It may also increase internal load, energetic cost, and sensitivity to unwanted high-frequency disturbances. Its effect is directional and task dependent.

The reduced biological bridge carries internal activation and force states through a preparation-to-delivery change; a changed excitation command need not reset those states. Two programs with the same net target can therefore retain different internal histories. Yet a state visible inside a simulation is not necessarily identifiable from an external measurement.

The [archived biological comparison](#) also contains a crucial numerical limit. Its reversal-minus-persistent torque-error impulse decreases from approximately 0.00277 to 0.000514 N m s as the timestep is refined from 1 to 0.05 ms. The sign is unchanged over that grid, but the effect magnitude is not established as converged. Calling the effect merely “small” would hide this uncertainty. It is not a reliable physiological effect size, a performance benefit, or proof that one preparation history is superior.

The reviewer would measure pretransition activation, grip force, force-rise delay, stiffness proxies, and posttransition task error together. Activation without force and stiffness data is not enough. The golfer gets a balanced interpretation: readiness may matter, but “more activation everywhere” is not a scientifically supported prescription.

### 30.2.11 “Why Not Build One Huge Human Model and Settle It?”

A large model can represent more anatomy while introducing more parameters, uncertainty, and opportunities for compensation. If muscle paths, activation dynamics, contact properties, and participant scaling are poorly identified, the model may fit observed motion through many incompatible internal stories. Size is not the same as explanatory power.

The model ladder assigns each rung a question. A double pendulum exposes coupling and geometry. A two-hand constrained model exposes resultant and differential contact modes. A moving base tests whether prescribed support hid the mechanism. A flexible shaft carries strain state. A spatial model checks out-of-plane effects and frame conventions. A musculoskeletal model tests candidate biological allocations. Cross-engine agreement catches implementation errors but does not convert assumptions into measurements.

The reviewer prefers the smallest model capable of failing the claim being tested, followed by a declared fidelity gate. The golfer benefits because each result remains interpretable rather than disappearing inside a black box.

### 30.2.12 “Can Inverse Dynamics Tell Us Who Did the Work?”

Inverse dynamics estimates net generalized moments under a chosen model and measured kinematics. It is powerful, but a net joint moment can result from many muscle-force combinations. In a closed two-hand chain, unmeasured internal grip forces add another ambiguity. Filtering, inertial parameters, joint-center definitions, and soft-tissue artifact affect the result.

Joint power pairs a net generalized force with its power-conjugate speed. For an ideal rotational joint, use the moment dotted with the distal segment’s angular velocity relative to the proximal segment. For several coordinates, use  $\tau^T \dot{q}$ , with the appropriate speed mapping when coordinate derivatives are not angular-velocity components. This net power is not metabolic effort, individual muscle work, or energy that can be tagged and followed into one distal segment. Biarticular muscles, tendon storage, and constraints complicate that attribution.

The reviewer asks whether the internal wrench was measured, whether residuals close, how uncertainty was propagated, and whether alternative models change the conclusion. The golfer can still learn from joint-power curves, provided they are read as model-dependent net quantities rather than photographs of individual muscle action.

### 30.2.13 “What Would Actually Prove the Proposed Mechanism Wrong?”

This is the question that turns an explanation into a scientific program. First state the prediction’s scope and the size of a meaningful disagreement. An adequately measured absence of a predicted loading difference challenges the specified persistent-loading hypothesis only if the intended allocation was actually changed. A noisy or unsuccessful manipulation does not test it decisively. Failure to reject zero is not an equivalence test.

For the two-hand model, nonconverging power residuals or an unstable moment sign invalidate the reported numerical evidence until resolved. A nonzero moment from forces applied exactly at the reporting point violates the moment definition. A directly applied free moment is a different quantity. If the effect disappears when prescribed support is replaced with a responsive base, the result is

limited to the earlier support assumptions; its extension to the new model has failed, while the earlier mathematical example may remain valid.

For the scapular pathway, a precise failed prediction weakens that particular mechanism. An alternative allocation fitting the same observations equally well instead shows that those observations do not uniquely identify the mechanism. Both mechanisms may remain possible. A universal sequencing claim can be contradicted by a valid counterexample within its declared population and task. A shaft-energy prediction must survive comparison with measured phase and magnitude, including measurement and model uncertainty.

Negative results should be published with the same care as positive ones. A failed mechanism can leave useful invariants, validated measurement tools, and better-bounded alternatives. The reviewer is not asking for certainty; the reviewer is asking the authors to name the observations that would force a revision.

### **30.2.14 “Why Make the Evidence Open?”**

An open paper is more useful when its figures, data summaries, scripts, event definitions, coordinate conventions, and residual gates can be inspected. Reviewers should be able to rerun a plot, change a parameter, and see whether a claim survives. Machine-readable manifests and hashes help distinguish a specific evidence snapshot from a later artifact.

Openness does not remove every barrier. Specialized engines, proprietary measurements, participant privacy, and computational cost can limit complete reproduction. The responsible response is to document those boundaries and publish the smallest sufficient evidence rather than implying that a static figure is the whole experiment.

For the golfer, open evidence means the explanation can improve without loyalty to a product or personality. For the reviewer, it means criticism can target a declared calculation instead of guessing how a result was made.

### **30.2.15 “What Can I Safely Take From This as a Golfer?”**

You can take a way of asking questions. A swing creates a state, not merely a pose. Geometry changes what forces and moments can do. Two hands can create resultant and differential force modes; either can be zero in a particular state. Negative torque is not automatically harmful. Compliance can help or hurt depending on the task. Peak sequence, joint power, EMG, and clubhead speed answer different questions.

You should not take a universal instruction to pull with one scapula, push with one arm, hold a wrist angle, maximize tension, or deliberately apply negative torque. The models have not established the safety or superiority of those instructions for an individual. Technique changes that load wrists, elbows, shoulders, or the spine belong with qualified coaching and clinical judgment.

The deepest practical lesson is that a good strategy must be judged as a whole: speed, face and path control, strike, robustness, effort, and load. A mechanism that sounds elegant in isolation may be a poor trade for a particular person.

**Practical Interpretation:** Use the framework to make coaching claims more specific and testable. Do not turn a model variable directly into a body cue.

### 30.2.16 Reader Lab: Hold Both Voices at Once

Choose a familiar claim such as “keep the club loaded,” “fire the hips,” or “let the club release itself.” Let the curious golfer explain what the phrase feels like and what visible outcome it tries to produce. Then let the reviewer rewrite it using a state, force, moment, power, stiffness, or task variable. Name the model tier that could test it. Name one negative control. Name one human measurement. Finish with a sentence beginning, “I would change my mind if ...”

If the scientific rewrite no longer resembles the original claim, that is valuable information. The phrase may have been a useful personal cue without being a literal mechanism. Cues and mechanisms can coexist, but they should not borrow each other’s evidence.

### 30.2.17 A Full Conversation From First Impression to Testable Claim

The framework becomes easier to use when we slow down and follow one claim all the way through it. Imagine that a golfer watches a slow-motion swing and says, “The arms stopped, so their energy must have gone into the club.” That sentence feels persuasive because two visible events occur together: the hands appear to slow and the clubhead appears to speed up. It is a reasonable observation. It is not yet a complete mechanical explanation.

The curious golfer should begin by saying what was actually seen. Which point on the hands was tracked? Was “speed” a linear speed in the laboratory frame, an angular speed about the shoulder, or an angular speed relative to the club? Did the apparent slowing happen before, during, or after the clubhead speed-up? How much did the club’s direction change? A camera view can make a point look slow because it moves partly toward the camera. Even excellent motion capture needs a coordinate convention before “slowing” has one stable meaning.

Now the skeptical reviewer asks what “their energy” means. The arms have translational kinetic energy, rotational kinetic energy, gravitational potential energy, elastic energy in tissues, and interactions with the torso, hands, and club. Those accounts are not little colored parcels that can be watched crossing a joint. A segment can lose kinetic energy while muscles add positive work, because an even larger amount leaves through joint reactions. It can also gain kinetic energy while its net joint moment does negative power, because other links and constraints contribute more. The correct ledger needs the declared system boundary and every power crossing it.

For a first model, use the planar double pendulum. It is small enough that every term can be inspected. Reproduce the observed arm and club state at one instant, then calculate the full acceleration. Next set the selected arm or wrist moment to zero without changing the state and calculate again. The difference is the instantaneous acceleration influence of that intervention under the model, with its constraints and other inputs held fixed. Repeating it along the observed trajectory produces a history of same-state comparisons. Integrating an assigned power along that history is a trajectory-conditioned accounting quantity; it is not automatically the work or energy difference of a swing that actually evolved with the moment removed.

To answer the latter question, branch at a declared time, inherit the complete state, specify the remaining controls or feedback policy, and integrate both futures. Activation, contact, and elastic states must be inherited when the model has them. Each future uses its own velocities to calculate work.

A constructed one-dimensional example makes the distinction concrete. A 1 kg mass starts at rest and is pushed for 1 s by two constant 1 N forces. Its baseline velocity is  $v_b(t) = (2 \text{ m/s}^2)t$ . Each force does 1 J of work, and final kinetic energy is 2 J. Remove one force at the start: the remaining force produces  $v_c(t) = (1 \text{ m/s}^2)t$ , does 0.5 J, and leaves 0.5 J of kinetic energy. The forward energy difference is 1.5 J, whereas the removed force's work along the baseline was 1 J. The surviving force's work also changed because the trajectory changed. The same distinction matters in a coupled swing.

The reviewer now separates three checks. A change of coordinates is an **invariance check**: transform the same physical motion, force, and moment consistently and preserve the total power. Individual signed components can legitimately change sign when an axis is reversed. Refining the timestep or solver tolerance is a **numerical convergence check**. Removing gravity or reversing a physical moment arm is a **mechanical intervention**: it changes the system, so the outcome need not remain unchanged. A negative control is a deliberately chosen intervention predicted to remove the particular mechanism being tested, with other changes and limitations declared.

Even a change of reporting point requires care. For two points on the same rigid body, let  $\mathbf{r} = \mathbf{r}_B - \mathbf{r}_A$ . Then

$$\mathbf{v}_B = \mathbf{v}_A + \boldsymbol{\omega} \times \mathbf{r}, \quad \mathbf{M}_B = \mathbf{M}_A - \mathbf{r} \times \mathbf{F}. \quad (30.3)$$

Substitution in Equation 30.1 makes the two added scalar triple products cancel. Total power stays the same although the force and moment contributions change. This is a re-expression of one physical velocity field relative to the same observer. Changing to an observer moving relative to the laboratory changes measured velocities and can change power; it is not merely a change of axes. For a flexible shaft, velocities at separated points need not satisfy this rigid-body relation.

Suppose the double-pendulum result survives. The claim has earned one rung, not the entire ladder. Add the two-hand constraint model. The club now receives a resultant force, a force-generated couple from the separation of the hands, and any directly represented grip moment. Ask whether the late negative couple is still present and whether it closes the club's moment balance. Collapse the hand points together at the reporting point as a geometric control, resolving a compatible initial state for that geometry. Their force-generated moment there must be zero. For separated points at  $\pm \mathbf{d}/2$  around their midpoint, the differential moment is

$$\mathbf{M}_{\text{diff}} = \frac{1}{2} \mathbf{d} \times (\mathbf{F}_R - \mathbf{F}_L). \quad (30.4)$$

Its magnitude is bounded by  $\|\mathbf{d}\|(\|\mathbf{F}_R\| + \|\mathbf{F}_L\|)/2$ , so it tends to zero with separation for bounded forces. Direct free moments remain separate. About another point, also include the resultant's moment arm; coincident hands away from that point can still exert a moment about it.

Next replace the rigid club with a flexible shaft. Some power can now enter strain energy rather than immediate rigid-body motion. Opposing handle-moment power can coexist with increasing shaft strain energy if other boundary work or decreasing kinetic energy supplies that storage. It does not itself prove that the shaft is loading. Energy extracted through the handle and energy dissipated within the shaft are separate terms. Only a strain-energy and dissipation ledger can distinguish those stories. The casual phrase “the shaft kicked” becomes several measurable questions: when did the shaft store

energy, how much returned before impact, in which bending or torsional mode, and how sensitive was the result to stiffness and damping?

Then let the base move. Prescribing the shoulder path can hide the reaction loads and work needed to create it. A moving-base model asks whether the same distal effect survives when the torso and support are allowed to respond. Finally move into three dimensions, where a moment that looks negative about one planar axis may coexist with positive work about another axis. At every rung, the reviewer asks the same question: did the mechanism survive, change meaning, or disappear when a previously hidden degree of freedom became available?

Human evidence enters after the model has produced distinct predictions. If two candidate strategies generate nearly identical club motion but different bilateral grip couples, measure the grip wrenches. If they predict different shaft histories, measure strain. If they predict a directionally persistent arm channel and a reversing wrist channel, estimate joint moments with the internal hand wrench constrained rather than guessed. EMG can help establish activation timing, but it cannot by itself supply the missing hand force or joint moment. The measurement package should be chosen because it separates the explanations, not because the instrument happens to be available.

If those measurements and checks succeed, the original sentence can be rewritten as the following reporting template, not as a result measured in this chapter: “During this interval, the measured hand point slowed in the laboratory frame while the clubhead accelerated. In the declared linked-segment model, arm-club constraint reactions were represented by the measured net grip wrench, which produced positive power on the club, while the modeled arm system lost kinetic energy. The energy balance closed within the stated residual, and the result survived the listed timestep, coordinate, and model-fidelity checks.” That sentence is longer than the slogan, but every phrase requires a specific piece of evidence. Count that grip interaction once, whether it is represented by contact reactions or an equivalent wrench. Simultaneous arm energy loss and positive grip power do not uniquely establish how much of the club’s gain originated in the arms: the arm subsystem also exchanges work with the torso and its own actuators.

The conclusion can still fail. The measured internal wrench may contradict the inverse-dynamics allocation. Shaft strain may show that energy returned after the useful impact window. A participant may create the same outcome with a different sequence. The result may vanish when shoulder motion is no longer prescribed. Those are not embarrassments. They identify exactly where a simple story stopped working and where a better one must begin.

### **30.2.18 A Full Conversation About “Keeping the System Loaded”**

Consider a second familiar claim: “Keep tension in the system through the transition so there is no slack.” The curious golfer may be describing a very real sensation of connectedness. Perhaps grip pressure remains steady, the arms feel organized against the torso, and the club changes direction without a little uncontrolled wobble. That sensation is valuable as a report. It still contains several possible mechanisms.

The reviewer replaces “tension” with candidate measurements. Is there a nonzero resultant grip force, a nonzero differential hand couple, forearm muscle activation, scapular-muscle activation, shaft strain, tendon force, or high tangent stiffness about a joint? These variables can move independently. Two opposing muscle groups can raise stiffness while their net joint moment stays near zero. Two hand

forces can create a large couple with a small resultant force. A shaft can remain strained while the hands briefly reduce their active moment. “Loaded” therefore needs both a location and a quantity.

Now compare the two preparation extremes developed earlier. In the first, arm-and-shoulder channels create more of the backswing push–pull tendency while the wrists resist and regulate the club. The broad channel directions persist into early downswing. In the second, wrist moments create more of the backswing turning tendency while the arms resist; around transition the arms take over as the positive driver and the wrists oppose release. These are proposed channel allocations, not uniquely identified human muscle programs. The archived allocation sweep matches an instantaneous club control-moment target at a declared state. Matching an entire club path under evolving activation and contact constraints is a stronger requirement and needs a forward test.

In the declared dead-zone model, persistent direction has an advantage because it does not cross the modeled unloaded zone. Complete role reversal must travel through that zone before torque rebuilds. This is a useful conditional result, not proof that a human wrist contains backlash. Set the dead zone to zero in the archived sensitivity sweep and the two programs’ net torque-error impulses become equivalent within the declared tolerance. That control locates the difference within this transmission family; it does not establish a human wrist dead zone or a global optimum.

A biological system can preserve readiness during a net reversal without literal continuous torque. Antagonistic preactivation may keep short-range stiffness high. Tendon or shaft strain may preserve a loaded elastic state. Contact forces may remain nonzero while their moment about a selected axis changes sign. Conversely, a person may report continuous effort while the task-relevant force mode briefly approaches zero. Feel and mechanics may be related, but their mapping must be calibrated rather than assumed.

The two strategies also have costs that a peak-speed-only model will miss. Persistent co-contraction can raise joint compression, metabolic cost, fatigue, and sensitivity to unwanted high-frequency loading. A role reversal can use different muscle groups, exploit changing leverage, and permit useful reorientation. Some compliance can store energy, absorb disturbances, and protect tissue. Too much compliance or a true contact gap can delay force rise. There is no reason to expect one point on this tradeoff surface to be best for every participant and every objective.

A decisive experiment therefore matches the visible task while manipulating allocation. Measure bilateral grip wrenches, grip pressure, wrist and shaft kinematics, shaft strain, transition-aligned force rise, and suitable forearm and shoulder-girdle EMG. Estimate tangent stiffness with an ethical bounded perturbation or system-identification method. Compare not only clubhead speed but face and path variability, peak joint demand, repeatability, and response to a small disturbance. Hold participants out when selecting a strategy so the test does not turn a group average into a universal prescription.

Interpret the outcomes in stages. If the instructions do not change measured allocation, the manipulation failed. If allocation changes but the predicted gap or stiffness difference does not, the dead-zone explanation weakens. If the mediator changes but performance does not, the mechanism may be real yet unimportant for the task. If performance changes without the predicted mediator, another pathway deserves attention. If participants split into repeatable subgroups, the scientifically useful result may be a conditional map rather than a winner.

The careful practical translation is modest: a golfer and coach can distinguish continuous engagement from maximum effort and can ask which measurable mode a cue is intended to change. They should

not infer that more tension is always better, that wrists should be passive, or that one scapular action uniquely creates the club couple. The framework earns better questions and experiments; it does not yet earn a universal body instruction.

### 30.2.19 How to Carry the Method Into a New Question

When a new swing explanation appears, first write the observation without a causal verb. Next define the system boundary, coordinate frame, interval, and quantity. Draw the smallest model that contains the proposed mechanism and a negative control capable of removing it. Record the state before applying the counterfactual so that history is not silently reset. Close force, moment, power, and energy balances at the level required by the claim.

Then name at least one serious alternative. Increase model fidelity only where that alternative exploits something the smaller model omitted. Derive predictions that differ in measurable variables, including an internal variable when external motion is not identifying. Specify the residual, sensitivity, and participant-held-out criteria before looking for a preferred result.

Finally, translate the result twice. The scientific translation states the model, intervention, observable, uncertainty, and falsifier. The ordinary translation uses a concrete picture and says where that picture breaks. If the two translations disagree, revise them until neither borrows certainty from the other. That repeated movement between intuition and audit is how an accessible explanation can remain genuinely scientific rather than merely simple.

### 30.2.20 Chapter Checkpoint

The curious golfer asks questions that matter. The skeptical reviewer makes their meanings stable enough to test. Together they replace simple slogans with a linked chain: define the system and quantity, choose the smallest adequate model, preserve state history, measure the predicted observables, run negative controls, compare alternatives, and declare what would refute the claim. That chain is the real framework of the project.

## 30.3 Where the Picture Breaks

A dialogue can make disagreements look cleaner than they are. Real experiments face noisy measurements, participant variation, coupled objectives, missing data, and ethical limits. Some questions cannot be settled by one study, and a null result can reflect weak manipulation or inadequate measurement rather than absence of a mechanism. The two voices also do not replace domain experts in biomechanics, motor control, instrumentation, statistics, coaching, or clinical care. They provide a disciplined entry point for those collaborations.

**Go Deeper:** Audit the claims against the [complete scientific monograph](#), use the open [reviewer workbench](#), and inspect the project-level [falsification matrix](#).

# Glossary

**Acceleration:** The rate at which velocity changes. It can change speed, direction, or both.

**Ablation:** A controlled model experiment in which one mechanism is removed or neutralized while the remaining conditions are held as comparable as possible.

**Constraint:** A rule limiting allowable motion, such as two linked segments sharing a joint or two hands remaining attached to one grip.

**Coriolis Term:** A cross-speed part of a motion equation under a declared coordinate convention. In the two-link example it requires simultaneous arm rotation and relative wrist motion; it is not a new external or muscle force.

**Centripetal and Centrifugal:** In an inertial description, centripetal names the inward requirement for curved motion. Centrifugal is the outward inertial interpretation in a rotating description. They are not two forces to add.

**Counterfactual:** A carefully stated “what if” comparison used to attribute a quantity or test whether a proposed mechanism persists.

**Drift:** The modeled zero-input part of the state derivative, or its acceleration component when that is the stated convention. Retained gravity, constraints, internal states, unselected inputs and prescribed boundary motion must be named. Zero declared input does not mean zero force or biological rest.

**Energy:** A scalar accounting quantity with specified forms, such as kinetic, gravitational, elastic or chemical energy. Values depend on the system and observer, and potential-energy zeros are conventional. Conservation of total energy does not require conservation of mechanical energy alone.

**Falsifier:** An observation or test result that would count against a claim.

**Identifiability:** Whether available observations can uniquely or usefully determine a model parameter, internal force, or mechanism.

**Impulse:** Force integrated over time. Angular impulse is moment integrated over time.

**Interaction Force:** A force transmitted at a connection between parts of a linked system.

**Moment or Torque:** The turning effect of a force or a direct rotational couple about a declared point and axis.

**Power:** The instantaneous rate of energy transfer or work, measured in watts.

**Preload:** A nonzero internal load already present before a command or event changes.

**Robustness:** Performance retained across a declared set of disturbances, uncertainties, participants, or model parameters.

**State:** The collection of variables needed by a model to predict its future with specified future inputs or input laws, commonly configuration, velocity, and internal states such as activation or deformation.

**Squared-Speed Term:** A velocity-dependent equation term containing one generalized speed squared. Its detailed allocation is coordinate-dependent.

**Wrench:** A combined force and moment expressed about a declared reference point and in a declared coordinate frame.

**Work:** The time integral of a declared mechanical power. Force power uses the material-point velocity; moment power uses the conjugate angular velocity. Joint power and external power on one body can therefore be different ledgers.

**Pointwise Zero-Torque Counterfactual (ZTCF) Force:** The generalized-force representation associated with the acceleration recomputed at a declared state after selected control removal. Its construction must name coordinates, mass mapping, retained loads and constraints. It is not an independently measured force, nor a universal tissue-force decomposition.

For the evidence-led technical treatment behind this tour, see the summary article [Proximal-to-Distal Energy Transfer in the Golf Swing](#), which connects the companion narrative to the monograph's evidence ledger.

# References

- Ball, Kevin A., and Russell J. Best. 2007. “Different Centre of Pressure Patterns Within the Golf Stroke i: Cluster Analysis.” *Journal of Sports Sciences* 25 (7): 757–70. <https://doi.org/10.1080/02640410600874971>.
- Balzerson, Daniel, Joydeep Banerjee, and John McPhee. 2016. “A Three-Dimensional Forward Dynamic Model of the Golf Swing Optimized for Ball Carry Distance.” *Sports Engineering* 19: 237–50. <https://doi.org/10.1007/s12283-016-0197-7>.
- Betzler, Nils F., Stuart A. Monk, Eric S. Wallace, and Steve R. Otto. 2012. “Effects of Golf Shaft Stiffness on Strain, Clubhead Presentation and Wrist Kinematics.” *Sports Biomechanics* 11 (2): 223–38. <https://doi.org/10.1080/14763141.2012.681796>.
- Bourgain, Maxime, Philippe Rouch, Olivier Rouillon, Patricia Thoreux, and Christophe Sauret. 2022. “Golf Swing Biomechanics: A Systematic Review and Methodological Recommendations for Kinematics.” *Sports* 10 (6): 91. <https://doi.org/10.3390/sports10060091>.
- Choi, Hyeob, and Sukyung Park. 2020. “Three Dimensional Upper Limb Joint Kinetics of a Golf Swing with Measured Internal Grip Force.” *Sensors* 20 (13): 3672. <https://doi.org/10.3390/s20133672>.
- Diaz, Rodolfo A., William J. Herrera, and Diego A. Manjarrés. 2009. “Work and Energy in Inertial and Noninertial Reference Frames.” *American Journal of Physics* 77 (3): 270–73. <https://doi.org/10.1119/1.3036418>.
- Featherstone, Roy. 2008. *Rigid Body Dynamics Algorithms*. Springer. <https://doi.org/10.1007/978-1-4899-7560-7>.
- Glazier, Paul S. 2011. “Movement Variability in the Golf Swing: Theoretical, Methodological, and Practical Issues.” *Research Quarterly for Exercise and Sport* 82 (2): 157–61. <https://doi.org/10.1080/02701367.2011.10599742>.
- Han, Ki Hoon, Christopher Como, Jemin Kim, et al. 2019. “Effects of the Golfer–Ground Interaction on Clubhead Speed in Skilled Male Golfers.” *Sports Biomechanics* 18 (2): 115–34. <https://doi.org/10.1080/14763141.2019.1586983>.
- Hu, Xiao, Wendy M. Murray, and Eric J. Perreault. 2011. “Muscle Short-Range Stiffness Can Be

- Used to Estimate the Endpoint Stiffness of the Human Arm.” *Journal of Neurophysiology* 105 (4): 1633–41. <https://doi.org/10.1152/jn.00537.2010>.
- Hume, Patria A., Justin Keogh, and Duncan Reid. 2005. “The Role of Biomechanics in Maximising Distance and Accuracy of Golf Shots.” *Sports Medicine* 35 (5): 429–49. <https://doi.org/10.2165/0007256-200535050-00005>.
- Joyce, Christopher, Angus Burnett, Jodie Cochrane, and Kevin Ball. 2013. “Three-Dimensional Trunk Kinematics in Golf: Between-Club Differences and Relationships to Clubhead Speed.” *Sports Biomechanics* 12 (2): 108–20. <https://doi.org/10.1080/14763141.2012.728244>.
- Kao, John T., Marilyn Pink, Frank W. Jobe, and Jacquelin Perry. 1995. “Electromyographic Analysis of the Scapular Muscles During a Golf Swing.” *The American Journal of Sports Medicine* 23 (1): 19–23. <https://doi.org/10.1177/036354659502300104>.
- Koike, Sekiya. 2016. “Force and Moment Exerted by Each Hand on an Instrumented Golf Club.” *34th International Conference on Biomechanics in Sports*. <https://ojs.ub.uni-konstanz.de/cpa/article/view/6828>.
- Lazic, Stanley E. 2010. “The Problem of Pseudoreplication in Neuroscientific Studies: Is It Affecting Your Analysis?” *BMC Neuroscience* 11: 5. <https://doi.org/10.1186/1471-2202-11-5>.
- Lynch, Kevin M., and Frank C. Park. n.d. “3.4. Wrenches.” Accessed October 2, 2026. <https://modernrobotics.northwestern.edu/nu-gm-book-resource/3-4-wrenches/>.
- MacKenzie, Sasho J., Matthew McCourt, and Luc Champoux. 2020. “How Amateur Golfers Deliver Energy to the Driver.” *International Journal of Golf Science* 8 (1). <https://www.golfsciencejournal.org/article/12640-how-amateur-golfers-deliver-energy-to-the-driver>.
- MacKenzie, Sasho J., and Eric J. Springs. 2009. “A Three-Dimensional Forward Dynamics Model of the Golf Swing.” *Sports Engineering* 11 (4): 165–75. <https://doi.org/10.1007/s12283-009-0020-9>.
- Marino, S., I. B. Hogue, C. J. Ray, and D. E. Kirschner. 2008. “A Methodology for Performing Global Uncertainty and Sensitivity Analysis in Systems Biology.” *Journal of Theoretical Biology* 254 (1): 178–96. <https://doi.org/10.1016/j.jtbi.2008.04.011>.
- Marsan, Thibault, Patricia Thoreux, Maxime Bourgain, Olivier Rouillon, Philippe Rouch, and Christophe Sauret. 2019. “Biomechanical Analysis of the Golf Swing: Methodological Effect of Angular Velocity Component on the Identification of the Kinematic Sequence.” *Acta of Bioengineering and Biomechanics* 21 (2): 115–20. <https://doi.org/10.5277/ABB-01318-2019-02>.
- McNally, William, and John McPhee. 2018. “Dynamic Optimization of the Golf Swing Using a Six Degree-of-Freedom Biomechanical Model.” *Proceedings* 2: 243. <https://doi.org/10.3390/proceeding2060243>.

- McPhee, John. 2022. “A Review of Dynamic Models and Measurements in Golf.” *Sports Engineering* 25. <https://doi.org/10.1007/s12283-022-00387-0>.
- Meister, David W., Amy L. Ladd, Erin E. Butler, et al. 2011. “Rotational Biomechanics of the Elite Golf Swing: Benchmarks for Amateurs.” *Journal of Applied Biomechanics* 27 (3): 242–51. <https://doi.org/10.1123/jab.27.3.242>.
- Milburn, Peter D. 1982. “Summation of Segmental Velocities in the Golf Swing.” *Medicine and Science in Sports and Exercise* 14 (1): 60–64. <https://pubmed.ncbi.nlm.nih.gov/7070260/>.
- Miura, Koryo. 2001. “Parametric Acceleration — the Effect of Inward Pull of the Golf Club at Impact Stage.” *Sports Engineering* 4 (2): 75–86. <https://doi.org/10.1046/j.1460-2687.2001.00071.x>.
- Morrison, Andrew, Denise McGrath, and Eric S. Wallace. 2016. “Motor Abundance and Control Structure in the Golf Swing.” *Human Movement Science* 46: 129–47. <https://doi.org/10.1016/j.humov.2016.01.009>.
- National Academies of Sciences, Engineering, and Medicine. 2019. *Reproducibility and Replicability in Science*. The National Academies Press. <https://doi.org/10.17226/25303>.
- National Institute of Standards and Technology. n.d. “Digital Signature.” Accessed October 2, 2026. [https://csrc.nist.gov/glossary/term/digital\\_signature](https://csrc.nist.gov/glossary/term/digital_signature).
- Nesbit, Steven M. 2005. “A Three Dimensional Kinematic and Kinetic Study of the Golf Swing.” *Journal of Sports Science and Medicine* 4 (4): 499–519. <https://pmc.ncbi.nlm.nih.gov/articles/PMC3899667/>.
- Nesbit, Steven M., and Ryan McGinnis. 2009. “Kinematic Analyses of the Golf Swing Hub Path and Its Role in Golfer/Club Kinetic Transfers.” *Journal of Sports Science and Medicine* 8 (2): 235–46. <https://www.jssm.org/volume08/iss2/cap/jssm-08-235.pdf>.
- Nesbit, Steven M., and Monika Serrano. 2005. “Work and Power Analysis of the Golf Swing.” *Journal of Sports Science and Medicine* 4 (4): 520–33. <https://www.jssm.org/volume04/iss4/cap/jssm-04-520.pdf>.
- Putnam, Carol A. 1991. “A Segment Interaction Analysis of Proximal-to-Distal Sequential Segment Motion Patterns.” *Medicine and Science in Sports and Exercise* 23 (1): 130–44. <https://pubmed.ncbi.nlm.nih.gov/1997807/>.
- Putnam, Carol A. 1993. “Sequential Motions of Body Segments in Striking and Throwing Skills: Descriptions and Explanations.” *Journal of Biomechanics* 26 (Suppl. 1): 125–35. [https://doi.org/10.1016/0021-9290\(93\)90084-R](https://doi.org/10.1016/0021-9290(93)90084-R).
- Raue, A., C. Kreutz, T. Maiwald, et al. 2009. “Structural and Practical Identifiability Analysis of Partially Observed Dynamical Models by Exploiting the Profile Likelihood.” *Bioinformatics* 25

- (15): 1923–29. <https://doi.org/10.1093/bioinformatics/btp358>.
- Robertson, D. Gordon E., and David A. Winter. 1980. “Mechanical Energy Generation, Absorption and Transfer Amongst Segments During Walking.” *Journal of Biomechanics* 13 (10): 845–54. [https://doi.org/10.1016/0021-9290\(80\)90172-4](https://doi.org/10.1016/0021-9290(80)90172-4).
- Robinson, Patrick G., Howie J. Carson, Jim Richards, Andrew Murray, Andrew D. Duckworth, and Doug Campbell. 2023. “What Differences Exist Between the Lead and Trail Wrist in Extensor Carpi Ulnaris Activity and Golf Swing Joint Kinematics in Sub-Elite Golfers?” *Journal of Sports Sciences* 41 (17): 1596–604. <https://doi.org/10.1080/02640414.2023.2285121>.
- Rockafellar, R. Tyrrell, and Stanislav Uryasev. 2002. “Conditional Value-at-Risk for General Loss Distributions.” *Journal of Banking & Finance* 26 (7): 1443–71. [https://doi.org/10.1016/S0378-4266\(02\)00271-6](https://doi.org/10.1016/S0378-4266(02)00271-6).
- Sharp, Robin S. 2009. “On the Mechanics of the Golf Swing.” *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences* 465 (2102): 551–70. <https://doi.org/10.1098/rspa.2008.0304>.
- Sprigings, Eric J., and Sasho J. MacKenzie. 2002. “Examining the Delayed Release in the Golf Swing Using Computer Simulation.” *Sports Engineering* 5: 23–32. <https://doi.org/10.1046/j.1460-2687.2002.00094.x>.
- Tinmark, Fredrik, John Hellström, Kjartan Halvorsen, and Alf Thorstensson. 2010. “Elite Golfers’ Kinematic Sequence in Full-Swing and Partial-Swing Shots.” *Sports Biomechanics* 9 (4): 236–44. <https://doi.org/10.1080/14763141.2010.535842>.
- Vigotsky, Andrew D., Israel Halperin, Gregory J. Lehman, Gabriel S. Trajano, and Taian M. Vieira. 2018. “Interpreting Signal Amplitudes in Surface Electromyography Studies in Sport and Rehabilitation Sciences.” *Frontiers in Physiology* 8: 985. <https://doi.org/10.3389/fphys.2017.00985>.
- Watson, Andrew, Andrew Murray, Alex Ehlert, et al. 2026. “Ground Reaction Force and Centre of Pressure During the Golf Swing and Associations with Clubhead Speed and Skill Level: A Systematic Review.” *Sports Medicine*, ahead of print. <https://doi.org/10.1007/s40279-025-02391-3>.
- Widnall, Sheila, and Jaime Peraire. 2008. *Lecture L12: Work and Energy*. MIT 16.07 Dynamics lecture notes, version 2.0. [https://ocw.mit.edu/courses/16-07-dynamics-fall-2009/6c3adbf48feaf3580731511ee9cb776\\_MIT16\\_07F09\\_Lec12.pdf](https://ocw.mit.edu/courses/16-07-dynamics-fall-2009/6c3adbf48feaf3580731511ee9cb776_MIT16_07F09_Lec12.pdf).
- Winter, David A. 2009. *Biomechanics and Motor Control of Human Movement*. 4th ed. John Wiley & Sons.
- Zajac, Felix E., and Michael E. Gordon. 1989. “Determining Muscle’s Force and Action in Multi-Articular Movement.” *Exercise and Sport Sciences Reviews* 17: 187–230. <https://pubmed.ncbi.nlm.nih.gov/2676547/>.

Zheng, Naiquan, Steve W. Barrentine, Glenn S. Fleisig, and James R. Andrews. 2008. "Kinematic Analysis of Swing in Pro and Amateur Golfers." *International Journal of Sports Medicine* 29 (6): 487–93. <https://doi.org/10.1055/s-2008-1038732>.